Tools to Design a Biomimetic Humanoid Robot Actuated by Pneumatic Artificial Muscles and a Synthetic Neural Network

by

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Abstract

Musculoskeletal humanoid robots enable controlled experiments in biological movement and motivate compliant locomotion in unstructured environments. Their development requires reliable actuator and joint models, physiological evidence, and tools for testing neural controllers with different mechanics. I address these needs through four connected contributions. First, characterization of Festo braided pneumatic actuator (BPA) across resting lengths, contractions, and pressures produced a resting-length-dependent maximum-force model and a nondimensional three-coefficient force surface. Joint-torque design factors then account for length offsets, effective series compliance, and wrapping-induced loss of usable actuator length. Incorporating these factors into BPA routing redesign enabled paired knee actuators to exceed selected human-muscle reference torques at measured configurations. I created the Sensory Afferent Database to connect experimental findings and model assumptions through searchable notes and citation navigation. Graphical neural blocks and simulation interfaces support controller transfer and testing on highly actuated musculoskeletal models, including Gait2392, in MuJoCo and Simulink/Simscape. Representative evaluations establish circuit responses, implementation agreement, and virtual-harness-assisted balance and walking behaviors. These contributions connect experimentally informed robot design with neuromechanical experiments and potential prosthetic and orthotic applications.

Dedication

*

Darian

azīzam, jānam,

nūr-e cheshmam — the light of my eyes

qorbānat beram

Bābā kheyli dūset dāre

↘ ↘ ↘

"Los corazones obreros que laten

por nuestra causa, felices serán.

Si entusiasmados y unidos combaten,

de la victoria, la palma obtendrán."

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Finally, I thank my family for their support throughout this long process.

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1 Introduction

1.1 Motivation

This research pursues two reciprocal goals. The first is to use principles from biomechanics and neuroscience to improve robot autonomy, balance, and adaptability in unstructured terrain [Frey et al., 2026, Ha et al., 2025, Ijspeert, 2020]. The potential for compliant actuation and neural control to improve locomotor energy economy is a further motivation for this approach.

The second goal is to use robots as controllable physical models of biological movement. Musculoskeletal humanoids and limb systems provide ways to investigate motion generation, internal forces, and neural control hypotheses [Asano et al., 2017, Hitzmann et al., 2018]. Robotic models permit repeatable changes to individual actuators or sensory pathways, including interventions that would be impractical or unethical in human participants. They complement animal studies and numerical models by allowing the mechanical and neural contributions to be manipulated together [Shin et al., 2018, Goldsmith et al., 2021]. Work with a salamander robot demonstrates this exchange: biological evidence informed a spinal-cord model, and robotic tests generated predictions that were examined in biological preparations [Ijspeert et al., 2007]. These goals form an iterative cycle: biological knowledge guides more biomimetic robots, robotic experiments generate biological questions, and the resulting understanding can guide further improvements in biomimicry. Prosthetic and orthotic devices are potential applications of this synthesis.

Bioinspired robotics draws broadly on biological principles; biomimetic robots reproduce selected biological structures or mechanisms. Musculoskeletal humanoid robots are a subset of biomimetic robots, using human-like skeletal arrangements and muscle-like actuation. Artificial muscles are a key enabling technology for this project. Among the available actuation technologies, McKibben-style braided pneumatic actuators (BPAs) combine low mass, high force-to-weight ratio, and intrinsic compliance with a force–length curve that qualitatively resembles that of skeletal muscle [Liang et al., 2020, Hunt et al., 2017a]. BPAs have been used in quadrupedal robots [Aschenbeck et al., 2006, Hunt et al., 2017a, Scharzenberger, 2019], musculoskeletal arms [Hitzmann et al., 2018, Kurumaya et al., 2016], and dexterous manipulation systems [Chen et al., 2019]. However, BPAs also differ from biological muscle in important ways: their maximum tensile force occurs at resting length rather than at an optimal fiber length, their maximum contraction ratio is substantially lower than that of human muscle, and their force output is a highly nonlinear function of internal pressure, contraction, and loading state [Hunt et al., 2017a, Sárosi et al., 2017, Liang et al., 2020]. My experience with 10 mm actuators made the design gap concrete: at the short resting lengths used in my leg designs, they delivered less force and contraction than I had anticipated from manufacturer specifications. Existing empirical models describe useful pressure–contraction relationships [Sárosi et al., 2017, Martens and Boblan, 2017, Hunt et al., 2017a], but selecting actuators for this robot required predictions across the resting lengths, pressures, and contractions of the tested Festo BPAs. Reliable force prediction is therefore the first requirement for designing the humanoid's actuated joints.

Actuator force alone, however, does not determine the torque a joint produces. A BPA-

actuated robot knee transmits actuator force to the skeleton through mounting brackets, artificial tendons, and paths that wrap over the joint, and each of these elements departs from the rigid-body idealization: brackets and fixtures deflect under load, tendons stretch, and a wrapping actuator loses usable length as the knee flexes. For this robot, the effects of compliance and bending and the usefulness of a common correction across knee configurations had not been established experimentally. Here, musculotendon routes are the combined BPA and artificial-tendon paths through attachment and wrapping points. The earlier robot actuator routes were selected from predicted muscle lengths, moment arms, and torque profiles [Bolen and Hunt, 2019]. The placement searches that I developed with Morrow and Hunt early in my doctoral research extended that design approach [Morrow et al., 2020], but their rigid-body predictions still required experimental validation at the knee. The later work in Chapter 4 uses those experiments to improve the prediction model and redesign the musculotendon routes.

Researchers implementing neural control models face two related problems. The first is access to physiological evidence. Sensory-afferent findings are distributed across species, preparations, and experimental protocols, so selecting a feedback pathway requires locating its measured properties and the conditions under which they were observed [Van de Crommert et al., 1998, Windhorst, 2007, 2022]. A second problem arises when a published neural circuit is coupled to a different mechanical model: its behavior depends on the actuator mechanics, sensory signals, support conditions, and numerical implementation as well as on the circuit itself. Published architectures and software supply essential components [Rybak et al., 2006, McCrea and Rybak, 2008, Markin et al., 2010, Nourse et al., 2023], but the behavior of their particular combination must be established.

Transferring a controller to different mechanics can require extensive hand tuning. Stepping in suspension alone does not establish that the coupled model can support locomotion on the ground. Efficient development therefore requires both graphical editing of neural circuits and automated evaluation of their interaction with human biomechanical models and BPA-actuated robot models.

I address these gaps through four connected contributions: BPA force characterization with resting length as a design variable, isometric knee-torque prediction and route design, the Sensory Afferent Database, and neuromechanical modeling tools. Together, they provide a basis for selecting actuators, predicting joint output, selecting sensory pathways, and testing neural controllers in coupled models, including virtual-harness-assisted locomotor evaluations.

1.2 The Biomimetic Humanoid Robot Project

This dissertation is part of a long-running effort at Portland State University to build a biomimetic humanoid robot representing the body from the lumbar spine down, actuated by pneumatic artificial muscles and controlled by a synthetic nervous system (SNS). The project connects mechanical design and neural-control development through successive physical prototypes and computational models, enabling controlled tests of how actuator mechanics and sensory feedback contribute to movement.

The present work builds directly on a series of studies by me and my collaborators. In my master's project, the kinematics and dynamics of a biomimetic robot leg were matched against the Gait2392 OpenSim model, musculotendon routes, including attachment and

wrapping locations, were determined for the robot legs, and MATLAB software was developed for evaluating muscle lengths, moment arms, and isometric joint torques over the full range of motion [Bolen, 2019, Bolen and Hunt, 2019]. Hand tuning the placement of artificial muscles was time consuming. The results of that work showed that simply transplanting human muscle attachment locations onto a BPA-actuated robot does not reproduce human torque profiles. The BPA routing optimization developed during this dissertation addresses this limitation and is introduced in Section 4.1.5. Steele designed and validated the biomimetic four-bar knee linkage with a migrating instantaneous center of rotation used in this robot leg [Steele et al., 2017, Steele, 2018, Steele et al., 2018]. Chapter 4 measures the isometric torque produced by BPAs acting on that knee.

1.3 Research Objectives

This dissertation investigates actuator force and knee torque experimentally, consolidates sensory-afferent evidence in the Sensory Afferent Database [Bolen et al., 2023, Bolen and Hunt, 2024], and develops neural-model and simulation interfaces with representative verification checks. The objectives are:

1. **Characterize the isometric force of Festo BPAs** with uninflated diameters of 10 mm and 20 mm over a wide range of resting lengths, contractions, and pressures; derive improved maximum-force and normalized force–strain–pressure models; and compare them against existing models (Chapter 3).

2. **Validate and improve the prediction of isometric knee torque** produced by BPAs on pinned-joint and biomimetic four-bar knees; identify and model losses caused by

constant length offsets, series compliance, and actuator bending; and compare the resulting torque envelopes with human capabilities (Chapter 4).

3. **Consolidate the sensory afferent literature into a searchable resource**, the Sensory Afferent Database, by auditing and curating the published record of afferent properties across animal species and their role in motor control, and by evaluating AI-assisted screening of candidate papers for expanding the collection (Chapter 5).

4. **Develop and demonstrate neuromechanical modeling tools** by implementing the Li–Deng walker, Gait2392-based models with alternative neural controllers and artificial-muscle actuation, SNS-Simscape blocks for neural pathways coupled with mechanical modeling, and staged tuning and experiment procedures; verify representative circuit responses, implementation agreement, and neural–mechanical signal exchange (Chapters 6 and 7).

1.4 Dissertation Organization

Chapter 2 introduces the robot, BPA actuation, musculoskeletal design, locomotor control, and the scientific rationale for robotic models. Chapter 3 presents the isometric force characterization and actuator models, adapted from my publication [Bolen et al., 2026]. Chapter 4 tests the joint-torque predictions, identifies compliance and bending-related corrections, and uses them to redesign actuator routes.

Chapter 5 presents the Sensory Afferent Database, its curation, and its use limits. Chapter 6 begins with attempts to transplant published controllers into a different mechanical model, identifies the development requirements exposed by those attempts, and presents

the tools and representative checks built to address them. Chapter 7 discusses the contributions together and develops the future work; Chapter 8 states the conclusions. Figure 1.1 summarizes these connections.

Appendices A–C provide the muscle calculations, neural equations, model geometry, and supporting mechanical results. Appendix D collects the larger simulation evaluations, and Appendix E provides the database coverage and pathway tables.

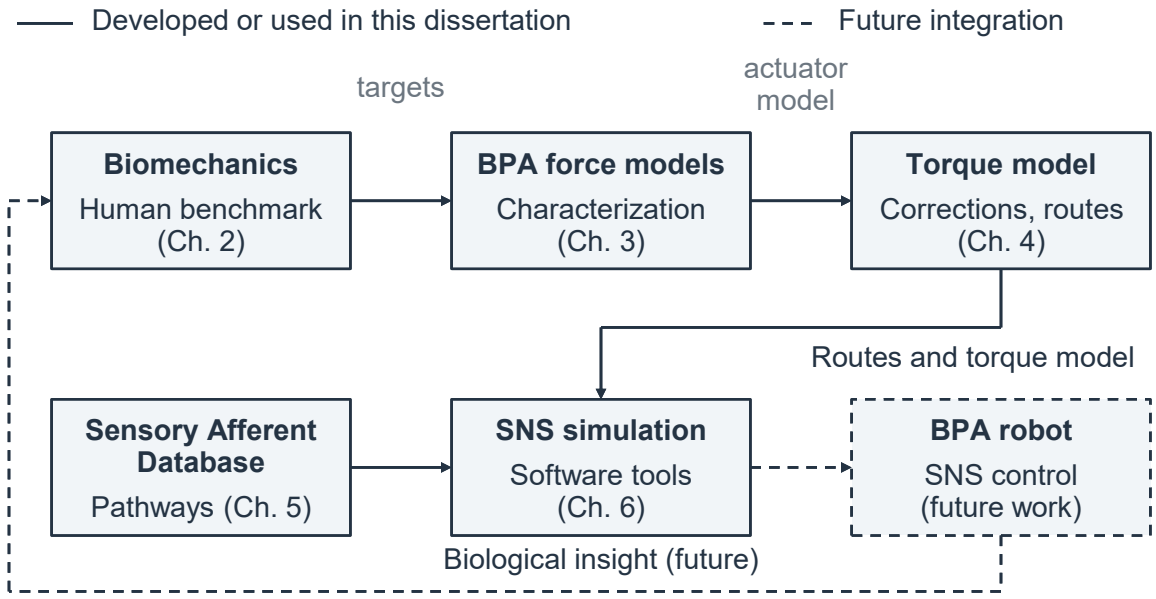


Figure 1.1: Dissertation roadmap. Biomechanical targets guide actuator characterization and route design; the sensory-afferent resource informs feedback selection; and simulation tools support controller evaluation. Dashed connections indicate future integration and the intended return of biological insight.

2 Background

I review the robotics, actuation, biomechanics, and neural-control background needed for this dissertation, followed by simulation tools and the ethical rationale for robotic experiments. The Sensory Afferent Database, my earlier AnimatLab walker [Bolen and Morrow, 2020], and the simulation interfaces I developed are described in Chapters 5 and 6.

2.1 Biomimetic and Musculoskeletal Robots

Biomimetic robots offer many potential applications, ranging from assisting in factories and environmental exploration to serving as physical models of human movement. By developing high-fidelity biomimetic robots, we can perform tests that help improve our understanding of both biomechanics and the underlying neuromechanical systems that control movement [Shin et al., 2018, Asano et al., 2019, Ijspeert, 2020, Goldsmith et al., 2021, Hunt et al., 2017b, Schilling et al., 2022]. A few such biomimetic robots exist, such as Kengoro and Kenshiro robots of Asano et al. [2017] and Asano et al. [2019], which have proportionally human bodies, anatomically correct musculotendon arrangements, rolling-sliding joints, and joint ranges and output power comparable with human-measured values; their artificial muscles are implemented with brushless DC motors, wires as tendons, gears, and integrated tension measurement. Kengoro, the more recent of the two, stands 1670 mm tall, weighs 56.5 kg, and carries 174 degrees of freedom when the end

effectors are included [Asano et al., 2017]. A dog-like robot developed by Aschenbeck et al. [2006] and actuated with Festo BPAs was designed from a planar kinematic model scaled to an adult greyhound, with three sagittal degrees of freedom per leg driven by antagonistic BPA pairs; its front-leg pair alone lifted a 13.5 kg payload, almost double the complete robot's predicted mass. That platform was used to show how central pattern generators provide robust locomotion patterns that are adaptive to perturbations in ground speed [Hunt et al., 2017b].

Robots built with artificial muscles thus provide opportunities to investigate how muscles are controlled to produce desired motions; however, artificial muscles do not exactly replicate muscle properties, and existing robots do not capture all of the degrees of freedom and joint torques of the animal limbs they mimic. The biomimetic leg developed and tested in this dissertation builds on this approach: a human-proportioned skeleton, muscle-like actuators, and the goal of human-matched joint torques.

Figure 2.1 illustrates the human-scale skeletal structure and tendon routing of this approach.

2.2 Robot Actuation Technologies

2.2.1 Overview

A wide range of actuation technologies have been explored in robotics, including electric motors, hydraulic systems, smart-material actuators, and artificial muscles [He and Gao, 2020, Liang et al., 2020]. Electric motors (Figure 2.2**(A)**) are widely used due to their controllability and maturity, but they are limited by thermal constraints, gearing requirements,

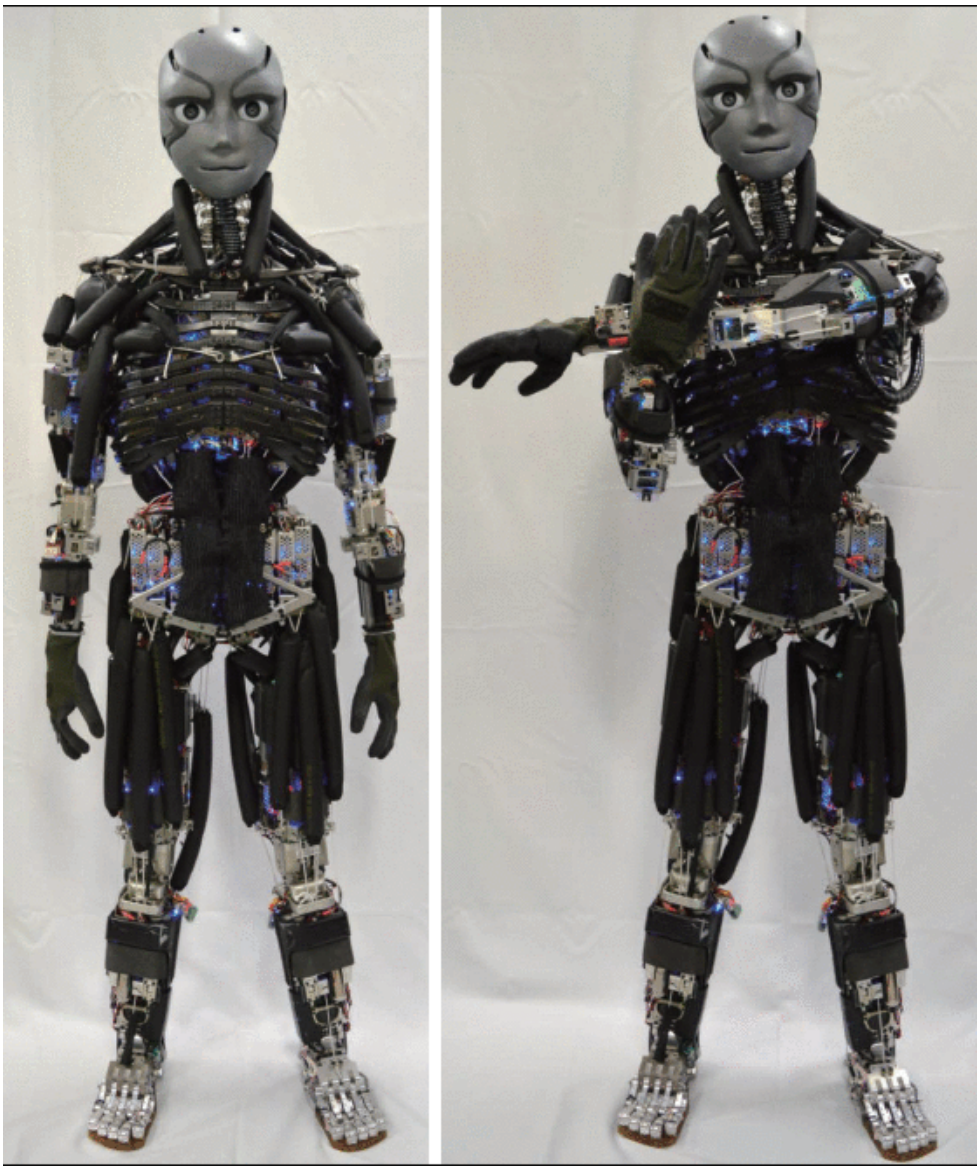


Figure 2.1: Kengoro human-mimetic musculoskeletal humanoid in neutral and upper-limb poses, illustrating its human-scale skeletal structure and densely routed tendon-driven musculature [Asano et al., 2017].

and reduced compliance compared to muscle-like actuators [Veale and Xie, 2016, Suzumori and Faudzi, 2018]. Hydraulic actuators (Figure 2.2**(C)**) offer exceptionally high power density and have enabled highly dynamic legged systems, but they remain relatively heavy, less efficient than biological muscle, and require a dedicated fluid infrastructure [Suzumori and Faudzi, 2018].

Among smart-material actuators, dielectric elastomer actuators (DEAs), a class of electronic electroactive polymers, contract by a different mechanism: an applied voltage squeezes a soft polymer film between compliant electrodes through electrostatic stress, and the thinning and radial spreading of the film produces strains that can exceed biological muscle (Figure 2.2**(D)**) [Madden et al., 2004]. Comparative studies report high specific work and good bandwidth for DEAs [Liang et al., 2020], but their typically kilovolt-range driving voltages, electromechanical instability, and compliant-electrode requirements complicate integration into a human-scale leg. Thermally activated artificial muscles reach the most extreme performance figures of any technology surveyed by Liang et al. [2020]: twisted-fiber muscles support loads more than one hundred times those of a biological muscle of the same length and weight, with output stresses of 22 MPa. Their activation route, however, is shared with shape-memory alloys, whose heating is fast but whose cooling is limited by heat transfer away from the element, and that asymmetry sets the cycle time.

Closer to muscle, series-elastic and pneumatic actuators trade peak force and bandwidth for compliance, a trade that wearable and rehabilitation robotics exploit because force control and backdrivability matter most when a machine interacts with a human [Aksöz et al., 2019, Leibach, 2020, Li et al., 2020]. A recent review of pneumatic artificial

muscle technology documents continued progress in actuator design and in assistive applications [Zhagiparova et al., 2025]. None of these technologies yet combines muscle-like force, strain, and compliance in one actuator, which is the combination a biomimetic humanoid leg demands and that braided pneumatic actuators approximate most closely.

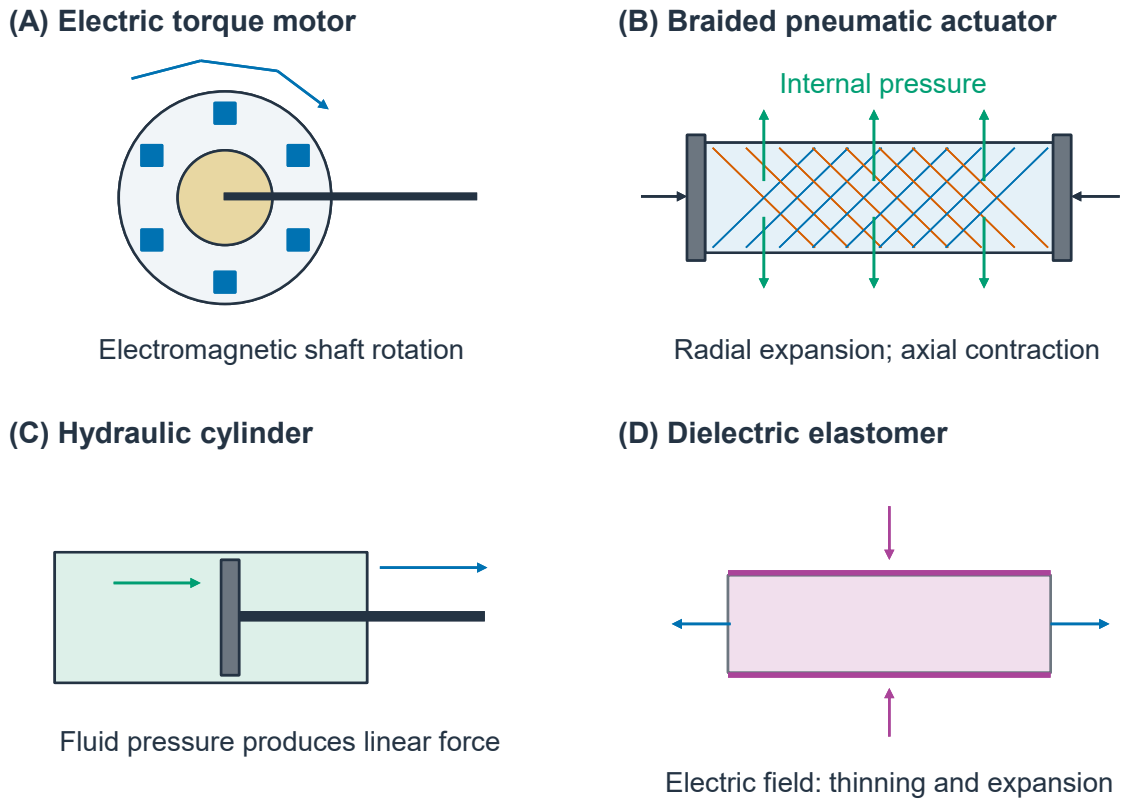


Figure 2.2: Representative actuator mechanisms: electric motor (A), braided pneumatic actuator (B), hydraulic cylinder (C), and dielectric elastomer (D) [Suzumori and Faudzi, 2018, Liang et al., 2020].

2.2.2 Braided Pneumatic Actuators

In contrast to the technologies above, McKibben-style braided pneumatic actuators (BPAs) combine low mass, high force-to-weight ratio, and intrinsic compliance with a force–length curve that qualitatively resembles that of skeletal muscle [Liang et al., 2020, Hunt et al., 2017a]. A BPA consists of an inflatable bladder inside a braided sleeve; pressurizing the bladder expands the bladder radially and contracts the sleeve axially, producing tensile force (Figure 2.2**(B)**). As shown by Chou and Hannaford, the force–length relationship of McKibben actuators is governed by geometric constraints imposed by the braided shell, leading to a nonlinear dependence of force on actuator length through the braid angle [Chou and Hannaford, 1996]: actuator force is proportional to pressure and to a nonlinear geometric term $(3\cos^2\theta - 1)$, which governs both the maximum contraction limit and the curvature of the force–length relationship. Predicting their force as a function of length and pressure is therefore the first improvement of the torque tools developed in this dissertation, and the state of that prediction is the subject of the next subsection.

Figure 2.3 shows the change in actuator shape with inflation.

However, BPAs exhibit several important differences from biological muscle. First, BPA maximum tensile force occurs at resting (i.e., uninflated) length (l_0) rather than at an optimal fiber length [Liang et al., 2020]. Second, their maximum contraction ratio is substantially lower than that of human muscle, presenting challenges for achieving biomimetic joint ranges of motion [Hunt et al., 2017a, Sárosi et al., 2017, Liang et al., 2020]. Third, BPAs have a highly nonlinear dependence on internal pressure (P), contraction (ϵ), and loading state [Hunt et al., 2017a, Martens and Boblan, 2017], which complicates



Figure 2.3: Festo BPAs in the test fixture: the inflated actuator is wider and shorter than the deflated actuator beside it.

control and motivates high-fidelity force–length–pressure modeling. Despite these differences, their muscle-like properties have made them attractive for biomimetic quadrupeds [Aschenbeck et al., 2006, Scharzenberger, 2019] and musculoskeletal arms [Hitzmann et al., 2018, Kurumaya et al., 2016, Chen et al., 2019].

2.2.3 BPA Force–Length–Pressure Models

Previous studies have generated empirical models of the force–length–pressure relationship of BPAs. Sárosi et al. [2017] discuss several high-fidelity BPA force models. They present a static force model with a 21-coefficient polynomial function for a Festo MAS-20-200N (i.e., $\varnothing 20\,\text{mm}$, $l_0 = 200\,\text{mm}$) with an impressive $R^2 = 0.9994$, as well as Sárosi's static force model for a Festo DMSP-20-400N-RM-RM (i.e., $\varnothing 20\,\text{mm}$, $l_0 = 400\,\text{mm}$), with six coefficients and a $R^2 = 0.9995$ [Sárosi, 2012, Sárosi et al., 2013]. Martens and Boblan present an even more accurate model, in terms of absolute error, using five coefficients for a DMSP-10-250 (i.e., $\varnothing 10\,\text{mm}$, $l_0 = 250\,\text{mm}$) and a DMSP-20-300 (i.e., $\varnothing 20\,\text{mm}$, $l_0 = 300\,\text{mm}$) [Martens and Boblan, 2017]. However, as demonstrated in this work, different resting lengths produce different force–length curves, preventing the use of these single-length models for designing systems with different resting lengths.

Hunt et al. [2017a] looked at six resting lengths of $\varnothing 10\,\text{mm}$ Festo BPAs, accounting for differences in maximum contractile percentages, and elucidated the force-length-pressure relationship. In particular, for a given contraction percent and force F (in Newtons), the scalar pressure P (in kPa) required to produce this force for a $\varnothing 10\,\text{mm}$ Festo artificial muscle can be determined by solving the equation:

$$P = P_0 + k_F F + k_S S + P_t \tan\left(c_a \left[\frac{\epsilon}{\epsilon_{\max} - k_E F} - c_b\right]\right) \tag{2.1}$$

where F is the required force, ϵ is the contraction, S encodes hysteresis ($S = 1$ shortening, $S = -1$ lengthening, $S = 0$ static), and the seven coefficients are defined in Table 2.1. The coefficients have been updated with corrected values, as those reported in Hunt et al. [2017a] contained typographical errors. The contraction ϵ and the maximum free-load contraction at 620 kPa, which appears as $\epsilon_{\max}$ in Equation (2.1), are defined in Section 3.1.1, together with the relative contraction $\epsilon^* = \epsilon/\epsilon_{620}$ on which the force surface developed in this dissertation is built. Because Equation (2.1) expresses pressure in closed form as a function of force and contraction, it can also be solved onto a grid of pressures and contractions to build a force–pressure–relative-contraction lookup table for design studies.

Table 2.1: Coefficients of the 10 mm Festo BPA pressure model of Equation (2.1), corrected from Hunt et al. [2017a].

Symbol	Value	Role
P_0	254 kPa	pressure offset
k_F	1.23 kPa N^{-1}	linear force-to-pressure gain
k_S	15.6 kPa	hysteresis offset (sign of S)
P_t	192 kPa	tangent-term amplitude
c_a	2.03	contraction-shape coefficient
c_b	0.46	contraction-shift coefficient
k_E	0.331×10^{-3} N^{-1}	force-dependent reduction of usable contraction

Festo provides a manufacturer tool, MuscleSIM, for predicting force as a function of pressure and contraction [Lang, Robert, 2005], but experimental comparisons indicate that it does not adequately account for variation in l_0 or actuator-specific contraction limits. Comparisons with experimental data indicate that none of the existing models effectively

take into account differences in maximum force due to initial actuator length, and the Hunt model was taken with low external forces ($\leqslant$106 N), so it is unclear how well it captures actuator behavior at higher forces. Chapter 3 compares these models with data collected for this dissertation and develops the improved characterization on which the torque and controller work of the later chapters depends.

2.3 Human Lower-Limb Biomechanics

Designing a biomimetic leg requires a quantitative benchmark of human muscle–joint behavior. This dissertation uses OpenSim [Delp et al., 2007, Seth et al., 2011, 2018] and the Gait2392 model, named for its 23 degrees of freedom and 92 musculotendon actuators. Gait2392 descends from the interactive lower-extremity model of Delp et al. [1990] and includes musculoskeletal geometry from Hoy et al. [1990] and planar knee kinematics from Yamaguchi and Zajac [1989]; related lower-limb models of the same family are characterized and validated by Arnold et al. [2010]. Muscle moment arms, force-generating properties, and functional roles in such models have been characterized extensively [Zajac and Gordon, 1989, Brand et al., 1982, Menegaldo et al., 2004, Millard et al., 2013, Sherman et al., 2013, Suderman and Vasavada, 2012], and standard reporting conventions for kinematics and joint mechanics are established by the International Society of Biomechanics [Wu and Cavanagh, 1995, Wu et al., 2002, 2005, Derrick et al., 2020].

Moment arms relate muscle force to joint torque. For a fixed-axis joint, the signed moment arm is $r_{\mathrm{m}} = \hat{\mathbf{a}} \cdot (\mathbf{r} \times \hat{\mathbf{u}})$, where $\hat{\mathbf{a}}$ is the joint-axis direction, $\mathbf{r}$ locates a point on the muscle's line of action relative to the joint, and $\hat{\mathbf{u}}$ is the force direction [Brand et al.,

1982]. The tendon-excursion method instead evaluates the sensitivity of musculotendon length to the joint coordinate, with sign determined by the coordinate and force conventions [Menegaldo et al., 2004]. For multiarticular muscles, the result also depends on adjacent-joint configurations [Zajac and Gordon, 1989, Young et al., 2019]. OpenSim uses length sensitivities, so the reference moment arms in this dissertation are properties of the Gait2392-derived geometry [Delp et al., 2007, Seth et al., 2018]. Chapter 4 compares the robot's resulting torque predictions with test-stand measurements.

Knee modeling requires particular care because the instantaneous center of rotation changes with flexion. Four-bar descriptions reproduce this behavior through linkage geometry [Romero et al., 2020, Steele, 2018]. Gait2392 instead prescribes planar translations as functions of knee flexion [Yamaguchi and Zajac, 1989]. These different kinematics produce different moment arms for the same nominal attachment locations. Joint-coordinate conventions must therefore be stated consistently when comparing models and measurements [Wu et al., 2002]. Gait2392 provides the human reference in this dissertation, while Steele's four-bar mechanism supplies the physical robot knee.

2.4 Biomimetic Knee and Bipedal Robot Model

Steele's biomimetic knee joint supplies the physical knee used in this dissertation [Steele et al., 2017, Steele, 2018, Steele et al., 2018]. It is a one-degree-of-freedom four-bar linkage whose crossed links reproduce the posterior migration of the human knee's instantaneous center of rotation (ICR) during flexion. The rolling-and-sliding behavior of the human joint therefore emerges from the link geometry itself rather than from cams or gear profiles, and

the design was calculated to reproduce an average sliding-to-rolling ratio of 0.079 over a 151° range of motion [Steele et al., 2018]. Because the ICR migrates, muscle moment arms about the knee change with flexion angle in a human-like way, and a torque prediction tool for this knee must capture that variation rather than assume a constant arm. The biomimetic knee tests of Chapter 4 therefore exercise the improved torque model against this geometry, and Figure 2.4 summarizes the mechanism.

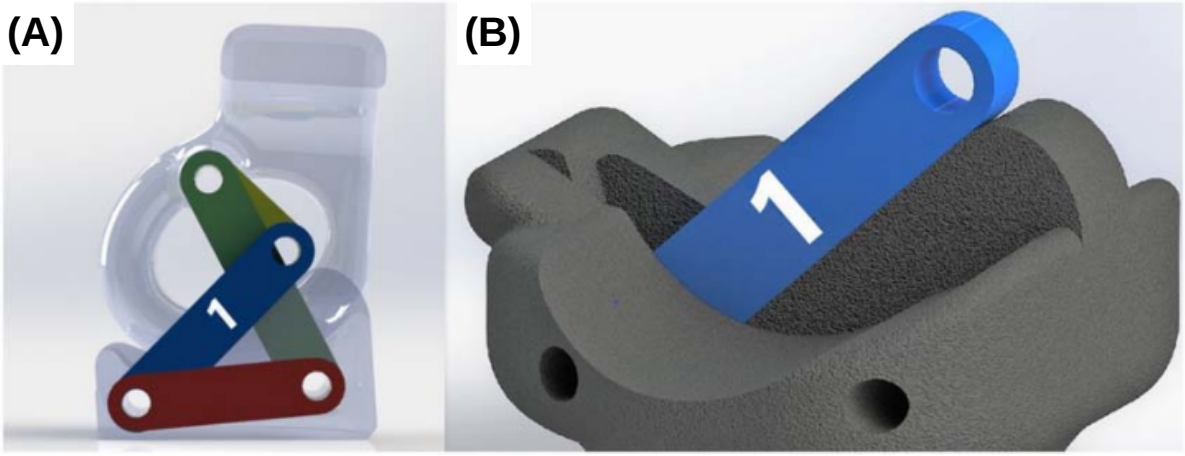


Figure 2.4: Biomimetic knee mechanism. **(A)** Sagittal cutaway at full extension showing the crossed four-bar linkage. **(B)** Exploded view of the mechanical stop. Reproduced from Steele et al. [2018, Fig. 6] under CC BY 4.0.

In my master's work, muscle lengths, moment arms, and isometric joint torques were computed over the range of motion for both the Gait2392 model and a PAM-actuated robot leg model, demonstrating that several robot muscle torque profiles could closely match the human model while many fell far short for part of the RoM [Bolen, 2019, Bolen and Hunt, 2019]. Hand placement of muscle origins and insertions was a painstaking and iterative task. This resulted in the creation of a placement optimization algorithm [Morrow et al., 2020], discussed in Chapter 4. Figure 2.5 places the laboratory skeleton beside the robot

PAM routes and human muscle paths used in that comparison.

(A) **(B)** **(C)**

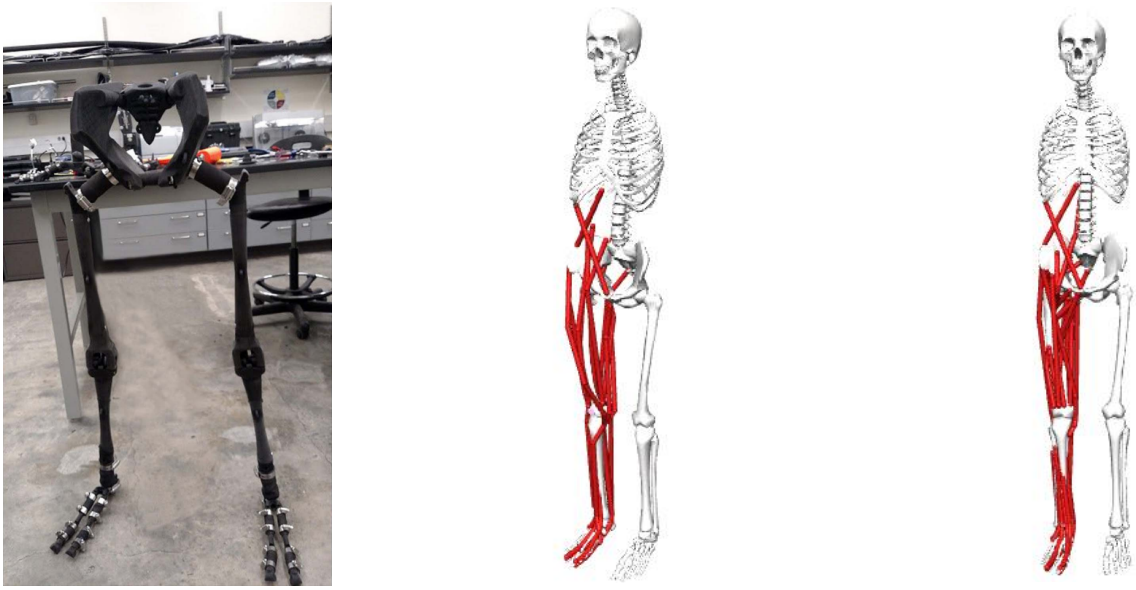


Figure 2.5: **(A)** Laboratory robot skeleton. **(B)** Robot PAM routes. **(C)** Human Gait2392 muscle paths used as design references [Bolen, 2019, Morrow et al., 2020].

The comparison evaluated muscle-path lengths, moment arms, and isometric torques across the human and robot models [Bolen, 2019]. I adjusted BPA origins, insertions, and via-point locations to respect usable contraction and avoid interference. The biceps femoris long head illustrates the remaining mismatch: the robot path and its knee- and hip-torque surfaces differ substantially from the human reference. These discrepancies motivated placement optimization [Morrow et al., 2020]. Figures 2.6 and 2.7 show the moment-arm construction and the torque comparison.

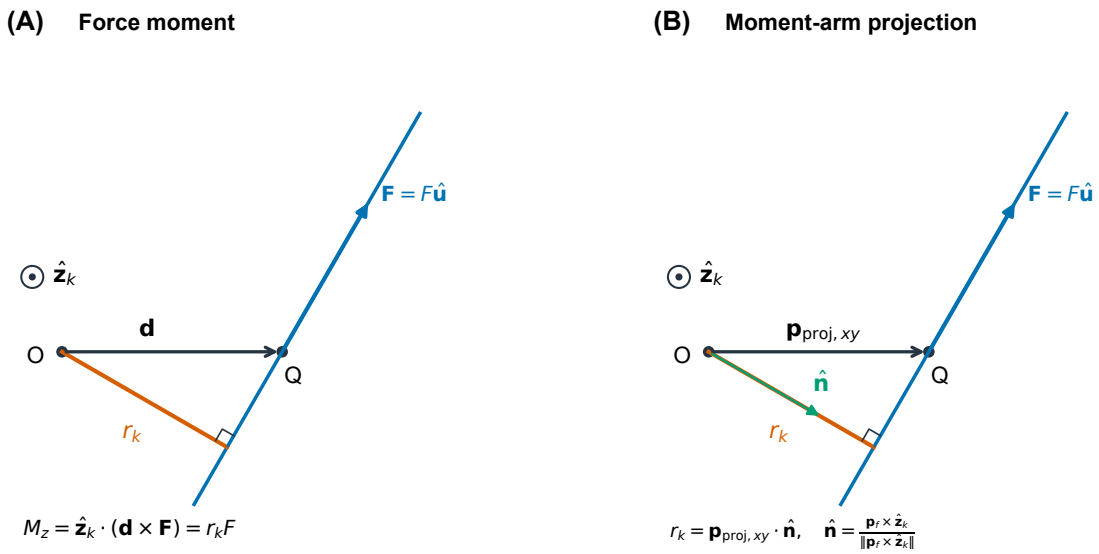


Figure 2.6: Equivalent fixed-axis constructions. **(A)** The cross product gives the moment of a force about the joint. **(B)** Projecting a joint-to-path vector onto the force-path normal gives the signed moment arm used in Equation (4.2). The joint axis points out of the page.

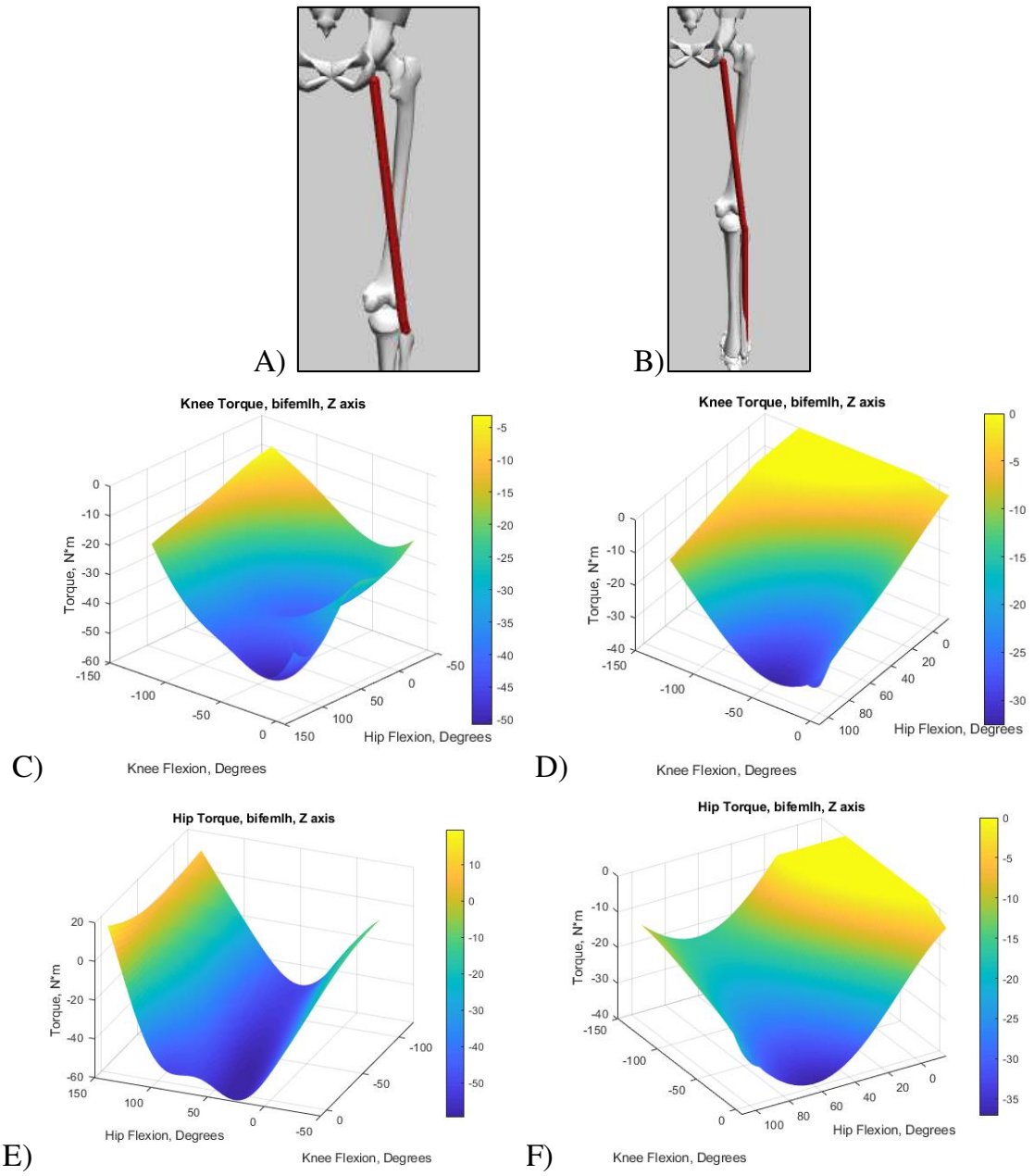


Figure 2.7: Biceps femoris long-head paths and isometric torque surfaces for the human OpenSim model and the robot model. **(A, B)** Muscle paths. **(C, D)** Knee torque. **(E, F)** Hip torque. The human model is shown in the left column and the robot model in the right column [Bolen, 2019, Fig. 8].

2.5 Neural Control of Locomotion: Central Pattern Generators

Spinal central pattern generators (CPGs) can produce alternating flexor and extensor activity without rhythmic input or proprioceptive feedback [Brown, 1911, Rybak et al., 2006]. Brown's half-center hypothesis attributes this alternation to reciprocally inhibitory central populations [Brown, 1911]. In the model of Deng et al. [2019], persistent sodium currents with slow inactivation support alternating bursts in the rhythm-generating neurons.

Experiments on the deafferented stick-insect thoracic nerve cord provide a direct example of central rhythm generation: application of pilocarpine induced alternating rhythmic activity in antagonistic leg motoneuron pools [Büschges et al., 1995]. This preparation separates rhythm production in the nervous system from the sensory feedback generated by movement.

The two-level model of Rybak et al. [2006] separates a rhythm generator (RG), which sets cycle timing, from a pattern-formation (PF) network, which regulates motor output (Figure 2.8). This distinction accounts for deletions of motoneuron activity that leave locomotor period and phase intact: an intervention can suppress pattern output without resetting the rhythm. Afferent inputs reach both levels. McCrea and Rybak [2008] reviewed the electrophysiological evidence for this organization, and Rybak et al. [2015] related it to genetically identified interneuron classes. Commissural pathways coordinate the left and right networks [Molkov et al., 2015, Danner et al., 2016], while the rhythmogenic core remains under experimental investigation [Rancic and Gosgnach, 2021].

The extension of Shevtsova et al. [2016] includes dedicated pattern-formation populations for the two-joint posterior biceps–semitendinosus and rectus femoris muscles

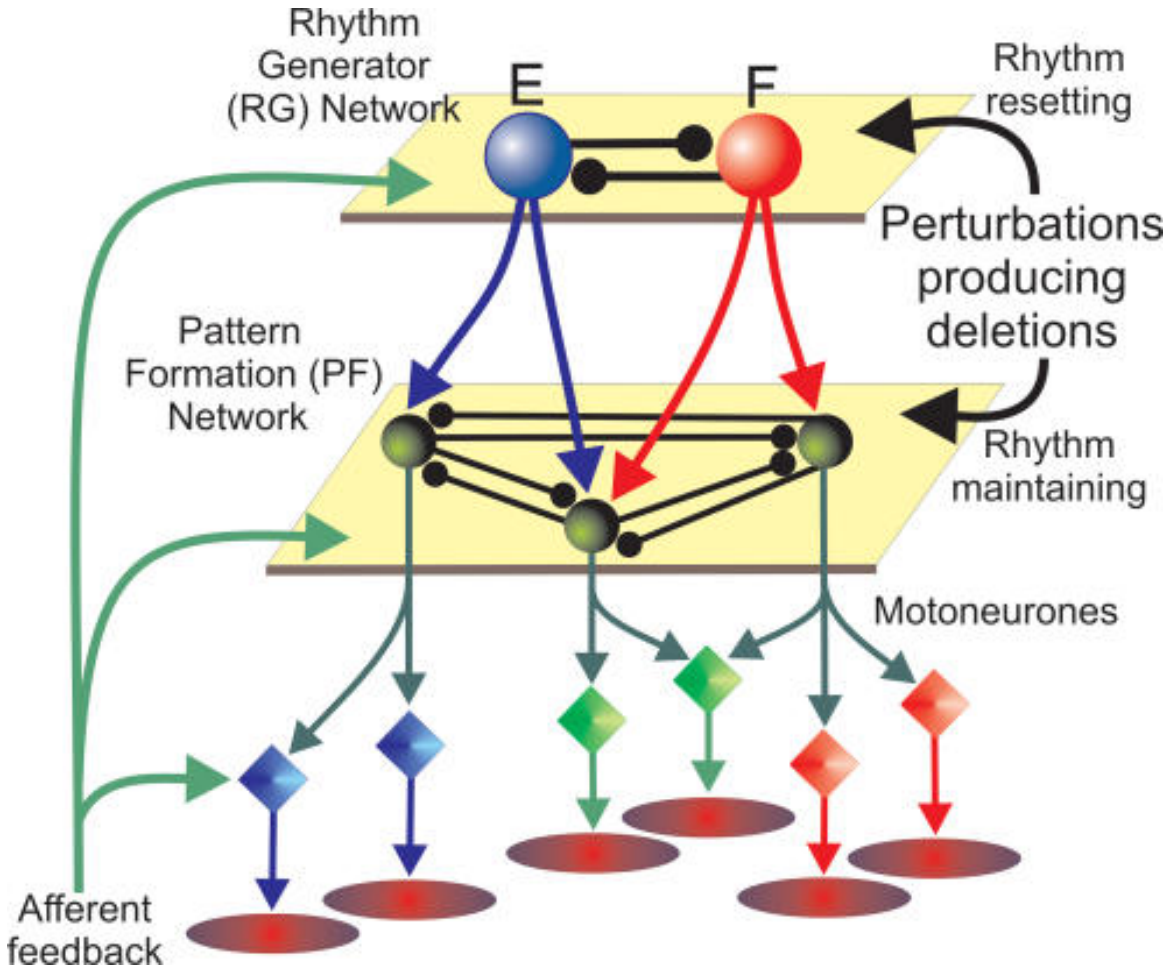


Figure 2.8: Two-level CPG: rhythm generation (RG), pattern formation (PF), motor output, and afferent feedback. Reproduced from Rybak et al. [2006, Fig. 1].

(Figure 2.9). Coordinating these muscle pools presents a convergence problem that also arises when a robot muscle crosses two joints. Section 2.6 shows how related circuit models incorporate afferent feedback; the controller developed in Chapter 6 uses this two-level organization.

2.6 Sensory Feedback in Locomotion

Sensory feedback is an integral part of locomotor pattern generation. Reviews of afferent control of the CPG [Van de Crommert et al., 1998] and muscle proprioceptive feedback and spinal networks [Windhorst, 2007, 2022] summarize the classical picture: group Ia afferents from muscle spindles signal muscle length and velocity, group Ib afferents from Golgi tendon organs signal force, and group II afferents signal length. Load-related feedback plays a role distinct from length-related feedback during locomotion [Dietz and Duysens, 2000].

Primary experiments show how the effect of a sensory signal depends on the preparation and step phase. In decerebrate, paralyzed cats undergoing fictive locomotion, Guertin et al. [1995] found that stimulating ankle-extensor group I afferents enhanced hip, knee, and ankle extensor activity during extension and could reset flexion to extension. Group I includes both Ia and Ib afferents, making the stimulation conditions important when interpreting this result.

Joint position also contributes to phase transitions. In chronic spinal cats, Grillner and Rossignol [1978] found that moving a held hindlimb backward could initiate swing, with lift-off depending on hip position and contralateral step phase. In walking decerebrate cats,

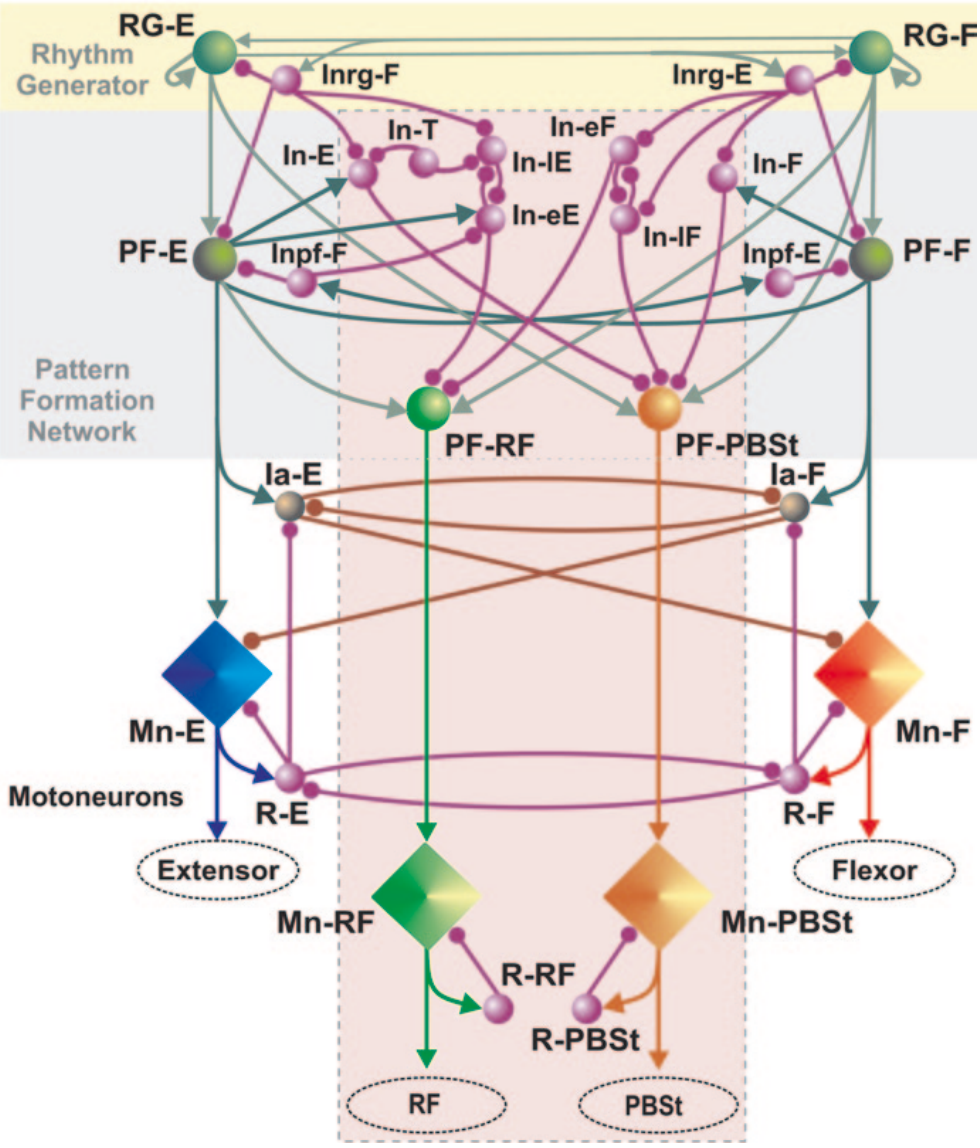


Figure 2.9: Extended CPG with pattern-formation and motor populations for the two-joint posterior biceps–semitendinosus (PBSt) and rectus femoris (RF). Reproduced from Shevtsova et al. [2016, Fig. 5.8].

assisting hip flexion advanced the onset of ankle-extensor activity at the swing-to-stance transition [McVea et al., 2005]. These experiments motivate testing position-dependent feedback while preserving the distinction between the two phase transitions.

Removing proprioceptive feedback degrades locomotor patterns in mice [Akay et al., 2014]; proprioceptive input regulates and adapts locomotor activity [Lam and Pearson, 2002]; and the relative contributions of proprioceptive and vestibular systems remain under study [Akay and Murray, 2021, Akay, 2020]. Coupled models show how feedback interacts with biomechanics to shape CPG output [Markin et al., 2010, Deng et al., 2022]. Tonic central and sensory stimuli can facilitate involuntary air-stepping in humans [Selionov et al., 2009]. The proprioceptive and contact feedback channels designed in Chapter 6 follow this afferent organization.

The rat hindlimb controller of Deng et al. [2019] combines RG/PF control with muscle-receptor pathways, interneurons, Renshaw cells, and motoneurons (Figure 2.10). Its sensory and central pathways converge on the motor output rather than acting as separate controllers.

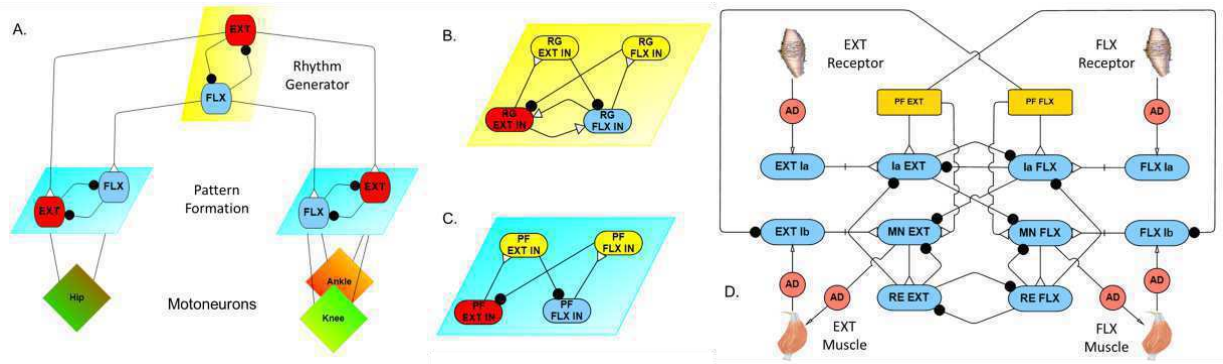


Figure 2.10: Synthetic nervous system of the rat hindlimb model: **(A)** single limb control hierarchy, **(B)** rhythm-generator configuration, **(C)** pattern-formation network, and **(D)** sensory–motor network. Reproduced from Deng et al. [2019, Fig. 4] under CC BY 4.0.

The human lower-limb model of Di Russo et al. [2023] represents Ia, II, and Ib reflex pathways and Renshaw cells for agonist–antagonist muscle pairs, with locomotor-pattern inputs reaching the same motoneuron pools (Figure 2.11).

The cat model of Shinohara et al. [2025] addresses interlimb coordination, coupling the two rhythm generators through V3 and commissural interneuron populations while force, length, and velocity feedback returns from the muscles (Figure 2.12). Together, these examples show how the neural populations and feedback paths are selected for the hypothesis being tested.

2.7 Computational Neuroscience

Computational neuroscience expresses hypotheses about neural circuitry as executable models. In conductance-based models, membrane voltages evolve through capacitance, leak, synaptic, and applied currents, and connections represent identifiable pathways [Nourse et al., 2023]. Model predictions can be compared with recordings or with responses

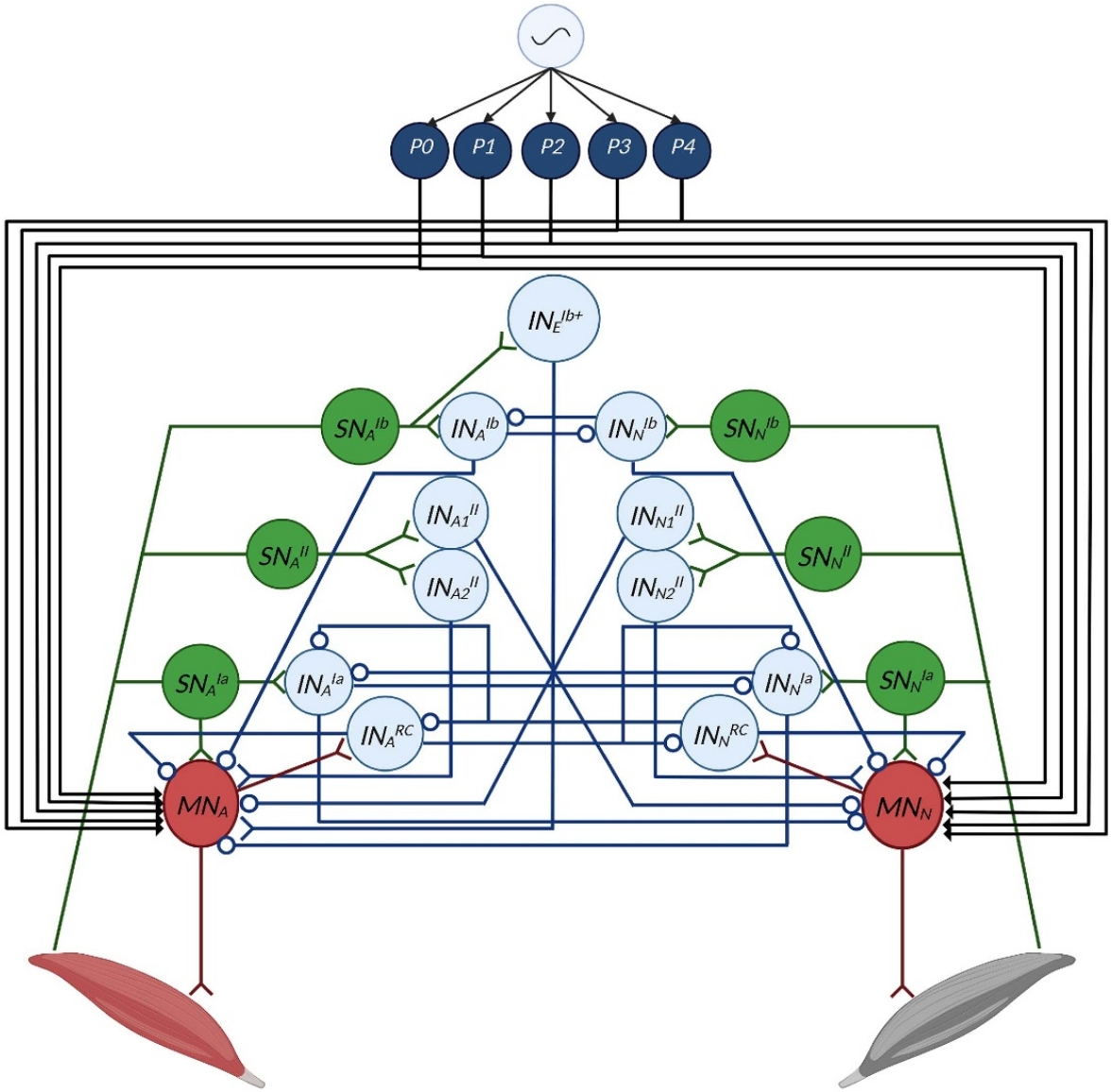


Figure 2.11: Antagonist-muscle spinal network with Ia, II, and Ib reflex pathways, Renshaw cells, and locomotor-pattern input. Reproduced from Di Russo et al. [2023, Fig. 5] under CC BY 4.0.

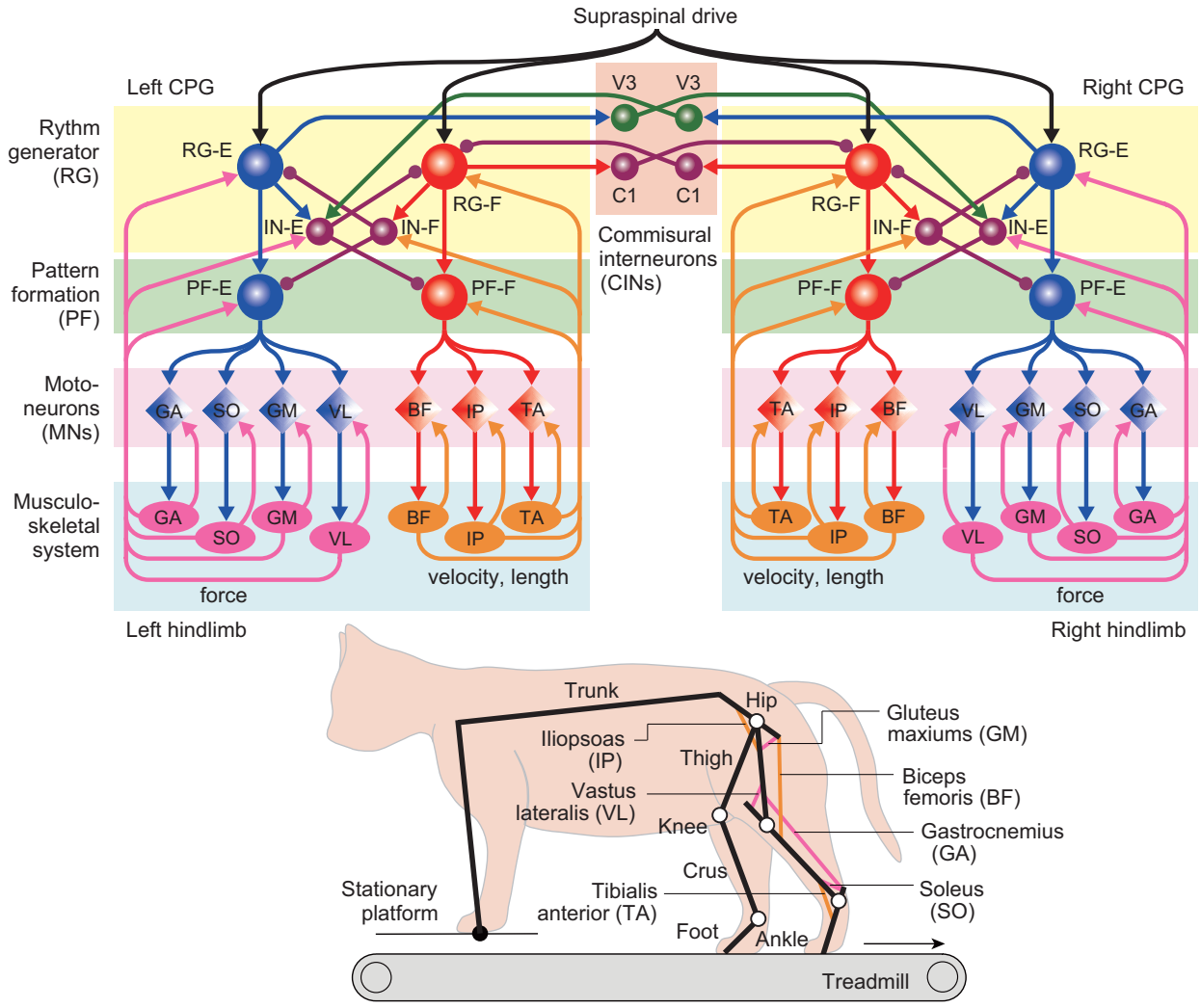


Figure 2.12: Neuromusculoskeletal cat model of Shinohara et al. [2025]: bilateral rhythm-generator and pattern-formation layers coupled by V3 and commissural interneurons, motoneuron pools for seven muscles per hindlimb, and force, length, and velocity afferent feedback. Reproduced from Shinohara et al. [2025, Fig. 1] under CC BY 4.0.

to stimulation and pathway removal. This provides a way to test a proposed mechanism rather than infer it from a wiring diagram alone. Synthetic nervous systems use this approach to control physical and simulated bodies.

2.8 Synthetic Nervous Systems

Synthetic nervous systems (SNSs) implement candidate neural circuits with conductance-based spiking and non-spiking neurons. Work in the Quinn laboratory at Case Western Reserve University and related laboratories, including my own, has applied this approach to pneumatic dog-robot control [Hunt et al., 2017b], human balance [Hilts et al., 2019], rat hindlimb mechanics [Young et al., 2019], the neuromechanical walker of Deng et al. [2019], and Muscle Mutt, the canine-inspired BPA robot developed at the Agile and Adaptive Robotics Laboratory (AARL) at Portland State University [Scharzenberger, 2019]. Related work includes sensory-entrained bipedal CPGs [Li et al., 2017], insect-inspired inter-leg coordination [Nourse et al., 2019], reinforcement-reflex hybrids [Goldsmith et al., 2021], and adaptive hindlimb walking controllers [Schilling et al., 2022].

The SNS-Toolbox [Nourse et al., 2023] makes such networks practical: it is an open-source Python package that simulates hundreds to thousands of spiking and non-spiking neurons in real time or faster, supports multiple software and hardware backends including GPUs and embedded platforms, and couples directly to physics simulators such as MuJoCo for musculoskeletal control. The field it serves is active and growing: the perspective review of Szczecinski et al. [2023] traces neuromorphic legged-locomotion control from its insect-robot beginnings through current platforms to open problems, and physical testbeds

in this lineage now extend from insect-scale robots to a robotic cat hindlimb [Armstead et al., 2025].

2.9 Neuromechanical Simulation

Neuromechanical simulation couples a musculoskeletal model to a neural controller. Three partially overlapping approaches are relevant here. The first encodes muscle reflexes directly: Geyer and Herr [2010] used muscle- and phase-specific pathways, including positive and negative force or length feedback, contact-dependent switching, and trunk-posture control, to produce human-like walking without a CPG. Hybrid work tests the contribution of feedback alongside other control signals: Haeufle et al. [2018] combined feed-forward muscle activations with neuronal reflex feedback and found that the robustness benefit of each depends on the muscle.

The second line replaces hand-designed circuitry with machine learning. Song et al. [2021] trained deep reinforcement learning policies on a physiologically realistic musculoskeletal model in OpenSim, reproducing human kinematics and muscle activations across locomotion tasks, with the learned policy re-usable for experiments that would be difficult in the laboratory. This approach inherits the fidelity of the musculoskeletal model but encodes control in opaque function approximations: the trained policy can reproduce behavior, but the individual "neurons" of the controller do not correspond to identifiable physiological circuits, so the class of scientific questions it can answer, such as lesions, deletions, and pathway stimulation, is narrower than its behavioral repertoire. Recent perspectives on predictive musculoskeletal simulation [De Groote and Falisse, 2021] place

both learned and optimal-control approaches in the same frame: the models are mature, and the open problems are the cost of solving them and the biological interpretability of the control they find.

The third approach uses physiological circuit models, including the CPG-plus-afferent-feedback models of Markin et al. [2010] and Shevtsova et al. [2016] (Sections 2.5 and 2.6). SNS-Toolbox supports conductance-based networks at speeds suitable for selected real-time robotic applications [Nourse et al., 2023]. Related work includes motor-module bipedal walkers [Ichimura and Yamazaki, 2019], a musculoskeletal macaque model [Nishizaki et al., 2026], and muscle-driven robotic hind legs [Fukuoka et al., 2021]. For the humanoid studied here, the methodological question is how to implement and verify candidate spinal circuits with human musculotendon mechanics and custom BPA actuation while distinguishing neural effects from conversion or coupling errors. Chapter 6 addresses that question through model implementations, graphical neural blocks, and controlled verification and feedback tests.

2.9.1 Simulation Tools

All three lines depend on a software substrate that is itself part of the method: elaborate musculoskeletal geometry and complex neural models. The tools used in this dissertation are the following.

OpenSim [Delp et al., 2007, Seth et al., 2018], distributed through the SimTK (Simulation Toolkit) repository, provides the biomechanical modeling side: musculoskeletal models with detailed bone geometry, muscle routing, and force-generating properties, along with

tools for analyzing moment arms and isometric torque. These models are approximations with known limits rather than exact anatomies: muscle paths are interpolations over discrete attachment and wrapping data [Bolen, 2019]. Their geometry is accurate for a purpose, namely moment arms and torque predictions over the functional range of motion, and that is the standard against which the robot model of this program is compared.

The same SimTK infrastructure hosts SCONE, an open-source package for predictive simulation and optimization of neuromuscular control of biological motion [Geijtenbeek, 2019].

AnimatLab [Cofer et al., 2010] provides a graphical environment in which physical bodies and synthetic neural systems are constructed and co-simulated; the AARL's prior walker (Section 6.1) was built in it.

MuJoCo [Todorov et al., 2012] provides the physics-engine side: fast, accurate simulation of rigid-body dynamics with contacts, along with muscle-like actuators that include activation dynamics.

MyoConverter bridges the biomechanical and physics-engine worlds by converting OpenSim musculoskeletal models to MuJoCo format with optimized muscle kinematics and kinetics [Caggiano et al., 2022, Wang et al., 2022].

SNS-Toolbox supplies the neuron and synapse models used to simulate the controller [Nourse et al., 2023].

I used these tools to develop a robotic OpenSim leg model, convert it to MuJoCo, and couple it to a synthetic nervous system. Chapter 6 describes the assembled interfaces and their evaluation. The coupled simulation supports repeated parameter, stimulation, and pathway-removal tests under controlled conditions. Whether those tests support a

biological hypothesis depends on the model's validation and the experimental comparison, as well as simulation speed.

2.10 Testing Neuromechanical Hypotheses on Robots: The Ethical Case

Destructive interventions, such as deleting a muscle or lesioning spinal circuitry, cannot be used as controlled experiments on healthy human volunteers. Tonic stimulation is a different case: human studies have used tonic central and sensory stimuli to facilitate air-stepping [Selionov et al., 2009]. Robotic and computational models additionally permit selective, repeatable pathway stimulation and removal without invasive procedures on a living subject. The same experiments in animals also carry real moral cost, and their justification is governed by the Three Rs framework: replacement of animal subjects with non-animal methods, reduction of the numbers required, and refinement of the remaining procedures [Russell and Burch, 1959, Tannenbaum and Bennett, 2015]. Robotic and computational models allow proposed neural interventions to be tested under controlled conditions before corresponding animal experiments are considered. The same model and initial conditions can be reused across interventions to isolate the effect of a changed pathway. Repeating a model does not create independent biological subjects, and its relevance to animal experiments depends on validation against biological observations.

The invasive preparations that motivate replacement are not always apparent in circuit schematics of decerebrate-cat locomotion, which emphasize the neural circuits under study but often leave the surgical preparation out of view. A simplified drawing of a walking cat can make the animal appear intact, alert, or even contented. That appearance should not

be confused with the condition of the experimental animal: a brainstem transection has disconnected the forebrain, while neural circuits below the transection can still generate treadmill stepping [Brown, 1911, Severin et al., 1967]. Figure 2.13, taken from archival footage, shows the physical preparation behind the circuit diagrams [WhiteCoatWaste, n.d.]. This distinction makes the ethical meaning of replacement concrete: the robotic experiments discussed here are intended, in part, to reproduce selected neural mechanisms without repeating these invasive preparations in living animals.

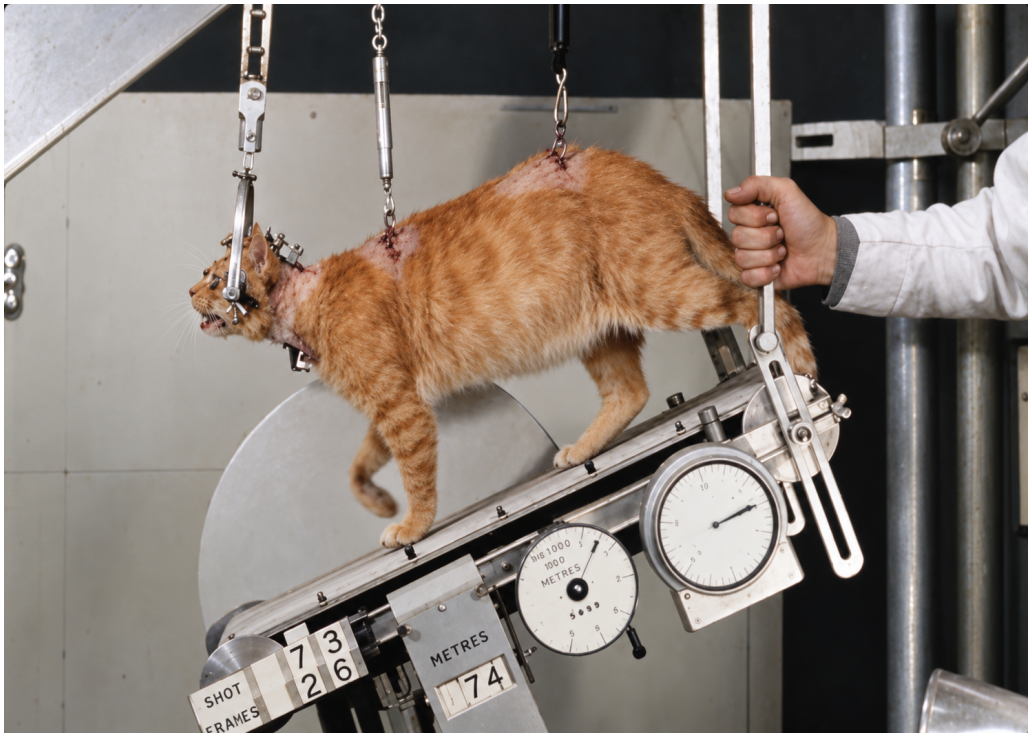


Figure 2.13: Archival decerebrate-cat treadmill experiment [WhiteCoatWaste, n.d.]. The frame is restored and machine colorized; color and fine detail are inferred. It provides documentary context, not gait evidence.

The procedures behind that record are documented in films that remain publicly

available. A 50 min classroom recording posted by the UTHealth Houston School of Dentistry first shows an intact cat playing with its handler, righting itself when lowered and placing its paws accurately on the edge of a table, and then documents the decerebration performed for the class [UTHealth Houston School of Dentistry, 2025]: the animal is anesthetized with open-drop ether, more than the horse in the same demonstration series received, the carotid arteries are ligated and a tracheal cannula installed, and the skull is opened with a trephine before the brainstem is transected between the superior and inferior colliculi. The treadmill preparations that established locomotor neurophysiology are likewise preserved in the historical literature: Stuart [2007] reproduce the arrangement of the high-decerebrate cat treadmill experiments, with electromyographic recording from the hindlimb muscles, from the original Russian-language reports [Severin et al., 1967].

The honest accounting is murkier than the Three Rs framing alone suggests, because the models a robot implements are themselves products of animal experimentation. The half-center organization, the two-level rhythm and pattern architecture, and the length, force, and load signaling of the afferents were each established in preparations like the ones described above [Brown, 1911, Severin et al., 1967, Rybak et al., 2006], and the databases of afferent properties that parameterize sensory feedback are compiled from animal recordings [Bolen et al., 2023]. A synthetic nervous system therefore replaces future experiments; it does not replace, and cannot repay, the animal work on which its architecture is built. Recognizing this inheritance sharpens rather than undermines the case for robotic subjects: the knowledge purchased at that cost is precisely what makes a robotic replacement scientifically meaningful, and the robot is the instrument that keeps that knowledge productive without adding to the ledger.

For a robot to serve as a replacement subject, however, it must first earn standing as a scientific model. Webb's influential analyses of bio-robotic methodology [Webb, 2001, 2002] argue that a robot is a good model of biological behavior when the mechanism it implements is tested at the same level of organization as the target phenomenon, matching dynamics, control structure, and sensory ecology rather than surface appearance. The validation chain of this dissertation is constructed to that standard: the actuator force model is validated against bench data (Section 3.2.1), the joint torque model against isometric measurements and human torque envelopes (Section 4.2), and selected controller responses are compared with published tonic stimulation and deletion phenomena [Selionov et al., 2009, Rybak et al., 2006], the verification step that precedes locomotion tuning in this program. Only then can a robot experiment be read as evidence about the hypothesized biological circuit rather than about the robot.

The argument is ultimately bidirectional, completing the loop described in Section 1.2. Experiments that are unethical, impractical, or statistically unreachable in humans and animals become routine on the robot; the resulting insights into neuromechanical control feed back into biology, into the design of the next robot iteration, and directly into prosthetic and orthotic devices that face the same problem of controlling muscle-like actuators for human users, whose central pattern generator controllers are an active topic in lower-limb exoskeleton research [Duan et al., 2024, Pery et al., 2026]. The ethical case and the scientific case are the same case: the robot is the instrument that makes these experiments possible.

3 Isometric Force Characterization of Braided Pneumatic Actuators

Publication note: Sections 3.1.1–3.1.2, Section 3.2.1, and Sections 3.3.1–3.3.5, with their associated figures and tables, appeared in Bolen et al. [2026] (Bolen, Elzein, Pang, and Hunt, *Actuators*, 2026). For that work, B.B. performed the conceptualization, methodology, software, validation, formal analysis, investigation, data curation, visualization, and original draft preparation; M.E. and L.P. contributed methodology, software, investigation, and data curation; and A.H. supervised the work, administered the project, and acquired funding. All co-authors contributed to writing, review, and editing.

This chapter develops and validates the actuator force models on which the rest of the dissertation builds, and it reads as one self-contained study, from methods through results to discussion. Sections 3.1.1 and 3.1.2 present the methods: the force equations used throughout this work, and the test jig and measurement procedure used to characterize Festo BPAs with 10 mm and 20 mm uninflated diameters over a wide range of resting lengths, contractions, and pressures. Section 3.2.1 reports the resulting maximum-force and normalized force–strain–pressure models and compares them with the manufacturer's tool. The remaining sections discuss the effects of resting length, the nondimensionalized force prediction, the comparison with other published BPA models, the implications for real-time control and simulation, and the analogy between these artificial muscles and biological muscle. The validated force model is the one used for the knee-torque predictions of Chapter 4 and for the BPA actuation of the neuromechanical simulations in

Chapter 6.

3.1 Methods

3.1.1 Actuator Force Model

The actuator force model used throughout this dissertation is described here. First, the amount of contraction ε in a BPA is calculated as

$$\varepsilon = \frac{(l_0 - l_{\mathrm{m}})}{l_0} \tag{3.1}$$

where l_0 is uninflated resting length of the BPA measured from hose clamp to hose clamp. Current muscle length is l_{m}.

Similarly, ε_{620} is the amount of contraction in a BPA without external load when inflated at 620 kPa, calculated as

$$\varepsilon_{620} = \frac{(l_0 - l_{620})}{l_0} \tag{3.2}$$

where l_{620} is the muscle length measured at 620 kPa and free from external loads.

The amount of relative contraction of a BPA ε^* can then be calculated by dividing equation (3.1) by (3.2), such that

$$\varepsilon^* = \frac{\varepsilon}{\varepsilon_{620}} \tag{3.3}$$

describes the BPA's current contraction as a normalized percentage of its total.

The amount of normalized isometric force F^* in a Festo BPA is then calculated as

$$F^*(\varepsilon^*, P^*) = \begin{cases} c_0 \cdot \left(\exp(-c_1 \cdot \varepsilon^*) - 1\right) + P^* \cdot \exp\left(-c_2(\varepsilon^*)^2\right) & \text{for } F^* > 0 \\ 0 & \text{for } F^* \leqslant 0 \end{cases} \tag{3.4}$$

where P^* is pressure P normalized by 620 kPa. BPA diameter dependent coefficients are determined by the actuator characterization experiments reported in Section 3.2.1 (Table 3.2).

The maximum force F_{max} in a BPA is calculated as

$$F_{max}(l_0, P_{\max}) = a_1 \cdot P \cdot \arctan\left(a_2 \cdot P \cdot (l_0 - 0.0075)\right) \tag{3.5}$$

with coefficients determined by the actuator characterization experiments reported in Section 3.2.1 (Table 3.1). When maximum system pressure is set to $P = 620\,\text{kPa}$, $F_{\max}(l_0, 620) = F_{620}(l_0)$. Multiplying (3.4) by (3.5) yields

$$F(\varepsilon^*, P^*, l_0) = F^*(\varepsilon^*, P^*) \cdot F_{620}(l_0) \tag{3.6}$$

as the equation for isometric force F in a Festo BPA.

3.1.2 BPA Isometric Force Characterization Experiment

To measure artificial muscle force as a function of length and pressure, we built a test jig from extruded aluminum and 3D printed parts. Festo BPAs with 10 mm and 20 mm diameter were tested with various resting lengths and different amounts of contraction.

These data are used to develop an improved model that more accurately predicts maximum muscle force based on l_0, current muscle length l_m, and pressure P. The results are compared with existing models and the new data are used to create an updated model.

A test jig was made of 80/20 brand 1515 series extruded aluminum (Figure 3.1). Artificial muscles were placed vertically in the jig one at a time. The upper end was attached to a load cell. Force data was collected using a CALT DYLY-103 100 kg S-shaped load cell for the ∅10 mm BPAs and a Loadstar RAS1-2HKS-S for the ∅20 mm BPAs. The load cell was used in conjunction with a HX711 Load Cell Amplifier. Pressure data was obtained from a Freescale MPX5700 GP 5 V pressure sensor. Building air supply pressure was controlled with two pressure regulators in series. The first was a Parker model 20R113GC $0 - 120$ psi pressure regulator. The second was a Husky 3/8 in. High Performance Air Regulator HDA72200. A Festo VTUG-10-MRCR-S1T-26V20-T516LA-UL-532S-8K valve manifold VTUG-G was used to deliver air from the pressure regulator to the actuator. This manifold is comprised of eight two-in-one bidirectional normally closed Festo VUVG-S10-T32C-AZT-M5-1T1L valves.

Pressure and load cell amplifier data were sent to MATLAB 2021b via an Arduino Uno style Sparkfun BlackBoard C microcontroller. The computers that used MATLAB were running either Windows 10 or Windows 11. During phases when the Arduino was collecting force and pressure data to send to MATLAB, the Arduino would also trigger (via MATLAB) an Onsemi 2N4401 NPN transistor to make the valve manifold open or close the valve. For other data collection, the valve was manually opened and closed.

The inner distance between the hose clamps on each BPA was measured with a FANUC tape measure to determine the muscle's length l and resting length (l_0). This is how

Festo AG & Co. KG [2026] defines l_0, although in Hunt et al. [2017a] it was measured to also include end cap length. We then inflated each BPA to $P_{620} = 620\,\text{kPa}$, with one end allowed to move freely in the axial Degree of Freedom (DoF), and measured the length l_{620} to calculate maximum contraction at 620 kPa using Equation (3.2). The distance between the crossmembers was then controlled to obtain different amounts of contraction, the muscles were inflated to various pressures, and the contractile force was recorded at the pressure–contraction pairs. This was done for ∅10 mm BPA resting lengths (l_0) of 112 mm–518 mm. For ∅20 mm BPAs, resting lengths (l_0) of 300 mm–509 mm were used.

3.2 Results

3.2.1 BPA Force Characterization

This section reports the results of the isometric force characterization experiments described in Section 3.1.2. First, maximum isometric force at 620 kPa is modeled as a function of resting length. The remaining data are then normalized to build a predictive force surface, and the resulting model is compared with the manufacturer's tool and with other published BPA models.

Maximum Force at 620 kPa

Data of the maximum force from BPA characterization tests of the 10 mm and 20 mm diameters at 620 kPa show a dependency on resting length (Figure 3.2). This is a previously unreported characteristic of these artificial muscles. Detailed analysis of the Festo

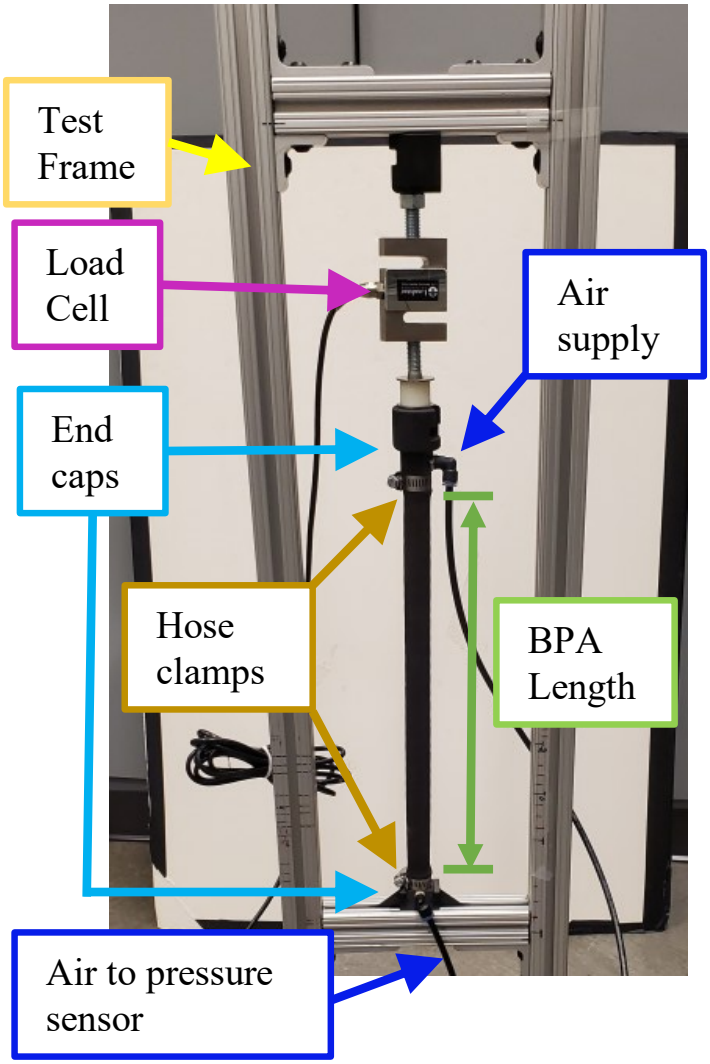


Figure 3.1: A picture of a BPA in the isometric force test stand with components labeled.

tool [Festo AG & Co. KG, 2026] does in fact predict a change in maximum force with the resting length; however, the Festo tool predicts increased force with shorter lengths, while our collected data indicate decreasing force with shorter lengths. The data show a force response resembling an arctan curve along the l_0 dimension. Using the Nonlinear Least Squares method and a Least Absolute Residual robustness, we fit an arctan curve to the data to get the maximum force at 620 kPa given a resting length as:

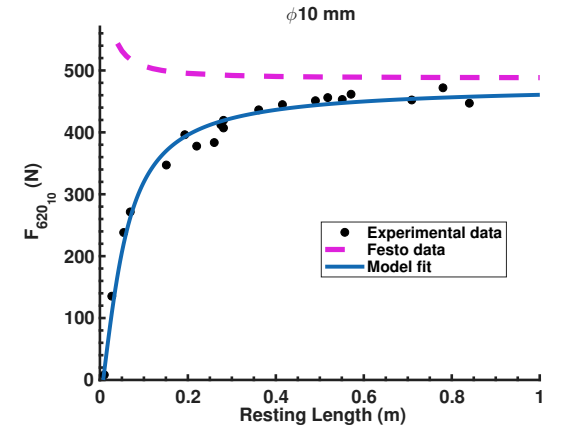
$$F_{620_{10}} = 303.5\,\mathrm{N} \cdot \arctan\left(19.03\,\mathrm{m}^{-1} \cdot (l_0 - 0.0075)\right) \tag{3.7}$$

$$F_{620_{20}} = 922.4\,\mathrm{N} \cdot \arctan\left(15.37\,\mathrm{m}^{-1} \cdot (l_0 - 0.013)\right) \tag{3.8}$$

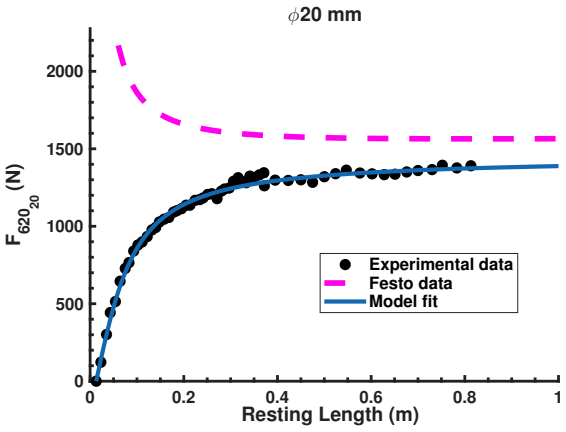
The length is offset by 0.0075 m and 0.013 m because solid modeling showed that the end caps contact each other at these lengths. At these lengths, the actuator would not be able to contract to produce force.

Force as a Function of Pressure and Resting Length

Data from BPA characterization tests of the 10 mm muscles resulted in force–pressure pairings for different muscle resting lengths (l_0) (Figure 3.3a). Similar to the maximum force data, these data show a force response resembling an arctan curve along the l_0 dimension with a more linear response to changes in pressure. Using the Nonlinear Least Squares method and a Least Absolute Residual robustness, we fit an arctan curve to the data to get the maximum force given a resting length and pressure as described by the model form given in Section 3.1.1, where $a_1 = 0.4895\,\mathrm{N\,kPa}^{-1}$ and $a_2 = 0.030\,68\,\mathrm{kPa}^{-1}\,\mathrm{m}^{-1}$



(**a**)



(**b**)

Figure 3.2: Results for finding the relationship between F_{620} and l_0. (**a**) $F_{620_{10}}$ as a function of l_0, at P_{620}. Dashed line is the $F_{620_{10}}$ data from Festo. Solid line is the fit from Equation (3.7). (**b**) $F_{620_{20}}$ as a function of l_0 at P_{620}. Dashed line is $F_{620_{20}}$ data from Festo. Solid line is the fit from Equation (3.8).

for the 10 mm actuator, and $a_1 = 1.49\,\mathrm{N\,kPa^{-1}}$, $a_2 = 0.0248\,\mathrm{kPa^{-1}\,m^{-1}}$ for the 20 mm actuator. Goodness-of-fit measures are given in Table 3.1; throughout this dissertation, goodness of fit is reported as the root-mean-square error (RMSE), the adjusted fraction of variance unexplained (FVU, for which $R^2 = 1 - \mathrm{FVU}$), and the maximum absolute residual, as defined alongside the reference implementation in Appendix A.

Maximum Contraction at 620 kPa

Figure 3.3(b) shows an attempt at a linear fit for maximum contraction at 620 kPa (ϵ_{620}) as a function of resting length (l_0). Error bars have been included to show the effect of ± 1 mm accuracy with measurements of l_0 and l_{620}. There was a large amount of variance in the data, with the linear fit giving an adjusted $\mathrm{R}^2 = 0.4124$ and an RMSE $= 0.0083$. Since there is no direct, predicable relationship between maximum contraction and l_0, this value should be recorded in each muscle used on a robot to best predict the force it may produce at different pressures and contraction.

Force as a Function of Pressure and Contraction

In addition to being a function of the pressure and resting length, the force produced by the actuator is also a function of the amount of actuator contraction, with less force being applied as the actuator contracts. To build this model, the collected data of force, pressure, and contraction are normalized by dividing by F_{620}, P_{620}, and ϵ_{620}, respectively. This had the effect of compressing the data into a 3D surface ranging from 0 to 1 on all axes. Normalized data are compared with pressure isoclines of force predicted by the

Table 3.1: Maximum force equation coefficient values with confidence intervals (CI) and goodness-of-fit measures. Equation (3.5) is compared to data taken at 620 kPa. Equations (3.7) and (3.8) are compared against maximum force from using the Festo tool. It was not necessary to fit an adjusted R^2 in these cases.

Equation	Coefficient	CI (95%)	Our Model			Festo Tool		
			Adj. R^2	RMSE	Max. Error	Adj. R^2	RMSE	Max. Error
(3.5)	$a_1 = 0.4895\,\mathrm{N\,kPa^{-1}}$ $a_2 = 0.030\,68\,\mathrm{kPa^{-1}\,m^{-1}}$	(0.4822, 0.4968) (0.0282, 0.03317)	0.9945	11.61 N	55.3 N	0.9854	14.7 N	30.9 N
(3.7)	$b_1 = 303.5\,\mathrm{N}$ $b_2 = 19.03\,\mathrm{m^{-1}}$	(300, 308) (17.48, 20.57)	0.9854	14.72 N	30.9 N	–	189.9 N	375.6 N
(3.8)	$b_1 = 922.4\,\mathrm{N}$ $b_2 = 15.37\,\mathrm{m^{-1}}$	(914.2, 930.7) (14.75, 15.98)	0.9945	23.83 N	62.1 N	–	668.4 N	1590.1 N

Table 3.2: Normalized isometric BPA force equation (Equation (3.4)) coefficient values and goodness-of-fit measures.

BPA	Coefficient	CI (95%)	Model			Validation		
			Adj. R^2	RMSE	Max. Error	Adj. R^2	RMSE	Max. Error
∅10 mm	$c_0 = 0.5682$	(0.5584, 0.578)	0.9998	0.005118	10.3%	0.9994	0.0245409	10.6%
	$c_1 = 4.254$	(4.126, 4.383)						
	$c_2 = 0.5597$	(0.5429, 0.5766)						
∅20 mm	$c_0 = 0.2579$	(0.2401, 0.2756)	0.992	0.02294	6.8%	0.9943	0.0231303	5.7%
	$c_1 = 6.477$	(5.558, 7.396)						
	$c_2 = 1.321$	(1.239, 1.403)						

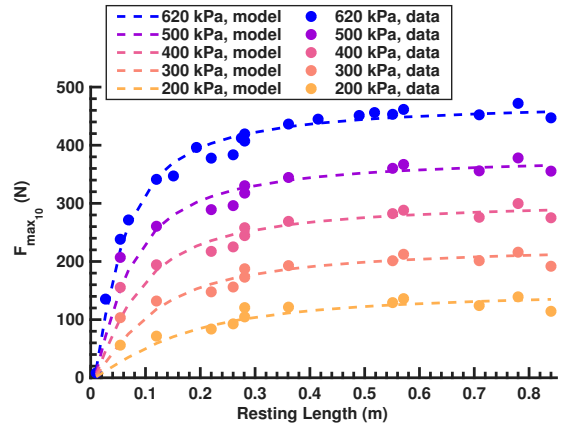
Festo tool divided by the maximum force equation described in the previous section in Figure 3.4. With this comparison, it is clear that the Festo tool over-predicts the expected force, especially at lower pressures, contraction, and resting lengths.

We therefore derived our own equation for isometric force in the BPA as a function of pressure and contraction. Visual analysis of the experimental data shows an exponential relationship between ϵ^* and F^*, and a linear relationship between P^* and F. We fit a surface to the data using Nonlinear Least Squares and a Least Absolute Residual robustness such that it is described by Equation (3.4). For the ∅10 mm BPAs, the result of the improved fit can be seen in Figure 3.4a. The adjusted $R^2 = 0.9998$, a RMSE $= 0.004537$, and a maximum absolute residual of 10.6%. Solving (3.4) for the ∅20 mm BPA, yields different coefficients, and results are seen in Figure 3.4b. Coefficient values and goodness-of-fit statistics are found in Table 3.2.

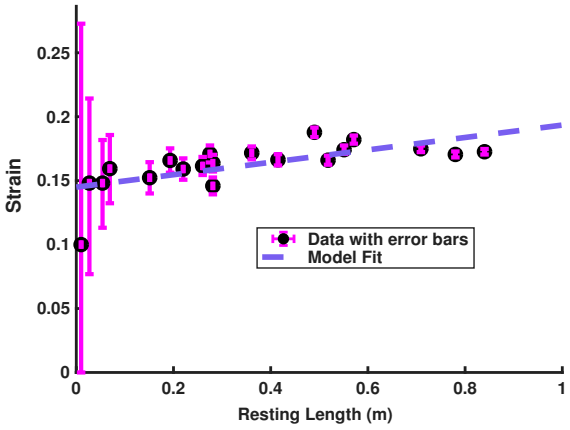
3.3 Discussion

3.3.1 Effects of Resting Length

We have developed new F_{620} equations for 10 mm and 20 mm diameter Festo BPAs that capture the change in maximum force produced by the actuators at 620 kPa as a function of their resting length. When examining Equation (3.7) and measured data in Figure 3.2b, it can be seen that as l_0 goes to infinity, $F_{620_{10}}$ goes to 470 N. When the Festo tool [Lang, Robert, 2005] was queried at $l_0 = 1$ m as an approximation of infinity, it predicts the ∅10 mm BPA can produce $F = 490$ N at P_{620}. This is within the 10% variability that Festo states may occur in manufacturing tolerances. Similarly, the Festo data predict a ∅20 mm

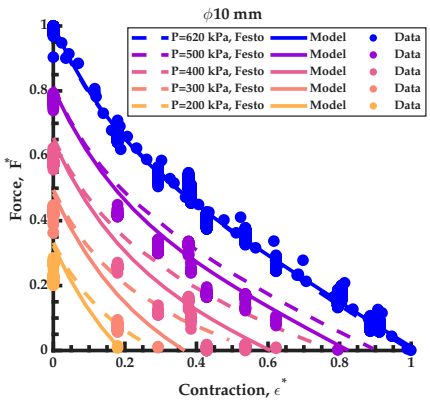


(**a**)

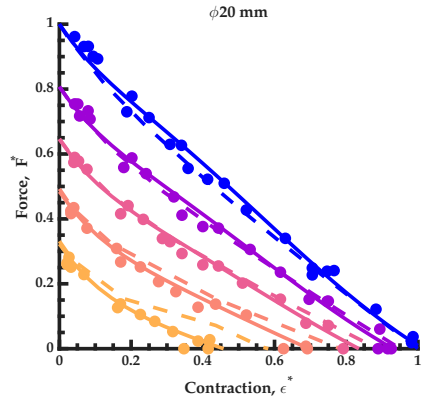


(**b**)

Figure 3.3: Results for finding the relationship between F_{620}, l_0, P_{620}, and $\epsilon_{620_{10}}$. (**a**) Isoclines of the surface fit for $F_{620_{10}}(l_0, P_{620})$. (**b**) $\epsilon_{620_{10}}$ versus l_0 at P_{620}. Although there is a general trend of longer resting lengths producing more contraction, no conclusive relationship between ϵ_{620} and l_0 could be deduced from this experiment. The error bars show the effect of being ± 1 mm in measuring l_0 and l_{620}.



(**a**)



(**b**)

Figure 3.4: Surface fit for $F^*(\epsilon^*, P^*)$. Solid lines are from our model. Dashed lines from Festo supplied data. In total, 537 data points were collected and used to fit the 3D surfaces (321 data points for ∅10 mm BPAs and 216 data points for ∅20 mm BPAs). In total, 80% of these data were used for training, and 20% were used for validation. For figure clarity, not all data is included and plotted circles represent collected data at ± 10 kPa of the stated pressure. (**a**) Fit data for ∅10 mm Festo BPAs. (**b**) Fit data for ∅20 mm Festo BPAs.

BPA will produce $F = 1570\,\text{N}$. Our results for Equation (3.8) show that $F_{620_{20}}$ goes to 1460 N as l_0 approaches infinity. Although the difference is much larger, it is also within the 10% manufacturing tolerance.

Our measured data differ significantly from the Festo model as l_0 goes to zero. For the measured data, as the length gets smaller, the force does as well, whereas the Festo tool predicts exponential force increase (see Figure 3.2). As a specific example, when specifying $l_0 = 112\,\text{mm}$ for the 10 mm diameter BPAs, the Festo predicted F_{620} is 498.6 N. However, using Equation (3.7), F_{620} is calculated to be 335.3 N. Actual force in the $l_0 = 112\,\text{mm}$ was measured at 350.9 N. Therefore, the error in predicted F_{620} for the ∅10 mm BPA is 4.5% for our model and 42.1% using the Festo tool. Researchers using BPA resting lengths under 300 mm should take note of these results.

Festo AG & Co. KG [2026] (p. 12) states that force will be reduced by approximately 10% for a minimum nominal length. In Festo AG & Co. KG [2018] (p. 14) they say the force can deviate by 10% based on factors such as production variation, material variation, and deviation from nominal length (l_0). Festo AG & Co. KG [2013] (p. 13) notes that testing was done with BPAs that are ten times the uninflated diameter, and that theoretical force can increase by up to 10%. In each datasheet, the characteristic curves appear to be the same. As demonstrated above, the 100 mm ∅10 mm BPA should have much less force than what is shown in the curve. This is our experience and the reason for the testing. It is unclear why the MuscleSIM tool [Lang, Robert, 2005] then predicts higher force as resting length decreases. It could be a reversed sign coefficient in their internal curve-fit model. We have shown here that as the nominal length goes to zero, so does force.

BPA models commonly predict higher forces than are measured experimentally because

they do not account for end effects in regions of the pressure vessel where those effects dominate. Nominally, the clamped ends fix the inner and outer diameter while allowing axial and radial displacement. In the isometric case, there is only radial displacement. The boundary conditions force steep gradients in stress and strain near the ends. Near the boundary conditions, axial, radial, and circumferential strains deviate from analytic solutions for a cylindrical model. For a constant pressure, the end effect region appears to have the same length. Therefore, in shorter BPAs, the end effects dominate. The axial length scale of the boundary section is comparable to l_0, and the active force differs substantially from that of an assumed perfect cylinder.

Manufacturing variations seem to play an important role in the behavior of these actuators. It remains to be seen how ϵ_{620} can be known a priori. We still suspect that it is a function of l_0, but it might also be a function of other factors such as product batch number or UV light exposure [Festo AG & Co. KG, 2026] (p. 15); [Festo AG & Co. KG, 2018] (p. 6); [Festo AG & Co. KG, 2013] (p. 13). We found there to be larger variations in the 10 mm BPAs than in the 20 mm BPAs. Modeling work on ideal McKibben actuators describes how the relationship between contraction and initial length is a function of the total number of twists and the angle of twist in the usable muscle area [Chou and Hannaford, 1996]. It is unclear if there are variations in the angle and number of twists in the Festo BPAs due to manufacturing. Uncovering the relationship between ϵ_{620} and l_0 for the Festo DMSP artificial muscles, if it exists in a meaningful way, will require additional controlled tests.

Similarly, it is not clear how the maximum force might change with manufacturing variations. This was reasonably corrected for by normalizing the equations to each individual's maximum contraction, but more nuanced details might emerge with more

data collection. Not knowing these relationships a priori means there is still a need for robots and other systems to go through an iterative design stage if it is discovered that the system does not produce the torque or have the RoM that the design team expects. However, the data and models provided here should enable faster redesigns with fewer additional measurements.

3.3.2 Nondimensionalized Force Prediction

We have also introduced the concept of nondimensionalized isometric force that is a function of relative strain and relative pressure (Equation (3.4)). This elegant equation has only three coefficients with low error (see Table 3.2). In the work presented here, we introduce the relative pressure term, $P^* = P/P_{620}$. Our data show that by normalizing force, contraction, and pressure, we are able to create a simplified force equation as a function of contraction and pressure that scales well with initial actuator length, as expressed by Equation (3.6) in Section 3.1.1.

For example, given $l_0 = 54\,\text{mm}$ and $P = 300\,\text{kPa}$, the measured force was $F_{actual} = 103\,\text{N}$. Force prediction using only the Festo tool predicts the force to be $F_{predict} = 270\,\text{N}$. This is 161% greater than the measured force. Using Equations (3.7) and (3.4) with Equation (3.6) gives $F_{predict} = F^*(0, 0.48) \cdot F_{620}(0.054) = 0.49 \cdot 230\,\text{N} = 112\,\text{N}$. This is an error of 8%, which is much more accurate than using the Festo tool only.

The presented work is capable of predicting force output for 10 mm and 20 mm muscles with different initial muscle lengths, internal pressure, and contraction amount with minimal initial data collection. However, it remains to be seen if these relationships

are maintained over the full lifecycle of the muscles. There are not yet any peer reviewed studies investigating the long-term behavior of these artificial muscles, though the Festo datasheet provides some indication of what can be expected [Festo AG & Co. KG, 2026] (p.19). They report that the 40 mm BPA can complete 1 million cycles at 6000 N (maximum rated load), or 10 million cycles at 4000 N. The service life of the other size BPAs is not reported; however, it is mentioned that thermal stressing may reduce the lifespan and they should be inspected every 500,000 cycles for signs of aging [Festo AG & Co. KG, 2018] (pp. 10–11). Future research should look at the behavior of these muscles over time.

In our work, $P_{620} = 620\,\text{kPa}$ is the maximum supply pressure for our system, and other users of this actuator may use a different supply pressure. However, this supply pressure is not required, and normalizing by this value was chosen as a method for improving the optimization techniques by reducing the sensitivity of the term associated with pressure. This equation will work for any pressure supplies below 620 kPa. We also anticipate the equation will work for pressures up to 800 kPa (the maximum rated pressure for Festo ∅10 mm BPAs); however, we do not have the data to confirm this.

3.3.3 Comparison with Other Published BPA Models

The proposed model was compared with the formulations by Sárosi [2012], Sárosi and Fabulya [2012], and Martens and Boblan [2017], to evaluate their predictive performance across actuator sizes and operating conditions (see Table 3.3 and Figure 3.5). For the ∅20 mm BPA, parameters from [Sárosi, 2012] were used, corresponding to actuators with resting lengths of 200 mm and 400 mm. These were compared against the experimental

data at $l_0 = 300\,\text{mm}$ and $l_0 = 450\,\text{mm}$, respectively. For the $\varnothing 10\,\text{mm}$ BPA with resting lengths of 257 mm and 233 mm, parameters from [Sárosi and Fabulya, 2012] were applied. In each case, the maximum contraction parameter (k_{max}) was determined numerically by solving $F(6.2\,\text{bar}, k_{max}) = 0$, ensuring consistency with the experimental maximum pressure condition.

While the Sárosi [2012] and Sárosi and Fabulya [2012] models capture general trends in force–contraction behavior, discrepancies were observed when applying parameters derived from different actuator lengths. In particular, the model tends to over-predict force at higher pressures and low amounts of contraction, suggesting sensitivity to geometric scaling effects not explicitly accounted for in the formulation.

The Martens and Boblan [2017] model is derived from a geometric and material-based framework and provides a more physically grounded representation of actuator behavior. However, its accuracy depends strongly on precise geometric parameters, including braid angle, wall thickness, and fiber length. Small deviations in these parameters can lead to significant differences in predicted force, particularly at higher pressures.

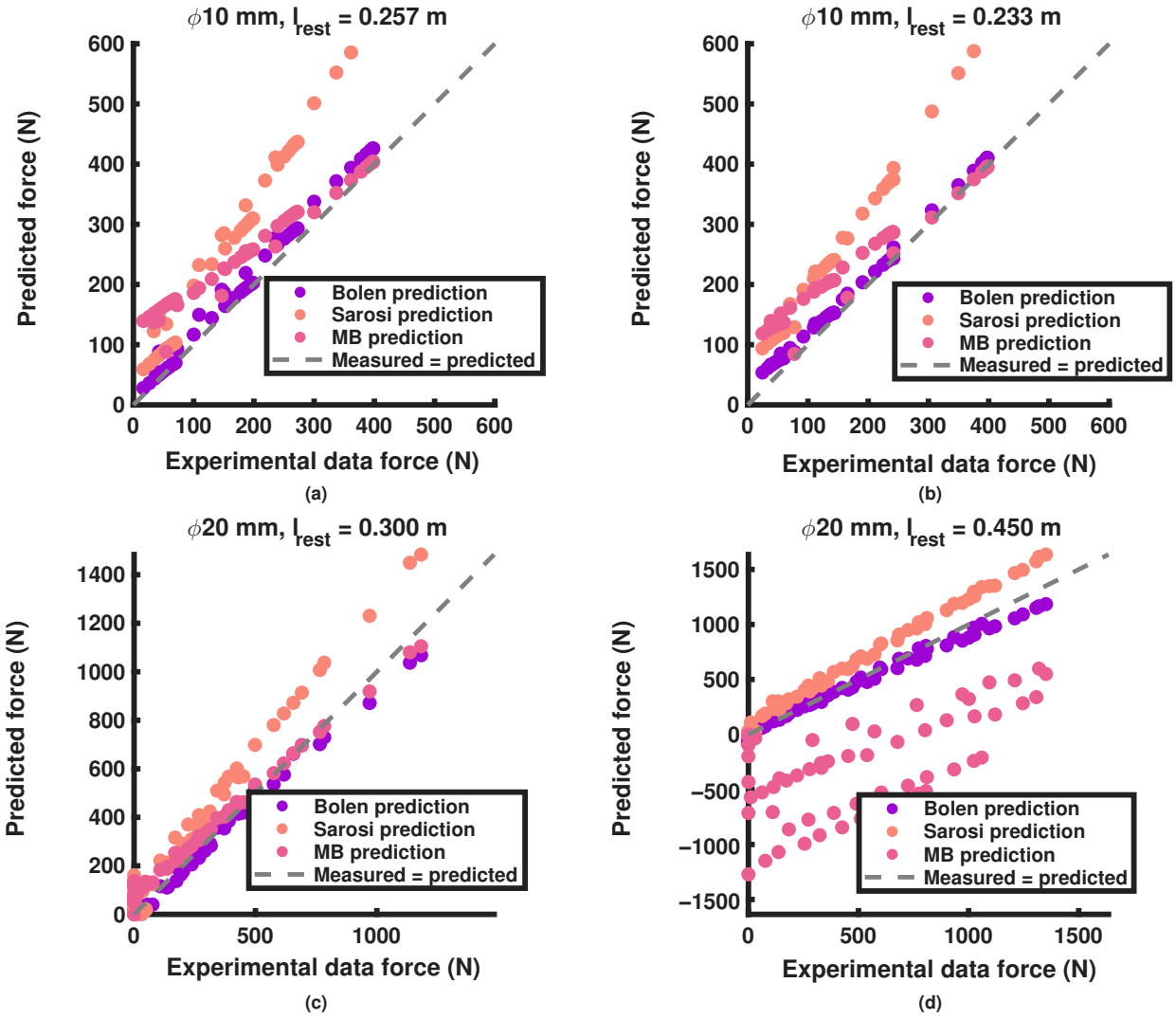


Figure 3.5: Comparison between measured and predicted BPA force at varying amounts of contraction and pressure. A $y = x$ perfect fit line is added for reference. The BPAs compared are: (**a**) ∅10 mm, $l_0 = 257$ mm, (**b**) ∅10 mm, $l_0 = 233$ mm, (**c**) ∅20 mm, $l_0 = 300$ mm, and (**d**) ∅20 mm, $l_0 = 450$ mm.

The Martens and Boblan model was first evaluated at the same pressure–contraction points as in their experimental data [Martens and Boblan, 2017] (Tables A1 and A2),

indicating that it reproduces the general published force map shape but is sensitive to the assumed initial geometry. Examining their ∅20 mm BPA, for example, using the explicitly stated $l_0 = 300\,\text{mm}$ and initial diameter $D_0 = 20\,\text{mm}$ significantly under-predicted the results compared to using the implied $l_0 = 296\,\text{mm}$ ([Martens and Boblan, 2017] (Table A1)) and $D_0 + H_0 = 21.8\,\text{mm}$, where $H_0 = 1.8\,\text{mm}$ is the material thickness. The diameter adjustment can be inferred from [Chou and Hannaford, 1996]. This adjusted diameter was also included for the ∅10 mm BPA to give a better fit. As can be seen in Figure 3.5 and Table 3.3, the highest accuracy is observed when ∅20 mm and $l_0 = 300\,\text{mm}$, which is at or very near their experimental measurements. For the ∅10 mm BPAs the accuracy decreases even though their experimentally measured $l_0 = 250\,\text{mm}$ is close to our $l_0 = 257\,\text{mm}$ and $l_0 = 233\,\text{mm}$. The largest error occurs at ∅20 mm and $l_0 = 450\,\text{mm}$ and the model starts to predict negative force. We have calculated that the $F_{PE} \cdot \frac{dD}{DL}$ term becomes very large compared to the others [Martens and Boblan, 2017] (p. 5). This suggests that the model is sensitive to geometric parameter selection and interpretation, particularly with respect to initial length and diameter definitions.

In comparison, the present normalized model demonstrates improved consistency across actuator lengths by incorporating experimentally derived scaling through F_{620} and ϵ_{620}. This normalization reduces sensitivity to geometric variation and enables a unified representation of actuator behavior, while maintaining predictive accuracy over the tested range of pressures and contractions.

3.3.4 Implications for Real-Time Control and Simulation

Although pneumatic muscle actuators exhibit hysteresis, thermodynamic effects, and other dynamic behavior, accurate and high-performance controller designs can still be achieved using a static force map approach when the underlying static force model is accurate. Martens and Boblan explicitly note that BPA dynamics are often dominated by the air dynamics inside the actuator and the mass flow behavior of the valve, such that precise modeling of the static force characteristic remains crucial for torque control accuracy [Martens and Boblan, 2017]. This view is consistent with more recent BPA control work, including a sensor-less torque control interface for BPA-driven joints [Martens et al., 2018], stable force/position/stiffness control of antagonistic pneumatic-muscle joints [Ugurlu et al., 2019, Hunt et al., 2017b], and sensor-less angle stiffness control of antagonistic BPA systems [Shin and Kogiso, 2022].

In this context, the present model is well suited for real-time use because it reduces the static isometric force prediction to a low-order normalized surface, $F^*(\epsilon^*, P^*)$, scaled by the resting-length-dependent term $F_{620}(l_0)$. This structure is computationally inexpensive to evaluate online and is also suitable for lookup-table implementation. While the present work does not model the full actuator–valve dynamics, it provides the static force relationship needed as a feedforward or inner-model component in real-time torque control, simulation, and preliminary design workflows. It is anticipated that improved real-time control for high-speed and high-precision systems will benefit more from improved modeling of airflow dynamics than from modeling actuator hysteresis, damping, and other dynamic behavior. Characterizing the actuators under dynamic operating conditions,

such as isokinetic (constant-velocity), isobaric, and isotonic (constant-load) contractions, is a natural next step beyond the isometric characterization reported here, for which the pressure–force dynamics model of Elzein et al. [2025] provides the modeling foundation.

3.3.5 Biomimetic Artificial Muscle Analogy

BPA artificial muscles are often said to be analogous to biological muscles because they have force–length curves and can produce force only in tension. This analogy is worth closer inspection. For example, the passive force–length curve has been described as an exponential function arising from connective tissue elements, while the active force–length relationship is typically modeled as a bell-shaped curve reflecting actin–myosin overlap [Winters, 1990]. These formulations emphasize that muscle force generation is governed by nonlinear geometric and material interactions rather than simple linear elasticity. McKibben-type pneumatic artificial muscles (of which Festo is an example) exhibit analogous nonlinear behavior, though arising from fundamentally different physical mechanisms. As shown by Chou and Hannaford, the force–length relationship of McKibben actuators is derived from geometric constraints imposed by the braided shell, leading to a nonlinear dependence of force on actuator length through the braid angle [Chou and Hannaford, 1996]. In their formulation, actuator force is proportional to pressure and a nonlinear geometric term $(3\cos^2\theta - 1)$, which governs both the maximum contraction limit and the curvature of the force–length relationship. The work presented here demonstrates that this nonlinear curve can be well approximated by an exponential as well.

However, the behavior of these artificial muscles differs distinctly from that of biological muscles. For example, the optimal fiber length of a biological muscle, l_{OFL}, is not equivalent to l_0 in an artificial muscle. Artificial muscles produce their maximum active force at their resting length, which is also their maximum length. In contrast, the maximum active force that a biological muscle can produce is somewhere in the middle of the total contractile length of the muscle.

Furthermore, biological muscles can adapt the optimal operating length through structural remodeling (e.g., sarcomere addition or tendon compliance) while BPAs have a fixed geometric configuration. This allows biological systems to self-optimize for the particular tasks the animal is subjected to. In order to re-optimize a robot for new tasks, new actuators would have to be swapped into the system. This distinction highlights a key limitation of BPAs in biomimetic applications, while also reinforcing the importance of accurate nonlinear modeling for control and design.

The analogy also suggests two design observations. First, the force–length penalty is shared by both families: force falls as the actuator operates away from its optimum, so a limb design must route each actuator such that its highest-force operating range coincides with the joint angles where torque is needed most, because the force lost at the extremes of the range of motion cannot be recovered by pressure alone. Second, the two families gain moment arm in different ways. Skeletal muscle attaches to the skeleton itself, through tendons or directly over broad areas of bone, so the moment arms of large animals follow from skeletal geometry. The robot's BPAs instead act through brackets that protrude from the limb segments; that hardware is mechanically exposed, and its series compliance is one of the loss mechanisms quantified in Chapter 4.

The analogy finally raises a question that this dissertation can pose but not settle. Every muscle-like actuator loses force as it operates away from its optimal length, and a soft actuator working against soft structure has no rigid skeleton from which to gain a moment arm. Given those two facts, it may be no accident that nearly all large land animals are built on rigid skeletons and that large soft-bodied land animals are essentially absent. Whether the force–length penalty contributes to that absence, or whether it reflects other constraints such as support and protection, is a question for comparative biomechanics rather than for a single actuator study. For the designer, the lesson holds regardless of the answer: some moment arm must come from somewhere, and on this robot it comes from protruding brackets whose behavior must be modeled rather than ignored.

Table 3.3: BPA force-model comparison. Diameter and resting length are in mm; RMSE and maximum absolute error are in N. FVU is dimensionless.

Diameter	Length	Model	RMSE	FVU	Max. error
10 mm	257	Bolen	22.9	0.0353	46.0
10 mm	257	S'arosi	143.8	1.3903	235.8
10 mm	257	Martens and Boblan	73.8	0.3668	122.9
10 mm	233	Bolen	16.8	0.0187	30.3
10 mm	233	S'arosi	132.7	1.1655	222.6
10 mm	233	Martens and Boblan	62.6	0.2590	101.2
20 mm	300	Bolen	35.6	0.0144	114.0
20 mm	300	S'arosi	132.1	0.1992	313.4
20 mm	300	Martens and Boblan	72.8	0.0605	137.8
20 mm	450	Bolen	67.6	0.0258	166.6
20 mm	450	S'arosi	177.7	0.1785	294.8
20 mm	450	Martens and Boblan	862.1	4.2022	1321.1

4 Isometric Knee Torque

Publication note: The lumbosacral actuator-placement summary at the beginning of Section 4.1.5 summarizes work published by Morrow et al. [2020]. For that published work, I contributed validation, formal analysis, investigation, resources, data curation, manuscript review and editing, and visualization. Connor Morrow contributed conceptualization, methodology, software, validation, formal analysis, investigation, resources, data curation, the original draft, manuscript review and editing, visualization, supervision, and project administration. Alexander J. Hunt contributed validation, resources, supervision, project administration, and funding acquisition. The remaining material in this chapter forms a manuscript in preparation by Bolen, Burgess, Morrow, Elzein, Pang, Scharzenberger, and Hunt [Bolen et al., 2025]. For the manuscript in preparation, B.B. performed the conceptualization, methodology, software, validation, formal analysis, investigation, data curation, visualization, and original draft preparation; M.E., L.P., L.B., and C.M. contributed methodology, software, investigation, and data curation; and A.H. supervised the work, administered the project, and acquired funding. All co-authors contributed to writing, review, and editing.

This chapter connects the actuator force characterization of Chapter 3 to torque about the robot knee. The earlier torque model assumed inextensible muscle paths on rigid links. Isometric tests on pinned and biomimetic knees show where those assumptions fail. The methods develop corrections for length offsets, attachment compliance, and wrapping loss,

then explain how actuator-placement searches use the improved prediction. Validation tests evaluate the correction terms, and physical tests assess whether the redesigned routes meet selected human-muscle torque targets.

4.1 Methods

I built a modular test stand to measure flexor and extensor torque with either a pinned knee or a biomimetic four-bar knee. The tests varied actuator diameter and resting length, joint contact, and tendon attachment. I first compared measured torque with predictions based on the characterized BPA force model. I then set the length, compliance, and wrapping corrections using separate training and validation tests, and used the improved model to redesign routes against human-muscle torque targets. Isometric tests evaluated the resulting hardware.

4.1.1 Robot Leg Model and Knee Test Stands

The primary components of the robot leg are the knee joint, femur, tibia, BPA assemblies, and artificial tendons (Figure 4.1**(A–C)**). The artificial bone components are 3D printed using a combination of Onyx, carbon fiber, and high-strength high-temperature fiberglass on Markforged Onyx One and Mark Two printers, as well as machined links for the knee mechanism. Onyx is a proprietary Markforged material that consists of chopped carbon fiber in nylon. Artificial tendons are made with Shimano bicycle brake cable.

The modular test stand accommodates either knee joint and both flexor and extensor configurations (Figure 4.1). This allows measured joint torque to be compared with

predictions while changing the joint geometry and actuator route. The extensor test at high flexion also exposes the actuator deformation caused by contact with the knee.

The leg assembly geometry and dimensions are based on bones scans of a 6 ft (1.83 m) tall person [Bolen and Hunt, 2019, Seth et al., 2018, Delp et al., 1990, 2007]. Two knee joints were tested, one pure rotational joint, and one biomimetic joint with rotation and translation. Each joint is driven by two antagonistic Festo actuators. Both robot joints have the same center of rotation in the femur frame at $\theta_{\mathbf{k}} = 0\,\text{deg}$.

Details on the biomimetic robot knee used in this study can be found in [Steele, 2018, Steele et al., 2017, 2018]. In brief, this 1-DoF joint (Figure 4.2) uses a four-bar linkage to change the effective moment arm during rotation. The intersection point of the internal crossed links define the instantaneous center of rotation (ICR). This causes the joint ICR location in the femur frame to translate posteriorly in the X direction and distally in the Y direction except for a brief proximal move near full flexion. Reference International Society of Biomechanics (ISB) coordinate frames are given by Wu and Cavanagh [1995], Wu et al. [2002]. This is similar to how the human knee behaves [Bolen and Hunt, 2019, Morrow et al., 2020, Seth et al., 2018, Delp et al., 2007, 1990]. The correct location of the ICR is important for both theoretical and experimental torque calculations.

Both knee types use uniarticular flexor and extensor actuators. Each knee joint type was tested with a single ∅10 mm Festo BPA per muscle. The biomimetic knee was additionally tested with ∅20 mm Festo BPAs in a paired configuration: the extensor with two BPAs in parallel, and the flexor with two BPAs whose pathways are mirrored about the Z axis. Each artificial muscle is pinned at the muscle origin location on the femur while the other end is either pinned or attached to the muscle insertion location on the tibia via an artificial

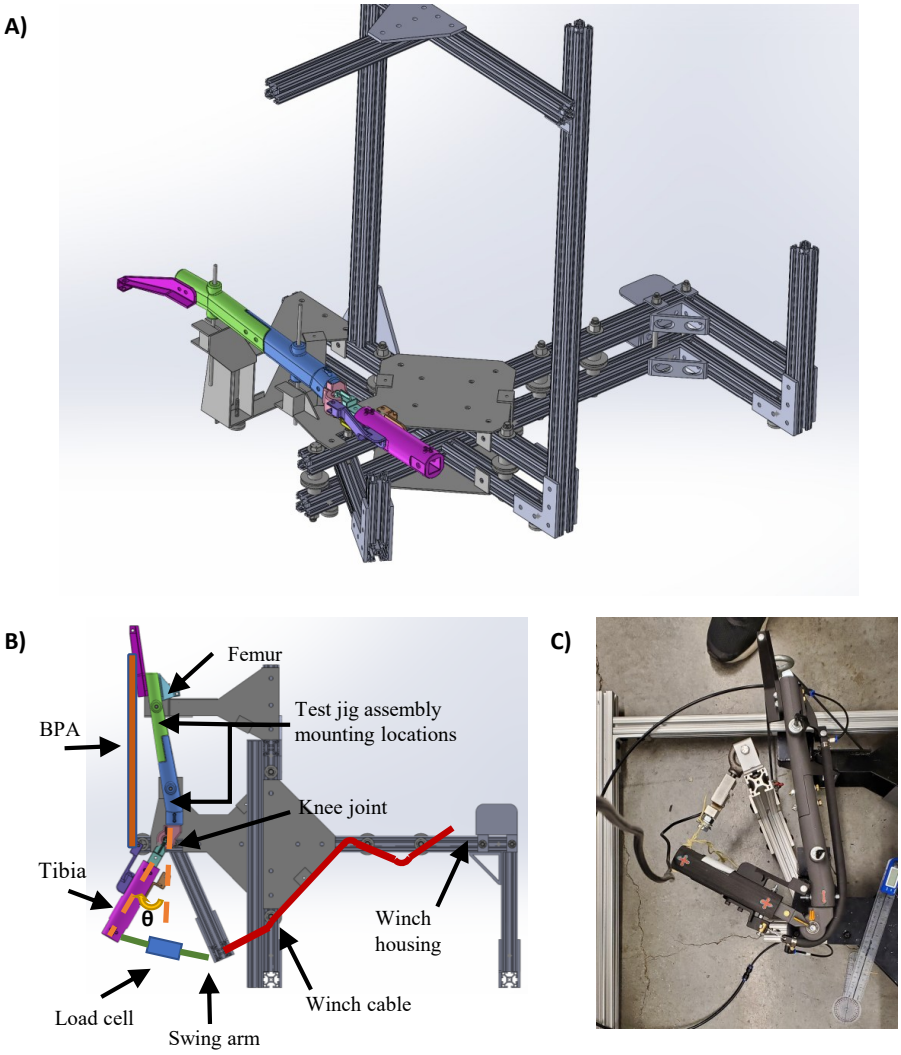


Figure 4.1: Pinned-knee test apparatus: CAD assembly (A), labeled mechanism at -30° (B), and extensor isometric test at -120° (C).

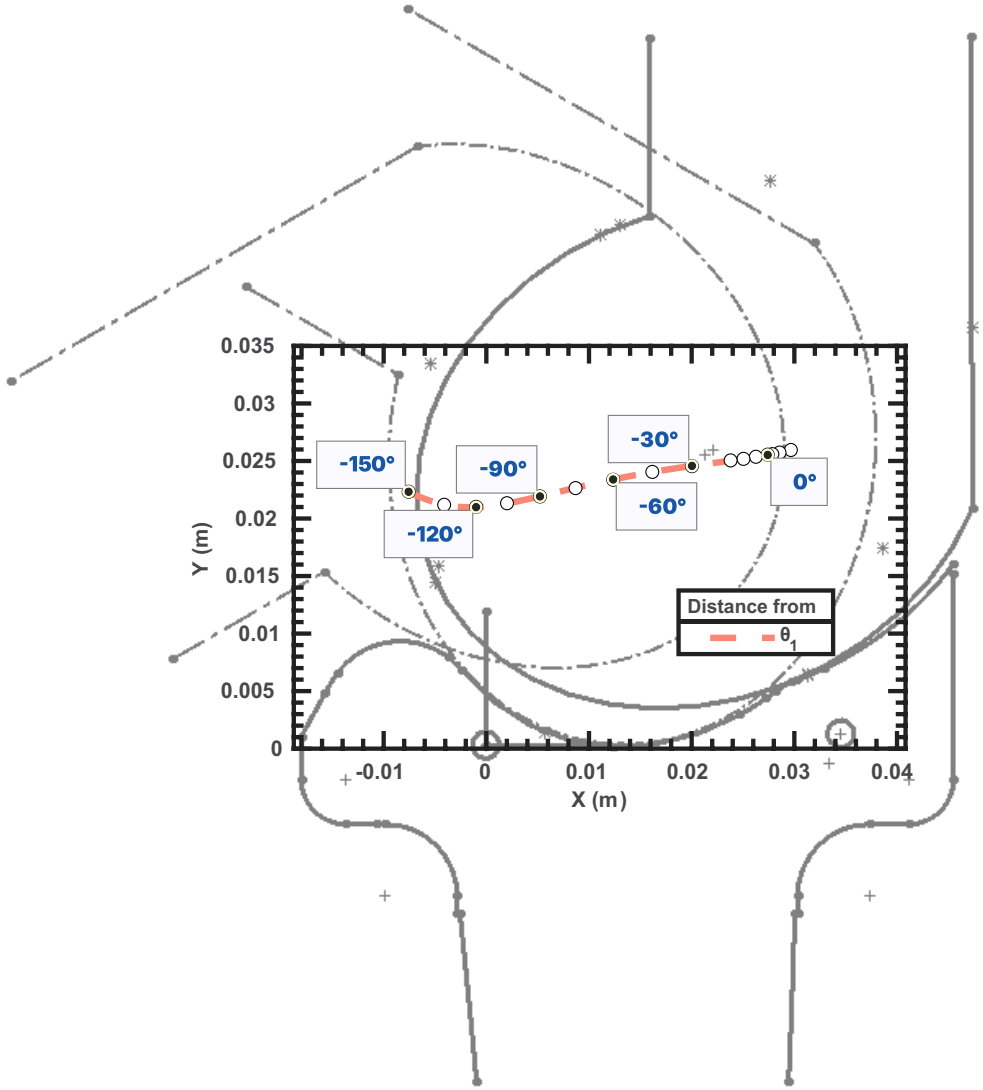


Figure 4.2: Biomimetic-knee instantaneous center of rotation relative to the tibial frame. White circles are measured points; knee outlines show 0° (solid), -60°, and -120° (dash-dot).

tendon.

The modeled geometry and torque calculation follow Bolen and Hunt [2019], Morrow et al. [2020]. Torque about the knee is generated by the moment of the BPA force about the joint's instantaneous center of rotation (ICR). Two vectors enter that moment: the position vector $\mathbf{d}$, drawn from the ICR to the point where the actuator force is applied, and the force vector $\mathbf{F} = \|\mathbf{F}\|\hat{\mathbf{u}}$, whose direction $\hat{\mathbf{u}}$ is the line of action of the actuator. The moment of the force about the joint is $\mathbf{M} = \mathbf{d} \times \mathbf{F}$, and the signed knee-axis torque is $M_{\mathbf{k}} = \hat{\mathbf{z}}_{\mathbf{k}} \cdot (\mathbf{d} \times \mathbf{F})$. For force acting in the plane normal to the knee axis, the moment-arm magnitude is

$$r = \|\mathbf{d} \times \hat{\mathbf{u}}\| \tag{4.1}$$

This planar moment arm is a scalar distance and can be computed from the vectors or measured directly. For an out-of-plane force, the knee-axis moment must instead use the axis projection above; the magnitude of the full three-dimensional moment is not generally the knee torque. In this study it is calculated using the method specified by Young et al. [2019]. The scalar moment arm $r_{\mathbf{k}}$ about the joint axis $\hat{\mathbf{z}}_{\mathbf{k}}$ (i.e., the Z axis in the knee frame) is calculated as

$$r_{\mathbf{k}} = \mathbf{p}_{proj,xy} \cdot \frac{\mathbf{p}_f \times \hat{\mathbf{z}}_{\mathbf{k}}}{\|\mathbf{p}_f \times \hat{\mathbf{z}}_{\mathbf{k}}\|} \tag{4.2}$$

A muscle path may be modeled as a single segment running from origin to insertion, or as a chain of segments joined at via points. The segment that crosses from one reference frame into the other, here from the femur frame to the tibia frame, is the free muscle segment: because its endpoints ride on different bodies, its length and its direction respond

directly to joint rotation, whereas a segment contained within one body's frame rotates rigidly with that body and changes neither. Here $\mathbf{p}_f$ is the force-path segment vector, and $\mathbf{p}_{proj,xy}$ locates a point on that line relative to the joint, projected onto the plane normal to the joint axis. The normalized cross product in Equation (4.2) is the in-plane normal to the force path. Its dot product with the projected position gives the signed moment arm of the projected force path (Figure 2.6). Multiplying this arm by the in-plane force magnitude gives the knee-axis torque; the full wrench calculation retains all three force components. Moment arm is calculated in the tibial body frame.

4.1.2 Knee Torque Data Collection and Equipment

A test jig was built to collect knee torque measurements at different knee angles over its range of motion. The test stand frame is made predominantly out of 80/20 1515 series extruded aluminum components. The knee joint is free to rotate the tibia while the femur is fixed to the frame. A force sensor has one end connected to the tibia and the other is connected to a swing arm. The swing arm is tied to a winch with 3/16 in Kevlar rope from Quality Nylon Rope routed over intermediate idler pulleys (see Figure 4.1**(B)** and Figure 4.1**(C)**). At each measured position of $\theta_{\mathbf{k}}$ the winch would be locked and the BPA would be inflated and then deflated. This test procedure kept the measured results as isometric as possible. The knee joint is co-axial with gravity to eliminate gravitational torque (see Figure 4.1**(A)** and Figure 4.1**(C)**).

Force data was collected using one of two different sensors. The first is a MODERN STEP 300 kg digital crane scale. The second force sensor is a CALT DYLY-103 100 kg S

shaped load cell. The load cell is used in conjunction with a HX711 Load Cell Amplifier. Pressure is measured with a Freescale MPX5700 GP 5 V pressure sensor. Building air supply pressure is controlled with two pressure regulators in series. The first is a Parker model 20R113GC 0 psi to 120 psi pressure regulator. The second is a Husky 3/8 in High Performance Air Regulator HDA72200. A Festo VTUG-10-MRCR-S1T-26V20-T516LA-UL-T532S-8K valve manifold was used to supply air to the actuator. Pressure and load cell amplifier data are sent to MATLAB via an Arduino Uno style Sparkfun BlackBoard C microcontroller.

The length measurements were made using a tape measure. When there was sufficient curvature in the BPA during knee extensor torque tests, a flexible piece of string was used to mark the axial length and then transferred to the tape measure. Angle measurements were taken with a Medigauge digital electronic goniometer or with MALENOO analog goniometers of 6 in, 8 in and 12 in lengths.

4.1.3 Hybrid Torque Calculation Method

The hybrid torque calculation measures the actual muscle length and angles of the robot to calculate force and joint torque, isolating which part of the torque equation did not achieve its predicted value. The classical mechanics way to calculate the torque about a joint is to take the cross product of the position vector **d** and the force vector **F**, both defined in Section 4.1.1,

$$\mathbf{M} = \mathbf{d} \times \mathbf{F} \tag{4.3}$$

Knee torque is the scalar component of this moment about the joint axis, written $M_{\mathbf{k}}$ to make its axis explicit. For the planar hybrid calculation, its magnitude is the measured perpendicular arm multiplied by the actuator-force magnitude,

$$|M_{\mathbf{k}}| = r\,F \tag{4.4}$$

where r is the scalar moment arm of Eq. (4.1), here measured directly with a tape measure as the perpendicular distance from the knee axis to the actuator's line of action, and l_{620} is the BPA length at 620 kPa. The scalar force magnitude F is calculated by measuring the BPA's P, l_0, l_{620} and l_{m} values and using the model fit equations determined in the previous chapter. The hybrid method therefore replaces the model-predicted moment arm $r_{\mathbf{k}}$ of Eq. (4.2) and the model-predicted muscle length with their measured counterparts, leaving the force model as the only predicted quantity. The flexor and extensor signs are then assigned using the adopted knee convention.

4.1.4 Improved Torque Model

The classical torque calculation treated links and attachments as rigid: the routed path depended on joint geometry but did not change under actuator load. Data collected during the first flexor experiments (Section 4.2) showed that this assumption does not hold for a pneumatic system: measured muscle lengths were consistently shorter than predicted, and measured torque deviated from prediction in a load-dependent manner. This section describes the corrections that were developed to account for a constant length offset, series compliance in the attachment hardware and artificial tendons, and the loss of

usable muscle length when a BPA wraps around the joint.

The correction terms represent the combined effects of the attachment brackets, fixtures, artificial tendons, and test-stand loading path. They are identified from measured torque because the torque prediction requires their aggregate effect on actuator length, rather than a separate structural model of every component. The constant term χ_0 corrects the unloaded length discrepancy, χ_1 and χ_2 describe effective stiffness in the local bracket frame, and χ_3 scales the wrapping-length loss. The compliance calculation couples force and displacement: force deflects the attachments, and the resulting change in muscle length changes force. The multiobjective procedure of Section 4.1.4 evaluates candidate terms on fitting and validation tests.

Length Offset and System Stiffness

Measured muscle length was consistently shorter than predicted at all joint angles, producing less muscle force, a shifted RoM, and less torque in the experimental setup. Because the discrepancy persists at $\varepsilon = 1$ and $F = 0$, where no compliance can act, the correction is a constant length offset: l_{m} was recalculated as

$$l_{\mathrm{m}} = l_{LMT} - 2\, l_{fitting} - \chi_0 \tag{4.5}$$

where l_{LMT} is the path length of the musculotendon from origin to insertion point, $l_{fitting}$ is the length of the end caps, and χ_0 is a constant length offset.

Actuator force deforms the attachments, changing effective BPA length and the predicted force. I approximate this load-dependent deformation with Hooke's law along the

bracket-frame principal directions. The fitted stiffnesses χ_1 and χ_2 represent the combined attachment system, not direct measurements of individual hardware components. Because fixtures and loading hardware contribute to that compliance, beam theory alone cannot determine the fitted values.

A local bracket frame was established at the attachment bracket. The $\hat{\mathbf{x}}_{br}$ axis was aligned with the vector from the bracket reference point to the BPA attachment point and represents the stiffest direction of the bracket. The $\hat{\mathbf{y}}_{br}$ axis was chosen to point in the posterior direction with respect to the tibia frame, corresponding to the primary cantilever bending direction (see Figure 4.3), while the $\hat{\mathbf{z}}_{br}$ axis completes the right-handed frame. In this frame the bracket is modeled as three independent linear springs along the principal axes, with stiffnesses k_x, k_y, and k_z in the x_{br}, y_{br}, and z_{br} directions.

Both attachment brackets, the insertion-side bracket on the tibia and the origin-side bracket on the femur, carry two-rotation frames: a yaw θ_z about the fixed space-frame axis $\hat{\mathbf{z}}_s$ followed by a pitch about the current $\hat{\mathbf{y}}_{br}$ axis aligns $\hat{\mathbf{x}}_{br}$ with the vector from the bracket origin to its attachment point, $\mathbf{p}_B$ on the insertion bracket and $\mathbf{p}_A$ on the origin bracket (Figure 4.4). Appendix C records both frames, their reference points, and their stiffness forms (Equation (C5), Tables C8 and C9). The resulting orientation and homogeneous transform are

$$\mathbf{R}_{sb} = \mathbf{R}_z(\theta_z)\mathbf{R}_y(\theta_y)^T. \tag{4.6}$$

$$\mathbf{T}_{sb} = \begin{bmatrix} \mathbf{R}_{sb} & \mathbf{p}_{br} \\ \mathbf{0}^T & 1 \end{bmatrix}. \tag{4.7}$$

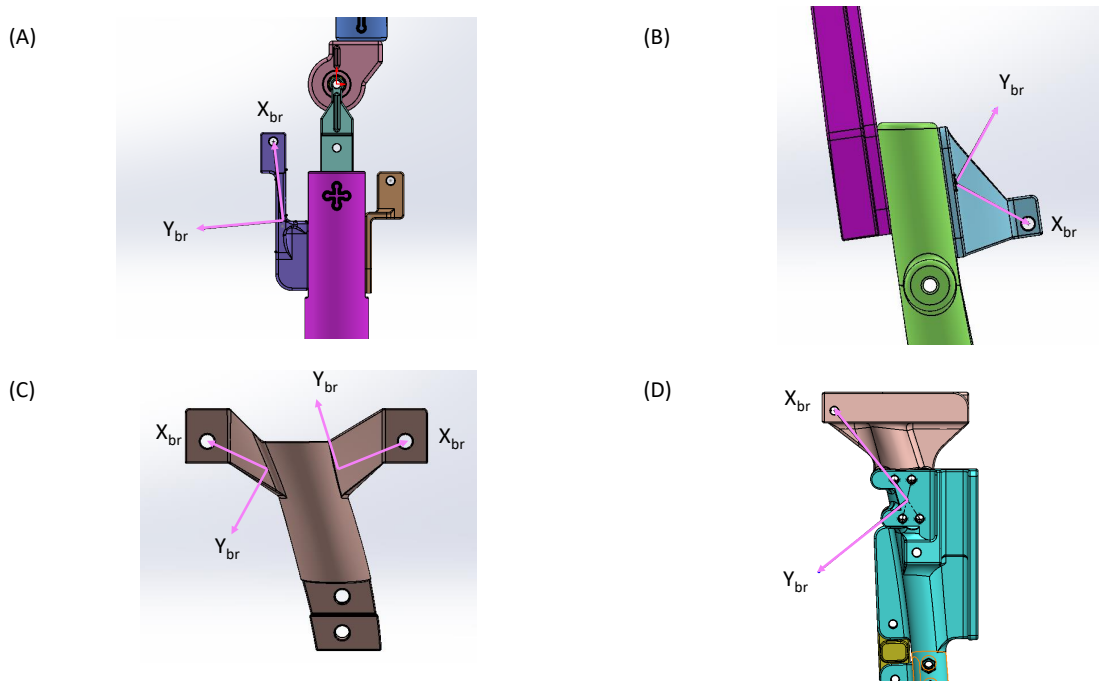


Figure 4.3: Bracket frame orientations for: **(A)** flexor configuration on the pinned knee, **(B)** extensor configuration on the pinned knee, **(C)** flexor and extensor configurations on the biomimetic knee with ∅10 mm BPAs, and **(D)** flexor configuration on the biomimetic knee with the ∅20 mm BPA.

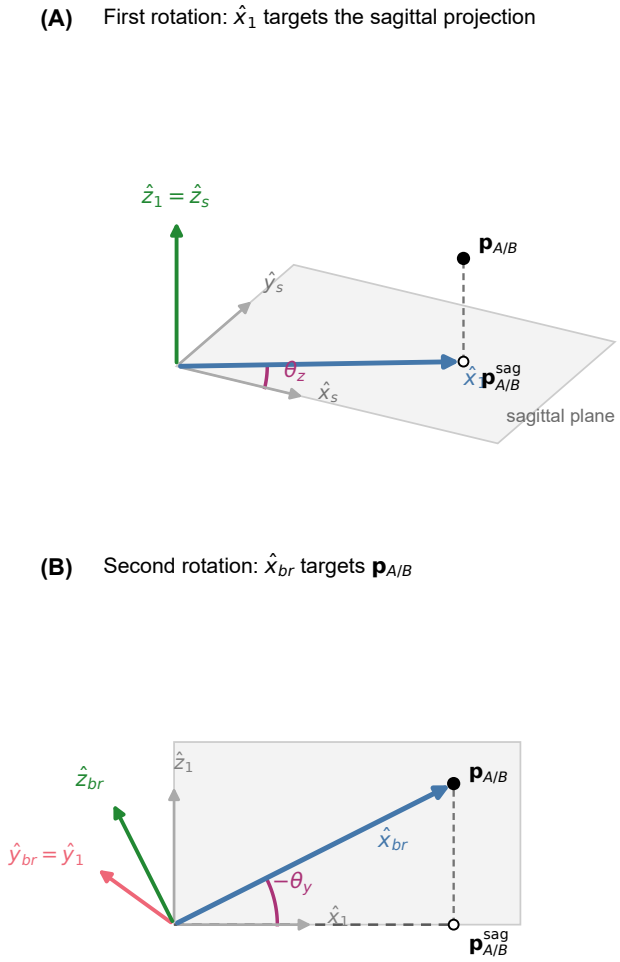


Figure 4.4: Bracket-frame construction from attachment point $\mathbf{p}_A$ or $\mathbf{p}_B$: yaw about fixed $\hat{\mathbf{z}}_s$ aligns the sagittal projection (A); pitch about current $\hat{\mathbf{y}}_{br}$ aligns the full point vector (B).

$$\mathbf{K}_{br} = \mathrm{diag}\big(k_x,\ k_y,\ k_z\big) \tag{4.8}$$

The fitted values χ_1 and χ_2 define the principal stiffnesses, with configuration-specific axis assignments in Table C9. The adopted pinned-flexor fit uses two brackets with two-rotation frames. For the insertion bracket, $k_x = k_z = \chi_1$ and $k_y = \chi_2$; this symmetric assignment makes rotation about $\hat{\mathbf{y}}_{br}$ immaterial to its compliance. The origin bracket uses $[\chi_1, \chi_1, \chi_2]$, for which that invariance does not hold. Appendix C records the remaining assignments. For the insertion bracket,

$$\mathbf{K}_{br} = \begin{bmatrix} \chi_1 & 0 & 0 \\ 0 & \chi_2 & 0 \\ 0 & 0 & \chi_1 \end{bmatrix} \tag{4.9}$$

The corresponding local compliance matrix is

$$\mathbf{C}_{br} = \mathbf{K}_{br}^{-1} \tag{4.10}$$

The force applied to the bracket was expressed in the bracket frame as $\mathbf{F}_{br}$. This force vector can be written in terms of its magnitude and unit direction as

$$\mathbf{F}_{br} = F\,\hat{\mathbf{u}} \tag{4.11}$$

where $F = \|\mathbf{F}\|$ is the scalar force magnitude of Eq. (4.4) and $\hat{\mathbf{u}} = [u_x, u_y, u_z]^T$ is the BPA force direction expressed in the bracket frame. The bracket deflection vector, $\boldsymbol{\epsilon}_{br}$, was calculated from Hooke's law as

$$\mathbf{F}_{br} = \mathbf{K}_{br}\,\boldsymbol{\epsilon}_{br} \tag{4.12}$$

Solving Eq. 4.12 for the bracket deflection gives

$$\boldsymbol{\epsilon}_{br} = \mathbf{K}_{br}^{-1}\,\mathbf{F}_{br} = \mathbf{C}_{br}\,\mathbf{F}_{br} \tag{4.13}$$

Substituting Eq. 4.11 into Eq. 4.13 gives

$$\boldsymbol{\epsilon}_{br} = F\,\mathbf{K}_{br}^{-1}\,\hat{\mathbf{u}} \tag{4.14}$$

For the diagonal stiffness matrix in Eq. 4.9, Eq. 4.13 is equivalent to the component-wise expression

$$\boldsymbol{\epsilon}_{br} = \begin{bmatrix} F\,u_x/\chi_1 \\ F\,u_y/\chi_2 \\ F\,u_z/\chi_1 \end{bmatrix} \tag{4.15}$$

Only the component of the three-dimensional deflection vector $\boldsymbol{\epsilon}_{br}$ along the BPA line of action changes the effective BPA length, so the scalar bracket deflection entering the force model is its projection onto $\hat{\mathbf{u}}$:

$$\delta_{br} = \hat{\mathbf{u}}^T \boldsymbol{\epsilon}_{br} \tag{4.16}$$

Substituting Eq. 4.13 into Eq. 4.16 gives

$$\delta_{br} = \hat{\mathbf{u}}^T \mathbf{K}_{br}^{-1} \mathbf{F}_{br} \tag{4.17}$$

Using Eq. 4.11, Eq. 4.17 becomes

$$\delta_{br} = F\, \hat{\mathbf{u}}^T \mathbf{K}_{br}^{-1} \hat{\mathbf{u}} \tag{4.18}$$

With $\mathbf{C}_{br} = \mathbf{K}_{br}^{-1}$, the scalar compliance along the BPA line of action is

$$c_{\hat{u},br} = \frac{\delta_{br}}{F} = \hat{\mathbf{u}}^T \mathbf{C}_{br} \hat{\mathbf{u}} \tag{4.19}$$

This projected-compliance form is used because the force is applied along $\hat{\mathbf{u}}$, while the bracket is allowed to deflect independently along its local $\hat{\mathbf{x}}_{br}$, $\hat{\mathbf{y}}_{br}$, and $\hat{\mathbf{z}}_{br}$ directions. The reduced-order Hooke's law expression for the bracket deflection along the BPA line of action is therefore

$$F = k_{eq}\, \delta \tag{4.20}$$

Here k_{eq} is the equivalent stiffness of every compliant element in the actuator path; with no tendon it reduces to $k_{eq} = c_{\hat{u},br}^{-1}$ of Equation (4.19). An artificial tendon is modeled as an axial spring with stiffness k_{ten}: bracket deflection and tendon stretch both shorten the effective BPA length along the actuator line of action, the two elements carry the same force, and their scalar compliances add in series.

The projected bracket compliance is $c_{\hat{u},br} = k_{\hat{u},br}^{-1}$, tendon compliance is $c_{ten} = k_{ten}^{-1}$, and the equivalent stiffness is $k_{eq} = c_{eq}^{-1}$.

$$c_{eq} = c_{\hat{u},br} + c_{ten} \tag{4.21}$$

The series model assumes quasi-static tensile loading, linear compliance, additive axial displacements, and a tendon without preload. Rotational compliance and force–moment coupling are neglected. Within each force-balance calculation the line of action is fixed; the attachment positions, force direction, and moment arm are then recalculated at the deflected geometry. Antagonistic loading that lengthens the actuator lies outside this treatment.

Since δ is the scalar compliance displacement along the BPA line of action, then the compliance-corrected active BPA length used in the force balance is

$$l_m(\delta) = l_m(0) - \delta \tag{4.22}$$

where $l_m(0)$ is the initial muscle length calculated from (4.5).

For a specified set of correction parameters, the following calculation solves the attachment-compliance displacement at each knee angle; it is distinct from the optimization that identifies those parameters from experimental torque data (Section 4.1.4). The corresponding relative contraction is $\varepsilon^*(\delta)$. Since the BPA force depends on contraction, and contraction depends on the compliance displacement, δ is the root of the nonlinear force-balance equation

$$F_{\max} \cdot F_{\mathrm{BPA}}(\varepsilon^*(\delta), P^*) - k_{eq}\,\delta = 0 \tag{4.23}$$

MATLAB `fzero` solves Equation (4.23) between zero displacement and the maximum

usable contraction, using its default tolerance. If the actuator is already at that contraction limit or the residual has no sign change, the evaluator returns $\delta = 0$. Solver tolerance is distinct from physical-model accuracy.

After solving Equation (4.23), I used the force to calculate vector bracket displacement, transformed the displaced attachment into the limb frame, and recalculated routed length, force direction, moment arm, and torque.

The actuator path can carry compliant hardware at both ends: the insertion-side bracket of the preceding development together with a second, origin-side bracket. Figure 4.5 distinguishes the three-dimensional displacement of each bracket from the scalar displacement along the BPA force path. For bracket j, the displacement resolves into components $\eta_{1,j}$, $\eta_{2,j}$, and $\eta_{3,j}$ along the local bracket-frame basis, whereas only its projection $\delta_{br,j}$ along the corresponding force direction $\hat{\mathbf{u}}_j$ changes the actuator length. The two bracket projections act in series with the tendon elongation δ_{tendon}:

$$\begin{aligned} \boldsymbol{\eta}_{br,j} &= \sum_{i=1}^{3} \eta_{i,j} \hat{\mathbf{e}}_i^{(j)}, \\ \delta_{br,j} &= \hat{\mathbf{u}}_j^T \boldsymbol{\eta}_{br,j}, \qquad j \in \{1,2\}, \\ \delta_{\text{series}} &= \delta_{br,1} + \delta_{\text{tendon}} + \delta_{br,2}. \end{aligned} \tag{4.24}$$

Actuator Bending

As the knee flexes, the extensor BPA contacts the anterior joint surface and bends around it (Figure 4.6). The rigid routed-length calculation does not account for the resulting change

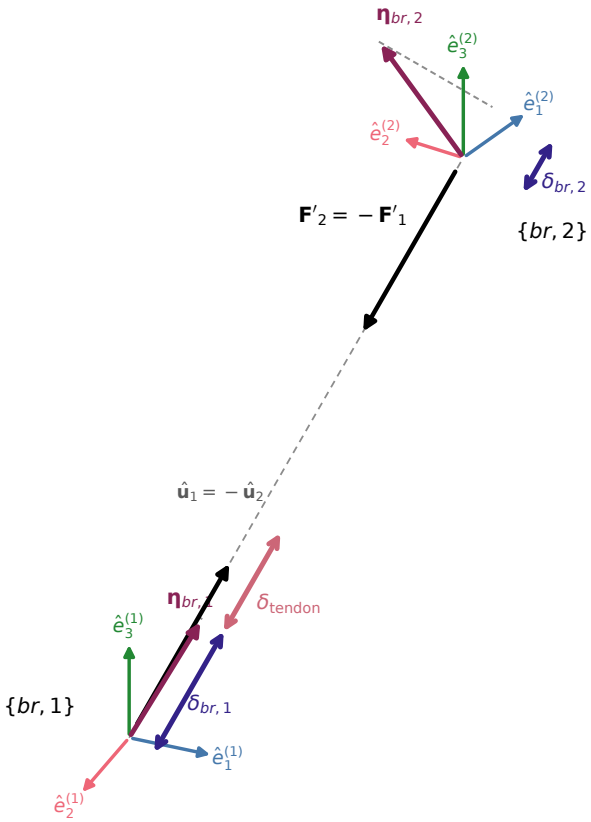


Figure 4.5: Projection of bracket displacement onto the actuator force path. Each bracket has a local basis $\hat{\mathbf{e}}_1^{(j)}$, $\hat{\mathbf{e}}_2^{(j)}$, and $\hat{\mathbf{e}}_3^{(j)}$. The displacement $\boldsymbol{\eta}_{br,j}$ has an axial projection $\delta_{br,j}$ along $\hat{\mathbf{u}}_j$; the two bracket projections and tendon elongation contribute to the change in actuator length. Displacements are exaggerated.

in usable contraction. I represent this effect with fictive strain: an empirical increase in the contraction seen by the force model, beyond the actuator's geometric contraction. The correction is phenomenological; it does not resolve wall stresses or the deformation of the braided actuator. A wrapping-loss term χ_3 scales the associated reduction in usable length:

$$\Delta l = \begin{cases} \chi_3 \; R \; |\theta_{\text{wrap}} - \theta_{\mathbf{k}}| \; \zeta^2 & \theta_{\mathbf{k}} \leqslant \theta_{\text{wrap}} \\ 0 & \theta_{\mathbf{k}} \geqslant \theta_{\text{wrap}} \end{cases} \tag{4.25}$$

where Δl is the change in usable length of the BPA, R is the wrapping radius, $|\theta_{\text{wrap}} - \theta_{\mathbf{k}}|$ is the angle through which the BPA is bent once contact begins, and θ_{wrap} is the knee flexion angle at which the BPA starts to bend over the knee joint, determined from solid modeling. ζ is the additive complement to relative strain, such that

$$\zeta = 1 - \varepsilon^* \tag{4.26}$$

so the fitted wrapping loss grows quadratically toward resting length and vanishes at the normalized free-contraction limit. Combining (4.26) into (4.25), and subtracting Δl from (4.5) yields

$$l_{\text{m}} = l_{LMT} - 2 \, l_{fitting} - \chi_0 - \Delta l \tag{4.27}$$

Figure 4.6 shows a sagittal schematic of an extensor BPA crossing a flexed robot knee, with its braided body following an anterior contact arc between the proximal attachment and distal tendon. Dashed radii bound the contact arc; R is the BPA-path wrapping

radius, and $\alpha = |\theta_{\text{wrap}} - \theta_{\mathbf{k}}|$ is the bend angle after contact begins, expressed in radians in Equation (4.25). The drawing illustrates the local circular-wrap approximation, not the full multi-contact route. The fitted usable-length loss Δl is not an arc-minus-chord length.

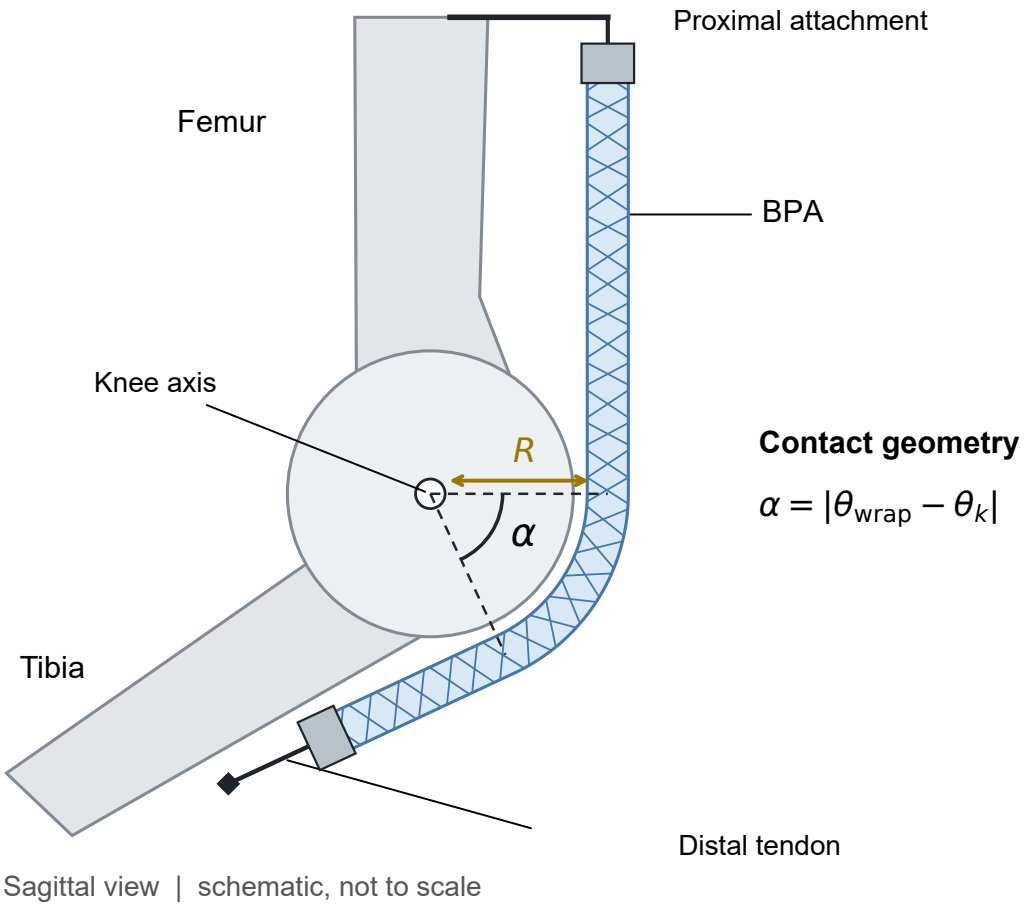


Figure 4.6: Extensor BPA wrapping around a flexed knee. Dashed radii bound the local contact arc, with wrapping radius R and bend angle α.

Identification of the Torque-Model Correction Terms

I set the correction terms in two stages using `gamultiobj` in MATLAB. The flexor stage fits (χ_0, χ_1, χ_2) with compliance at both attachments. The selected training/validation split trains on the $l_0 = 48.5$ cm and 41.0 cm tests and validates on the 45.7 cm and 40.6 cm tendon tests. The extensor stage locks (χ_1, χ_2) to that same flexor-front member and fits only (χ_0, χ_3). Its selected training/validation split trains on the 43.6 cm, 43.2 cm-tendon, and 45.7 cm tests and validates on the 40.0 cm plain, 40.0 cm-tendon, and 46.5 cm tests. The extensor length offset is allowed to be negative. Appendix C records the test constants, bounds, stiffness-axis assignments, and adopted values in Tables C1, C4, C10, C9, and 4.9.

Each candidate is evaluated using RMSE, FVU, and maximum absolute residual. These metrics jointly measure torque error; none isolates a unique source such as level offset, waveform shape, or angular misalignment.

The objective vector of the identification is defined explicitly. For a candidate correction vector $\boldsymbol{\chi}$, (χ_0, χ_1, χ_2) in the flexor stage and (χ_0, χ_3) on the locked stiffness pair in the extensor refit, the predicted torque curve of test i is interpolated to its n_i measured knee angles θ_{ij}, giving residuals $e_{ij} = M_{ij} - \hat{M}(\theta_{ij}; \boldsymbol{\chi})$ against the measured torque M_{ij}, and the three goodness-of-fit components of that test are

$$\mathrm{RMSE}_i = \sqrt{\frac{1}{n_i} \sum_j e_{ij}^2} \tag{4.28}$$

$$\mathrm{FVU}_i = \frac{n_i - 1}{n_i} \cdot \frac{\sum_j e_{ij}^2}{\sum_j (M_{ij} - \bar{M}_i)^2} \tag{4.29}$$

$$e_i^{\max} = \max_j |e_{ij}|. \tag{4.30}$$

FVU expresses the residual sum of squares relative to the sum of squared deviations of the measured torque from its mean $\bar{M}_i$, with the factor $(n_i - 1)/n_i$ carrying the small-sample adjustment of the evaluators; RMSE and the maximum absolute residual carry torque units, whereas FVU is dimensionless. For a given test, FVU is proportional to squared RMSE, so the two metrics are not independent measures of shape and level, and across tests their different scaling changes the relative weighting of the errors. The solver minimizes the objective vector

$$\mathbf{g}(\boldsymbol{\chi}) = \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} \begin{bmatrix} g_{1,i}(\boldsymbol{\chi}) / g_{1,i}^{(0)} \\ g_{2,i}(\boldsymbol{\chi}) / g_{2,i}^{(0)} \\ g_{3,i}(\boldsymbol{\chi}) / g_{3,i}^{(0)} \end{bmatrix}, \tag{4.31}$$

where $g_{1,i}$, $g_{2,i}$, and $g_{3,i}$ are the RMSE, FVU, and maximum absolute residual of Equations (4.28)–(4.30), the superscript (0) marks the zero-correction baseline value of the same component, the same kinematics and force model with every correction term disabled, and $\mathcal{T}$ is the set of tests used for training. The mean of the baseline-normalized components makes the three objectives dimensionless and the training tests commensurate. The search carried no constraints beyond the parameter bounds of each stage, and the returned Pareto fronts (the candidate parameter sets for which improving one objective requires worsening another) were filtered to members scoring at or below the zero-correction baseline on every component of every test in that stage's identification pool.

MATLAB `gamultiobj` uses NSGA-II to return the Pareto front [Deb et al., 2002, The MathWorks, Inc., 2025]. Candidate predictions are checked on the training and valida-

tion tests. The search uses bounded, normalized parameters, with logarithmic stiffness coordinates. Population, mutation, crossover, and stopping settings are retained with the archived run records.

4.1.5 Actuator Placement and Route Redesign

The placement search varies actuator attachments to control moment arm and muscle excursion over the joint range of motion [Bolen and Hunt, 2019, Morrow et al., 2020]. For the erector spinae, internal oblique, and external oblique at the lumbosacral joint, the targets are Gait2392 torque surfaces sampled at 100 positions along each of two joint coordinates.

At each scale the search perturbs both attachments by $\pm\epsilon$ offsets along the Cartesian coordinates (eight candidates per attachment, 64 candidate pairs), centers on the lowest-cost pair, and halves ϵ when no candidate improves, from an initial $\epsilon = 0.1\,\mathrm{m}$ down to 1 mm [Morrow et al., 2020].

The original cost combined the summed absolute torque mismatch over the sampled poses, actuator length, displacement of the attachments from their starting locations, and a diameter-dependent penalty. The published weights for the first three terms were 10^{-4}, 10^{-2}, and 1, respectively; the penalties for 10 mm, 20 mm and 40 mm BPAs were 1, 5, and 15. These were numerical weights chosen to balance differently scaled cost terms. The geometric terms discouraged impractical placements but did not impose collision avoidance: attachments could lie away from the modeled bones, and routes could intersect them. This study established the placement-search approach; it did not establish physical

torque accuracy or satisfaction of the human torque requirement at every pose.

A subsequent knee implementation restricted candidates to mesh points on the femur and tibia: for the biceps femoris short-head replacement, every origin–insertion pair was evaluated over 100 knee poses from approximately -120° to 10° at a fixed 20 mm diameter, candidates outside the allowable contraction range were rejected, and the lowest-cost usable pair was retained. Like the earlier cube-perturbation search, this discrete mesh search used rigid attachment geometry without the experimentally identified compliance corrections.

The knee-flexor mesh search also used a different objective from the original torque-mismatch cost. It rewarded torque surplus in the human torque's direction while penalizing disagreement between the human and robot torque-vector directions:

$$C_{\text{mesh}} = -\frac{1}{3N}\sum_{j=1}^{N}\left[w_{\mathbf{k}}\, D_{\mathbf{k},j} + \sum_{a\in\{x,y\}} w_a\, D_{a,j}\right] + \frac{w_u}{N}\sum_{j=1}^{N}\left(1 - \hat{\mathbf{M}}_j^{\mathrm{H}}\cdot\hat{\mathbf{M}}_j^{\mathrm{R}}\right). \tag{4.32}$$

Here N is the number of evaluated poses, and $D_{a,j} = M_{a,j}^{R} - M_{a,j}^{H}$ when $M_{a,j}^{H} \geqslant 0$, with the subtraction reversed otherwise; Table 4.1 records the weights and their meaning. A zero robot-torque vector is assigned the opposite human direction in the direction term, which assumes a nonzero human torque vector at the evaluated pose. Torques are supplied in newton meters; this is a numerically weighted score, rather than a sum of normalized physical errors, and minimizing it favors torque capacity in the required direction without guaranteeing that the requirement is met at every pose.

Table 4.1: Terms of the knee-flexor mesh-placement cost. The cost is minimized; a surplus reduces it, while a direction mismatch increases it.

Term	Weight	Meaning
Knee torque surplus	5	Signed surplus $D_{\mathbf{k},j}$, summed with the off-axis terms and divided by $3N$; enters with a negative sign.
Off-axis torque surplus	0.25 each	Signed surplus $D_{x,j}$ and $D_{y,j}$, using the human torque's sign on each axis; enters with a negative sign.
Direction mismatch	10	Mean $1 - \hat{\mathbf{M}}_j^H \cdot \hat{\mathbf{M}}_j^R$; zero for aligned vectors and two for opposing vectors.

4.1.6 Constrained Knee-Route Redesign

The dissertation's route redesign extends the placement search as a constrained continuous optimization over the design vector $\mathbf{x} = [\mathbf{p}_1,\ \mathbf{p}_{\text{end}},\ l_0,\ l_{\text{tendon}}]$, the BPA origin $\mathbf{p}_1$, insertion $\mathbf{p}_{\text{end}}$, resting length, and tendon length (Appendix C), so that the flexor and extensor configurations meet or exceed human knee torque capability over the range of motion. Because the route prediction (contact activation, wrapping, and compliance correction) is expensive to evaluate and only moderately smooth, the redesign minimizes a single scalar cost assembled from weighted penalty terms. The cost has the form

$$J = w_w\, J_{\text{worst}} + w_t\, J_{\text{torque}} + \textstyle\sum_a J_a + J_{\text{geom}} + J_{\text{len}}, \tag{4.33}$$

where J_{worst} is the largest fraction by which robot torque falls short of the human target at any angle across the range of motion, weighted heavily ($w_w = 10^5$); J_{torque} is the mean-squared shortfall fraction ($w_t = 10^3$), weighted two orders of magnitude below the worst-angle term so that no shortfall at a single angle can be traded away for a better average;

the J_a are configuration-specific auxiliary penalties; and J_{geom} and J_{len} are regularizers that keep the endpoints and actuator lengths near their seed values. The two leading terms are built from the normalized shortfall

$$s_j = \frac{\max\Big(0,\ (1+m_r)|M_j^{\text{H}}| - |M_j^{\text{R}}|\Big)}{\tau}, \qquad J_{\text{worst}} = \max_j s_j, \qquad J_{\text{torque}} = \frac{1}{N}\sum_j s_j^2, \quad (4.34)$$

where $m_r = 0.05$ is the required torque margin and $\tau = \max(1, \max_j |M_j^{\text{H}}|)$ is the human torque scale. The flexor and extensor configurations carry the remaining terms of Equation (4.33) as

$$\begin{aligned} J_{\text{flx}} = {} & 10^5 J_{\text{worst}} + 10^3 J_{\text{torque}} + 10^{-2} J_{\text{shape}} + J_{\text{off}} + J_{\text{rest}} + J_{\text{strain,hi}} \\ & + J_{\text{strain,lo}} + J_{\text{wrap}} + J_{\text{geom}} + J_{\text{len}} + J_{\text{anchor}}, \end{aligned} \quad (4.35)$$

$$\begin{aligned} J_{\text{ext}} = {} & 10^5 J_{\text{worst}} + 10^3 J_{\text{torque}} + J_{\text{over}} + J_{\text{rest}} + J_{\text{strain,hi}} \\ & + J_{\text{strain,lo}} + J_{\text{contr}} + J_{\text{geom}} + J_{\text{len}}. \end{aligned} \quad (4.36)$$

Tables 4.2–4.4 define the normalized penalties and their weights. A candidate is rejected if its route cannot be constructed or its torque or strain calculation is not finite.

Human torque targets are computed from the Gait2392 lower-extremity model: biceps femoris short head for the flexor and vastus medialis for the extensor [Delp et al., 2007, Seth et al., 2018]. At each knee angle, the OpenSim moment arm is multiplied by the muscle's maximum isometric force [Delp et al., 1990, Hoy et al., 1990, Sherman et al., 2013]. Appendix A gives the model parameters and exported values. The search also enforces bone clearance, feasible wrapping, actuator contraction limits, and tendon geometry; a

Table 4.2: Torque penalties in the constrained knee-route search. FLEX and EXT identify configuration-specific terms.

Term	**Weight**	**Meaning**
J_{worst} worst-angle shortfall	10^5	Largest fraction by which predicted torque magnitude falls short of the required envelope (human curve plus margin m_r) at any angle; dominates so that no single-angle shortfall can be traded for a better average.
J_{torque} mean-square shortfall	10^3	Mean squared shortfall fraction across all angles.
J_{shape} shape agreement (FLEX)	10^{-2}	Small mean-square tracking error of the full torque curve beyond the requirement.
J_{over} overshoot (EXT)	10^{-3}	Mean-square fraction by which torque exceeds the plain human curve, discouraging over-designed moment arms.
J_{off} off-axis excess (FLEX)	10^3	Worst and mean squared excess of the off-axis (x, y) torques beyond the original single-BPA route's off-axis torques, each axis normalized separately.

Table 4.3: Feasibility and geometric penalties in the constrained knee-route search.

Term	**Weight**	**Meaning**
J_{rest} rest-length feasibility	10^6	Squared violation of resting length plus tendon plus fittings covering the longest path over the range of motion.
$J_{\text{strain,hi}}$ upper strain limit	FLEX: 10^7 EXT: 10^4	Squared violation of the maximum admissible relative strain and contraction.
$J_{\text{strain,lo}}$ stretch limit	FLEX: 10^7 EXT: 10^4	Squared violation of the minimum strain, forbidding a stretched (slack-free) actuator.
J_{contr} contraction cap (EXT)	10^6	Squared violation of the measured maximum usable contraction; also enforced as a hard constraint.
J_{wrap} wrap release (FLEX)	10^3	Penalty when the moving wrap contact fails to release during the extension sweep, scaled by the remaining turn of the wrap.
J_{geom} endpoint regularizer	10^{-2}	Keeps origin and insertion near their seed values; the flexor scales each axis separately, allowing large proximal motion of the origin.
J_{len} length regularizer	10^{-3}	Keeps the resting length near its seed value (40 mm norm).
J_{anchor} second-BPA anchor (FLEX)	10	Keeps the second actuator's derived endpoints near the build line of the loaded design (10 mm norm); a score term only, never a constraint.

Table 4.4: Additional cam-profile penalties and rejection values.

Term	Weight	Meaning
J_{arm} moment-arm reward (CAM)	-10^2, capped	Negative reward on the worst-angle margin against the required envelope, capped at +100% of the torque scale, so the solver keeps buying moment arm instead of stopping at the requirement.
J_{armStep} arm-step penalty (CAM)	10^3	Penalizes adjacent-pose moment-arm steps beyond the accepted release discontinuity of 2.2 mm.
J_{reg} cam regularizer (CAM)	10^{-2}	Tie-break toward the first-cut cam neighborhood (25.4 mm norm).
Infeasible sentinels	10^{11}, 10^{12}	Non-finite predictions, or a route the predictor cannot build, are rejected outright.

candidate that violates these requirements cannot be accepted.

Both configurations use `surrogateopt` followed by `patternsearch` refinement [The MathWorks, Inc., 2025, Audet and Dennis, 2006]. The surrogate/refinement evaluation budgets are 3,000/15,000 for the direct extensor, 1,000/15,000 for the flexor, and 1,000/1,500 for the cam variant. Search can terminate earlier when its step-size and feasibility tolerances are satisfied.

The cam redesign exists because the direct route met the torque envelope at actuator forces that broke the printed and machined parts during testing; increasing moment arm to reduce actuator force is the design response. Its objective adds three terms to the extensor cost,

$$J_{\text{cam}} = J_{\text{ext}}(\mathbf{x}_{\text{full}}) + J_{\text{arm}} + J_{\text{armStep}} + J_{\text{reg}}, \qquad J_{\text{arm}} = -10^2 \, \min\left(1, \, \min_j m_j^{\text{req}}\right), \quad (4.37)$$

where $m_j^{\text{req}} = (|M_j^{\text{R}}| - (1 + m_r)|M_j^{\text{H}}|)/\tau$ is the margin against the required envelope of

Equation (4.34), with the moment-arm step and cam regularizer of Table 4.4. The capped reward favors additional margin after the required envelope is met, while the step penalty limits abrupt changes in the contact-release schedule.

The flexor uses continuous bone-clearance constraints and penalties for off-axis torque and incomplete wrap release. The extensor uses the contact-elimination rule of Appendix C.7 and penalizes excessive torque and contraction. Both require torque magnitude of at least 1.05 times the human target; the prescribed 5 % allowance is a design requirement, not an experimentally established error bound. The two flexor BPA paths are evaluated separately, whereas the extensor scales the force of its shared route by actuator count. The final flexor uses a staggered-offset pair rather than a mirrored origin arrangement that conflicts with the pelvis.

The redesign retains the objective of obtaining adequate torque from anatomically constrained routes, but changes both the prediction and the search formulation. Actuator and tendon lengths become independent design variables, geometry is constrained continuously, and pointwise torque shortfall is penalized using the improved force model. Section 4.3.3 quantifies the redesign, and Section 4.3.4 tests it.

4.2 Results

The pinned-flexor results first compare the length-dependent actuator model of Hunt et al. [2017a] with the improved BPA force model. Subsequent comparisons use the improved model before and after design-parameter optimization. Hybrid calculations substitute

measured muscle length and moment arm to help distinguish force-model error from geometric error on the pinned knee.

4.2.1 Flexor on Pin Joint

The pre-optimization comparisons distinguish force-model error from errors in the predicted muscle geometry (Figure 4.7). The hybrid calculation uses measured length and moment arm, whereas the other predictions use the routed geometry. Flexion angles and torques are negative; the caption identifies the individual tests.

Both pre-optimization force models overpredicted measured flexor-torque magnitude. On the common domain where both predictions are defined, the improved BPA characterization gives the smaller errors (Table 4.5). Those scores use a different domain from the full-test improved-model scores in Table 4.6. The hybrid calculation, using measured length and moment arm, follows the measurements more closely, identifying the predicted geometry as an important source of torque error.

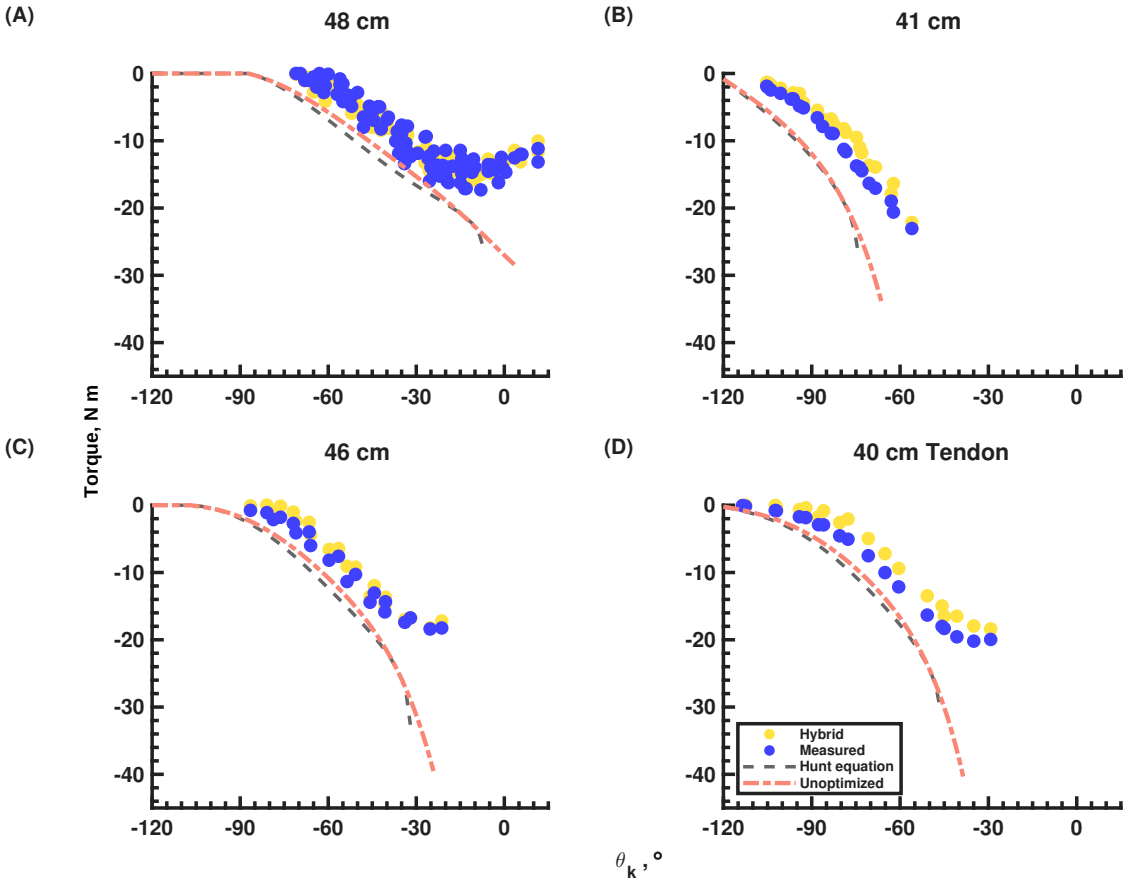


Figure 4.7: Pinned-knee flexor torque before design-parameter optimization: **(A)** $l_0 = 48.5\,\text{cm}$, **(B)** $l_0 = 41.0\,\text{cm}$, **(C)** $l_0 = 45.7\,\text{cm}$, and **(D)** $l_0 = 40.6\,\text{cm}$ with a 35 mm tendon. Panels **(A)** and **(B)** are training tests; **(C)** and **(D)** are validation tests. The legend distinguishes measurements, the Hunt model, the improved BPA model, and the hybrid calculation.

Table 4.5: Lengths l_0 (resting BPA) and l_{ten} (artificial tendon) are both in cm. Pre-optimization pinned-knee flexor torque-prediction goodness of fit per test, comparing the three pre-optimization torque predictions of Figure 4.7: the Hunt-equation rigid-body prediction, the unoptimized prediction of the improved BPA characterization, and the back-calculated (hybrid) torque. RMSE and maximum absolute residual (Max.) are in N m, and FVU denotes the fraction of variance unexplained.

		Hunt equation			**Improved BPA**			**Back-calculated (hybrid)**		
l_0 (cm)	l_{ten} (cm)	**RMSE**	**FVU**	**Max.**	**RMSE**	**FVU**	**Max.**	**RMSE**	**FVU**	**Max.**
48.5	–	5.779	1.1761	9.798	4.871	0.8354	9.865	1.635	0.0991	3.842
41.0	–	6.633	3.2313	11.409	6.074	2.7090	9.262	2.187	0.1184	4.243
45.7	–	5.315	0.9408	9.991	4.465	0.6640	9.536	1.246	0.0409	2.275
40.6	3.5	3.945	0.6750	7.527	3.226	0.4512	7.376	2.085	0.0755	3.073

The flexor identification of record is the two-bracket fit of Section 4.1.4 on the four cross-validation tests; the optimized model follows the measured torque across the range of motion, including the loss of torque near full extension that the rigid-body prediction does not produce (Figure 4.8, Tables 4.9 and 4.6). The complete five-test comparison before and after correction is provided in Table C11.

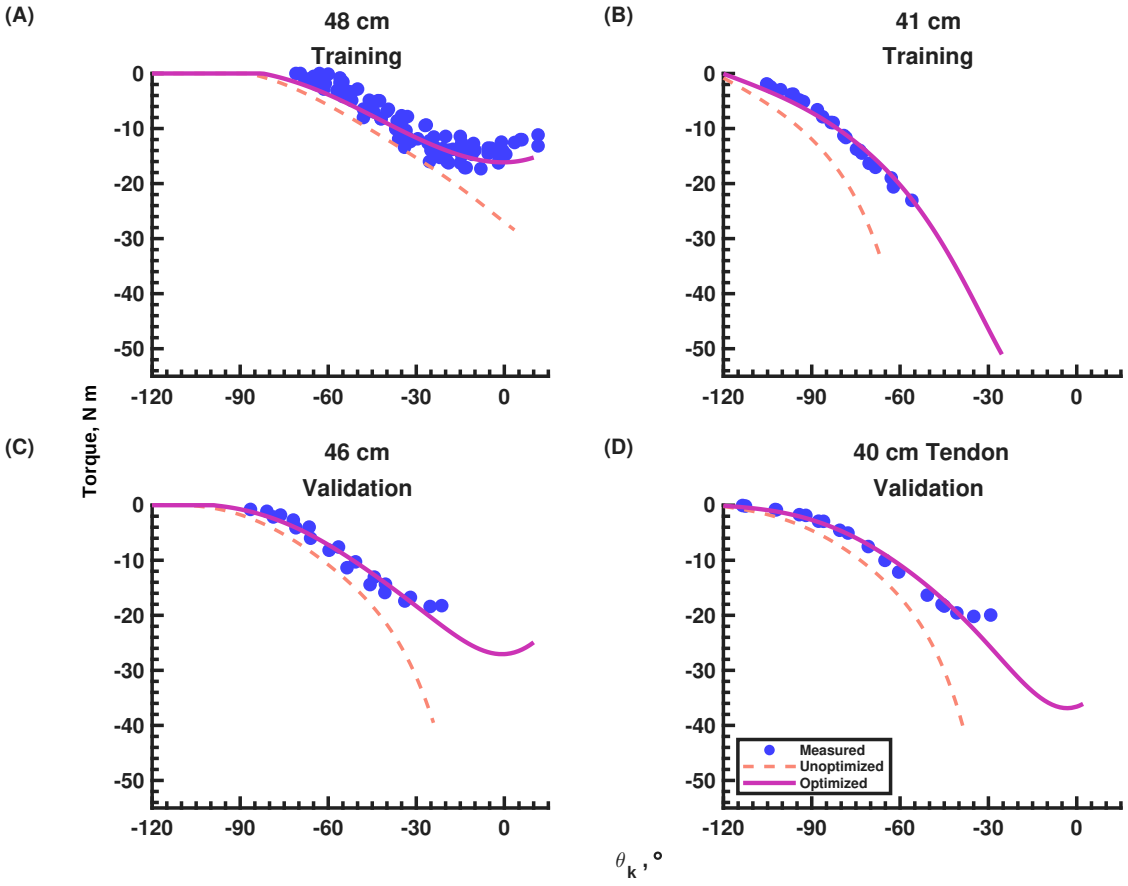


Figure 4.8: Optimized pinned-knee flexor torque: **(A)** $l_0 = 48.5\,\text{cm}$, **(B)** $l_0 = 41.0\,\text{cm}$, **(C)** $l_0 = 45.7\,\text{cm}$, and **(D)** $l_0 = 40.6\,\text{cm}$ with a 35 mm tendon. Panels **(A)** and **(B)** are training tests; **(C)** and **(D)** are validation tests. Circles show measurements, dashed curves the improved BPA model, and solid magenta curves the optimized model.

Table 4.6: Lengths l_0 (resting BPA) and l_{ten} (artificial tendon) are both in cm. Pinned-knee flexor goodness of fit before and after correction, using the identified terms. RMSE and maximum absolute residual (Max.) are in N m; FVU is dimensionless.

		Improved BPA			**Optimized model**		
l_0 (cm)	l_{ten} (cm)	**RMSE**	**FVU**	**Max.**	**RMSE**	**FVU**	**Max.**
Training							
48.5	–	7.617	2.1278	25.531	2.165	0.1718	5.467
41.0	–	9.694	2.1149	21.076	1.084	0.0277	1.766
Mean		8.656	2.1213	23.303	1.624	0.0998	3.617
Validation							
45.7	–	8.803	1.9380	26.264	1.368	0.0468	3.614
40.6	3.5	6.373	0.5935	16.859	1.611	0.0427	5.891
Mean		7.588	1.2657	21.562	1.490	0.0447	4.752

The improved model reduces mean torque-prediction error in both the training and validation tests, with the optimized curves tracking the measurements across the tested range (Figure 4.8 and Table 4.6). The baseline uses the same kinematics and force model with the correction terms disabled; Figure 4.7 separately compares the original rigid-body prediction.

4.2.2 Extensor on Pin Joint

The correction terms reduce mean pinned-knee extensor torque-prediction error in both the training and validation tests (Table 4.7). Figure 4.9 compares the measured and predicted torque across knee angle. By inspection, the zero point of the measured torque was shifted to the right of the original model, consistent with an offset opposite in sign to the χ_0 term in the previous section. The BPA in Figure 4.9**(C)** illustrates the pattern: the hybrid-method calculated torque correlates well with the originally predicted torque, while the actual measured torque diverges. The fitted wrapping term χ_3 enters the usable-length correction in Equation (4.25). Table 4.7 reports the improvement of the fit. Extensor-specific terms (χ_0, χ_3) were specified while stiffness terms (χ_1, χ_2) were kept the same as the previous section (see Table 4.9).

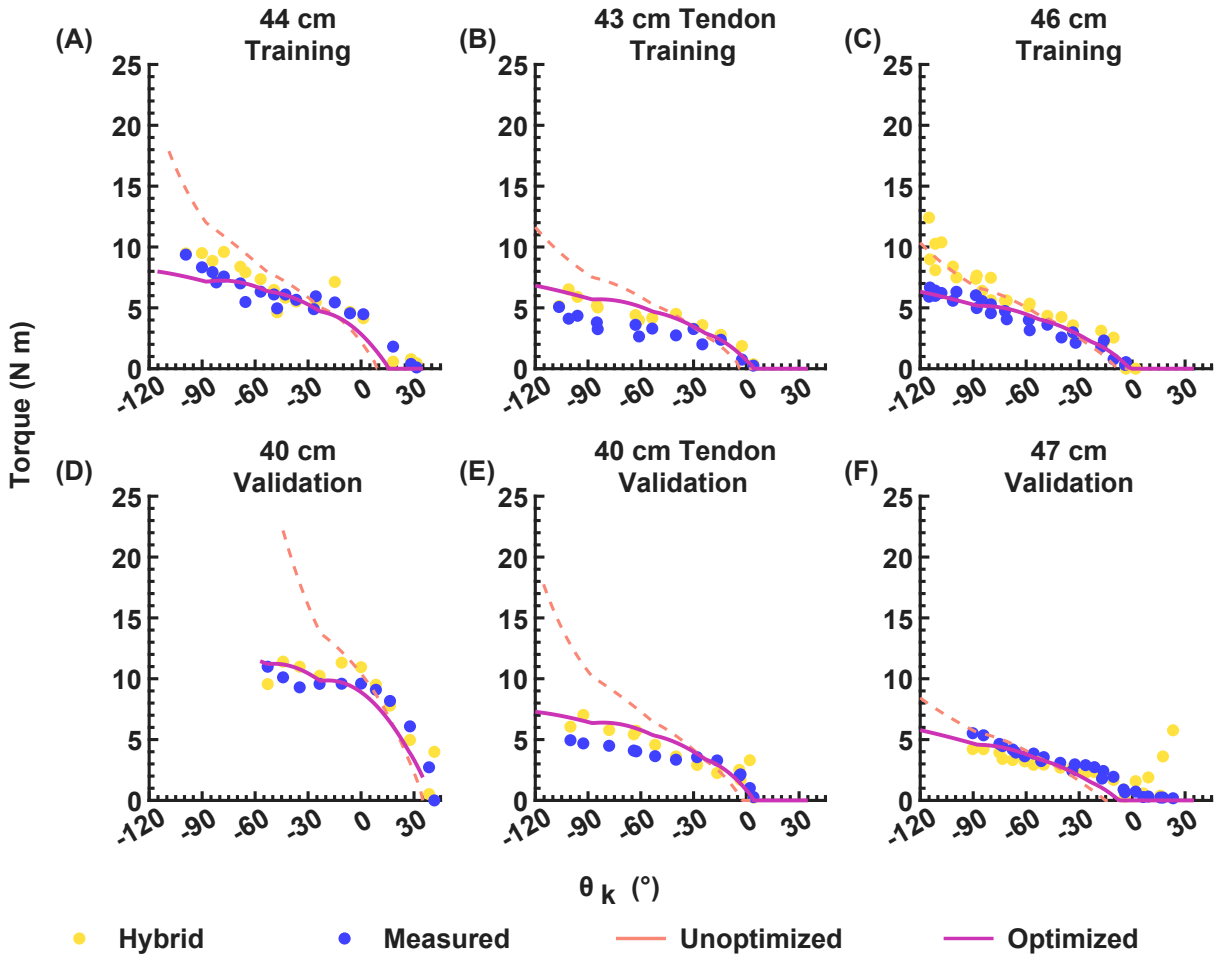


Figure 4.9: Optimized pinned-knee extensor torque: **(A)** $l_0 = 43.6\,\text{cm}$, **(B)** $l_0 = 43.2\,\text{cm}$ with a 22 mm tendon, **(C)** $l_0 = 45.7\,\text{cm}$, **(D)** $l_0 = 40.0\,\text{cm}$, **(E)** $l_0 = 40.0\,\text{cm}$ with a 40 mm tendon, and **(F)** $l_0 = 46.5\,\text{cm}$. Panels **(A)**–**(C)** are training tests; **(D)**–**(F)** are validation tests. The legend applies to all panels.

Table 4.7: Pinned-knee extensor goodness of fit before and after correction, with stiffness shared with the flexor. RMSE and maximum absolute residual (Max.) are in N m; FVU is dimensionless.

		Improved BPA			**Optimized model**		
l_0 (cm)	l_{ten} (mm)	**RMSE**	**FVU**	**Max.**	**RMSE**	**FVU**	**Max.**
Training							
43.6	–	2.437	0.9618	4.284	0.978	0.1634	1.818
43.2	22	2.878	4.7304	4.817	1.525	1.3270	2.481
45.7	–	2.178	1.0586	4.974	0.490	0.0535	1.206
Mean		2.498	2.2503	4.692	0.997	0.5146	1.835
Validation							
40.0	–	3.092	0.5638	4.842	1.244	0.1303	2.196
40.0	40	3.622	6.3919	7.845	1.350	0.8878	1.890
46.5	–	0.949	0.3152	2.272	0.699	0.1714	1.559
Mean		2.554	2.4236	4.986	1.098	0.3965	1.882

4.2.3 Extensor on Biomimetic Knee

I tested the extensor configuration on the biomimetic knee with a ∅10 mm BPA at $l_0 = 51.8$ cm. The measured torque fell well below the original prediction (Figure 4.10). Applying the extensor terms identified on the pinned knee improved the prediction: at those terms, the optimized model reached an RMSE of 1.1702 N m, an FVU of 0.6681, and a maximum absolute residual of 2.2088 N m, against 3.0322 N m, 4.4856, and 5.5870 N m for the original rigid-body prediction.

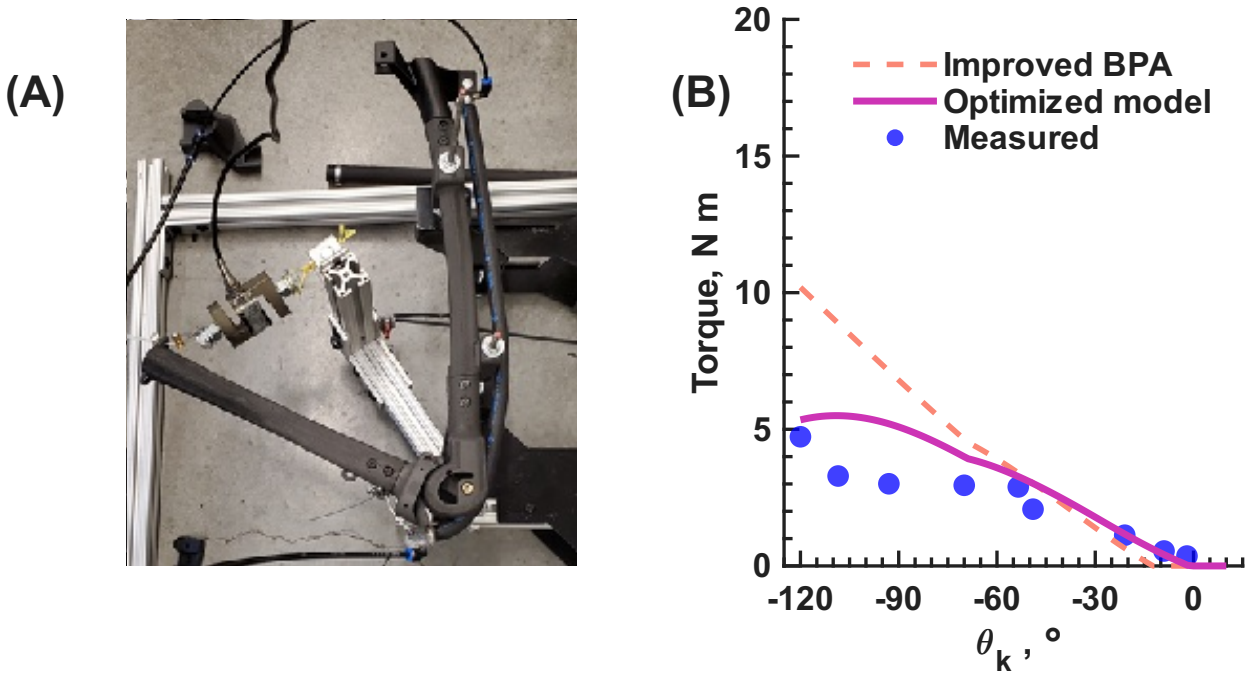


Figure 4.10: Biomimetic-knee extensor test: **(A)** apparatus with the BPA passing between the femoral condyles; **(B)** measured torque, the improved BPA baseline, and the optimized model at the identified terms for the 10 mm-diameter, 51.8 cm BPA.

4.2.4 Flexor on Biomimetic Knee

The shared correction terms reduce torque-prediction error for both biomimetic-knee flexor configurations (Table 4.8). Figure 4.11 compares these predictions with the measurements; the bracket frame for the ∅10 mm BPA is shown in Figure 4.3**(C)**, and for the ∅20 mm BPA in Figure 4.3**(D)**. Figure 4.11**(A)** presents the ∅10 mm actuator at $l_0 = 41.5$ cm with a 21 mm artificial tendon (∅1.5 mm Shimano bicycle brake cable wrapped four times), pressurized to 604 kPa; the optimized model reached an RMSE of 2.8002 N m, an FVU of 0.8835, and a maximum absolute residual of 4.1201 N m. Figure 4.11**(B)** presents the ∅20 mm actuator at $l_0 = 41.5$ cm with a 14 mm artificial tendon made out of the same brake cable, wrapped three times so that six cable segments share the load, each wrap contributing two tendon lengths and cross-sectional areas. The configuration in Figure 4.11**(B)** was tested at 620 kPa, for which the optimized model reached an RMSE of 2.2070 N m, an FVU of 0.0748, and a maximum absolute residual of 3.7320 N m.

The torque of the single ∅20 mm flexor actuator shown in Figure 4.11**(B)** did not exceed the magnitude of the maximum human isometric flexion torque between 60° and 100° of knee flexion, and within a ±20% band about the human mean it still fell short of human-level torque between 78° and 103° of flexion. Because operating at 620 kPa across the full desired range of motion would load the printed femur beyond its ultimate strength, single points were also recorded at the reduced pressures 560 kPa, 421 kPa, 325 kPa and 281 kPa, which show that the desired range of motion is achievable at reduced pressure. This shortfall in flexion torque defines the target that the attachment-route redesign of Section 4.1.5 addresses.

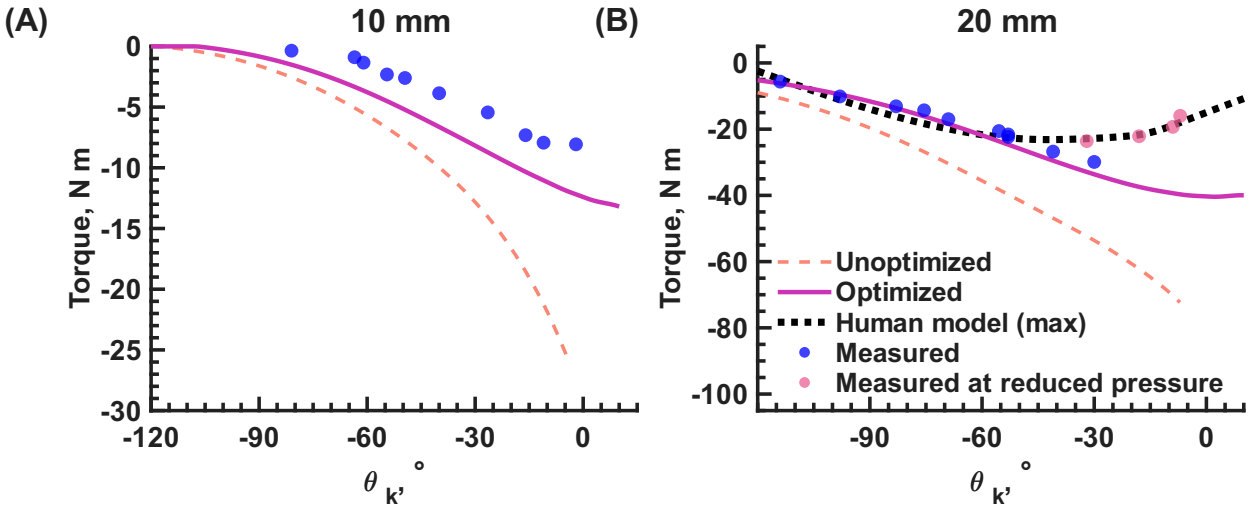


Figure 4.11: Biomimetic-knee flexor torque: **(A)** a single 10 mm BPA; **(B)** a single 20 mm BPA. The legend in **(B)** identifies the measurements, predictions, and human reference.

Table 4.8: Biomimetic-knee goodness of fit before and after design-parameter optimization. RMSE and maximum absolute residual (Max.) are in N m; FVU is dimensionless.

	Improved BPA			Optimized model		
Configuration	**RMSE**	**FVU**	**Max.**	**RMSE**	**FVU**	**Max.**
Extensor, ∅10 mm	3.0322	4.4856	5.5870	1.1702	0.6681	2.2088
Flexor, ∅10 mm	7.4513	5.5612	13.3163	2.8002	0.8835	4.1201
Flexor, ∅20 mm	14.8064	3.3672	23.7218	2.2070	0.0748	3.7320

4.3 Discussion

4.3.1 Reasons for the Correction Terms

The correction terms of Section 4.1.4 exist because the mechanical connection between the actuator and the joint is compliant and changes under load, and each term names one family of physical reasons. A constant length offset χ_0 exists because of manufacturing and positioning tolerances: the printed bone segments, the machined knee links, and the end-cap fittings each carry dimensional tolerance, and the accumulated discrepancy appears as a length that is consistently shorter than the predicted path at every joint angle, including at zero load where no compliance can act. The stiffness terms χ_1 and χ_2 exist because loaded components displace: the isometric system is not rigid, and there is compliance in the winch cable of the test stand, the artificial tendons, and the cantilevered attachment brackets. The wrapping-loss term χ_3 exists because a BPA that bends over the joint loses usable contraction length that no straight-line model accounts for; at high flexion the same loading was observed to stretch and compress shorter-l_0 actuators and to slip the BPA off the anterior head of the femur bolt by up to ± 20 mm, changing both the force vector and the torque.

Wrapping a BPA around the knee remains an approximation with inherent error (Figure 4.10): high actuator force and a short moment arm make the torque sensitive to attachment deflection and small geometric errors [Young et al., 2019]. The pinned-knee tests therefore separate actuator-force error from attachment-compliance and routed-geometry error, and applying the identified terms to the biomimetic knee tests them where the moment arm and the instantaneous center of rotation move with knee angle. Isometric

testing establishes a static capacity envelope; velocity dependence, damping, and behavior under controlled speed and load remain unidentified.

4.3.2 Design-Factor Optimization

The results figures use the two-stage identification of Section 4.1.4. Relative to the zero-correction baseline, the optimized model reduces mean RMSE by 81 % on the flexor training tests and 80 % on the flexor validation tests (Table 4.6), and by 60 % on the extensor training tests and 57 % on the extensor validation tests (Table 4.7); on the biomimetic extensor it reduces FVU by 85 % (Table 4.8). Table 4.9 records the identified terms: the stiffness pair is shared, and the wrapping factor was identified on the extensor and applied to wrapping in the redesigned flexor. The genetic multiobjective solver is retained for the identification, where the Pareto front is required for holdout validation, and the surrogate-plus-pattern-search pipeline serves the route redesign (Section 4.1.5); the search bounds are recorded in Appendix C (Table C10).

Table 4.9: Adopted correction terms. The effective stiffness pair is shared; χ_3 was identified on the extensor and is applied to wrapping in the redesigned flexor.

Configuration	χ_0 (m)	χ_1 (N m^{-1})	χ_2 (N m^{-1})	χ_3
Flexor	0.00394	3.998×10^4	1.473×10^4	–
Extensor	−0.00638	3.998×10^4	1.473×10^4	0.158

The original placement methods selected routes by predicted torque under rigid-attachment assumptions; the isometric tests show those predictions require the compliance corrections before they can guide hardware selection, so the redesigned routes evaluate every candidate through the force balance with the identified terms. On the biomimetic

knee the shared terms transfer at an FVU of 0.8835 for the 10 mm flexor and 0.0748 for the 20 mm flexor (Table 4.8); the larger error for the smaller actuator limits transfer across configurations, and the redesigned two-BPA routes provide the further physical test of Section 4.3.4.

4.3.3 Redesigned Routes for the Biomimetic Robot Model

The redesigned routes combine the existing biomimetic robot geometry with human-muscle torque references. The surrogate-plus-pattern-search procedure of Section 4.1.6 uses the correction terms in Table 4.9, two 20 mm BPAs per route, a 620 kPa pressure ceiling, and a maximum usable contraction of 0.255. Both direct routes meet the required torque envelope (Table 4.10).

Table 4.10: Predicted direct-route torque capacity on the biomimetic robot model. Both margin rows use the selected human-muscle torque magnitude as their denominator.

	Extensor	**Flexor**
Predicted torque (N m)	35.45 to 109.4	-81.23 to -3.864
Human reference torque (N m)	20.25 to 50.32	-23.16 to -2.461
Minimum predicted margin	5.02 %	5.01 %
Required design allowance	5 %	5 %
Angle of minimum margin	-20.2°	-106.9°

The cam-assisted extensor allows the tibial route points to trace a profile rather than remain fixed (Figure 4.12). Table 4.11 summarizes the resulting capacity and geometry. The flexor uses two separately evaluated BPA paths and the wrapping factor identified on the extensor. Each route family minimizes its own weighted penalty sum; objective values are not directly comparable. These are mechanical design predictions; incorporation into

the converted neuromechanical models remains future work.

The cam summaries use two different normalizations. For the evaluated knee-angle grid Θ, let $M_h(\theta)$ be the human-reference torque, $M_{h,\max}$ its peak magnitude, and $M_{\mathbf{k}}(\theta)$ the predicted knee-axis torque. The minimum surplus above the required envelope and the minimum relative margin above the human reference are

$$\begin{aligned} s_{\text{req}} &= \min_{\theta\in\Theta} \frac{|M_{\mathbf{k}}(\theta)| - 1.05|M_h(\theta)|}{M_{h,\max}}, \\ m_h &= \min_{\theta\in\Theta} \frac{|M_{\mathbf{k}}(\theta)| - |M_h(\theta)|}{|M_h(\theta)|}. \end{aligned} \tag{4.38}$$

Table 4.11: Predicted cam-extensor capacity and routing geometry for the biomimetic robot model. The torque summaries are defined in Equation (4.38).

Quantity	**Value**
Minimum surplus / peak human torque, s_{req}	$+30.83\,\%$
Minimum pointwise margin above human, m_h	$+39.77\,\%$
Pressure to meet required envelope	269 kPa to 507 kPa
Pressure floor used in model	150 kPa
Tibial bend radius	96.9 mm
Femoral-condyle clearance at 10°	43.1 mm
Moment-arm range	59.4 mm to 102.8 mm
Mean moment arm	72.4 mm

Pressure headroom permits the required torque below the supply ceiling. In my operating experience, the present manifold has an approximately 200 kPa lower operating limit, which can restrict minimum controllable nonzero torque magnitude while the BPAs are taut. The saved design calculation used a 150 kPa pressure floor.

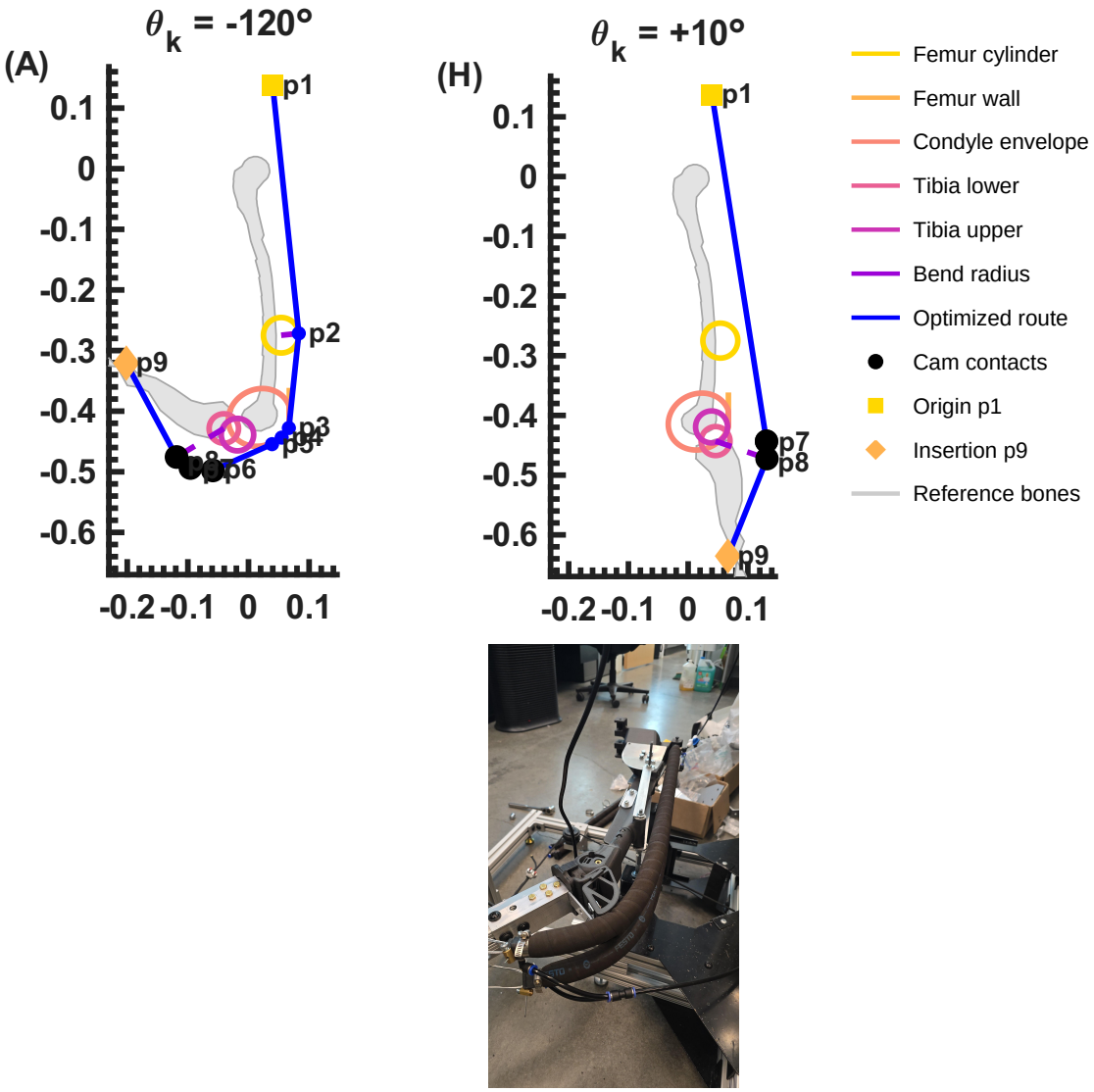


Figure 4.12: Cam-extensor geometry at -120° (A) and 10° (H), with the assembled two-BPA hardware below. Panel letters match the complete sequence in Figures C4 and C5.

4.3.4 Exceeding Human Torque Magnitude

Both redesigned routes met or exceeded their selected human-muscle reference torques at the measured knee angles. The cam-assisted extensor was tested at nine isometric configurations and the two-BPA flexor at fifteen, spanning -119° to 5°. The analysis transforms the measured load-cell wrench to the moving knee center and evaluates the model at each measured angle and pressure. Table C7 in Appendix C records the measured per-actuator dimensions and distinguishes them from the shared prediction inputs. Figures 4.13 and 4.14 compare these results with the selected human-muscle targets and the optimized 620 kPa predictions.

The extensor measurements meet or exceed the vastus medialis reference torque at the measured angles. Measured extension torque ranges from 29.02 N m to 51.17 N m, with margins of 0.24 N m to 3.52 N m, at pressures of 256 kPa to 488 kPa. The measured torque exceeds the pressure-specific prediction at eight angles, by up to 13.75 N m; at -86.5°, the measurement exceeds the human reference and is 2.06 N m below the pressure-specific prediction. This prediction uses the recorded angle and pressure with the modeled route. It is not the hybrid calculation, which additionally requires measured muscle length and moment arm.

The flexor measurements meet or exceed the biceps femoris short-head reference magnitude, with margins from 0.42 N m at -105° to 17.0 N m at -14°. Agreement with the pressure-specific prediction remains uneven. These measurements support torque capacity at the sampled configurations; variable test pressure and discrete sampling do not establish a measured full-pressure envelope between every pair of angles.

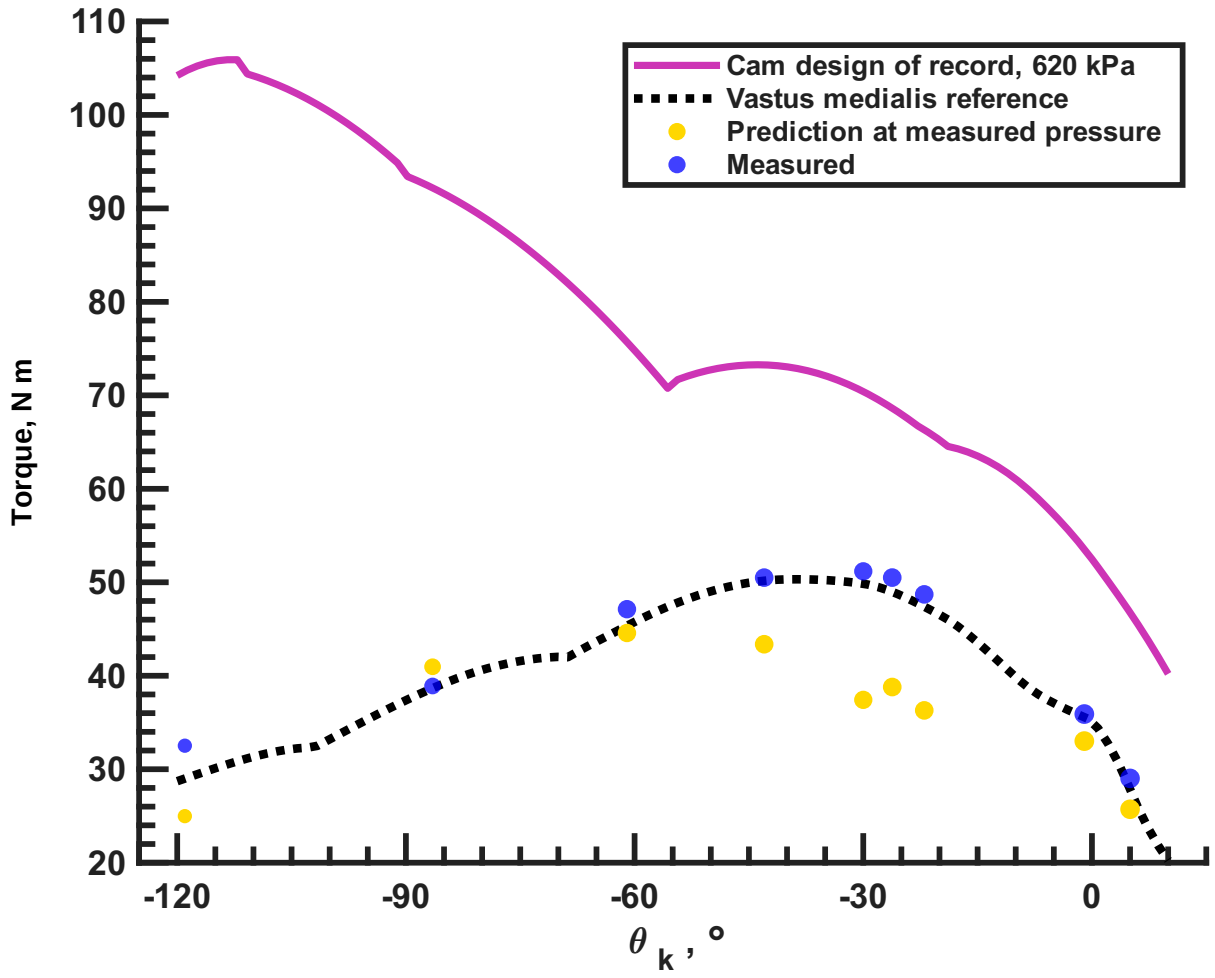


Figure 4.13: Cam-extensor measurements, pressure-specific predictions, the optimized 620 kPa curve, and the vastus medialis reference. Marker area scales with pressure. Series follow the plot legend.

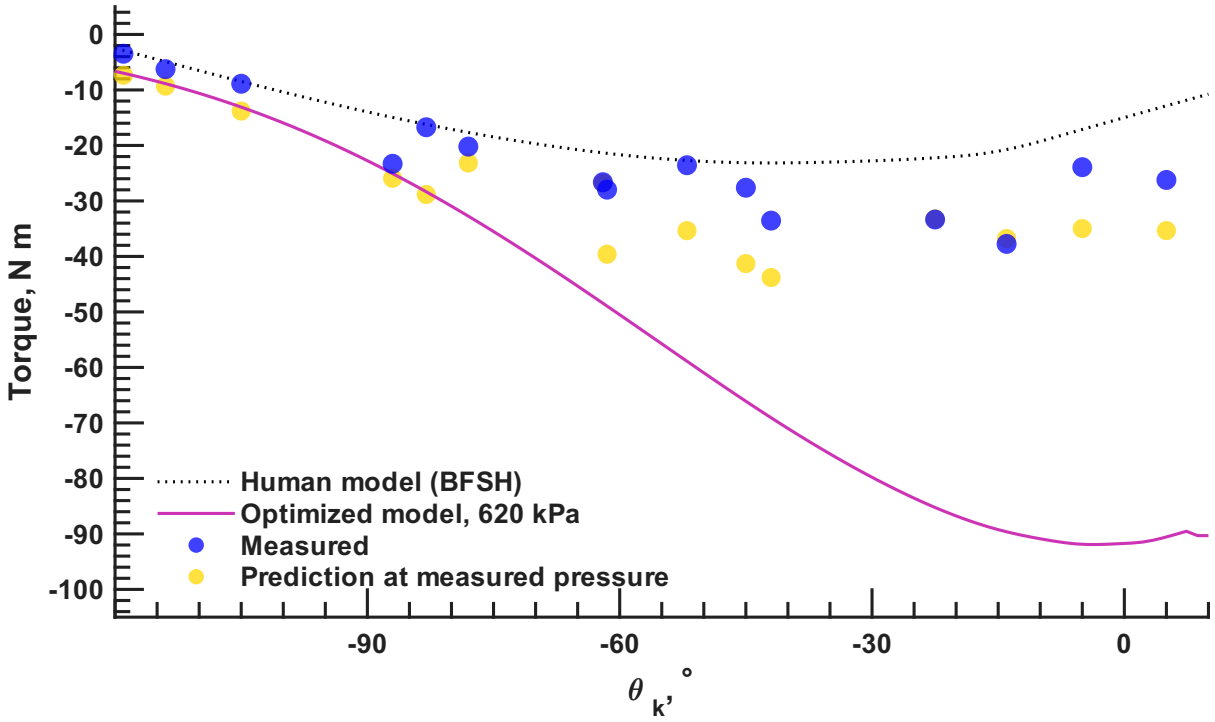


Figure 4.14: Two-BPA flexor tests (nominal 562 mm plot label); the installed resting lengths were 561 mm and 560 mm. Blue points are measured torques; gold points are model predictions at the measured angles and pressures. The solid magenta curve is the optimized 620 kPa prediction, and the black dotted curve is the human biceps femoris short-head target.

4.3.5 Tendon Reeving and Parallel BPAs

Reeving the artificial tendon through a reverse block-and-tackle trades transmitted force for tendon travel (Figure 4.15). Let F_{BPA} be the total input force, n the number of cable parts supporting the moving block, $\Delta\ell_{\mathrm{BPA}}$ the BPA contraction, and Δs_t the tendon travel. For an ideal massless cable and frictionless pulleys,

$$F_t = \frac{F_{\mathrm{BPA}}}{n}, \qquad \Delta s_t = n\Delta\ell_{\mathrm{BPA}}. \tag{4.39}$$

The illustrated two-BPA input is $F_{\mathrm{BPA}} = 2F_{20}$; with $n = 2$, $F_t = F_{20}$ and tendon travel doubles. The BPA-origin reaction magnitude is $F_{R,\mathrm{origin}} = F_{\mathrm{BPA}}$. Reeving, rather than the number of pulley wheels alone, determines n; the travel multiplier used in the route calculations is $G = n$.

This trade can extend the RoM over which a route meets its torque requirement. At $G = n = 1$, force and travel ratios relative to the combined BPA input are unity, recovering direct transmission. The design calculations retain internal-load and mount-deflection effects in their stiffness-aware force balance. Appendix C.8 gives the specific optimized knee-route comparisons and the five-site transmission study (Figure C3); its biarticular evaluation planes hold the adjacent joint fixed, and physical verification remains future work.

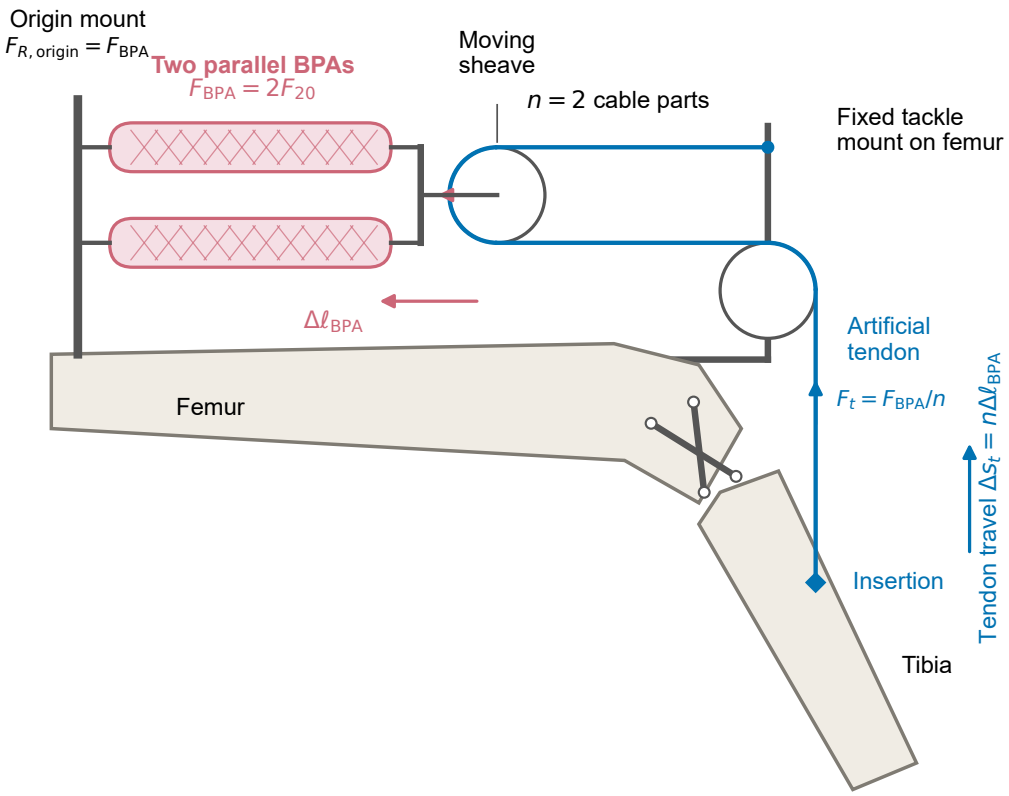


Figure 4.15: Example of two parallel BPAs actuating a biomimetic knee through a two-part reverse tackle.

4.3.6 Whole-Leg Actuator and Tackle Capability

As a proof of concept for the whole tool chain, I applied the actuator and tackle search to the robot-leg muscle routing of my master's model [Bolen and Hunt, 2019]: the expanded design selects the best transmission configuration at each of the 27 actuator sites (Table C13) and re-solves actuator and tendon lengths on the inherited routing. Each site is evaluated against its same-named Gait2392 muscle [Delp et al., 2007, Seth et al., 2018] on a 5×5 grid sampling combinations of two joint coordinates at 620 kPa [Bolen et al., 2026]: a sampled configuration meets the requirement when the taut BPA's torque has the human torque's sign and at least its magnitude, and the reported coverage is the equal-site mean of angular-domain fractions, estimated with trapezoidal area weights (Appendix C). The baseline substitutes four parallel 20 mm BPAs at the eight nominal 40 mm sites. Under matched ideal pressure, braid geometry, and contraction, maximum force scales with diameter squared, giving this parallel group the same theoretical maximum force as one 40 mm BPA. The comparison combines transmission, actuator sizing, and the diameter substitution; it does not isolate any one contribution.

Mean area-weighted human-torque coverage improves from 37.35 % to 65.81 %, improving at 23 sites and tying at four, while mean active-force coverage (positive insertion tension, regardless of torque) rises from 93.00 % to 99.31 %; positive tension alone does not meet the torque requirement. Table 4.12 summarizes the means, and Table C14 gives the per-site coverage.

With actuator force characterized, knee torque tested, and routes redesigned to meet selected human-muscle torque targets, the dissertation turns to the sensory side: Chapter 5

Table 4.12: Mean angular-domain torque and active-force coverage across 27 actuator sites at 620 kPa.

Configuration	**Human torque met (%)**	**Active force (%)**
Baseline	37.35	93.00
Selected configurations	65.81	99.31

assembles the measured afferent responses that the synthetic nervous system of Chapter 6 draws on.

5 The Sensory Afferent Database

I developed the Sensory Afferent Database (SADb), a curated literature resource for studying sensory feedback in locomotion and posture. It links findings from animal and human studies to afferent classes, sensory afferent pathways, and computational models. Researchers can use these records to compare individual feedback connections or whole neural circuits before implementing a neuromechanical model. This work forms part of the AARL's NeuroNex Communication, Coordination, and Control in Nervous Systems (C3NS) effort.

Presentation credits: The SADb resource was presented as an abstract and poster at Neuroscience 2023 [Bolen et al., 2023]. The AI-assisted curation study was presented as an abstract and poster at Neuroscience 2024 [Bolen and Hunt, 2024].

SADb addresses the need to recover measured afferent properties, feedback pathways, and experimental conditions from a dispersed literature. A researcher performing neuromechanical modeling needs to know what was measured, how a pathway was stimulated or represented, and the locomotor or postural state in which it was studied. SADb brings individual studies into a searchable collection so that researchers can compare findings, locate evidence relevant to a candidate sensory afferent pathway, and follow each record to its source.

The primary deliverable is SADb Explorer, a browser application containing 1,393 literature and related records. It has four main tabs (Figure 5.1, callout (A)): Table searches

individual records, Pivot compares classifications, Bubble map explores citation links among papers, and Search (online) finds candidate additions through OpenAlex, an open bibliographic index. The records, notes, and citation graph are embedded in the application, so the first three tabs can be used without an internet connection. Search (online) requires an internet connection and lets the user save candidate records for later curation. The application preserves both the records retained for active curation and archived records from the collected material.

The database was curated in Airtable, an online database system, before being transferred to SADb Explorer. Linked tables connect papers to feedback pathways and computational models, with separate records for reviews and recording methods. Paper records contain bibliographic information, curation notes, and classification tags; pathway and model records link these sources to candidate feedback mechanisms and their implementations.

5.1 Curation Methods

SADb is a curated research collection, rather than a registered systematic review or an exhaustive search of every relevant publication. I first collected records from selected AARL and personal Zotero collections used for previous papers on locomotion and postural balance, together with a few records already curated in Airtable by my advisor, Dr. Hunt. Duplicate Zotero entries describing the same publication were combined into a single record using DOI and title matching. I expanded the collection through keyword searches, references cited by relevant papers, and searches for subsequent papers citing

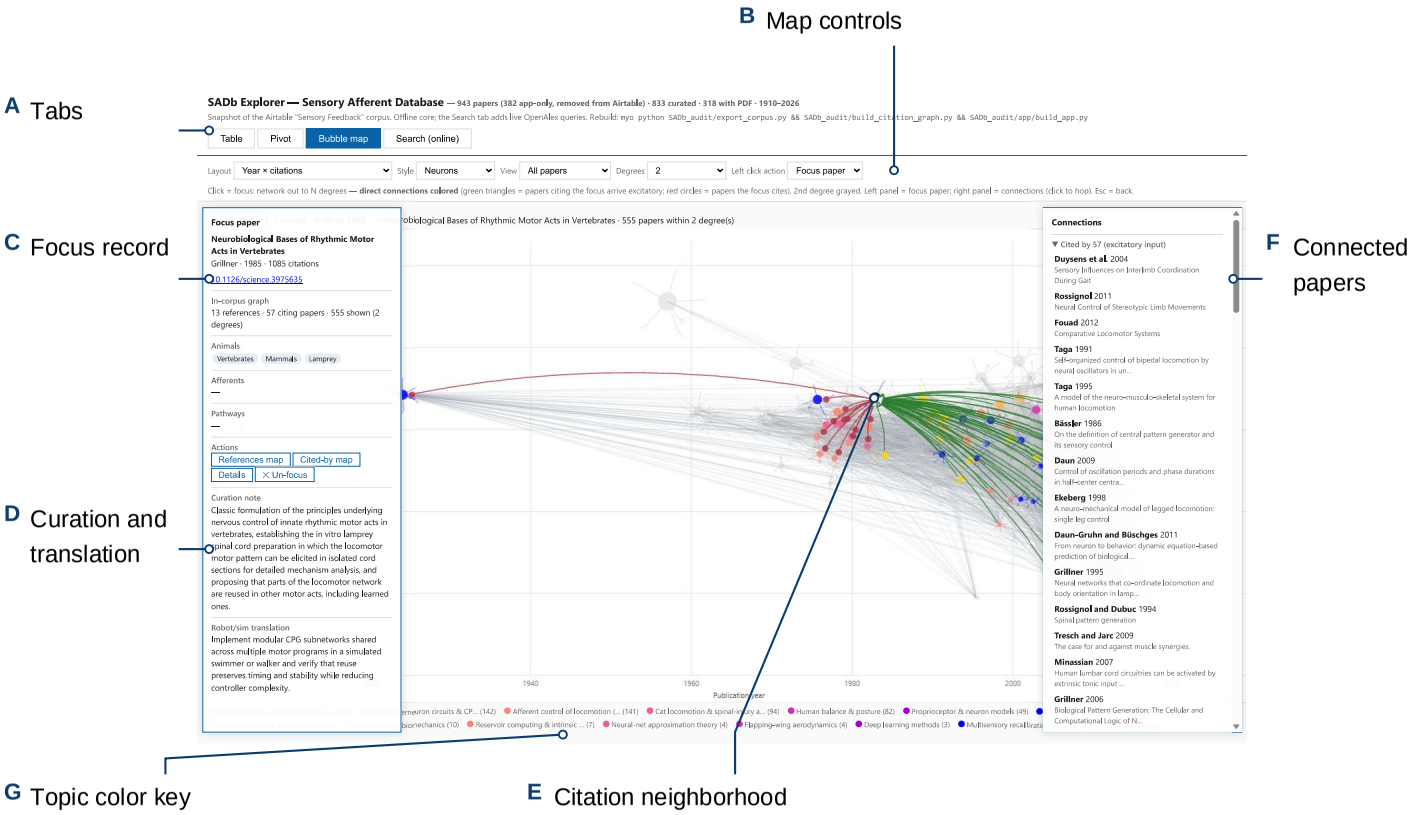


Figure 5.1: SADb Explorer with Grillner (1985) selected. Callouts identify the four main tabs (A), map controls (B), selected record (C), curation and translation notes (D), citation neighborhood (E), connected papers (F), and the topic-cluster color key (G).

them.

I initially imported the broader Zotero collections as groups, then narrowed the working base to sensory feedback in locomotion and posture. I retained records reporting sensory-afferent measurements, investigating feedback in locomotion or posture, or modeling a relevant sensory pathway. I removed records outside this scope from the working base. I also flagged records whose source text I could not locate, including conference posters found through ResearchRabbit, and removed them from that base. The application preserves the broader collection as archived records, which users can distinguish with the archive filter. Figure 5.2 summarizes the curation sequence. For retained papers, I recorded the animal or human study context where applicable, afferent class, feedback pathway, and the mechanism described by the study.

I implemented a monthly literature watch that queries OpenAlex and Crossref for candidate additions matching selected authors and keywords. Its search terms include "Ia afferent," "Ib afferent," "muscle spindle," "proprioceptive feedback," and "central pattern generator." The script removes DOI and version duplicates, verifies shortlisted DOIs through Crossref, and writes a shortlist for manual curation. Source-access limitations are recorded with the shortlist; inclusion and classification decisions remain manual.

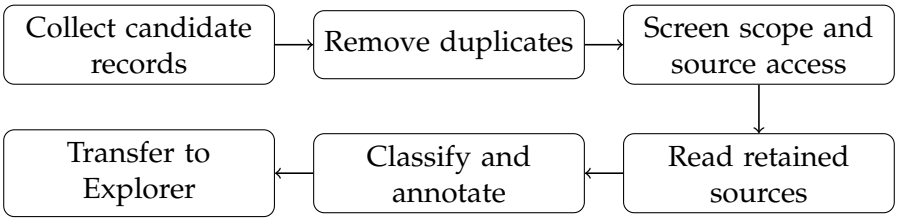


Figure 5.2: SADb curation workflow. The monthly skimmer supplies candidate records; source review, inclusion, classification, and annotation remain manual.

5.2 Results: Application and Corpus Coverage

The application contains 1,393 entries in five record categories: 943 Papers, 247 Models, 157 Review Papers, 42 Feedback pathways, and 4 Recorders. The searchable table includes the complete 1,393-entry collection. The Pivot and Bubble map tabs use the 943 entries in the Papers category. Archived records remain searchable in the application.

Within the application's Papers category, containing 943 entries, 833 carry curation notes, 606 have tags for the animals or human participants studied, 316 carry animal-study assessments, 119 link to feedback pathways, 95 carry an afferent classification, and 363 carry robot-or-simulation translation notes. DOI identifiers are present for 892 papers, and 318 have attached-PDF indicators. Within the Papers category, 730 papers are connected by 11,130 directed citation pairs. The Louvain algorithm identifies citation communities: groups of papers with comparatively dense citation connections. The application labels these communities as 18 named topic clusters (Figure 5.1, callout (G)), which help users navigate related studies in the Bubble map tab.

The Table tab combines text search with filters for animal or human study category, pathway, afferent class, record type, archive status, year, and note or PDF availability (Figure 5.3). Selecting a record opens its source information, curation notes, and notes on how the reported findings could inform a robot or simulation. The Search (online) tab queries OpenAlex for candidate additions and marks DOIs already represented in SADb.

The Bubble map tab offers two selectable layouts (Figure 5.1). The year–citations layout places publication year on the horizontal axis and citation count on the vertical axis. The topic landscape instead draws papers as nodes joined by citation links. A

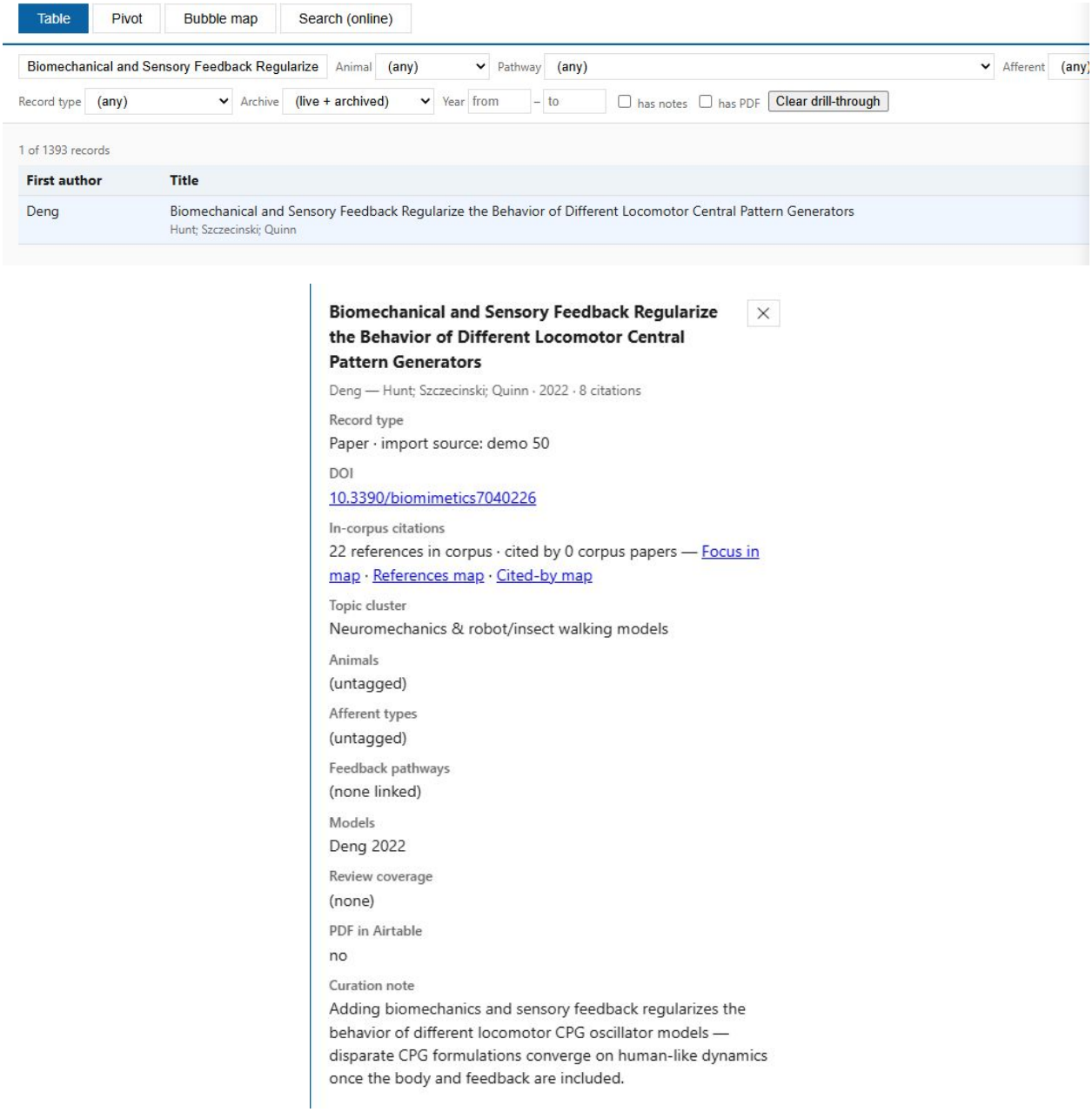


Figure 5.3: SADb Table view after a text search (top) and the selected record's detail view (bottom), showing its source information, linked records, and curation note.

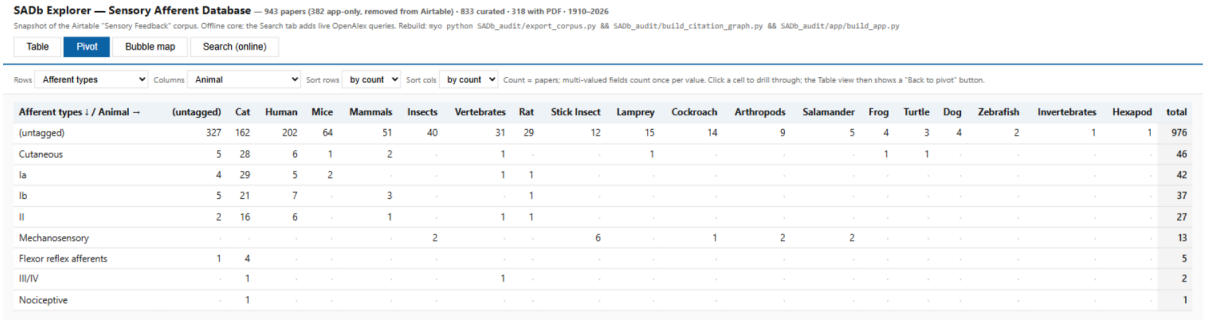
spring-layout algorithm balances attraction along links with repulsion between nodes to arrange connected groups together. Its coordinates arrange the network; neither axis measures publication year or citation count. The user controls the layout, display style, record grouping, citation-neighborhood depth, and click action (Figure 5.1, callout (B)).

Selecting a paper makes it the reference for the displayed citation neighborhood; the application calls this the focus paper. Its source information appears in the left record area (Figure 5.1, callout (C)), together with its curation note and notes on possible applications to a robot or simulation (Figure 5.1, callout (D)). The neighborhood selector displays papers up to one, two, or three citation links from the focus paper, following either its references or papers that cite it (Figure 5.1, callout (E)). A two-link neighborhood adds one level beyond the directly connected papers. Directly connected papers retain color, while papers reached through additional links are gray. In the illustrated neighborhood of the Grillner (1985) focus paper, 57 in-corpus papers cite it, it cites 13 in-corpus papers, and 555 papers lie within two links. Green triangles identify papers that cite the focus paper, and red circles identify papers cited by it.

The connection list separates papers citing the focus paper from its references (Figure 5.1, callout (F)). Selecting a listed paper makes it the new focus paper, allowing a reader to follow the development of an idea without repeatedly searching for individual titles. The reference-only and cited-by views isolate either direction, and the detail action opens the underlying record. A researcher can therefore start from a known experiment, inspect earlier studies supporting it, follow subsequent models or experiments citing it, and read the associated translation notes before choosing a candidate pathway for simulation.

For example, a researcher may ask whether type Ia feedback excites the muscle from which it originates or inhibits its antagonist. SADb contains separate pathway records for "Ia monosynaptic excitation" and "Ia inhibitory." These describe complementary connections: direct excitation of the agonist motor pool and inhibition of an antagonist through an inhibitory interneuron. The researcher can inspect the linked studies and model records to compare their target muscles, intervening neurons, and task conditions before choosing how to implement the sensory afferent pathways. The afferent class alone does not specify the sign of every connection.

The Pivot tab counts papers by two selected dimensions, including topic cluster, decade, primary author, animal or human study category, feedback pathway, and afferent classification. Figure 5.4 crosses afferent classes with categories of animal and human studies: each cell counts papers carrying both tags. Rows and columns can be ordered alphabetically or by count. Selecting a cell opens the corresponding records in the Table tab, and a return control restores the pivot. The pivot is computed over the 943 entries in the Papers category, including archived entries, rather than all 1,393 records. Because a paper can carry several animal or afferent tags, it can contribute to several cells. The counts describe the distribution of curated records, not independent experimental observations or the strength of evidence for a pathway.



Afferent types ↓ / Animal →	(untagged)	Cat	Human	Mice	Mammals	Insects	Vertebrates	Rat	Stick Insect	Lamprey	Cockroach	Arthropods	Salamander	Frog	Turtle	Dog	Zebrafish	Invertebrates	Hexapod	total
(untagged)	327	162	202	64	51	40	31	29	12	15	14	9	5	4	3	4	2	1	1	976
Cutaneous	5	28	6	1	2	·	1	·	·	1	·	·	·	1	1	·	·	·	·	46
Ia	4	29	5	2	·	·	1	1	·	·	·	·	·	·	·	·	·	·	·	42
Ib	5	21	7	·	3	·	·	1	·	·	·	·	·	·	·	·	·	·	·	37
II	2	16	6	·	1	·	1	1	·	·	·	·	·	·	·	·	·	·	·	27
Mechanosensory	·	·	·	·	·	2	·	·	6	·	1	2	2	·	·	·	·	·	·	13
Flexor reflex afferents	1	4	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	5
III/IV	·	1	·	·	·	·	1	·	·	·	·	·	·	·	·	·	·	·	·	2
Nociceptive	·	1	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	1

Figure 5.4: SADb Explorer pivot table by afferent classification and animal or human study category. Each cell counts papers carrying both tags and links to those records.

The feedback-pathway records extend beyond the mammalian core of Table E2: paired excitation and inhibition records cover the chordotonal organs and trochanteral hair plates of insect legs, a load-feedback record covers the campaniform sensilla, and further records hold fictive locomotion without sensory feedback (12 linked papers), group III/IV fatigue, and one paper each for somatosensory, vestibular, and visual contributions to postural control. The controller of Chapter 6 implements the mammalian subset.

5.2.1 AI-Assisted Literature Triage

I evaluated AI-assisted literature identification, summarization, and categorization, judging the outputs against how well and how quickly I could complete these tasks myself. This work was presented at Neuroscience 2024 [Bolen and Hunt, 2024]. The comparison includes ResearchRabbit, Litmaps, txyz, SciSpace, and MirrorThink. Some tools helped identify related papers or summarize straightforward material, but categorization remained inadequate: none received a good rating, and only txyz was rated adequate. Complex models and nuanced physiological findings required substantial human correc-

tion. The tools therefore offered limited help with the central task of accurately assigning papers to database categories. I made the final inclusion, classification, and removal decisions.

5.3 Discussion

5.3.1 Uses of the Database

SADb helps researchers connect an account of sensory feedback to the experiments and models supporting it. Searchable notes, linked records, and citation neighborhoods provide complementary routes through a literature spread across animal studies, human studies, and modeling disciplines. A reader can begin with a familiar paper, follow its references and later citing studies, and compare the conditions under which a feedback mechanism was observed. This makes the resource useful for learning a research area and for identifying questions that require a closer reading of the primary evidence.

The database also supports the design of neural models for locomotion and balance. Table E1 in Appendix E summarizes afferent classes, receptors, and encoded signals for selecting model inputs. Table E2 gives representative feedback pathways, their actions in the step cycle, and the studies linked to those records. These source links let a modeler examine the experimental basis for a candidate connection and the assumptions needed to implement it.

SADb supplies records and source connections through which a researcher can compare experimental evidence with model assumptions. Ting and Macpherson [2005] linked muscle-activation synergies to limb endpoint-force vectors in cats responding to support-

surface translations. Latash et al. [2020] compared cat stepping patterns with a mathematical model to investigate how rhythm-generator operation could change with control drive under tied-belt and split-belt conditions. The first study connects measured muscle activity to postural output; the second tests a candidate circuit explanation for locomotor phase patterns. Following the source papers exposes these differences in task, preparation, and explanation.

Comparing model records helps a researcher identify how sensory afferent pathways depend on the mechanical model and task. Geyer and Herr [2010] generated walking in a human musculoskeletal model using phase-dependent muscle reflexes. In a rat neuromechanical model, Deng et al. [2022] found that biomechanical entrainment and sensory feedback reduced behavioral differences associated with alternative CPG parameter choices. Model records provide starting points for comparing these approaches and two-level rhythm-generation and pattern-formation architectures [Rybak et al., 2006]. For the controllers in Chapter 6, pathway selection is followed by explicit choices and tests of receptor dynamics, gains, delays, and species-specific assumptions.

The balance, posture, and multisensory-recalibration clusters provide entry points for the vestibular and visual extensions proposed in Section 7.2.4. Searches of existing animal studies let users compare the conditions and tasks behind a proposed feedback rule before testing it in a robotic or computational model. Reusing this evidence for model-based tests supports the aim of reducing reliance on further animal experimentation (Section 2.10).

SADb supports selecting sensory afferent pathways and comparing whole neural circuits for neuromechanical models. The researcher can trace candidate mechanisms to their source studies and compare species, task conditions, and modeling assumptions.

Their encoding, connection signs, gains, and interaction with the body must then be implemented and tested. Robotic experiments can examine these neural mechanisms under controlled mechanical conditions and generate questions for the biological literature. Chapter 6 develops the models and simulation tools needed for that next step.

6 Tools for Neuromechanical Simulation

This chapter follows the development of tools for transferring neural controllers between mechanical models. An initial AnimatLab walker motivated the move to Gait2392, graphical neural construction, and accelerated simulation. These capabilities were then combined with staged controller training and BPA-actuated robot models.

6.1 Motivation: The AnimatLab Story

For a course project in ME 610: Neuromechanical Modeling, Morrow and I coupled the bipedal mechanics of Li et al. [2017] with the two-level central pattern generator (CPG) of Deng et al. [2019] in AnimatLab [Cofer et al., 2010]. The model had bilateral thigh, shank, foot, and toe segments, sagittal-plane joints, and antagonist Hill-type muscles at the hip, knee, and ankle. I added a passive toe joint. A shared rhythm generator (RG) supplied bilateral pattern-formation (PF) and motor-neuron (MN) pathways, with afferent feedback at the MN level. We adjusted currents, synaptic weights, and muscle parameters by hand and presented the suspended stepping result at the 2021 NIH BRAIN Initiative Investigators Meeting [Bolen and Hunt, 2021].

In a saved graded-neuron trial, alternating activity propagated through the RG, PF, and MN layers while the hip, knee, and ankle cycled at approximately 2.2 Hz. The aligned recording is retained in Appendix D (Figure D4). When released toward the ground, the model lost support while the neural rhythm continued (Figure 6.1). The controller

schematic is shown separately in Figure 6.2. This contrast between suspended motion and ground loading motivated a workflow for retuning a circuit when its mechanics and sensory conditions change.



Figure 6.1: **(A)** Original AnimatLab walker developed by Morrow and Bolen. **(B)** The model tipping during the ground test.

6.2 Knowledge Gap and Framing

The central problem was how to transfer an identifiable neural controller to a different mechanical model and train it efficiently. Changing muscle geometry, joint motion, inertia, or actuator type changes both the force produced by a neural command and the sensory signals returned to the circuit. The target was a detailed Gait2392 human model, followed by a robot counterpart actuated by BPAs. Comparing controllers therefore required consistent command and sensory interfaces, explicit mechanical checks, and repeatable training conditions.

I developed a workflow that combines graphical circuit construction with automated experiments. OpenSim and MyoConverter supply musculoskeletal mechanics, SNS-Toolbox supplies neural formulations, and Simulink/Simscape supplies graphical neural and me-

chanical modeling [Delp et al., 2007, Caggiano et al., 2022, Todorov et al., 2012, Nourse et al., 2023]. My contributions are the model implementations, custom blocks, actuator interfaces, and experiment drivers that connect these tools. The following sections build these capabilities around the Gait2392 reference before combining them in the supported robot example.

6.3 Gait2392 and Controller Transfer

I converted the Gait2392 human reference from OpenSim to MuJoCo with MyoConverter and connected it to SNS-Toolbox. The robot counterpart required separate treatment: its BPA paths could not simply be combined with unchanged human-muscle parameters. I therefore retained biological-muscle actuators in the human reference and implemented routed BPA forces for the robot counterpart.

The converted human model retains 92 actuator entries. Six small hip-rotator entries, the quadratus femoris, gemelli, and piriformis on each side, are disabled and contribute no force. The knee retains OpenSim's flexion-negative convention. Before interpreting a controller's motion, I checked muscle attachments, joint-action directions, force scale, and intended foot contact. Appendix D.6 records the conversion repairs and force comparison. Neural and mechanical signals are exchanged at a 2 ms solver time step.

For artificial-muscle simulations, I implemented an actuator-force calculation that evaluates the quasi-static BPA model from Chapter 3 at the current simulated actuator length and commanded pressure. The calculated tensile force acts along the model's specified tendon route. Length, velocity, force, and foot-contact measurements are then returned

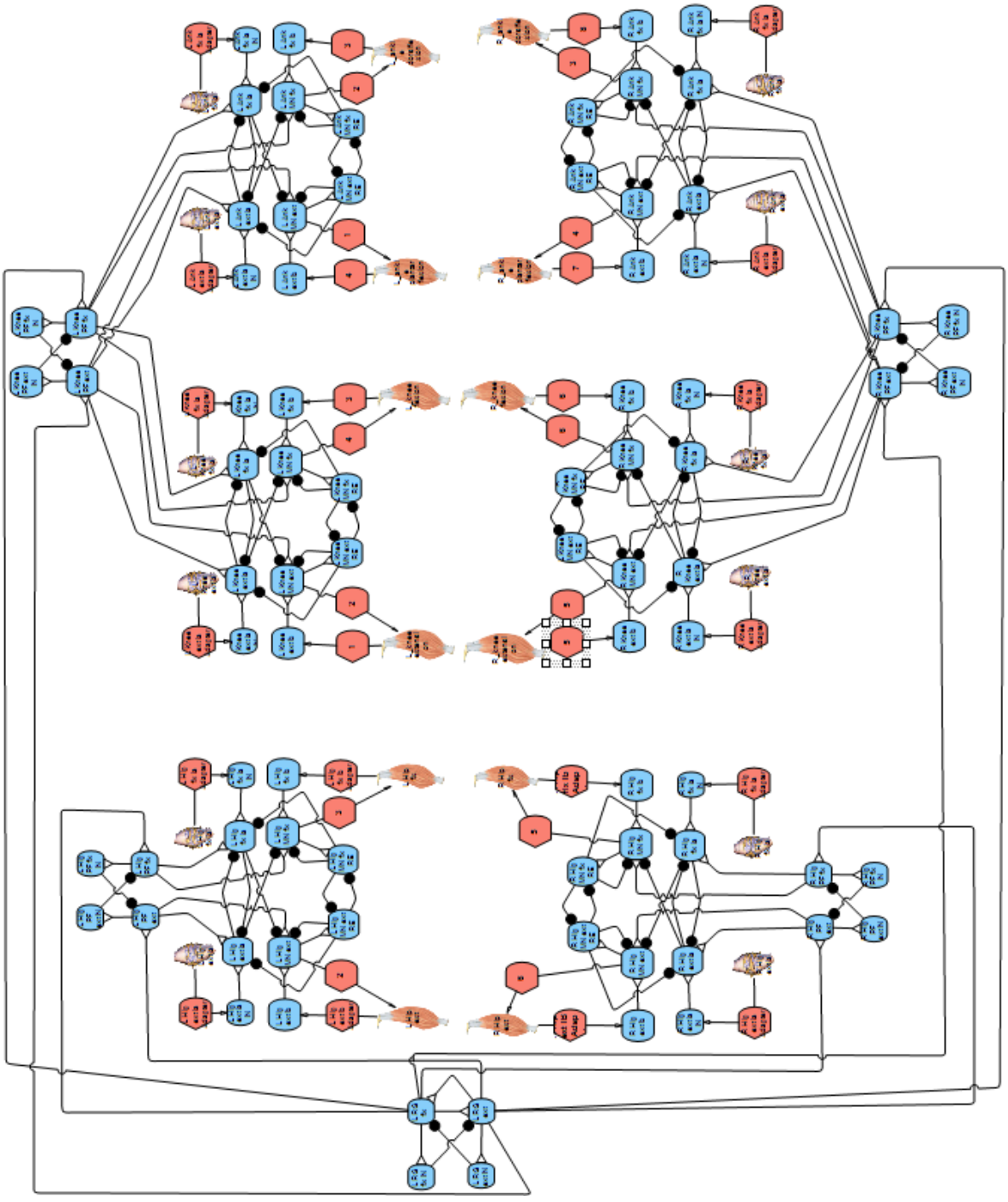


Figure 6.2: Original shared-rhythm-generator controller. The four-neuron RG supplies bilateral hip and knee/ankle pattern-formation groups.

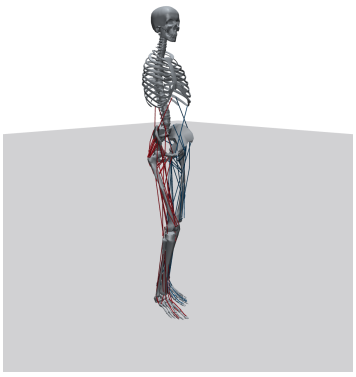


Figure 6.3: Converted Gait2392 human reference with bilateral muscle routes in a static flexed-knee pose. The geometry supports the neural-controller and actuator comparisons; this view is separate from the CAD-derived robot assembly.

to the neural controller through sensory channels. This closes the neural-command–actuator-force–sensory-feedback loop. The calculation represents the measured static force dependence; it does not add valve-flow or pressure-transient dynamics. Routing geometry and dynamic actuator loading therefore require their own checks beyond the isometric force fit.

The SADb of Chapter 5 supplies the physiological context for selecting Ia, II, Ib, and contact pathways. A database link identifies the supporting observation; it does not by itself determine a numerical gain or guarantee the same function in another model. The implemented encoding, connection sign, phase gating, and tested response therefore remain explicit parts of the controller specification.

Table 6.2 distinguishes the model bodies, actuation, and representative tests. A controller organization, such as the W2L layout, can be applied to more than one body; its name does not identify the mechanical model. Figure D2 contrasts the converted Gait2392

robot with the reduced-order knee model used for block demonstrations.

Single-actuator and matched-pose tests checked the converted mechanics before closed-loop controller tests. These checks separated conversion faults from changes caused by neural control.

At matched poses and OpenSim static-optimization activations (Figure D11; Table D9), eight of twelve sampled muscles had force-prediction R^2 between 0.87 and 0.98 and mean forces within 10 % of the reference. Rectus femoris and biceps femoris short head had negative R^2, indicating errors greater than a constant mean-force prediction. This comparison checks force predictions in the repaired MyoConverter-derived human model while identifying remaining muscle-specific differences. Because the comparison sets pose and activation frame by frame without reproducing trajectory velocities or tendon dynamics, it does not establish dynamic equivalence between the muscle formulations. That distinction matters when interpreting feedback based on muscle force.

These controller comparisons use a common mechanical reference and the same sensory and recording interfaces. Pathway removal tests how a controller uses feedback, while inverse-kinematics overlays show the resulting joint timing and amplitude relative to human motion. Appendix D.3.1 retains the controller comparisons, and Appendix D.3.2 retains the direct trajectory comparison. The next requirement was to make these experiments practical both through graphical circuit editing and through automated simulation.

6.4 Building the Neuromechanical Simulation Workflow

The Gait2392 reference defines the detailed mechanical target. The workflow also retains the simpler W2L model as a test case for checking circuit calculations, sensory pathways, and controlled perturbations before interpreting full-body results. Table 6.1 summarizes the capabilities, and Table 6.2 distinguishes controller organization from mechanical body and actuator type.

Graphical blocks support circuit construction, subnetwork inspection, and interactive stimulus application. Headless execution runs the same experiment without opening its graphical interface, allowing parameter searches and repeated perturbation tests to be automated. An application programming interface (API) exposes commands for constructing, controlling, and recording experiments; a software development kit (SDK) supplies supporting libraries and tools. Saved configurations and traces connect these automated runs to graphical review.

Figure 6.4 shows how the model-construction routes and controller interfaces connect. Chapter 7 places them within the broader simulation-to-hardware workflow (Figure 7.4). Detailed configurations and evaluation sets are retained in Appendix D.

6.4.1 Li–Deng/W2L Walker and Controller

I replaced the original shared RG with one RG on each side, linked through commissural pathways. The bilateral PF and MN organization was retained, while afferent feedback was extended to the RG and PF layers and heel- and toe-contact channels were added. This later controller is referred to as Walker Two-Layer (W2L).

Table 6.1: Models and tools developed for neuromechanical experiments.

Model or tool	**What it is used for**
Li–Deng/W2L walker and controller	Combine walker mechanics and neural control in AnimatLab and MuJoCo with SNS-Toolbox; compare matched implementations of the contact-driven neural core in NumPy and Simulink.
Gait2392-based models	Compare alternative neural controllers on the human reference model and a robot counterpart with BPA routing and the force model from Chapter 3.
SNS-Simscape block library and editor	Build neural pathways graphically from sensory, neuron, synapse, activation, and muscle-force blocks, including custom BPA blocks.
SOLIDWORKS-derived robot assembly	Couple a CAD-derived Simscape Multibody mechanism to a neural controller and BPA actuators using the force model.
MuJoCo–Simulink integration	Run an SNS-Simscape controller with a MuJoCo mechanical model through the MathWorks bridge, exchanging actuator commands and sensory measurements.
Curriculum and experiment drivers	Tune controllers in stages, vary support and sensory pathways, apply neural stimuli or mechanical perturbations, and record neural and mechanical responses.
Literature circuits and spiking variants	Compare selected published neural mechanisms and the behavior of graded and spiking implementations.
Accelerated execution and actuator options	Run neural and muscle calculations with compiled kernels, compare execution time, and select biological-muscle, compliant-tendon, or BPA force implementations.

Table 6.2: Model bodies, actuator formulations, support conditions, and representative evaluations.

Mechanical model	**Controller and actuation**	**Support**	**Representative evidence**
Gait2392 human	Alternative graded or spiking SNS; biological muscles	Virtual harness in ground tests; suspension in air tests	Reference-trajectory overlays and pathway-removal comparisons
Gait2392 robot counterpart	SNS; routed BPA force model	Test-specific virtual support	Artificial-muscle force and sensory-signal interfaces
Li–Deng/W2L walker	Two-level controller; source-model muscle interfaces	Suspension or virtual-walker support	Air stepping, implementation agreement, and feedback tests
CAD robot in Simscape	SNS-Simscape; BPA model	Virtual harness	Integrated standing example
CAD robot in MuJoCo	Neural timing and force allocator; reference-tension actuators	Specified mechanical assistance	Supported stance and abductor-control comparison

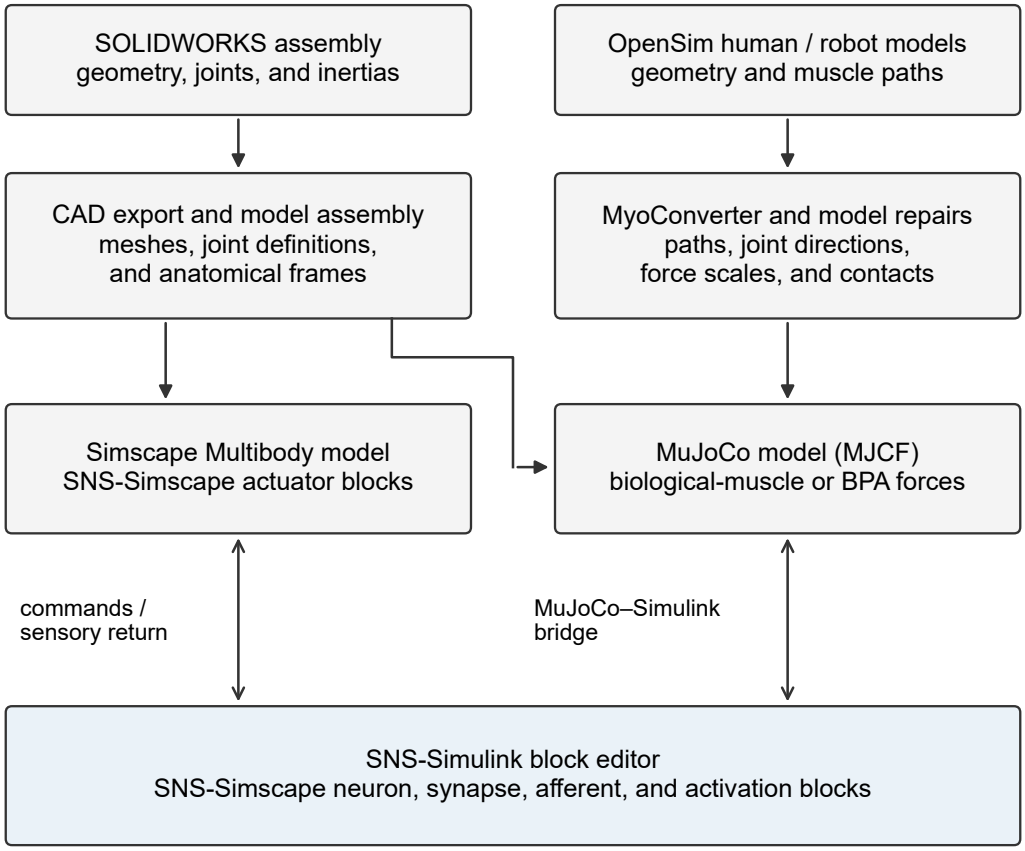


Figure 6.4: Model construction and neural-controller interfaces. SOLIDWORKS-derived geometry supplies Simscape Multibody and separately assembled MuJoCo models; OpenSim models follow the MyoConverter and repair route. The SNS-Simulink block editor assembles controllers from SNS-Simscape library blocks. Commands and sensory measurements couple these controllers to Simscape or, through the MuJoCo–Simulink bridge, to MuJoCo.

The newer bilateral W2L implementation contains 108 graded neurons and 216 chemical connections. Separate type-II relays serve flexor and extensor muscles at both hips, knees, and ankles, alongside retained phase-related feedback and distinct heel/toe contact channels. Figure 6.5 illustrates the selected left RG and hip organization from the bilateral W2L reference drawing. The aligned stimulus recordings in Appendix D.2.3 connect this circuitry to joint motion.

For cross-implementation checks of the contact-driven W2L rhythm core, I used the same neuron definitions, synaptic connections, and parameter values, with identical drive and contact inputs in NumPy and Simulink. Closed-loop experiments held a saved controller fixed while changing selected sensory pathways, drive, or support. Appendix B.5 documents the bilateral pathways and conductances of the independently replayed W2L configuration.

Under these matched inputs, burst periods matched at 1.000 s, all ten expected bursts were reproduced, and voltage-trace correlation was at least 0.99996 (Figure D3). This matched-input comparison verifies the rhythm-core implementation.

Removing the phasic heel-contact inputs caused the circuit to latch in both implementations. The alternating activity therefore depends on the specified sensory drive.

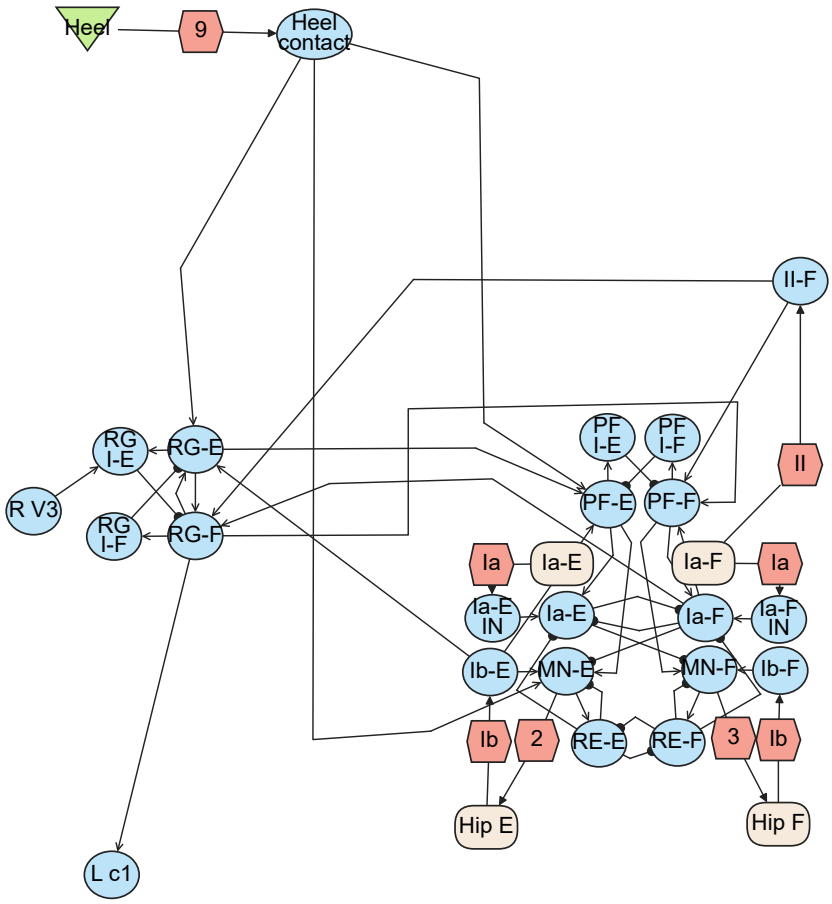


Figure 6.5: Selected left RG and hip subnetwork of the bilateral W2L reference. Connections leaving the selection are omitted. Muscle and spindle images are replaced with labeled vector symbols. E/F denote extensor/flexor; Ia, II, and Ib identify sensory pathways, and RE denotes Renshaw interneurons.

The saved W2L controller was tested with all saved effective gains delivered to the compiled network. With controller, drive, and support settings fixed, disabling all sensory feedback shifted median bilateral contact phase from 0.498 to 0.223 cycles. A phase of 0.5 cycles indicates alternating foot loading; the supported-stepping criterion accepts 0.35–0.65 cycles. Forward progression and joint cycling persisted, but the feedback-disabled condition failed this criterion for supported stepping (Figure 6.6). Over the 4.5 s to 30 s scored window, forward displacement was 2.239 m with feedback and 1.996 m without it. The comparison identifies a sensory-feedback contribution to bilateral contact coordination. Additional pathway deletions and stimulus tests are summarized in Appendix D.5.2 (Figure D10).

Anatomical action checks, initial-pose alignment, and bilateral recruitment adjustments produced nine left and eight right qualifying swings over ten seconds in the newer supported MuJoCo implementation (Figure 6.7). Passive rails and guides retained controllable support while the legs alternated.

These controlled comparisons establish a reusable test case. To construct and alter the same kinds of neural pathways graphically, I next developed the SNS-Simscape blocks and editor.

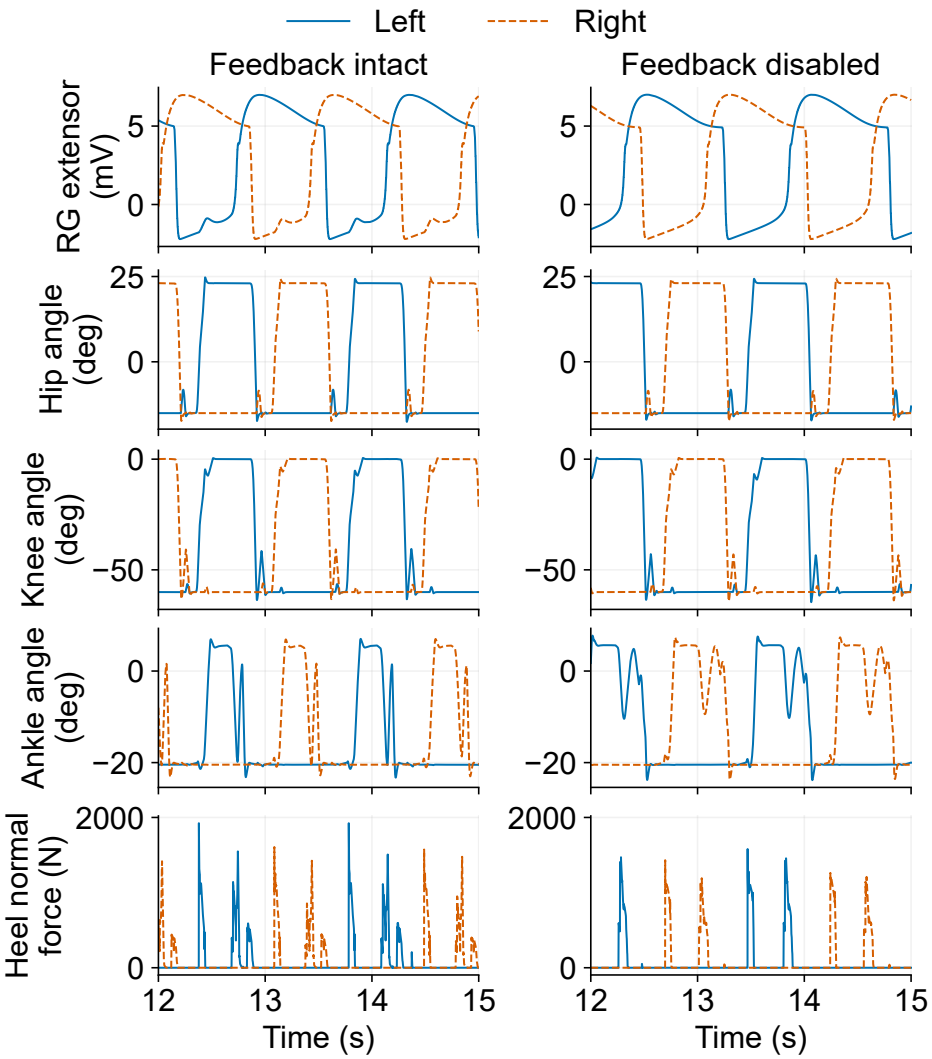


Figure 6.6: Sensory-feedback comparison with fixed W2L conductances, drive, and support settings. Extensor-RG voltages, hip, knee, and ankle angles, and heel normal forces are shown over the same 12 s to 15 s interval, with feedback intact (left) and all sensory feedback disabled (right). Blue solid and orange dashed curves denote the left and right legs. Cycling persists in both conditions. Over the full scored window, median bilateral contact phases are 0.498 and 0.223 cycles; the latter falls outside the 0.35–0.65 acceptance interval.

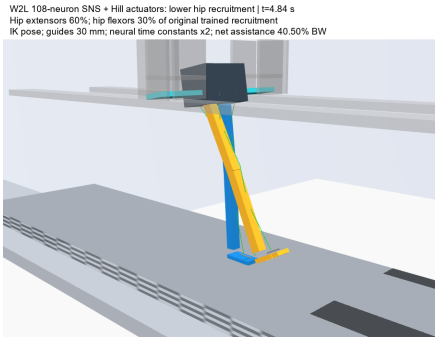


Figure 6.7: Newer W2L model stepping on the MuJoCo treadmill with passive rails and guides; mean net assistance is 40.5% of body weight.

6.4.2 SNS-Simscape Block Library and Editor

I used MATLAB for graphical controller experiments, using Simulink for the neural controller and Simscape Multibody for the mechanical model. Motor-neuron outputs passed through activation and muscle-force blocks to act on the mechanical model; muscle and joint measurements returned through afferent-encoder blocks as sensory input. Importing a mechanism and obtaining neural oscillation separately checked individual components. Testing the closed loop was also necessary because the timing, magnitude, and sign of those exchanged signals determine the coupled response.

The MATLAB work included both mechanical-model import and neural-response comparisons, followed by coupled simulations. Selected neural implementations reproduced their reference responses, and mechanical import provided models that could be assembled and simulated. Longer comparisons identified useful calibration targets: a full-network export agreed with its reference during an early transient and subsequently diverged, while one constant-drive Simulink realization reached a steady state and its NumPy reference oscillated (Appendix D.2). These checks guided separate verification of neural dynamics, actuator action, and sensory coupling before full-model tests.

I subsequently developed both block-programmed and scripted implementations, allowing controller experiments through SNS-Simscape or MuJoCo with SNS-Toolbox.

The SNS-Simscape library implements neuron, synapse, afferent-encoder, activation, and muscle-force elements as editable Simulink blocks. The library retains the conductance-based neuron and synapse formulation supplied by SNS-Toolbox; it does not introduce a new neuron model. Verification uses selected circuits under matched inputs. Figure 6.8

shows the core blocks and graded-neuron signal flow.

I used reduced-order reflex demonstrations to check the graphical blocks in a coupled model before assembling a full locomotor controller. In these examples, the sensory pathway, neural response, and mechanical correction can be inspected directly.

The library assembles Ia stretch excitation, reciprocal inhibition, and Ib autogenic inhibition in a reduced-order antagonist knee model (Figure 6.9**(C)**). After the model reaches a steady operating point, a 2 nA, 30 ms current pulse at the extensor Ia sensory-neuron input evokes 1.09° of extension, peaking 111 ms after pulse onset. The knee then returns toward the matched no-stimulus trajectory, meeting the specified recovery thresholds after 6.355 s. The displayed configuration broadens the MN and Ia synaptic saturation ramps; the original narrow-ramp settings maintained oscillations. This is a transient test of the implemented feedback and actuator equations, rather than a physiologically calibrated tendon-tap model. Appendix D.2.1 gives the configuration and numerical checks, with the complete response in Figure D5.

A second demonstration tests feedback in an arm holding a beverage container as it is loaded (Figure 6.10). With the reflex enabled, arm sag was 2.00°, compared with 7.85° when it was disabled. As the forearm sags, biceps stretch feedback excites the biceps motoneuron and reciprocally inhibits the triceps motoneuron. The separate Ib pathways provide autogenic inhibition of each muscle. The reduced sag demonstrates a corrective effect of the implemented sensory pathways in the coupled neural and actuator blocks.

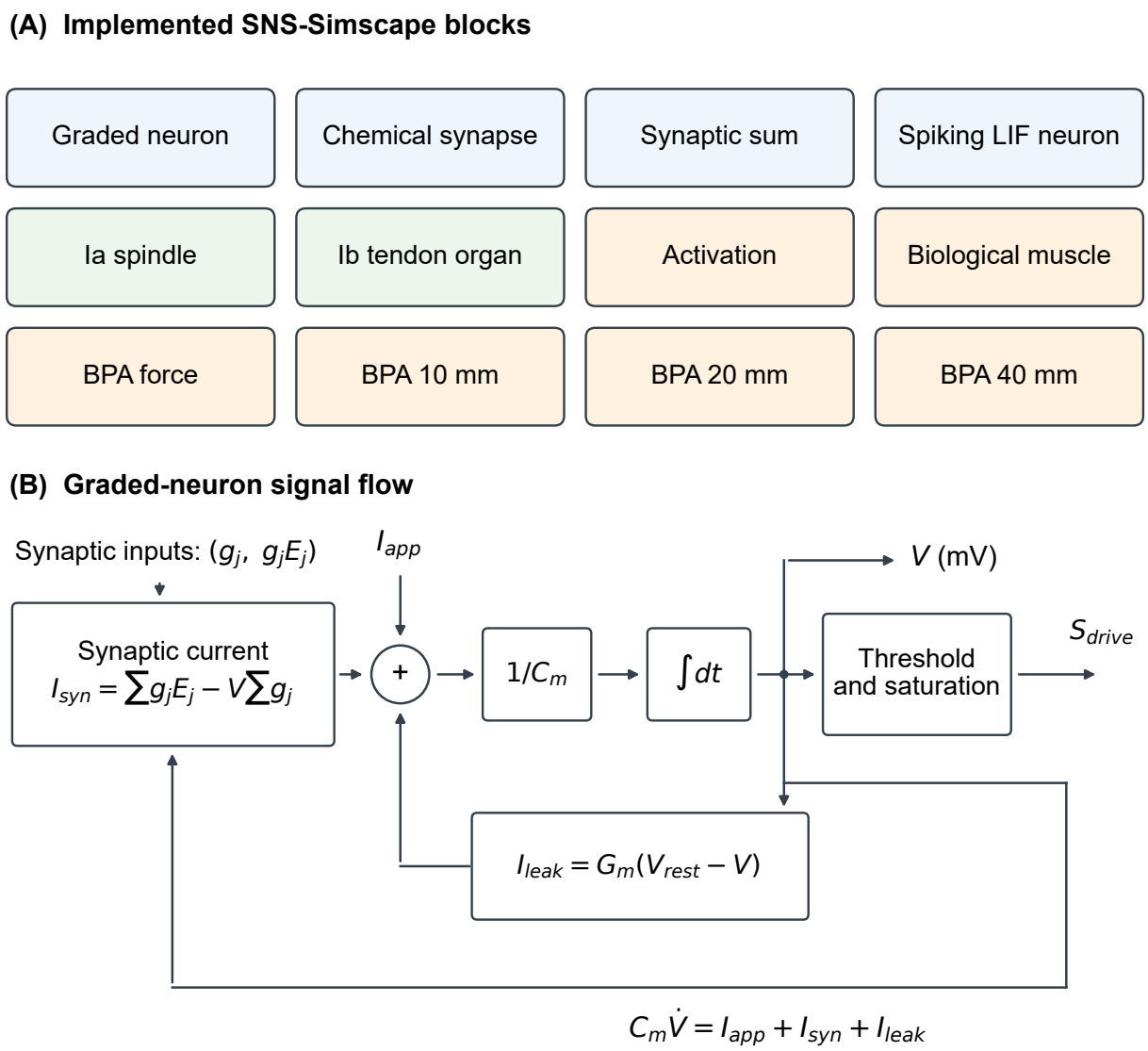


Figure 6.8: **(A)** The twelve implemented SNS-Simscape library elements, redrawn as a functional block catalog. **(B)** Signal flow of the implemented graded neuron: synaptic conductance and reversal-potential products form the synaptic current, which is combined with applied and leak currents before membrane integration. Membrane voltage and normalized drive are separate outputs.

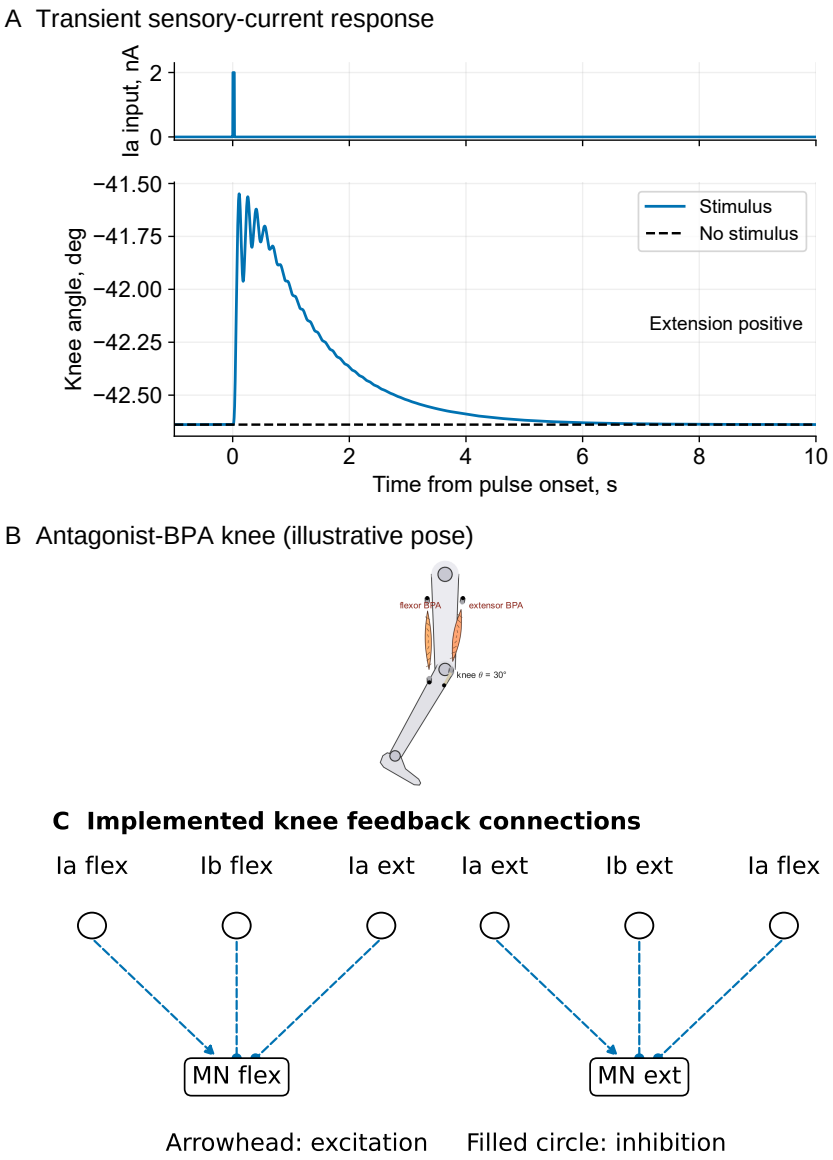


Figure 6.9: Transient knee response to a 2 nA, 30 ms sensory-current pulse after settling (A), antagonist-BPA leg schematic in an illustrative pose (B), and the six implemented feedback connections (C). Flexion is negative. The graded-response configuration uses broadened MN and Ia synaptic saturation ramps; its response is compared with a matched no-stimulus control.

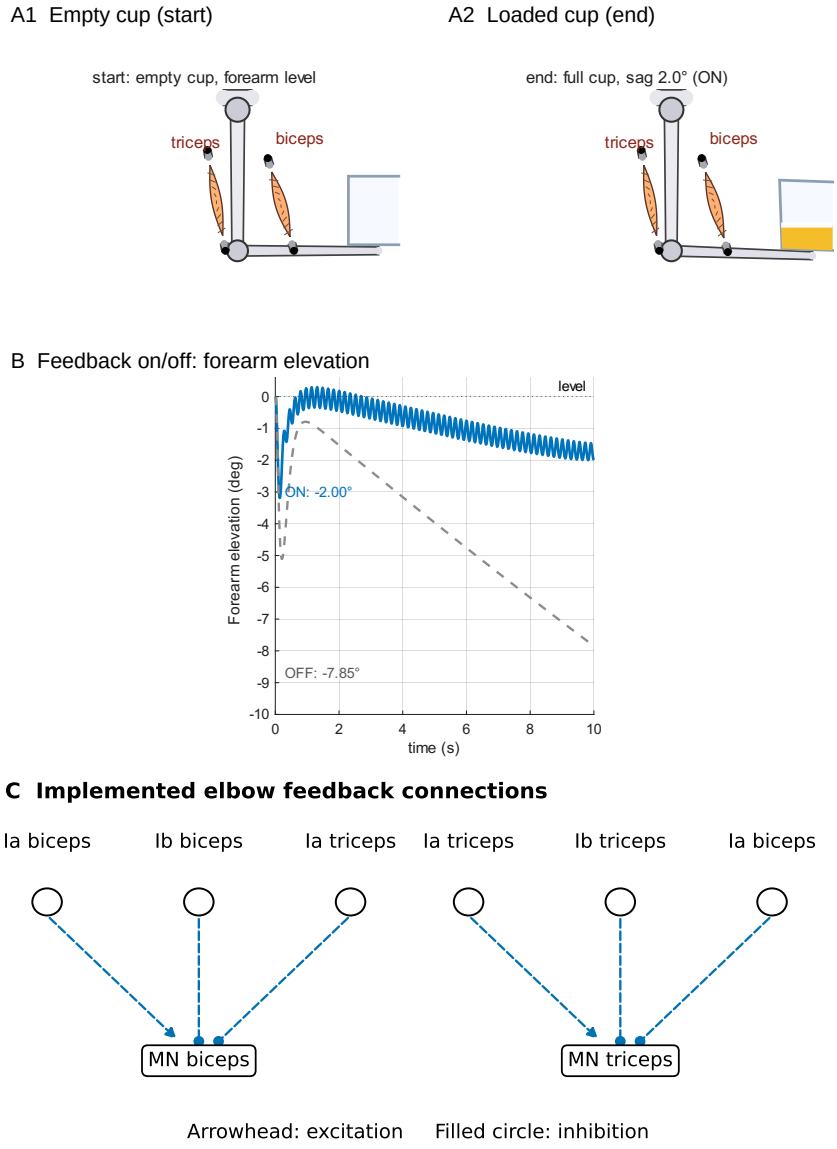


Figure 6.10: Loaded-arm demonstration: empty and loaded cup positions (A1, A2), feedback-on/off kinematics (B), and all six implemented feedback connections (C).

6.4.3 SOLIDWORKS-Derived Robot Assembly

I imported OpenSim-derived and SOLIDWORKS-derived mechanisms into Simscape Multibody. For the robot assembly, SNS-Simscape blocks connect the neural controller to BPA actuators using the force model, while joint and actuator measurements return through sensory blocks. This provides a graphical route from CAD geometry to a neuromechanical simulation. A separately assembled MuJoCo CAD model supplies additional integration and control tests (Appendix D.5.4).

A newer CAD-model comparison tests whether additional abductor control prevents the legs from crossing (Figure 6.11). Under the same assistance settings, muscle-mediated separation feedback changed reversed ankle ordering from 67 % of sampled states to zero. The model uses a layered SNS for timing together with an engineering force allocator. The allocator sets MN current targets for posture and separation, while the SNS supplies movement timing. This comparison demonstrates supported model integration and a separation-control effect. Appendix D.5.4 gives the matched measures and foot-mount sensitivity.

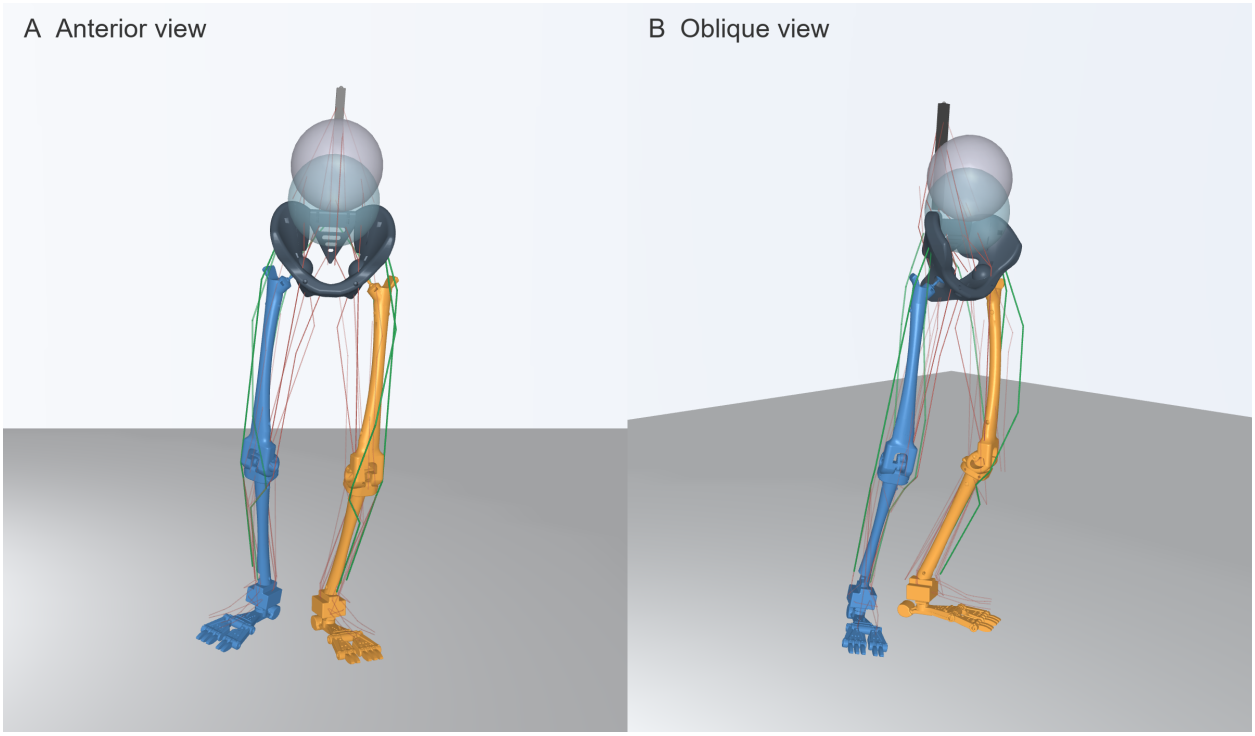


Figure 6.11: Anterior (A) and oblique (B) views of the same recorded CAD-model state. Right and left legs are blue and gold; selected hip-abducting routes are green. The 75 kg model uses 54 reference-tension actuators with hybrid neural and postural control. Vertical assistance supplies 80% of body weight in this interval; lateral and angular assistance remain active. Attachments, the welded foot mount, and passive knee stops are provisional.

6.4.4 MuJoCo–Simulink Integration

The MathWorks MuJoCo–Simulink bridge couples a block-programmed neural controller to a MuJoCo mechanical model. SNS-Simscape outputs supply actuator commands, and muscle length, velocity, force, and joint measurements return to the controller. A controller can therefore be edited graphically in Simulink and tested with MuJoCo mechanics. I implemented and tested the neural and actuator signal connections supplied through this bridge.

To test this connection, I paired the bridge with SNS-Simscape and compared single-muscle activation, different control rates, and sensor readback against a Python integration. Readback was bit-exact over the scored window. An exported neural network also drove all 92 actuator channels during a 5 s run without afferent return; a separate closed-loop reflex experiment exercised both commands and sensory measurements. These tests verify signal exchange through the interface. Under constant neural drive, one Simulink circuit realization reached a steady, nonoscillating state while its NumPy reference oscillated, showing why successful signal exchange must be followed by circuit-specific dynamical checks before transferring tuned parameters.

6.4.5 Curriculum and Experiment Drivers

The training workflow combines the four neural-training operations presented by Hunt [2019] with my five-stage physical curriculum (Table 6.3). The neural operations establish intrinsic rhythm, tune its entrainment to expected afferent signals, fit feedforward motor activation, and refine that activation with sensory feedback. The physical stages then test

the coupled model in deafferented air motion, afferented air motion, standing, supported ground stepping, and refinement with reduced assistance. A controller must be assessed at each physical stage rather than advancing because its neural activity alone appears rhythmic.

For the gain searches, I used a tree-structured Parzen estimator to propose parameter combinations from earlier evaluations. Each candidate is tested through simulation, and saved configurations allow a selected controller to be replayed before the next condition is introduced.

The experiment drivers then hold a selected controller fixed while changing sensory pathways, drive, mechanical support, or external perturbations. Finalists are also replayed under small parameter perturbations. Gait scoring requires detected cycles before a score is interpreted as movement quality; unsuccessful runs cannot be promoted by a reserved numerical failure value. The reference-library comparison tests transfer beyond the tuning reference. These procedures provide a repeatable way to identify where a controller works and which changes disrupt it, rather than making a single favorable trajectory the endpoint of tuning. Detailed scoring and configuration records are in Appendix D.3.

A separate training implementation follows the four-step approach presented by Hunt [2019]: select a CPG whose undriven period is approximately twice the desired locomotor period; fit afferent input connections using expected feedback to recover the target timing; fit CPG–PF outputs to motor-neuron activation peaks; and fit Ia and Ib feedback to refine the activation pattern. These parameter-training operations are distinct from the physical evaluation tasks in Table 6.3. With the Gait2392 human model, one trained configuration passed deafferented and afferented air-motion checks and assisted standing; these checks

provide the starting configuration for training load-bearing steps.

Table 6.3: Five-stage controller-tuning curriculum. Each selected controller seeds the next stage. Air-stepping stages are suspended; ground stages retain mechanical support. The table defines the physical tasks and evaluation measures; it does not imply that every controller has passed every stage.

Stage	Training task	Condition introduced	What is evaluated
1	Deafferented air stepping	Legs suspended; sensory feedback disabled	Rhythm generation and swing depth
2	Afferented air stepping	Heel, toe, and load feedback enabled	Rhythm and swing with sensory feedback
3	Standing balance	Standing posture and push perturbations	Ability to maintain posture during a push
4	Harness-supported ground stepping	Ground contact with measured external support	Alternating stance load, swing-foot unloading and clearance, forward progression, and joint kinematics
5	Contact-driven ground walking	Contact-driven pathway variant	Joint kinematics against a human reference cycle

A separate W2L configuration reproduced its supported-stepping trajectory in four independent replays, and all twelve reruns with 1 % parameter perturbations retained supported stepping. This demonstrates that a saved controller can be replayed and tested in its local parameter neighborhood. The repeatability tests and the pathway-removal panels use separately specified support settings; their event counts are therefore reported separately. Appendix D.5.1 retains the event counts, bilateral phase, displacement, speed, and support accounting.

The support protocol combines a constant weight-support term with position guidance,

providing controllable loading conditions during training. These components can be reduced progressively as the controller adapts to greater stance loading.

6.4.6 Literature Circuits and Spiking Variants

The circuit diagrams in Chapter 2 provide the reference architectures for these tests: the Rybak RG/PF hierarchy (Figure 2.8), the two-joint PF populations in the Shevtsova model (Figure 2.9), and the Deng sensory–motor organization (Figure 2.10). The Di Russo reflex network and Shinohara bilateral network (Figures 2.11 and 2.12) provide further comparisons for afferent routing and interleg coordination. The experiments here test selected implemented mechanisms from this literature.

Biarticular mechanics require explicit consideration when organizing the pattern-formation layer. Deng et al. [2023] tested a suspended rat-hindlimb model and compared active, passive-only, and disabled biarticular muscles during perturbations. Their results motivate explicit multi-joint pattern-formation groups and perturbation tests when adapting controllers to the W2L and Gait2392 models.

Selected literature-derived circuits were tested with matched inputs and layer-specific stimuli before interpreting coupled motion. The reduced Rybak circuit was tested without mechanics; the Deng-derived circuit drove a suspended W2L model. Spiking variants retained a continuous motor-activation interface while changing the neural dynamics. Appendix B gives their equations and calibration, and Appendix D.4 gives the comparisons.

A complementary neural-only test distinguishes mechanisms that can produce a missing motor burst. In the reduced Rybak implementation, pattern-formation stimulation

suppresses flexor motor output while the rhythm generator continues; rhythm-generator stimulation shifts the subsequent phase. Reproducing that distinction demonstrates the use of controlled interventions to test circuit organization before adding mechanical coupling (Appendix D.2.2).

A missing motor burst can arise from a disturbance of the rhythm generator or from suppression farther downstream. I applied stimuli separately to the RG, PF, and MN layers of a reduced Deng-derived circuit [Deng et al., 2019, Nourse et al., 2023]. A resetting deletion suppresses expected motor activity and shifts the subsequent rhythm; a non-resetting deletion suppresses activity while preserving the original schedule. Matched baseline and stimulus runs distinguish the rhythm-generator clock from the motor output.

Stimuli lasting two baseline periods suppressed two expected flexor motor bursts at each layer. RG stimulation shifted both RG and motor timing by 0.211 cycles, whereas PF stimulation and direct MN inhibition preserved the original schedule. A separate 1 s PF stimulus shifted motor timing by 0.208 cycles while leaving the RG clock unchanged. The comparison separates motor-pattern timing from rhythm-generator resetting. The smaller-current sweep produced no observable deletions. Appendix D.2.3 gives the protocols and all stronger-current controls. A separate bilateral W2L test shows how brief hip-PF inhibition changes motor and joint responses while RG activity continues (Figure D6).

Spiking-neuron comparisons preserved a continuous motor-activation interface while changing rhythm timing and, in some runs, limb coordination (Appendix D).

A public biarticular rat-hindlimb model was also imported for circuit inspection. It contains bilateral hindlimb mechanics, 120 neurons, sixteen muscles, and native RG, PF, and biarticular subnetworks (Figure 6.12). This public precursor retains a fixed pelvis;

it is distinct from the reduced fourteen-neuron test and from the final published 2023 implementation.

6.4.7 Execution Speed and Actuator Implementations

The accelerated workflow stores circuit topology and connection indices once, evaluates only existing synapses, reuses preallocated state and recording arrays, and compiles repeated neural, sensory, and muscle calculations with Numba where supported. Reference and accelerated implementations are compared under the same inputs, integration steps, and mechanical conditions before using the faster backend for training. Table 6.4 distinguishes complete coupled-loop timings from neural-only timings after warm-up. Loading, compilation, file output, and animation rendering are excluded.

Table 6.4: Execution times on easteregg2 after warm-up. The Gait2392 reference uses native SNS-Toolbox; W2L uses the same ported equations in Python. Neural-only rows exclude mechanical integration. The equation-kernel row compares a Di Russo-inspired Python implementation with its compiled Numba implementation.

Computation	Workload	Reference (s)	Fast (s)	Speedup
Coupled MuJoCo models				
Gait2392 balance	8 s	122.120	13.253	9.21×
W2L80 suspended	4 s	17.208	0.970	17.75×
Neural calculations only				
SNS-Toolbox, 1,027 neurons	500 updates	24.534	0.245	100.17×
Same benchmark, quiet repeat	500 updates	14.306	0.253	56.55×
Python equation kernel	1 update	1.978×10^{-3}	3.385×10^{-6}	584.33×

The 1,027-neuron benchmark accelerated fixed-conductance neural updates by 100.17 times, with a maximum voltage discrepancy of 3.55×10^{-15} mV. A quiet-machine repeat

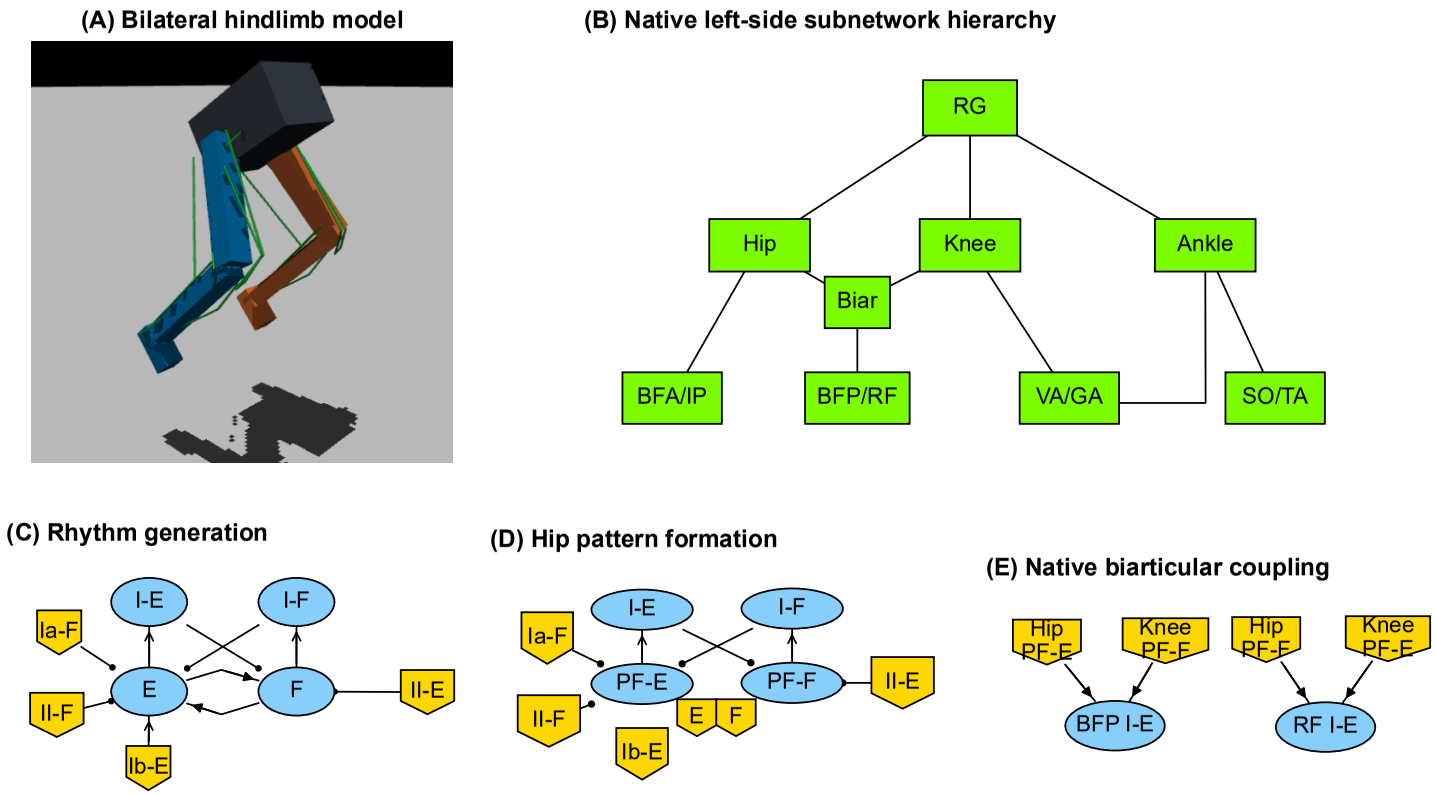


Figure 6.12: Imported public biarticular hindlimb precursor: model geometry and selected native hierarchy, rhythm-generation, hip pattern-formation, and biarticular coupling pages. Native neuron positions and internal connection routes are retained. BFP and RF identify biceps-femoris posterior and rectus-femoris pathways. The static model view shows geometry; it is separate from the W2L stepping experiments.

measured 56.55 times acceleration. The separately compiled Di Russo-inspired neural equations reached 584.33 times acceleration over their Python implementation. Mechanical integration, contact, and muscle calculations account for the smaller speedups of complete coupled simulations.

The saved Gait2392 positions, controls, center-of-mass, and neural arrays were identical between backends. The W2L comparison had a maximum force difference of 7.94×10^{-11} N. Its accelerated loop ran approximately 4.13 times faster than real time. The newer 108-neuron ground example advanced ten simulated seconds in 4.265 s, approximately 2.34 times real time. Matched whole-model AnimatLab and SNS-Simscape timings were not measured for these comparisons; their graphical construction and inspection roles complement headless training. Additional backend checks are retained in Appendix D.

Actuator mechanics introduce a separate trade-off. The implementations include the converted biological-muscle models and the quasi-static BPA force calculation; a compliant-tendon prototype adds dynamic fiber and tendon equilibrium. In an isolated benchmark, 92 compliant muscle units advanced 16 s of simulated time in 3.55 s after compilation, approximately 4.51 times real time. The separately timed muscle bank therefore advances faster than real time. The prototype also includes fiber damping; separate joint-damping tests checked transient response. The additional equilibrium calculation increases cost relative to rigid-tendon actuation. These alternatives allow actuator fidelity and execution cost to be examined separately when selecting a model for controller training or eventual hardware control.

A separate BPA test interface combines the static Festo force surface with chamber-pressure dynamics and a synthetic hysteresis term based on filtered shortening and length-

ening velocity. This history-dependent term provides a controlled actuator-model discrepancy for neural-compensation tests; identifying its parameters from dynamic BPA measurements is a direction for future research.

The integration culminates in the SOLIDWORKS-derived robot assembly coupled in SNS-Simscape to a synthetic neural controller and BPAs using my force model, with a virtual harness assisting standing. This example combines graphical neural construction, CAD mechanics, artificial-muscle forces, and sensory return in one model. It provides a starting point for studying the transition from supported posture to coordinated movement.

6.5 Discussion: What the Tools Establish

The tools connect graphical neural construction, detailed mechanics, alternative actuator laws, and automated experiments. Matched neural traces establish implementation agreement, and command and sensor exchange through the bridge couples block-programmed controllers to MuJoCo models. The W2L comparison identifies a sensory-feedback contribution to bilateral contact coordination, while the loaded-arm example demonstrates a corrective response in the coupled neural and actuator blocks.

These simulations provide a platform for investigating neural coordination with experimentally characterized BPA forces. Motor-neuron outputs are converted to actuator excitation or pressure commands, and sensory feedback allows the controller to respond to the resulting movement and loading. Future researchers can extend this simulated BPA control through improved dynamic characterization and closed-loop hardware experi-

ments, including pressure response, loading-dependent behavior, and neural compensation for actuator-model errors. Chapter 7 connects these opportunities with the actuator characterization, torque models, SADb, and robot design.

7 Discussion

7.1 Overview

The research program connects two goals: using biology to improve robots and using biomimetic robots to investigate biology. Actuator mechanics and neural control must be considered together if a legged robot is to adapt to uneven terrain, disturbances, and changing loads. A sufficiently characterized robot also permits controlled changes to mechanics and sensory pathways that would be impractical or unethical in human experiments. Prosthetic and orthotic devices provide a further application of the resulting design and control knowledge.

This dissertation advances that program through four contributions: BPA force characterization, isometric joint-torque design factors, the Sensory Afferent Database, and tools for neuromechanical experiments. They address different sources of uncertainty between a proposed biological mechanism and a robotic test. The force and torque models support first-pass static sizing of the tested actuator and knee configurations; the database connects controller choices to published evidence; and the simulation tools make the resulting coupled systems available for controlled evaluation. Figure 7.1 places these contributions within the broader development process.

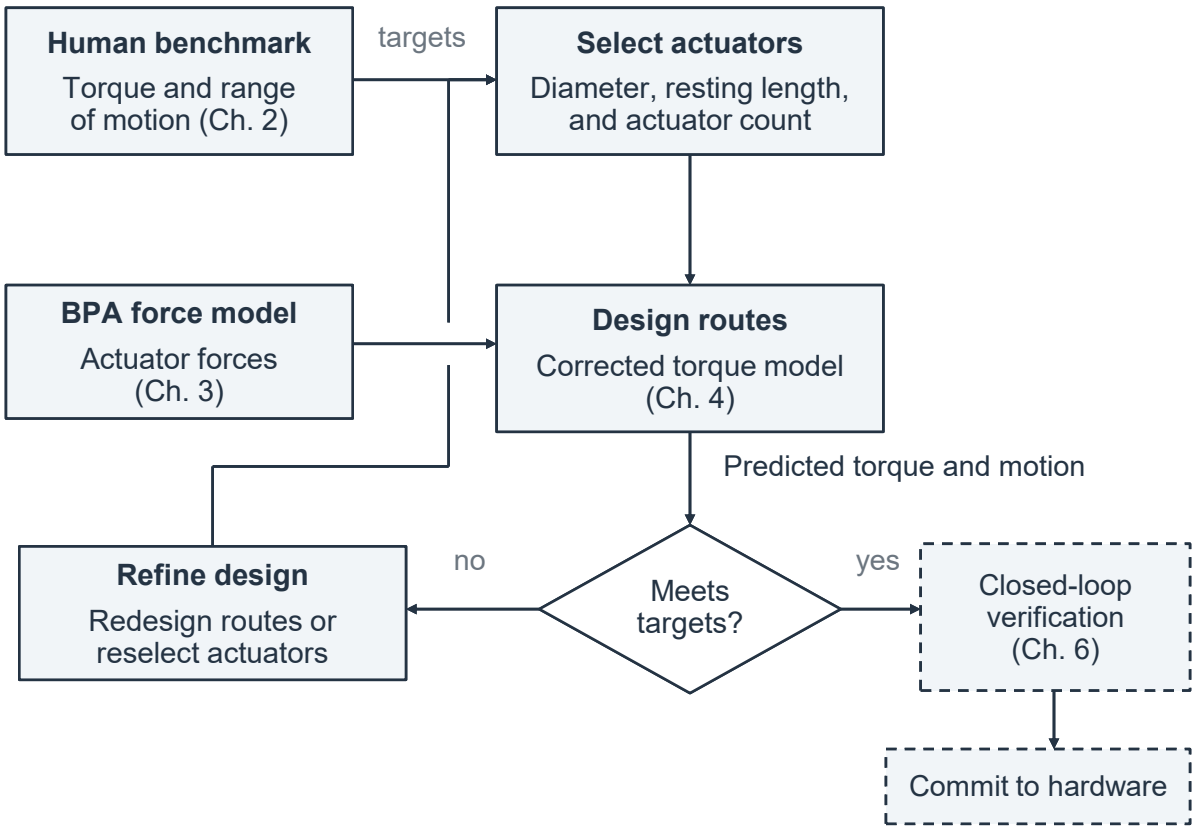


Figure 7.1: Proposed workflow connecting human torque targets, BPA characterization, route design, neuromechanical simulation, and hardware development.

7.2 Research Contributions and Future Work

7.2.1 BPA Actuator Characterization

Chapter 3 establishes that the maximum isometric force of Festo BPAs depends on resting length, a design variable not represented adequately by the manufacturer's tool or the compared force models. The resting-length-dependent maximum-force model and three-coefficient nondimensional force surface provide a basis for selecting actuator length and predicting force across the measured contractions and pressures. This closes an actuator-sizing gap; it does not identify the complete dynamic behavior of a pressurized actuator.

Future Work

The maximum-contraction function remains a useful next target. A predictive relationship between ε_{620} and resting length was not established, so each actuator's free contraction must still be measured. Tests across product batches and the 10 mm, 20 mm and 40 mm diameters could distinguish a repeatable length dependence from batch variation. The characterized 20 mm actuators came from a single batch.

Dynamic characterization should distinguish active pressurized-braid force, passive resistance, and contraction dynamics. Biological muscle models provide a useful organizational framework for these components [Thelen, 2003, Millard et al., 2013]; Appendix A gives their constitutive relations. Constant-speed, constant-load, and constant-pressure experiments would extend the isometric measurements. The pressure–force model of

Elzein et al. [2025] supplies a complementary starting point for pulse-modulated actuation.

Tensile tests with the Onyx and aluminum end caps, under pressurized and unpressurized conditions, could test the proposed end-effect explanation for resting-length-dependent force. A finite-element model of the rubber layers, crossed aramid braid, end constraints, and contact interfaces would connect that explanation to the construction described in the Festo patent literature [Festo AG & Co. KG, 2026]. The proposed geometry in Figure 7.2 provides a starting point for comparing predicted deformation with measured behavior.

7.2.2 Isometric Joint Design Factors

Chapter 4 shows why actuator force and rigid-body moment arms alone are insufficient to predict the tested knee torques. Length offsets, effective series compliance, and wrapping-induced loss of usable actuator length affect the transmitted force. The identified corrections improve prediction across the tested configurations and support the redesigned flexor and extensor routes. Their measured capacity meets or exceeds the selected human-muscle references at the tested knee angles. These results extend the earlier attachment-placement work [Bolen, 2019, Bolen and Hunt, 2019, Morrow et al., 2020] from geometric prediction toward experimentally informed joint design.

Future Work

The next mechanical questions concern transfer and packaging. The staggered flexor paths need physical clearance, load-sharing, and off-axis-load checks before extension to a

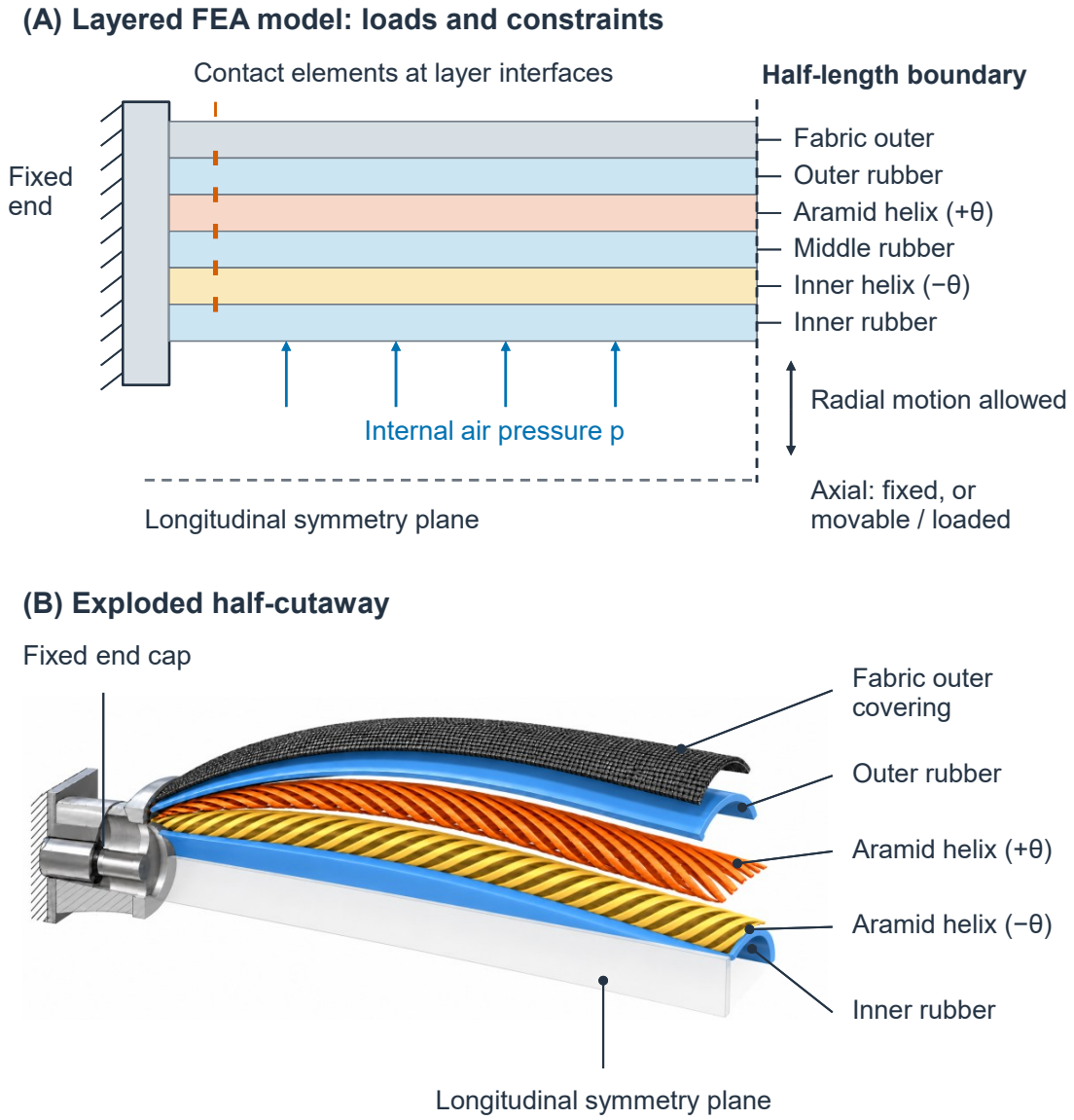


Figure 7.2: Proposed layered BPA finite-element model: contact and boundary conditions (A), and illustrative material cutaway (B). Geometry and layer spacing are schematic; the middle rubber layer is separated only in A.

complete leg. Reeved routes require space as well as sufficient torque, and the broader route search still contains incomplete configurations (Appendix C.8). A cam or artificial patella offers another way to control moment arm and protect an actuator from joint contact, but friction and contact transitions must be included when assessing such a design. The cam result supports further investigation; it does not establish the performance of an artificial patella.

The correction set also leaves a mechanical hypothesis to test: an additional force-path stiffness, χ_4, may explain effects presently absorbed into the orientation-dependent stiffness pair. This can be investigated through the existing identification framework without treating the current effective stiffnesses as literal bracket material properties.

Controlled-speed and controlled-load knee tests would extend the static capacity measurements to movement. Loading–unloading sequences would characterize hysteresis, and pressure-step tests would characterize the pneumatic response. A torque-controlled motor and a less compliant loading connection would reduce winch slack and permit these tests on the existing rig. These experiments would establish where the isometric model remains useful and where velocity dependence, damping, or pressure regulation needs an additional model.

7.2.3 Sensory Afferent Database

Chapter 5 addresses a different gap: selecting sensory feedback requires evidence distributed across animal studies, human studies, and computational models. SADb makes that evidence searchable through curation notes, linked records, classifications, and cita-

tion neighborhoods. It supports tracing a proposed pathway to its source and comparing the conditions under which a response was observed. The contribution is the research collection and its navigation tools, rather than a claim that every record is fully classified or every proposed feedback rule transfers directly to a robot.

Future Work

The implemented monthly literature watch produces candidate shortlists for manual curation (Section 5.1). A community-aware Bubble map layout could place papers from the same citation community into visually separated clusters, complementing the current community colors and citation-driven spring layout. Continued work should complete useful classifications and connect model assumptions to the primary experiments that support them. Each pathway record should identify its sensory signal, connection sign, task dependence, and source; each model record should identify the pathways it implements. Search and summarization tools can assist this process, while inclusion and interpretation remain research judgments [Bolen and Hunt, 2024].

Vestibular, visual, and multisensory contributions to posture provide useful areas for expansion. Better links between their experimental conditions and modeled mechanisms would support the sensory extensions below. Source availability should be checked during curation; records whose sources cannot be located should be removed from the working research base rather than treated as verified evidence. This is a curation rule, not a claim that a particular number of existing records has already failed that check.

7.2.4 SNS Tools and Neuromechanical Experiments

Chapter 6 begins with a controller-transfer problem: placing an existing neural controller on a different biomechanical model and tuning it by hand produced stepping-like motion in the air, but not walking. That experience motivated tools for constructing, inspecting, tuning, and comparing more complex neural–mechanical systems. The implemented models, graphical blocks, simulation interfaces, and curriculum support those tasks. Matched-input tests establish selected implementation agreement, reflex demonstrations show corrective feedback, and fixed-controller pathway removal tests sensory dependence in supported stepping.

The later virtual-harness-assisted evaluations establish coupled balance and locomotor behaviors while retaining their stated support conditions. Human-model stepping persists after removal of externally supplied neural drive, in place under the support rig; the robot model progresses under its harness with reduced drive (Section D.5). Neural-drive dependence and mechanical-support dependence are different experimental questions. The balance-platform studies provide a separate hardware precedent for BPA-actuated balance with pseudo-vestibular feedback [Bolen et al., 2024b,a,c].

Future Work

The next controller-transfer experiments should compare defined neural and mechanical configurations under matched stimuli and tuning conditions. Stimulation at the rhythm-generator, pattern-formation, and motoneuron layers can distinguish resetting the neural clock from suppressing motor output (Section D.2.3). Physiological and modeling studies

supply reference phenomena [Selionov et al., 2009, Rybak et al., 2006, Markin et al., 2010, Shevtsova et al., 2016]. Pathway additions and removals should retain the other settings so that an observed change can be associated with the intervention. Spiking and non-spiking controllers require the same care when comparing their responses.

Sensory and Higher-Level Control Vestibular feedback could extend the balance-platform work using estimates of body orientation and angular velocity [Hilts et al., 2019, Bolen et al., 2024b]. Visual measurements could complement those estimates and help correct inertial drift. Their integration should follow the task-dependent sensory weighting established in the relevant literature, rather than assuming that one channel always dominates [Akay and Murray, 2021]. Figure 7.3 separates these proposed additions from the implemented spinal and body-model components.

Cerebellar and basal-ganglia models offer further questions about adaptation, coordination, and action selection. Multilevel synthetic nervous systems for insect-like legs [Grummer et al., 2024] and cortico-cerebellar-spinal models of adaptive step length [Chen et al., 2025] provide starting points. Such extensions should be evaluated for a defined change in behavior, rather than added solely to make the controller more anatomically elaborate.

Hardware Transfer Figure 7.4 places the implemented simulation interfaces within the proposed hardware workflow. The model-construction and controller-editing routes are detailed in Figure 6.4. Computer-in-the-loop testing would connect the actual embedded controller to the simulated robot before physical deployment; valve-control interfaces and treadmill deployment remain integration tasks.

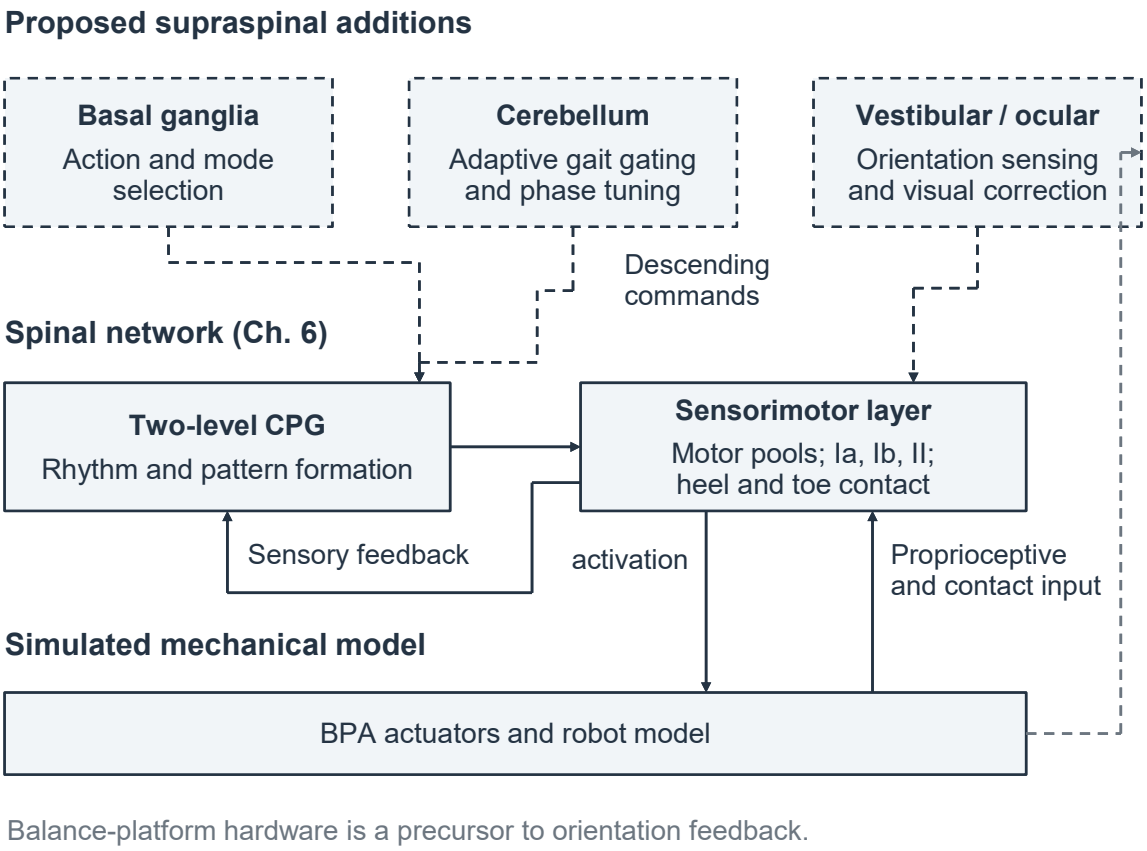


Figure 7.3: Control hierarchy connecting spinal pathways and the BPA-actuated model to proposed supraspinal and sensory extensions. Dashed elements indicate future work.

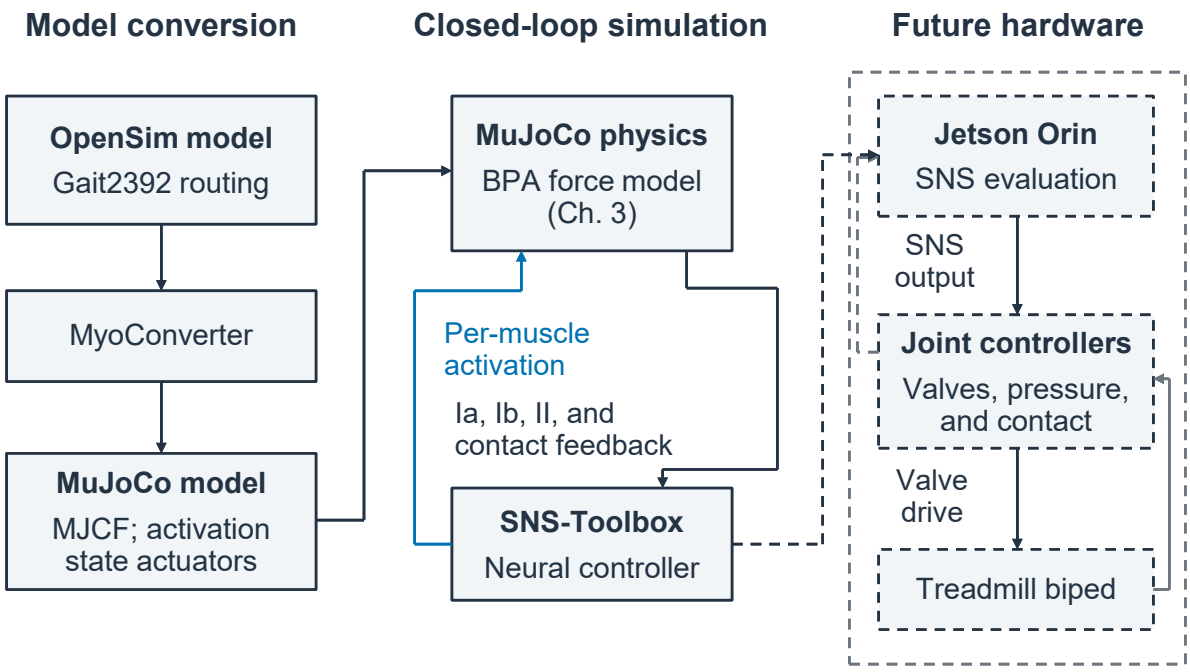


Figure 7.4: Model conversion, closed-loop simulation, and proposed hardware deployment. Simulation interfaces exchange actuator commands and afferent signals. The embedded valve-control and network-evaluation system and treadmill deployment are future work.

The hardware objective is low-latency joint control on Teensy-class microcontrollers and neural-network evaluation on an NVIDIA Jetson Orin. Encoders, strain gauges, pressure sensors, contact sensors, and IMUs would supply the sensory channels. Length and velocity estimates are needed for Ia/II feedback, while Ib force estimates must account for actuator length and loading direction. The static force and torque maps could supply feedforward commands, complemented by pressure–force dynamics [Elzein et al., 2025].

Physical PAM-biped control [Li et al., 2024] and the transfer of Muscle Mutt's SNS controller from MuJoCo to hardware [Lutz, 2025] support further investigation of that route. They do not validate the present biped. Each new body model, actuator configuration, and controller combination needs its own geometry, interface, and behavioral checks.

Dynamic Whole-Body Maneuvers Beyond steady locomotion, the program aims at gait transitions, reactive balance, and dynamic whole-body maneuvers. My longer-term aspiration is a robot that can perform a kickflip, requiring coordinated extension, aerial orientation, and controlled landing. That goal connects the actuator and routing questions to contact sensing, spinal coordination, vestibular estimation, and higher-level adaptation.

7.3 From Design Tools to Neuromechanical Experiments

Together, the four contributions connect mechanical designs and neural hypotheses to controlled tests. Dynamic characterization, sensory integration, and hardware transfer could extend this foundation toward autonomous robots and more informative models of biology. Chapter 8 summarizes the established findings and their implications.

8 Conclusion

This dissertation moved the biomimetic humanoid robot project from an unvalidated design concept to a quantitatively characterized actuator set and experimentally tested knee-torque models, and it defined the path from those experimental foundations to neuromechanical control.

Chapter 3 produced equations for isometric force in Festo 10 mm and 20 mm-diameter BPAs. The resting-length dependence $F_{620}(l_0)$ corrects the manufacturer tool's opposing trend at 620 kPa, and the nondimensional surface $F^*(\epsilon^*, P^*)$ uses three diameter-dependent coefficients. Together, these fits support actuator selection and route evaluation across resting lengths before prototyping, within the tested isometric conditions.

At the joint level (Chapter 4), the BPA model was combined with design factors for effective length, series compliance, and actuator bending. Identification on the pinned knee improved torque prediction, and the shared correction terms also improved predictions on the biomimetic four-bar knee. The improved model supports route redesign: the optimized two-BPA flexor and cam-extensor configurations are predicted to meet their human-scaled torque targets across the design range. The isometric tests demonstrate that the redesigned configurations meet or exceed their selected human-muscle reference torques at all fifteen flexor and nine extensor measurement points. This establishes capacity at the sampled angles and pressures, while agreement with pressure-specific predictions remains uneven.

The SADb application in Chapter 5 makes physiological evidence searchable by sensory

pathway, animal or human study context, and modeling relevance. Linked records, curation notes, and citation navigation connect findings to their sources. This resource helps researchers compare evidence and select pathways for neuromechanical models.

The tools in Chapter 6 turn selected pathways into testable neural–mechanical models: the Li–Deng/W2L and Gait2392 implementations, SNS-Simscape blocks, MuJoCo–Simulink integration, and staged tuning and intervention procedures. Demonstrations establish matched-circuit agreement, corrective feedback, command and measurement exchange, and pathway-dependent stepping under specified support. These tests provide a repeatable framework for comparing neural and mechanical hypotheses. General controller transfer and independent balance remain open.

These actuator and joint models, the sensory-afferent application, and the neuromechanical tools make design assumptions testable: force against measurement, routing against torque requirements, and sensory pathways against controlled changes in movement. The future studies in Chapter 7 extend these tests toward dynamic actuation, independent balance, and hardware. This work provides an experimental foundation for biomimetic robot design and for studying how neural control interacts with mechanical structure.

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APPENDICES

Appendix A: Muscle Force and Joint Torque Calculations

This appendix documents the reference implementation of the force, torque, and compliance equations used in Chapters 3 and 4. When present in MATLAB code, file names are given so that any number in this dissertation can be traced to the code that produced it.

A.1 Human muscle force calculations

A.1.1 OpenSim Thelen2003 and Millard2012 Formulations

The Thelen2003 and Millard2012 equilibrium muscle models describe active and passive fiber force, pennation, and a series tendon [Thelen, 2003, Millard et al., 2013]. Their common force balance is

$$F_T(l_T) = F_M(l_M, \dot{l}_M, a)\cos\alpha, \qquad l_{MT} = l_T + l_M\cos\alpha, \tag{A1}$$

where a is activation, α is pennation, and l_M, l_T, and l_{MT} are fiber, tendon, and musculotendon lengths. The cosine projects fiber force onto the tendon direction. When tendon compliance is enabled, fiber length is a state determined by the equilibrium calculation rather than the full musculotendon path length.

In the damped Millard formulation, fiber force has the form

$$F_M = F_0\left[a f_L(\tilde{l}_M) f_V(\tilde{v}_M) + f_{PE}(\tilde{l}_M) + \beta\tilde{v}_M\right], \tag{A2}$$

with normalized fiber length $\tilde{l}_M$, normalized velocity $\tilde{v}_M$, maximum isometric force F_0, and damping coefficient β. The active, passive, and tendon curves and the treatment of singular configurations differ between the Thelen and Millard models. The OpenSim implementation can also disable tendon compliance or activation dynamics, so the model class alone does not establish which dynamics were exercised in an analysis [Millard et al., 2013].

Thelen activation and contraction dynamics

The following seven relations correspond, in order, to Equations (1)–(7) of Thelen [2003]. They specify the activation state, passive and active force–length curves, tendon force, and fiber velocity. Bars denote normalization by maximum isometric fiber force F_0^M or optimal fiber length L_0^M, as appropriate.

Activation follows

$$\frac{da}{dt} = \frac{u - a}{\tau_a(a, u)}, \tag{A3}$$

with an activation-dependent time constant

$$\tau_a(a, u) = \begin{cases} \tau_{\text{act}}(0.5 + 1.5a), & u > a, \\ \tau_{\text{deact}}/(0.5 + 1.5a), & u \leqslant a. \end{cases} \tag{A4}$$

Here $u \in [0, 1]$ is neural excitation, a is muscle activation, and τ_{act} and τ_{deact} set the activation and deactivation time scales.

The normalized passive fiber force and active force–length multiplier are

$$\bar{F}^{PE} = \frac{\exp\left[k^{PE}(\bar{L}^M - 1)/\varepsilon_0^M\right] - 1}{\exp(k^{PE}) - 1}, \tag{A5}$$

$$f_l = \exp\left[-\frac{(\bar{L}^M - 1)^2}{\gamma}\right]. \tag{A6}$$

The tendon relation is

$$\bar{F}^T = \begin{cases} \dfrac{\bar{F}^T_{\text{toe}}}{\exp(k_{\text{toe}}) - 1}\left[\exp\left(\dfrac{k_{\text{toe}}\varepsilon^T}{\varepsilon^T_{\text{toe}}}\right) - 1\right], & \varepsilon^T \leqslant \varepsilon^T_{\text{toe}}, \\ k_{\text{lin}}(\varepsilon^T - \varepsilon^T_{\text{toe}}) + \bar{F}^T_{\text{toe}}, & \varepsilon^T > \varepsilon^T_{\text{toe}}. \end{cases} \tag{A7}$$

The fiber-velocity relation and its shortening/lengthening denominator are

$$V^M = (0.25 + 0.75a)V^M_{\max}\frac{\bar{F}^M - af_l}{b}, \tag{A8}$$

$$b = \begin{cases} af_l + \bar{F}^M/A_f, & \bar{F}^M \leqslant af_l, \\ \dfrac{(2 + 2/A_f)(af_l\bar{F}^M_{\text{len}} - \bar{F}^M)}{\bar{F}^M_{\text{len}} - 1}, & \bar{F}^M > af_l. \end{cases} \tag{A9}$$

Here $\bar{F}^M$ is the normalized active fiber force entering the force–velocity relation, $V^M = \dot{L}^M$ is negative during shortening, and $V^M_{\max}$ is the maximum shortening-speed scale. Table A1 defines the curve parameters. A shape factor is a dimensionless parameter controlling a curve's curvature or width; it is not a stiffness in $\mathrm{N\,m^{-1}}$. For example, increasing γ broadens the active force–length curve while preserving its peak at $\bar{L}^M = 1$. Changing k^{PE} changes the passive curve's shape while retaining unit force at strain ε^M_0.

Table A1: Length, strain, and shape parameters in the Thelen formulation.

Symbol	Definition
$\bar{L}^M = L^M / L_0^M$	Fiber length relative to optimal fiber length.
ε_0^M	Passive fiber strain at force F_0^M.
k^{PE}	Passive force–length exponential shape factor (5 in the paper).
γ	Active force–length width parameter (0.45 in the paper).
ε^T	Tendon strain relative to tendon slack length.
ε_0^T	Tendon strain at force F_0^M.
$\varepsilon_{\text{toe}}^T$	Strain at the end of the nonlinear tendon region; $0.609\varepsilon_0^T$.
$\bar{F}_{\text{toe}}^T$	Normalized force at that transition; 0.33.
k_{toe}, k_{lin}	Tendon exponential shape factor (3) and linear slope ($1.712/\varepsilon_0^T$).
A_f	Force–velocity shape factor (0.25 in the paper).
$\bar{F}_{\text{len}}^M$	Maximum normalized lengthening-force factor.

The passive expressions describe tensile loading; they do not imply that a slack muscle or tendon transmits compression. These are the original paper's constitutive relations. Numerical implementations must also handle slack states, minimum activation, pennation limits, and the inversion of the force–velocity relation. Consequently, the original equations alone do not specify every numerical safeguard in an OpenSim implementation.

Millard equilibrium and damping

The equilibrium relation of Millard et al. [2013] can be written in normalized form as

$$\left[a f_L(\tilde{l}_M) f_V(\tilde{v}_M) + f_{PE}(\tilde{l}_M)\right] \cos\alpha - f_T(\tilde{l}_T) = 0. \tag{A10}$$

The damped formulation adds $\beta\tilde{v}_M$ inside the brackets, as in Equation A2. The functions f_L, f_V, f_{PE}, and f_T describe active force–length, force–velocity, passive fiber force, and

tendon force. Millard's smooth characteristic curves and damping address numerical behavior while retaining the series force balance. Figure 3 of that paper illustrates the equilibrium, damped-equilibrium, and rigid-tendon formulations. A rigid-tendon model removes tendon stretch as a state; it does not simply set the passive fiber force to zero.

A.1.2 OpenSim Moment Arms and Human Torque Targets

The human torque targets of Section 4.1.5 are OpenSim tendon-excursion moment arms: for each muscle, the moment arm about the knee is the partial derivative of muscle-tendon length with respect to the knee coordinate, with positive tensile force and torque positive in the coordinate direction, $r(\theta_k) = -\partial l_{\mathrm{MT}} / \partial \theta_k$, evaluated across the range of motion. The exported analysis tables archived with the test data (`OpenSim_Vasti_MomentArm.txt`, `OpenSim_Vasti_Results.txt`, `OpenSim_Bifem_MomentArm.txt`, `OpenSim_Bifem_Results.txt`) carry the per-muscle moment arms and the corresponding maximum-isometric torque curves (moment arm multiplied by each muscle's maximum isometric force) for vastus intermedius, lateralis, and medialis, and for the biceps femoris short head; the optimization loads the vastus medialis and biceps femoris short-head columns as its targets (`buildKneeExtContext20mm.m` lines 119–140; `Knee_Flexor_Data_20mm.m` line 546). For an unconstrained planar rotation, this derivative definition and the geometric perpendicular-foot construction of Equation (A14) below are equivalent for a straight muscle path crossing a planar revolute joint; the constrained and moving-path-point geometry is evaluated by OpenSim rather than by substituting the robot-side straight-path construction.

A.2 Converted MuJoCo Muscle Formulation

A.2.1 MuJoCo Muscle Actuator

The converted model uses MuJoCo muscle actuators, which combine activation-dependent active force with passive tensile force. Let L denote the routed musculotendon length, L_T the constant biological-tendon length, and L_0 the optimal fiber length inferred from the actual and normalized operating ranges. The normalized muscle length is $\bar{L} = (L - L_T)/L_0$. The signed actuator force is

$$F_{\text{act}} = -F_0 \left[a f_L(\bar{L}) f_V(\dot{L}/(L_0 v_{\max})) + f_P(\bar{L}) \right]. \tag{A11}$$

Here $v_{\max}$ is the maximum shortening speed in optimal fiber lengths per second. The negative sign denotes a pulling actuator force; both active and passive terms contribute to tension. The rigid tendon has a constant length, but no elastic state. MuJoCo also omits pennation, so its differences from the OpenSim equilibrium formulations are not limited to tendon compliance [MuJoCo Developers, 2023].

A.3 BPA Force Model Implementation

A.3.1 Strain Measures

The musculotendon path length l_{LMT} is the sum of straight segments between route via points, transformed into a common frame at each frame crossing. The contractile strain used by the force model subtracts the artificial tendon and both end-cap fittings from the

path length before normalization by the resting length,

$$\varepsilon \;=\; \frac{l_0 - \left(l_{LMT} - l_{ten} - 2\,l_{fitting} - \chi_0\right)}{l_0}, \tag{A12}$$

where χ_0 is the constant length offset of Section 4.1.4 (zero in the uncorrected model). [Provenance: `MonoPamDataExplicit.m` get.Contraction, lines 210–217; `minimizeFlxPin.m` lines 415–426. End-cap fitting length $l_{fitting} = 25.4\,\text{mm}$ is a constructor default, `MonoPamDataExplicit.m` line 69.]

The maximum free-load contraction at 620 kPa is $\varepsilon_{620} = (l_0 - l_{620})/l_0$, where l_{620} is the BPA length at 620 kPa with no external load (code variable `KMAX`; `MonoPamDataExplicit.m` line 278). The relative strain is $\varepsilon^* = \varepsilon/\varepsilon_{620}$ (line 281).

A.3.2 Normalized Force Surface

The normalized isometric force surface of Equation (3.4) is implemented in closed form in `Functions/f_festo.m` (lines 1–26),

$$F^* = c_0\left(e^{-c_1\varepsilon^*} - 1\right) + P^*\,e^{-c_2(\varepsilon^*)^2}, \qquad F^* \leqslant 0 \;\Rightarrow\; 0, \tag{A13}$$

with $P^* = P/620\,\text{kPa}$. The zero-clamp $F^* \leqslant 0 \Rightarrow 0$ is applied by the callers (e.g. `Functions/festo4.m` line 35), not inside `f_festo.m` itself. The coefficients were fitted with `fittype` nonlinear least squares and least-absolute-residual robustness (`Testing_Data/bpaFits.m`, lines 45–60 and 115–140, lower bounds $[0, 0, 0]$) and are reported in Table 3.2. The code additionally holds a 40 mm fit not used in this dissertation ($c_0 = 0.1224$, $c_1 = 10.47$,

$c_2 = 2.023$; `f_festo.m` lines 17–20).

At runtime the torque model may alternatively evaluate a spline interpolation of the manufacturer's datasheet curves (`Functions/festo4.m`, backed by `FestoLookup.mat`), with force clamped to zero for $\varepsilon^* > 1$ or $F^* < 0$. Equation (A13) replaced this lookup in the improved model.

A.3.3 Maximum Force

The resting-length-dependent maximum force of Equation (3.5) is implemented in `Functions/maxBPAforce.m` (10 mm: line 46; 20 mm: line 57; duplicate for 20 mm in `Functions/Fmax20.m`) with the fixed 7.5 mm end-contact offset. The fitted 20 mm coefficients carry confidence intervals $a_1 \in [1.4745, 1.5011]\,\mathrm{N\,kPa^{-1}}$ and $a_2 \in [0.0238, 0.0258]\,\mathrm{kPa^{-1}\,m^{-1}}$ (`Fmax20.m` lines 26–27). Total force is the product $F = F^*(\varepsilon^*, P^*)\, F_{620}(l_0)$, resolved into a vector along the force direction (`MonoPamDataExplicit.m` lines 283–287).

A.4 Muscle Path Length and Moment Arm

Path length accumulates the Euclidean norm of each segment between via points, applying the body-frame transformation whenever the path crosses between the femur and tibia frames (`MonoPamDataExplicit.m` lines 78–95, total at lines 122–133).

The vector moment arm is implemented as the perpendicular foot of the joint origin onto the line of action,

$$\mathbf{r} \;=\; \mathbf{p} \;-\; \hat{\mathbf{u}}\,(\hat{\mathbf{u}} \cdot \mathbf{p})\,, \tag{A14}$$

where $\mathbf{p}$ is the vector from the joint origin to the muscle-path crossing point and $\hat{\mathbf{u}}$ is the unit force direction (`MonoPamDataExplicit.m` lines 156–168; identical implementations in `MonoMuscleData.m` lines 111–123, the `Mom` function of `minimizeFlxPin.m` at line 539, and `MonoPam_mult.m` lines 186–201). This is the same quantity as the cross-product projection of Equation (4.2); the planar scalar moment arm about the joint axis is the magnitude of the in-plane components, computed as `hypot(mA(:,1), mA(:,2))` (`predictKneeFlexor20mm.m` line 217). Torque about the joint z-axis is $\mathbf{m} = \mathbf{r} \times \mathbf{f}$, evaluated only where the BPA is in tension (values are discarded for $\varepsilon < -0.02$, the excessive stretching region; `MonoPamDataExplicit.m` line 308).

A.5 Compliance Model Implementation

The compliance-corrected muscle length entering Equation (4.5) is computed as $l_m = l_{LMT} - \chi_0 - l_{ten} - 2\,l_{fitting}$ (`minimizeFlxPin.m` lines 415–426; identically `minimizeExtX3.m` line 345). In the two-bracket fit of record, the insertion-bracket stiffness matrix and the projected scalar stiffness along the force direction are

$$\mathbf{K}_{br} = \mathrm{diag}(\chi_1,\, \chi_2,\, \chi_1), \qquad k_{\hat{u},br} = \left(\hat{u}^{\,T}\mathbf{K}_{br}^{-1}\hat{u}\right)^{-1}, \tag{A15}$$

matching Equation (4.9). The origin-side bracket contributes an additional scalar compliance c_{b2} in series. Both bracket frames use the two-rotation construction of Equation (C5). Their stiffness arrays are $[X_1, X_2, X_1]$ at insertion and $[X_1, X_1, X_2]$ at origin, as listed in Table C9.

The artificial tendon is modeled as an axial spring with stiffness

$$k_{ten} \;=\; \frac{n\,A_c\,E}{L_{ten}}, \tag{A16}$$

with $n = 2$ parallel cable strands ($\times 6$ for the 20 mm assembly), $A_c = 1.51 \times 10^{-6}\,\mathrm{m}^2$ (19-strand cable), and $E = 193 \times 10^9$ Pa (`MonoPam_mult.m`, function `Spr()`, lines 627–637; the evaluators obtain the spring rate through `Spr(bpa_i)`, `minimizeFlxPin.m` line 156). Series combination with the projected bracket compliance gives $k_{eq} \;=\; \left(k_{\hat{u},br}^{-1} + k_{ten}^{-1} + c_{b2}\right)^{-1}$ (`minimizeFlxPin.m` lines 394–409).

The scalar compliance displacement δ of Equation (4.23) is found numerically with `fzero` on the bracket $[0,\, l_m - l_{620}]$,

$$F_{620}(l_0)\,F^*(\varepsilon^*(\delta), P^*) - k_{eq}\,\delta = 0, \tag{A17}$$

with $\delta \equiv 0$ when the uncorrected relative strain already exceeds unity (`minimizeFlxPin.m` lines 434–441). The vector bracket deflection is then $\boldsymbol{\epsilon}_{br} = \mathbf{K}_{br}^{-1}\,(\|\mathbf{f}\|\,\hat{\mathbf{u}})$ (line 461), the cable stretch is $\|\mathbf{f}\|/k_{ten}$ (line 478), and the attachment point, path, moment arm, and torque are recomputed. For the two-BPA flexor grouping, the coupled balance shares one common force between both muscles and is solved analogously on $[0, F_{620}]$ (`MonoPam_mult.m` lines 556–612).

A.6 Wrapping-Loss Implementation

For the pinned-knee extensor, Equation (4.25) is implemented segment-wise over the wrap arc with radii $R_1 = 12\,\text{mm}$ and $R_2 = 40\,\text{mm}$ and segment boundaries $\theta_1 = 27^\circ$ and $\theta_2 = -23^\circ$ (`minimizeExtX3.m`, nested `Contraction`, lines 327–370). For the biomimetic-knee 20 mm cell, the arc-length factor is the routed bend measure $b(\theta_k)$ (arc length swept by the wrap, built with the `wRap` bend radius in `buildKneeFlexorRoute20mm.m` lines 265–268), giving $\Delta l = \chi_3\, b(\theta_k)\,(1-\varepsilon^*)^2$ (`MonoPam_mult.m` lines 398–441), with a hardcoded-geometry fallback for degenerate routes (lines 428–434). The loss enters only the force-producing strain; the kinematic (prediction) strain excludes it (`minimizeExtX3.m` lines 236–240).

Appendix B: Neuron Model Equations

This appendix defines the neuron and synapse equations and records the parameters used in the paired graded/spiking comparisons. The archived AnimatLab and 410-neuron Gait2392 comparisons are distinct from the independently replayed W2L configuration in Chapter 6. The final section documents that W2L configuration directly from its compiled network.

B.1 Spiking Neuron Model

Each spiking neuron is a conductance-based leaky integrator with threshold and adaptation,

$$C_m \dot{V} \;=\; -\,g_{\text{leak}}\,(V - E_{\text{rest}}) \;+\; I_{\text{syn}} \;+\; I_{\text{app}}, \tag{B1}$$

where V is the membrane potential, C_m the membrane capacitance ($g_{\text{leak}} = C_m/\tau_m$ with membrane time constant τ_m), E_{rest} the resting potential, I_{syn} the summed synaptic current, and I_{app} any applied (tonic stimulation) current. A spike is emitted when V exceeds the firing threshold, after which V resets and an after-hyperpolarization (AHP) of amplitude A_{AHP} with time constant τ_{AHP} is applied; the threshold itself accommodates with the firing history (relative accommodation r_{acc}, time constant τ_{acc}). Optional calcium channels are defined but disabled in the archived AnimatLab model ($G_{\text{max,Ca}} = 0$).

The parameter values shared by the neurons in the walker model are given in Table B1.

Table B1: Spiking neuron parameters in the AnimatLab bipedal walker model (`Biped_2xCPG_wSubs`). All neurons share these values except the motor neuron threshold noted below.

Parameter	Symbol	Value
Resting potential	E_{rest}	$-60\,\text{mV}$
Membrane time constant	τ_m	$5\,\text{ms}$
Initial firing threshold	V_{th}	$-55\,\text{mV}$ (motor neurons: $-59\,\text{mV}$)
Relative accommodation	r_{acc}	0.3
Accommodation time const	τ_{acc}	$10\,\text{ms}$
AHP amplitude	A_{AHP}	$1\,\text{mV}$
AHP time constant	τ_{AHP}	$3\,\text{ms}$

B.2 Non-Spiking (Graded) Chemical Synapses

Pattern-formation and motoneuron-drive connections in the walker use graded chemical synapses, whose conductance is a clipped linear ramp of the presynaptic potential,

$$g_{\text{syn}}(V_{\text{pre}}) \;=\; g_{\max}\;\text{clamp}\!\left(\frac{V_{\text{pre}}-V_{\text{th}}}{V_{\text{sat}}-V_{\text{th}}},\,0,\,1\right), \qquad I_{\text{syn}} = g_{\text{syn}}\left(E_{\text{syn}}-V_{\text{post}}\right), \tag{B2}$$

Here inward current is positive, consistently with Equation (B1); an excitatory reversal potential above the postsynaptic voltage therefore produces depolarization. The quantities V_{th} and V_{sat} are the synaptic threshold and saturation potentials and E_{syn} the reversal potential. The walker model instantiates three non-spiking synapse types:

- **Nicotinic ACh (excitatory drive):** $E_{\text{syn}} = -10\,\text{mV}$, $g_{\max} = 0.5\,\mu\text{S}$, $V_{\text{th}} \to V_{\text{sat}} = -65\,\text{mV} \to -20\,\text{mV}$.
- **Hyperpolarizing IPSP (inhibition):** $E_{\text{syn}} = -70\,\text{mV}$, $V_{\text{th}} \to V_{\text{sat}} = -55\,\text{mV} \to -40\,\text{mV}$.

- **Depolarizing IPSP:** $E_{\text{syn}} = -55\,\text{mV}$, $V_{\text{th}} \rightarrow V_{\text{sat}} = -60\,\text{mV} \rightarrow -40\,\text{mV}$.

Spiking chemical synapses and electrical synapses (rectifying coupling weight 0.1, non-rectifying 0.2) are also defined in the model and are available for the sensory afferent pathways of the MuJoCo/SNS implementation.

B.3 Relation to the SNS-Toolbox Implementation

SNS-Toolbox implements corresponding non-spiking and spiking conductance-based neurons and synapses [Nourse et al., 2023]. The AnimatLab parameters initialized a 410-neuron SNS-Toolbox network with per-leg rhythm-generator half-centers, phase-shifted pattern-formation groups, motoneuron pools for all 92 muscle-tendon actuators, and Ia, Ib, and II afferent populations. Its non-spiking equations have the forms of Equations (B1) and (B2), with spiking threshold and accommodation removed. Afferent resting-tone offsets are zero, so Ia and Ib encode activity without a tonic offset.

The rhythm-generator layer oscillated under tonic drive with the legs in antiphase (Section D.4). The layer-specific tests apply constant I_{app} at MN, PF, or RG cells. The matched circuit checks in Chapter 6 concern selected implementations and do not establish equivalence of the larger networks.

B.4 Spiking-Mirror Implementation, Calibration, and Verification

The mirror builder and calibration script are `Code/MuJoCo_SNS/spinal/build_network_spiking.py` and `calibrate_spiking.py`. Equations, calibration results, verification scripts, and logs are archived under `Code/MuJoCo_SNS/spinal/reports_spiking_20261002/`.

The corresponding Simulink blocks and tests are `Code/Matlab/SNS_Simscape/sns_build_spiking.m` and `sns_units_test_spiking.m`.

B.4.1 Adapting Integrate-and-Fire Cells and Hybrid Synapses

Between spikes, every spiking cell of the MuJoCo mirror obeys the membrane equation

$$C_m \dot{V} \;=\; -\,g_m\,(V - V_{\text{rest}}) \;+\; I_{\text{syn}} \;+\; I_{\text{bias}}, \tag{B3}$$

A spike occurs at threshold θ, after which V resets to V_{reset} and θ increases by $\Delta\theta$. Between spikes, θ decays toward its initial value with time constant τ_θ, bounded by the adaptation floor. Tonic bias I_{bias} places the equilibrium below threshold. Table B2 gives the values used by SNS-Toolbox's `SpikingNeuron`; this archived implementation uses $g_m = 1\,\mu\text{S}$ throughout to retain consistency with its specified current units.

The rhythm-generator adaptation time constant is inherited from the analog persistent-sodium inactivation value, $0.35\,\text{s}$ at the rhythm test's operating point. In the contact-driven analog cores, that slowly inactivating inward current terminates bursts. Its inactivation time constant is fixed in the tested implementations because the voltage-dependent form quenched termination and left the network latched.

Spiking chemical synapses, in both the toolbox class used in the MuJoCo mirror and the Simulink `SpikingSynapse` block, carry a conductance that decays between events and jumps by a fixed increment per presynaptic spike, saturating at its maximum,

$$\dot{g} \;=\; -\,g/\tau_{\text{syn}} \;\text{ between spikes,} \qquad g \leftarrow \min(g_{\max},\, g + g_{\text{inc}}) \;\text{ per spike,} \tag{B4}$$

Table B2: Spiking cell parameters of the MuJoCo mirror (real millivolts; `build_network_spiking.py`, `SPIKE` table, with the rhythm-generator increment superseded by the calibration of record). All cells rest at -70 mV with membrane conductance 1 µS and the tonic bias of Table B3.

Cell class	θ (mV)	V_{reset} (mV)	$\Delta\theta$ (mV/spike)	τ_θ (s)
Rhythm generator E/F	-50	-60	4.0 (calibrated)	0.35 (analog value)
Pattern-formation	-50	-60	0.3	0.15
Interneurons, commissurals	-50	-60	—	—
Afferent encoders	-52	-58	—	—
Renshaw	-52.5	-58	—	—

with $I_{\text{syn}} = g\,(E_{\text{syn}} - V_{\text{post}})$ as for the graded synapse of Equation (B2). The Simulink `HybridSpikingSynapse` implements the same dynamics with the spike detected from the presynaptic voltage crossing a detection threshold that must sit strictly between the presynaptic resting and firing potentials (a detection threshold equal to the firing potential fires nothing, because the presynaptic voltage resets at threshold). Reversal potentials are 0 mV (excitation, above rest) and -70 mV (inhibition, shunting at rest) in the spiking frame, and 8 mV/-5 mV in the motor neurons' 0 mV to 5 mV frame. The graded-to-spiking command pathways (descending drive, posture, and bias surrogates) keep the graded synapse of Equation (B2).

B.4.2 Calibration of Record

Table B3 lists every calibration factor produced by `calibrate_spiking.py` (`spiking_calibration.json`), with the measured residual of each. The spike-to-motor-neuron

factors are bisected so that a steady 40 Hz presynaptic train reproduces the non-spiking steady-state motor-neuron voltage; the spike-to-spiking factors are rate-calibrated (a 40 Hz source drives a relay cell at 40 Hz, and a calibrated inhibitory factor silences a 40 Hz-driven cell); the readout-tap factor maps a 50 Hz train to the full-scale analog level. The rhythm-generator threshold increment was set by an isolated-pair sweep followed by a full-network sweep over alternation quality and period target.

Table B3: Calibration factors for the spiking mirror (`spiking_calibration.json`). Residuals are the measured deviation at each factor's calibration target.

Factor	**Value**	**Calibrated against**	**Residual**
Graded saturation reference	0.9567	analog network measurement	—
Spike-to-MN, excitatory ($k_{sn,exc}$)	5.4087	MN steady voltage	1.0×10^{-4} mV
Spike-to-MN, inhibitory ($k_{sn,inh}$)	1.2284	MN steady voltage	2.6×10^{-4} mV
Graded-to-spiking drive (k_{ns})	0.35	relay firing rate	-0.67 Hz
Spike-to-spiking, excitation	1.8754	40 Hz rate transfer	0.0 Hz
Spike-to-spiking, inhibition	0.9996	silences 40 Hz cell	0.0 Hz
Readout tap (g_{inc})	1.7093	50 Hz → full scale	5.00 mV check
RG threshold increment	4.0 mV/spike	pair + full-net sweeps	period target
Tonic bias current	16 nA	equilibrium -54 mV	—
Synaptic time constants	5 ms (exc), 20 ms (inh)	analog values	—

B.4.3 Verification Gates of the MuJoCo Mirror

Table B4 records the complete gate set of Section D.4 with verdicts and headline numbers. The two failures are reported as failures: the constant-drive period misses its tolerance

with the tuning knobs identified, and the air-regime cadence clause misses while every structural clause passes. Gate scripts, logs, and machine-readable outputs are archived in the reports folder (`tools/`, `logs/`, and the `json` metric files respectively).

B.4.4 AnimatLab Conversion Variants

Table B5 lists spiking copies of the AnimatLab exports made with `tools/make_spiking.py`. Unreachable positive thresholds were replaced by $-55\,\text{mV}$, or $-59\,\text{mV}$ for motor neurons, in v1. The v2 copies use measured graded peaks: $-53.5\,\text{mV}$ for RG, PF, and interneurons, $-44\,\text{mV}$ for motor neurons, and $-44.5\,\text{mV}$ for Renshaw cells. Relative accommodation was zeroed. Each graded chemical synapse became a per-spike synapse with its identifier and amplitude retained, 10 ms decay, and maximum conductance $\max(5,\ 2 \times \text{amplitude})$.

The per-connection class tag was also changed to the spiking type; changing the synapse type alone prevented loading. Structural checks confirmed that neuron, connection, synapse-type, chart, and stimulus sets were retained and every connection class matched its synapse type.

Figure B1 compares the graded baseline (A1), full spiking conversion (B1), and threshold-only conversion with graded synapses retained (C1). Joint oscillation persists in C1 but not B1, identifying the synapse conversion as the breaking change in this two-layer comparison (Table D6).

The spiking conversions were verified through simulations run without opening the graphical model editor. These tests establish the reported simulated responses, but do not verify how the converted projects appear in that editor. The mirror work also scoped, but

Table B4: Complete verification-gate table of the spiking mirror. Baseline scores reproduce digit for digit; "split" denotes a gate whose clauses pass and fail in part.

Gate	Verdict	Headline numbers
Pipeline compile	PASS	runner, curriculum, parameters, and network builder all compile.
Defaults gate	PASS	(410, 376, 1186) populations, inputs, and connections after the selector edit; variant build (888, 382, 7430).
Non-spiking bit-identity	PASS	complete numerical baseline outputs identical before and after the edit for all three tested configurations.
Topology mirror	PASS	directed edge multisets identical in three gain configurations (1,186 / 7,908 / 7,762); inputs and motor-neuron maps identical; readout taps 12 / 12 / 16.
Constant-drive rhythm	SPLIT	sustained bilateral antiphase (E 4.9 Hz, F 3.2 Hz per leg; swing 5.13 mV in every window) = PASS; period 0.621 s versus analog 0.399 s ($+55.6\,\% > 20\,\%$) = FAIL.
Perturbation basin	PASS	12/12 trials under 1 % jitter on every scalar; final swing minimum 4.58 mV (margin $0.89\times$); zero explosions.
Air-regime smoke	FAIL (cadence)	finite, 53 envelope rises, knee range 106.0° of twin's 117.9°, hip 50.4° of 71.1°; period 0.215 s versus 2.108 s.
Untuned ground evaluation	ran	score -236.35; bilateral (8 right / 9 left cycles); cycle knee minimum -79.8°; no non-finite states; stayed up. Single run, no statistics.
Simulink regression	PASS	all committed demonstrations and units tests reproduce exactly with the spiking blocks in the library (two-neuron test 6.07×10^{-4} mV).
AnimatLab baselines	PASS	five baseline simulations completed without errors, with periods and antiphase reproducing the records (0.443 s, 0.462 s, 1.305 s).

Table B5: AnimatLab spiking-conversion variants of the spiking mirror. "Connections" counts flipped per-connection class tags; the ablation converts thresholds only. All copies validate structurally and complete their configured simulation durations; rhythm outcomes as in Table D6.

Variant	**Thresholds**	**Neurons**	**Synapse types**	**Connections**	**Rhythm outcome**
Two-layer v1	−55 mV/−59 mV	80	17	150	no sustained rhythm (2 spikes / 5 s)
Two-layer v2	calibrated	80	17	150	no sustained rhythm
Two-layer v3	v1, 30 ms decay	80	17	150	no sustained rhythm
Bilateral v1	−55 mV/−59 mV	96	20	208	sustained unilateral firing, one side 442 Hz–450 Hz
Bilateral v2	calibrated	96	20	208	no sustained alternation
Project export v1	−55 mV/−59 mV	122	50	243	synchronous 11.8 Hz co-bursting
Project export v2	calibrated	122	50	243	passive relaxation without rhythm
Ablation	spiking, graded synapses	80	0	0	oscillates, near-baseline ranges
2023 reference (baseline)	spiking from birth	56	3 (spiking)	84	1.305 s bursts, antiphase +0.494 cycle

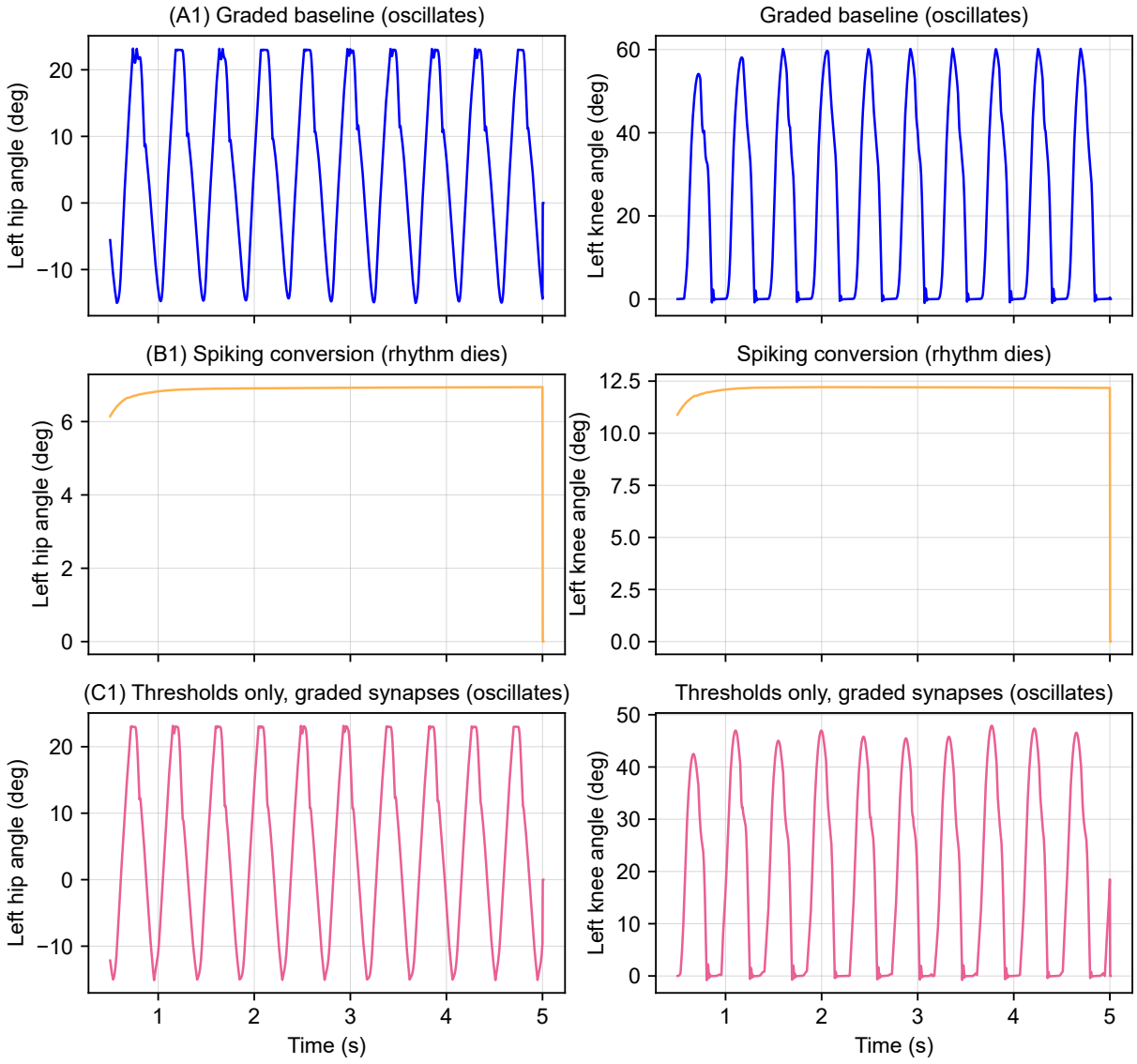


Figure B1: AnimatLab left hip (left column) and knee (right) angles over 0.5 s to 5 s: graded baseline (row A1), full spiking conversion (row B1), and spiking thresholds with graded synapses retained (row C1).

did not attempt, two further conversions: the full 410-population Simulink spinal network (whose every one of 1,392 gains is tuned for graded voltages, making a spiking conversion a full re-tune rather than a parameter port) and the spiking mirrors of the alternative MuJoCo network variants (separate architectures whose fresh tuning is future work).

B.5 Connectome of the Verified W2L Model

Figures B2 and B3 document the W2L configuration used for the independent replays in Section D.5.1. The corrected bilateral template compiles to 96 neurons, 204 chemical synapses, and 22 input channels. The node and connection lists were read from the compiled SNS-Toolbox network using the saved configuration `w2l_verified_config.json`. This is a different network from the 410-neuron Gait2392 model used in the earlier spiking comparisons; their neuron counts and results are not interchangeable.

The two sides each contain 48 neurons. RG denotes rhythm-generator cells; RG-IN, their local interneurons; PF, pattern-formation cells; PF-IN, their local interneurons; MN, motor neurons; RC, Renshaw cells; Ia, Ib, and II, muscle-afferent encoders; and Ia-IN, reciprocal-inhibition interneurons. The c1 and V3 commissural interneuron populations follow the bilateral organization of Shinohara et al. [2025]: c1 mediates inhibitory coupling between the flexor rhythm generators, while V3 excites contralateral extensor interneurons. Contact comprises heel and toe encoders. The motor outputs drive six actuator channels per side. External inputs and actuator outputs are interfaces, not additional neurons.

Figure B2 groups individual connections by their presynaptic and postsynaptic populations. A symbol indicates at least one connection of that sign; a population pair can contain

both excitatory and inhibitory connections. The diagonal quadrants contain within-side pathways, and the other quadrants show interleg pathways. The matrix exposes the sensory projections to both RG and PF populations as well as to motor circuitry.

Figure B3 shows the actual individual maximum conductance values, grouped by postsynaptic population. Equal values overlap. This representation retains the spread of synaptic strengths without implying that every connection in a pathway family shares a single gain. Conductance values alone do not predict the closed-loop contribution of a pathway; that requires the controlled interventions described in Chapter 6.

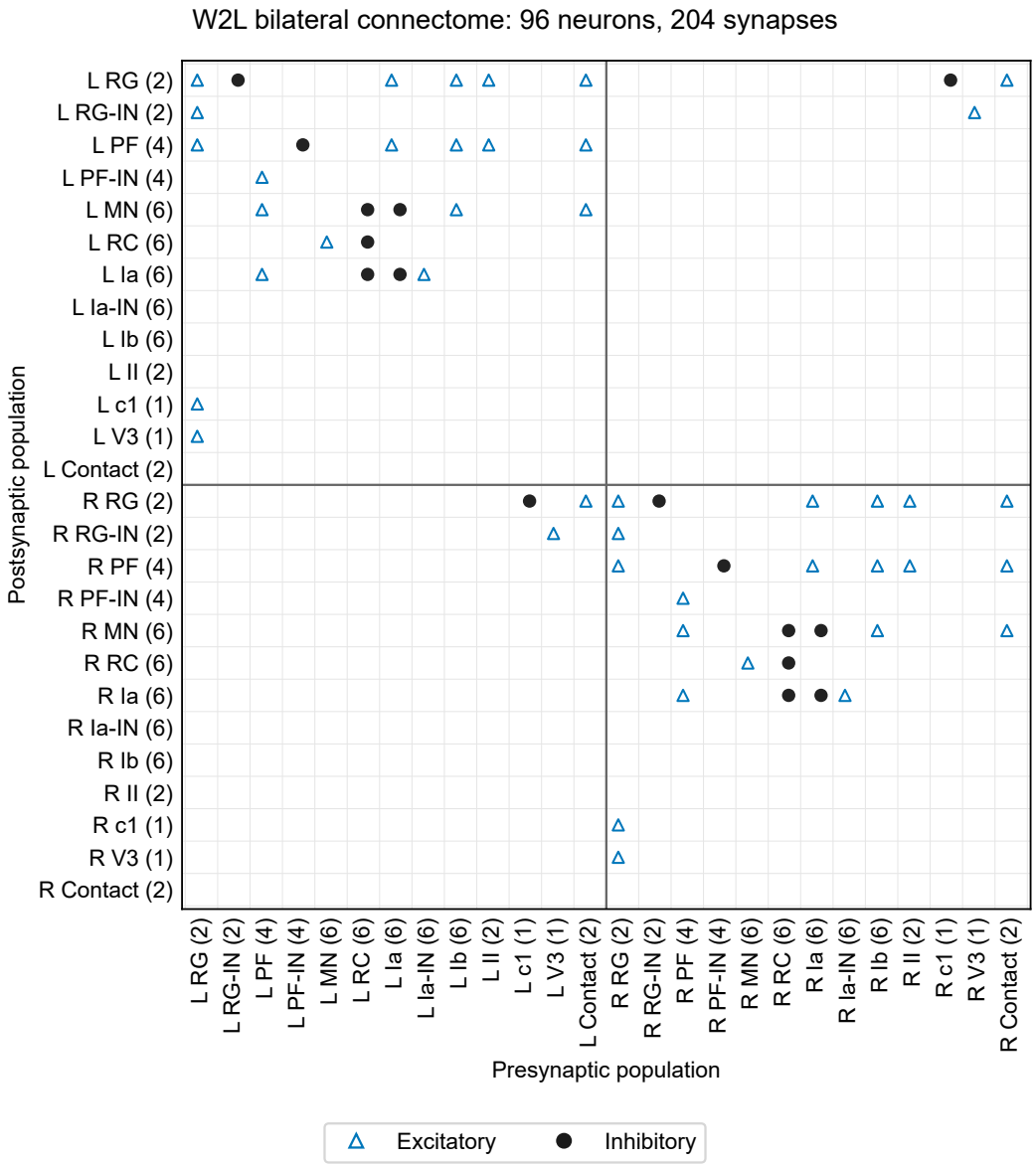


Figure B2: Population connectivity of the verified W2L network. Parentheses give neuron counts. Open triangles denote excitation, and filled circles inhibition.

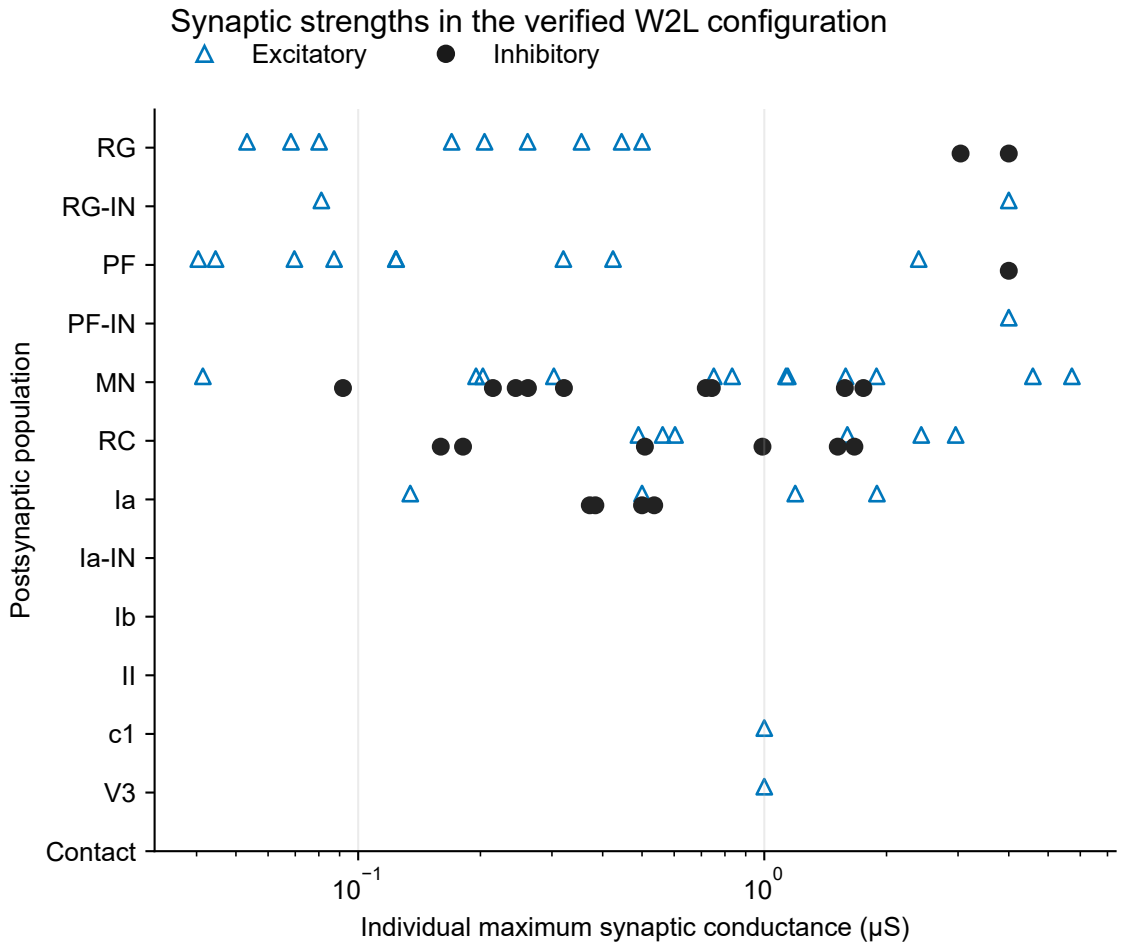


Figure B3: Synaptic conductances in the verified W2L configuration, grouped by target population. The horizontal axis is logarithmic; coincident values overlap.

Appendix C: Model Geometry, Bracket Frames, and Compliance Parameters

This appendix documents the numeric geometry and compliance parameters behind the force and torque models of Chapters 3 and 4 as they exist in the project's MATLAB code, with file names given so that any value can be traced to the code that produced it. Route and bracket coordinates are coded in meters and reported here in millimeters; stiffnesses are coded and reported in newtons per meter. The bracket reference points are interpretation choices for the 3D-printed (Onyx FDM) attachment hardware, set from CAD by me; moving a reference point changes the identified stiffness values, so the points are recorded here alongside the values they produced.

C.1 Static BPA Pathway Data

The static BPA model routes each actuator as a straight-segment polyline through via points stored in a per-test `Location` array of dimensions $M \times 3 \times N$, where N is the number of joint frames in the motion sweep and M the number of via points; the first row in the distal link frame is marked by a cross-point index. There are no binormal or surface-frame quantities: at each frame the force direction is the normalized vector from the cross-point row to the previous row transformed into the crossing frame,

$$\mathbf{d} = \mathbf{R}_{ft}^{\top}(\mathbf{p}_{C-1} - \mathbf{t}_{ft}) - \mathbf{p}_{C}, \qquad \hat{\mathbf{u}} = \frac{\mathbf{d}}{\|\mathbf{d}\|}, \tag{C1}$$

where $\mathbf{p}_C$ is expressed in the tibia frame, $\mathbf{p}_{C-1}$ in the femur frame, and $\mathbf{T}_{ft} = [\mathbf{R}_{ft}, \mathbf{t}_{ft}; \mathbf{0}^\mathsf{T}, 1]$ maps tibia coordinates to femur coordinates. The inverse transform includes both rotation and translation, as in the implemented point transformation [Provenance: `minimizeFlxPin.m` lines 160 and 524–537 (`UD`) and 201–235 (`computeForceVector`); `minimizeExtX3.m` lines 687–700; BPA struct fields, `minimizeFlxPin.m` lines 28–35]. The pinned-knee test routes are archived as `FlxPinBPASet.mat` (5 flexor tests, $M = 2$, knee angle -120° to 10°) and `ExtPinBPASet.mat` (9 extensor tests, $M = 7$, knee angle -125° to 35°).

Table C1 lists the five pinned-knee flexor tests; their routes are two points, femur-frame origin $\mathbf{p}_1 = [-75.0,\ 100.0,\ 32.8]$ mm and tibia-frame insertion $\mathbf{p}_2 = [-50.1,\ -45.0,\ 32.3]$ mm (test 1; `FlxPinBPASet.mat`, kf(1), frame 1).

Table C3 lists the extensor pinned-knee route of test 1 at $\theta_\mathbf{k} = -125^\circ$; rows 1–6 are femur-frame and row 7 is tibia-frame, and as the knee extends, rows 2–6 release onto $\mathbf{p}_1$ (wrap released at 35°).

Table C4 lists the per-test constants of the nine extensor tests; the correction terms of Section 4.1.4 used tests {1, 2, 5, 6, 7, 8}, of which the adopted fit trained on {5, 6, 7} and used tests 1, 2, and 8 for validation; tests 3, 4, and 9 were not used in identification; their retrospective predictions are reported below [Provenance: validation-test columns of `minimizeExt10mmX3_results.mat`; `Dig_crossPredict.m` line 46].

Table C1 lists pinned-knee flexor test BPA constants, read directly from `FlxPinBPASet.mat` (kf(1)–kf(5)). Labels are the laboratory shorthand used in the fitting logs (`minimizeFlxPin10mm_2brk.m` line 51); l_0, tendon, and pressure are the values used by the model. All tests use ∅10 mm BPAs; the 47 cm test's measured angles carry the 5.3° encoder correction at load time (`minimizeFlxPin.m` lines 77–80). The test was not used in the

two-bracket identification, and the improved-model fit reported for it in Table C11 is a post hoc evaluation; of the remaining four, the adopted fit trained on kf(1) and kf(5) and used kf(2) and kf(4) for validation.

Table C1: Pinned-knee flexor test constants from `FlxPinBPASet.mat`. All actuators have 10 mm diameter; test names use laboratory shorthand.

Test	**Label**	l_0 (m)	Tendon (m)	P (kPa)	F_m (N)	**Use**
kf(1)	48 cm	0.485	0	605.7	443.4	training
kf(2)	46 cm	0.457	0	614.2	441.4	validation
kf(3)	47 cm	0.479	0	623.2	443.0	not used
kf(4)	40 cm-tendon	0.406	0.035	623.4	436.9	validation
kf(5)	41 cm	0.410	0	615.4	437.3	training

The seed in Table C2 follows my femoral-condyle ray construction. The saved direct-route elimination schedule releases p5 at -89.8°, p4 at -55.7°, p6 at -47.8°, p7 at -20.2°, p3 at 6.1°, and p8 at 7.4°. These values come from `routeInfo.eliminatedAngleD` in `Vas_Pam_20mm_Result_20260925.mat`; the seed construction is in `buildKneeExtContext20mm.m`, lines 431–440.

Table C4 lists pinned-knee extensor test BPA constants, read directly from `ExtPinBPASet.mat` (ke(1)–ke(9)). Labels are the laboratory shorthand used in the fitting logs (`Dig_resultsTable.m` line 32); l_0, tendon, and pressure are the values used by the model; the shorthand rounds several lengths (e.g. "42 cm" is $l_0 = 0.415$ m and "43 cm-tendon" is $l_0 = 0.432$ m). All tests use ∅10 mm BPAs. The correction terms of Section 4.1.4 used tests $\{1, 2, 5, 6, 7, 8\}$: the adopted fit trained on $\{5, 6, 7\}$ and used tests 1, 2, and 8 for validation; tests 3, 4, and 9 were not used in identification; their retrospective predictions are included in Table C12.

Table C2: Direct-route seed for the 20 mm biomimetic-knee extensor. Points p1–p5 use the femur frame; p6–p9 use the tibia frame.

Point	x (mm)	y (mm)	z (mm)	**Frame**	**Role**
p1	40.00	35.00	0	femur	BPA origin (attachment)
p2	83.90	−274.76	0	femur	BPA contacts mounting base
p3	64.06	−426.80	0	femur	femoral contact
p4	52.83	−441.88	0	femur	femoral condyle contact
p5	37.77	−452.14	0	femur	femoral condyle contact
p6	69.85	34.33	0	tibia	tibial contact (upper)
p7	73.50	19.13	0	tibia	tibial contact (mid)
p8	71.98	−12.23	0	tibia	tibial contact (lower)
p9	34.06	−66.87	0	tibia	patellar ligament ring (distal, pEnd)

Table C3: Pinned-knee extensor BPA route of test 1 at $\theta_{\mathbf{k}} = -125^{\circ}$ (`ExtPinBPASet.mat`, ke(1), frame 1; mm).

Point	x (mm)	y (mm)	z (mm)
p1	30.00	−50.00	0
p2	65.00	−350.00	0
p3	42.40	−408.50	0
p4	28.20	−421.60	0
p5	10.50	−425.20	0
p6	−8.60	−422.80	0
p7	42.50	−75.90	0

Table C4: Pinned-knee extensor test constants from `ExtPinBPASet.mat`. All actuators have 10 mm diameter; test names use laboratory shorthand.

Test	**Label**	$\boldsymbol{l_0}$ (m)	Tendon (m)	P (kPa)	F_m (N)	**Use**
ke(1)	40 cm	0.400	0	624.9	436.3	validation
ke(2)	40 cm-tendon	0.400	0.040	618.4	436.3	validation
ke(3)	42 cm	0.415	0	605.2	437.8	not used
ke(4)	42 cm-tendon	0.415	0.040	605.2	437.8	not used
ke(5)	43 cm	0.436	0	621.8	439.7	training
ke(6)	43 cm-tendon	0.432	0.022	622.9	439.3	training
ke(7)	46 cm	0.457	0	605.9	441.4	training
ke(8)	47 cm	0.465	0	605.9	442.0	validation
ke(9)	48 cm	0.480	0	605.9	443.1	not used

The single-BPA biomimetic flexor tests in Figure 4.11**(A)** and Figure 4.11**(B)** use the femur-frame origin $\mathbf{p}_1 = [-50.0, 35.0, 50.0]$ mm and tibia θ_1 insertion $\mathbf{p}_2 = [-12.24, -8.87, 27.87]$ mm. The insertion is mapped first from the tibia θ_1 frame into the knee-ICR frame and then into the femur frame at each knee angle. These endpoints also seed the later two-BPA redesign, with $l_0 = 415$ mm and a 25 mm design tendon parameter. The redesigned flexor route adds a moving tibial wrap point on a standoff surface of radius `bpaRb`, with its z coordinate on the projected origin–insertion line. The extensor seed route and contact-release schedule are given in Table C2.

The biomimetic-knee evaluator inputs are listed in Tables C5 and C6. These tables report the saved model dimensions loaded by `minimizeFlx` and `minimizeExt`; the laboratory test names are nominal labels. The reduced-pressure flexor row identifies the reference model used with the separate pressure measurements.

Table C5: Biomimetic-knee flexor inputs loaded by `minimizeFlx`. Tendon length is the effective value stored in the saved actuator model.

Model	Diameter (m)	l_0 (m)	Tendon (m)	P (kPa)	F_m (N)
10 mm	0.010	0.4150	0.0110	603.5	437.8
20 mm, full pressure	0.020	0.4150	0.0140	620.0	1302.9
20 mm, reduced pressure	0.020	0.4150	0.0140	325.0	1302.9

Table C6: Biomimetic-knee extensor inputs loaded by `minimizeExt` from the saved 10 mm actuator model.

Model	Diameter (m)	l_0 (m)	Tendon (m)	P (kPa)	F_m (N)
10 mm extensor	0.010	0.5200	0.0380	596.4	445.7

Table C7: Measured dimensions of the four BPAs used in the final physical tests. Resting lengths and tendon lengths are recorded separately for each actuator in the extensor and flexor test worksheets. All four BPAs have 20 mm diameter.

Actuator	Resting length (mm)	Tendon length (mm)
Flexor 1	561	28
Flexor 2	560	30
Extensor 1	828	92
Extensor 2	826	85

The prediction inputs are distinct from the per-actuator measurements in Table C7. The flexor predictions use a shared resting length of 561.49 mm, with the nominal 562 mm label retained on the physical-test plot. The saved route tendon parameter and the effective prediction input are related by

$$\begin{aligned} l_{\text{ten,model}} &= l_{\text{ten,design}} - 20\,\text{mm} = 5\,\text{mm}, \\ l_{m,\text{geom}}(\theta) &= L_{\text{route}}(\theta) - l_{\text{ten,model}} - 2l_{\text{fitting}}. \end{aligned} \tag{C2}$$

Here $l_{\text{ten,design}} = 25\,\text{mm}$ is the route parameter and L_{route} is the total routed length. The geometric BPA length $l_{m,\text{geom}}$ precedes the offset, compliance, and wrapping corrections. The 20 mm input adjustment is distinct from the 21 mm end-fitting length l_{fitting} subtracted at each end. The measured cable lengths in Table C7 are hardware dimensions, not replacements for this shared effective prediction input. The extensor predictions retain the shared resting length of 827.74 mm and approximate 92.49 mm tendon parameter. These original prediction inputs are retained; no new evaluation with the measured dimensions is implied.

C.2 Joint Transformation Matrices

The knee transform is planar: every joint frame is a rotation about z by the knee angle $\theta_{\mathbf{k}}$ composed with a translation fitted to robot CAD tables of the migrating instantaneous center of rotation. With $\mathbf{R} = \mathbf{R}_z(\theta_{\mathbf{k}})$, the robot (femur→ICR) transform is $\mathbf{T}_{\text{Pam}} = [\mathbf{R},\ [f_3(\theta_{\mathbf{k}}),\ f_4(\theta_{\mathbf{k}}),\ 0]^\top;\ \mathbf{0}^\top\ 1]$ with f_3, f_4 cubic-spline fits (ICR x: 23.30 mm to 10.47 mm; y: -416.65 mm to -395.66 mm over $\theta_{\mathbf{k}} = 0.17$ rad to -2.62 rad), and the ICR→tibia transform is the pure translation $[f_{13}(\theta_{\mathbf{k}}),\ f_{14}(\theta_{\mathbf{k}}),\ 0]^\top$ (x: 29.66 mm to -7.58 mm; y: 25.97 mm to 22.33 mm), composed as $\mathbf{T}_{t1\to f} = \mathbf{T}_{t1\to\text{ICR}}\ \mathbf{T}_{\text{Pam}}^{-1}$ over 100 frames spanning $\theta_{\mathbf{k}} \in [-120^\circ, 10^\circ]$ [Provenance: `buildKneeFlexorContext20mm.m` lines 34–80; `buildKneeExtContext20mm.m` lines 30–78; `Functions/ModernRobotics/RpToTrans.m`]. In the archived test sets the per-frame transform maps tibia to femur/ICR and takes the form

$$\mathbf{T}_{\mathbf{k}}(\theta_{\mathbf{k}}) \;=\; \begin{bmatrix} \cos\theta_{\mathbf{k}} & -\sin\theta_{\mathbf{k}} & 0 & d_x(\theta_{\mathbf{k}}) \\ \sin\theta_{\mathbf{k}} & \cos\theta_{\mathbf{k}} & 0 & d_y(\theta_{\mathbf{k}}) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \tag{C3}$$

with rotation block $\mathbf{R}_z(\theta_{\mathbf{k}})$ and translation components d_x, d_y in meters. In the archived flexor test set the translation is constant across the sweep, $[d_x,\ d_y,\ 0] = [4.5,\ -395.8,\ 0]$ mm, and the rotation sweeps from $\mathbf{R}_z(-120^\circ)$ at full flexion to $\mathbf{R}_z(10^\circ)$ at full extension (`FlxPinBPASet.mat`, kf(1)).

C.3 Bracket Reference Points and Frames

Table C8 lists the bracket reference points. Each compliant bracket carries an instantaneous body-fixed frame anchored at its reference point, constructed by two rotations (yaw about z, then pitch about the yawed y). With $\mathbf{p}_b$ the vector from the reference point to the bracket-to-BPA attachment target, the yaw and pitch angles are

$$\theta_{br} = \operatorname{atan2}(p_{by},\, p_{bx}), \qquad \phi_{br} = \operatorname{atan2}(p'_{bz},\, p'_{bx}) \quad \text{with} \quad \mathbf{p}'_b = \mathbf{R}_z(\theta_{br})^\top \mathbf{p}_b, \tag{C4}$$

and the bracket-frame pose is

$$\mathbf{R}_{kbr} = \mathbf{R}_z(\theta_{br})\, \mathbf{R}_y(\phi_{br})^\top, \qquad \mathbf{T}_{kbr} = [\mathbf{R}_{kbr},\ \mathbf{p}_{br};\ \mathbf{0}\ \ 1]\,, \tag{C5}$$

The frame's x axis follows the nominal load path, with angles calculated from the bracket geometry. Both brackets in the pinned-flexor model and the pinned extensor use the two-rotation construction of Equation (C5). Forces are transformed into each bracket frame, and displaced attachment points are transformed back into the limb frame. The symmetric insertion stiffness array is invariant to the additional rotation; the origin array is not (Table C9).

C.4 Stiffness Arrays and Compliance

Every configuration models bracket compliance as a diagonal Hooke-law spring in the bracket frame, $\mathbf{K}_{br} = \operatorname{diag}(k_1, k_2, k_3)$ newtons per meter. The force-path projection and

Table C8: Bracket reference points (mm, as coded; divided by 1000 in code). All are author-set CAD interpretation points; frame denotes the frame the vector is expressed in.

Point	x	y	z	**Configuration / meaning**
Pbri	−48.11	−107.81	13.80	pinned-knee flexor, insertion bracket (knee-ICR frame)
Pbr2	−52.61	0	75.06	pinned-knee flexor, origin-side 2nd bracket (hip frame)
Pbr	−3.84	−46.44	62.50	pinned-knee extensor, rib midpoint, medial side (hip frame)
Pbr	8.38	20.75	25.10	biomimetic extensor 10 mm, cantilever centroid
Pbr	9.48	−33.38	33.90	biomimetic flexor 20 mm, bolt-pattern centroid
Pbr	−19.00	22.00	27.60	biomimetic flexor 10 mm, cantilever centroid

series combination of the compliances are defined in Chapter 4, Equations (4.9)–(4.19).

When an artificial tendon is present, its axial spring ($\times 2$ cables, effective steel area $1.51 \times 10^{-6}\,\mathrm{m}^2$, $E = 193\,\mathrm{GPa}$) adds in series. The force balance of Equation (4.23) is solved at each pose by `fzero` on the contraction interval. Table C9 gives the configuration-specific stiffness arrays.

Table C9 assigns the fitted variables X_1 and X_2 to the bracket-frame axes, in $\mathrm{N\,m^{-1}}$. The two pinned-flexor bracket frames follow Equation (C5); the final column identifies the stiffness-array assignments in the source, rather than a separate frame convention.

The wrapping-loss term X_3 (Equations 4.25–4.27) is evaluated as $X_3\, R\, |\Delta\theta|\, \zeta^2$ per contact arc. On the pinned extensor the two physical contact arcs use $R_1 = 12\,\mathrm{mm}$ (27° to -23°) and $R_2 = 40\,\mathrm{mm}$, with arc-length normalizers $27.75/(ang_1 - ang_2)$ and $92.46/(ang_2 + 120^\circ)$; on the biomimetic configurations the bend measure is $w_{\mathrm{rap}}|\theta_{\mathrm{turn}}|$ with $w_{\mathrm{rap}} = 25\,\mathrm{mm}$ [Provenance: `minimizeExtX3.m` lines 333–359; `buildKneeFlexorRoute20mm.m` line 268; `MonoPamDataExplicit_balanceX3.m` lines 489–494]. Table C10 records the optimizer parameterization and bounds of the adopted identification: the two-bracket flexor fit solves

Table C9: Bracket-frame stiffness arrays by configuration, in $\mathrm{N\,m^{-1}}$. The fitted variables X_1 and X_2 are assigned to the axes shown.

Configuration	$\boldsymbol{K}$ array	**Stiffness-array source**
Pinned flexor, single bracket (legacy)	$[X_1,\ X_2,\ X_1]$	`minimizeFlxPinX3.m` line 267
Pinned flexor, insertion bracket (two-rotation frame)	$[X_1,\ X_2,\ X_1]$	`minimizeFlxPin.m` line 387
Pinned flexor, origin bracket (two-rotation frame)	$[X_1,\ X_1,\ X_2]$	`minimizeFlxPin.m` line 400
Pinned extensor, 2trans (with X_3)	$[X_1,\ X_2,\ X_2]$	`minimizeExtX3.m` line 575
Biomimetic extensor	$[X_1,\ X_2,\ X_1]$	`minimizeExt.m` line 352
Biomimetic flexor	$[X_1,\ X_2,\ X_2]$	`minimizeFlx.m` line 314

all three terms, and the extensor refit locks (X_1, X_2) to the adopted flexor-front values (the extensor front was generated with the pair fixed to the adopted values of Table 4.9 and only (X_0, X_3) searched), enforcing a single stiffness pair across configurations.

The physical correction terms X_i used elsewhere in this appendix correspond to χ_i; the lowercase x_i below are the optimizer coordinates. The initial flexor population in Table C10 used $X_0 \in [5, 15]$ mm, $X_1 \in [5 \times 10^4, 5 \times 10^5]\,\mathrm{N\,m^{-1}}$, and $X_2 \in [7 \times 10^3, 1.5 \times 10^4]\,\mathrm{N\,m^{-1}}$.

[Provenance: `minimizeFlxPin10mm_2brk.m` lines 82–83 (bounds) and 424–432 (parameterization); `minimizeExt10mmX3.m` lines 23–26 (lock load) and 67–68 (bounds); `Collect_ExtPinX3_sweep.m` lines 59–60.]

Table C10: Optimizer coordinates and equivalent physical correction-term bounds. The numerical optimizer variables are x_i; length offsets χ_0 are reported in millimeters, stiffnesses in $\mathrm{N\,m^{-1}}$, and χ_3 is dimensionless. The extensor refit fixes the stiffness pair at the adopted flexor values.

Identification	Variable	Optimizer bounds	Physical bounds / mapping
Pinned flexor	x_0	$[0, 2]$	$\chi_0 = 10x_0 \in [0, 20]\,\mathrm{mm}$
	x_1	$[\log_{10}(3 \times 10^4), 6]$	$\chi_1 = 10^{x_1} \in [3 \times 10^4, 1 \times 10^6]\,\mathrm{N\,m^{-1}}$
	x_2	$[\log_{10}(5 \times 10^3), \log_{10}(2 \times 10^4)]$	$\chi_2 = 10^{x_2} \in [5 \times 10^3, 2 \times 10^4]\,\mathrm{N\,m^{-1}}$
Extensor refit	x_0	$[-2, 0.5]$	$\chi_0 = 10x_0 \in [-20, 5]\,\mathrm{mm}$
	x_3	$[0, 1]$	$\chi_3 = x_3 \in [0, 1]$

C.5 Adopted Correction Terms and Complete Pinned-Knee Results

The adopted correction terms are listed in Table 4.9. The flexor uses row 77 of `filtered_results` in `minimizeFlxPin10_results.mat`; the extensor uses row 32 in `minimizeExt10mmX3_results.mat`. The extensor identification holds (χ_1, χ_2) at the flexor values and solves (χ_0, χ_3), preserving one effective stiffness pair across both configurations.

Tables C11 and C12 report every recorded pinned-knee test before and after applying the adopted corrections. "Before" uses the characterized BPA force model with zero length and wrapping corrections and rigid attachments; "after" uses the adopted correction terms. Both predictions use the same measured angles and test inputs. RMSE and maximum absolute residual have units of N m, and FVU is dimensionless. The sample-size convention for FVU is given in Chapter 4, Equation (4.29).

The adopted flexor identification uses tests 1 and 5 for training and tests 2 and 4 for

validation. Test 3 (47 cm) was not used in identification. The extensor identification uses tests 5, 6, and 7 for training and tests 1, 2, and 8 for validation. Tests 3, 4, and 9 (42 cm, 42 cm with tendon, and 48 cm) were excluded from identification. Results for unused tests are retrospective predictions, not validation results.

Table C11: Complete pinned-knee flexor test results before and after correction. Test names follow the laboratory shorthand.

		Before			After		
Test	Use	RMSE	FVU	Max.	RMSE	FVU	Max.
1: 48 cm	Training	7.617	2.1278	25.531	2.165	0.1718	5.467
2: 46 cm	Validation	8.803	1.9380	26.264	1.368	0.0468	3.614
3: 47 cm	Unused	7.670	1.8463	21.650	2.287	0.1641	7.818
4: 40 cm+tendon	Validation	6.373	0.5935	16.859	1.611	0.0427	5.891
5: 41 cm	Training	9.694	2.1149	21.076	1.084	0.0277	1.766

Table C12: Complete pinned-knee extensor test results before and after correction. Test names follow the laboratory shorthand.

		Before			After		
Test	Use	RMSE	FVU	Max.	RMSE	FVU	Max.
1: 40 cm	Validation	3.092	0.5638	4.842	1.244	0.1303	2.196
2: 40 cm+tendon	Validation	3.622	6.3919	7.845	1.350	0.8878	1.890
3: 42 cm	Unused	3.528	2.2190	7.665	1.299	0.3418	2.700
4: 42 cm+tendon	Unused	2.910	6.2250	6.367	1.289	1.2209	2.003
5: 43 cm	Training	2.437	0.9618	4.284	0.978	0.1634	1.818
6: 43 cm+tendon	Training	2.878	4.7304	4.817	1.525	1.3270	2.481
7: 46 cm	Training	2.178	1.0586	4.974	0.490	0.0535	1.206
8: 47 cm	Validation	0.949	0.3152	2.272	0.699	0.1714	1.559
9: 48 cm	Unused	0.578	0.1085	1.569	0.784	0.1996	2.282

The adopted corrections reduce all three errors on every test used in identification. The unused 48 cm extensor test has larger errors after correction, as shown in Table C12.

The 43 cm-tendon extensor test retains an FVU above unity despite improvement over its baseline.

The values were evaluated with `minimizeFlxPin.m` and `minimizeExtX3.m`. On the biomimetic extensor test ($l_0 = 51.8$ cm), the shared adopted terms give RMSE 1.17 N m, FVU 0.67, and maximum residual 2.21 N m, compared with the rigid-model values 3.03 N m, 4.49, and 5.59 N m. Table 4.8 in Chapter 4 collects the biomimetic comparisons at the adopted terms.

C.6 Redesigned Extensor Route Geometry

The direct redesigned extensor route is preserved in `Vas_Pam_20mm_Result_20260925.mat`; the cam-assisted route supersedes it as the design of record. Figure C2 shows the direct route at full flexion, immediately before each contact release, and at full extension. The plotted event pose precedes the first inactive grid pose used to report each release angle in Table C2. All route points, including p8, are present in the saved geometry; plotting-title metadata do not define the route.

The optimized origin is $\mathbf{p}_1 = [40.0, 137.0, 0]$ mm in the femur frame, and the insertion is $\mathbf{p}_{\text{End}} = [34.4, -197.1, 0]$ mm in the tibia frame. Resting length is 827.7 mm, and tendon length is 36.8 mm. The displayed envelopes and bend-radius checks explain the route's geometric constraints.

Figure C1 first locates the optimized route on the OpenSim lower-body bone-surface point clouds (`Open_Sim_Bone_Geometry`), then enlarges the knee region at exactly zero knee angle. The femur, tibia, pelvis and hip, sacrum, lumbar spine, and foot have distinct colors.

The black BPA–tendon route and gray patterned human vastus paths distinguish mechanical routing from anatomical context. The optimized route carries a total BPA-plus-tendon path length of 0.784 m at this pose. Animated sweeps of both redesigned routes over the full flexion range are archived with the optimization results (`Knee_Extensor_20mm.gif`, 2.3 MB, and `Knee_Flexor_20mm.gif`, 4.7 MB, in `Code/Matlab/Mesh_Optimization/Results`).

C.7 Contact and Release Angles of the Wrapped Route

The route-elimination rule that schedules the wrapped extensor route is a route-topology state machine rather than a shortest-path solver: the fully flexed route carries every candidate contact, and each contact is released at the first more-extended knee angle at which the straight chord between its surviving neighbors can replace it without entering the exclusion envelopes. This section states the calculation in generic terms, so that it applies to any wrapped actuator route on a rotating joint pair; the extensor route of Table C2 is the worked example, and the release schedule the calculation produces is the one quoted there.

The route is stored as a fixed number of rows $p_1, \ldots, p_9$: rows p_1 through p_5 are femur-frame points, rows p_6 through p_9 are tibia-frame points, and the segment from p_5 to p_6 crosses from one body frame into the other. The endpoints p_1 and p_9 are the actuator origin and insertion, and they are always active. Each interior row is either active, a genuine wrap contact, or eliminated; an eliminated femur-side row repeats the preceding active femur row, and an eliminated tibia-side row repeats the neighboring active tibia row, so the route keeps a fixed size and its segment chain crosses the joint exactly once. Let

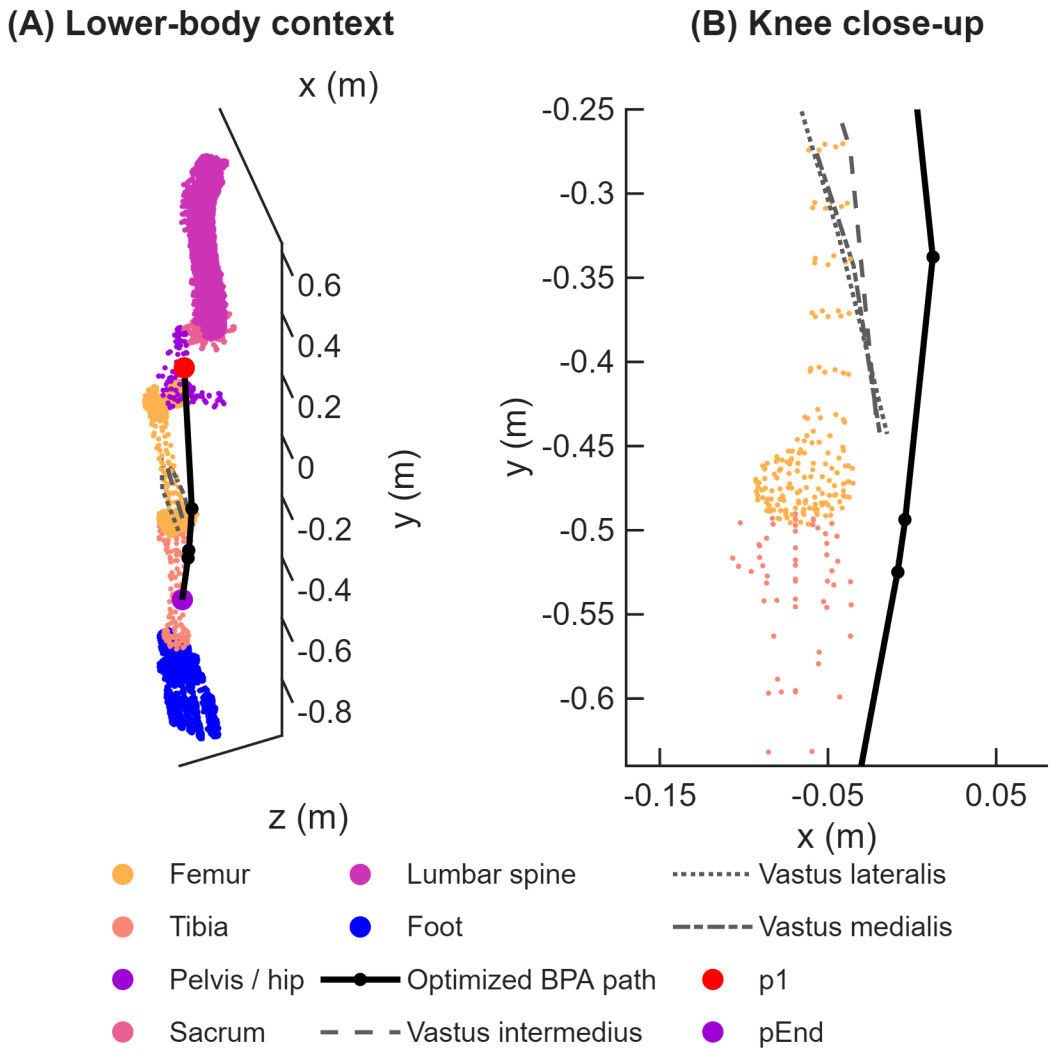


Figure C1: Optimized extensor route at 0°: **(A)** lower-body context and **(B)** knee close-up. Black shows the BPA–tendon route; gray patterned paths show human vastus references.

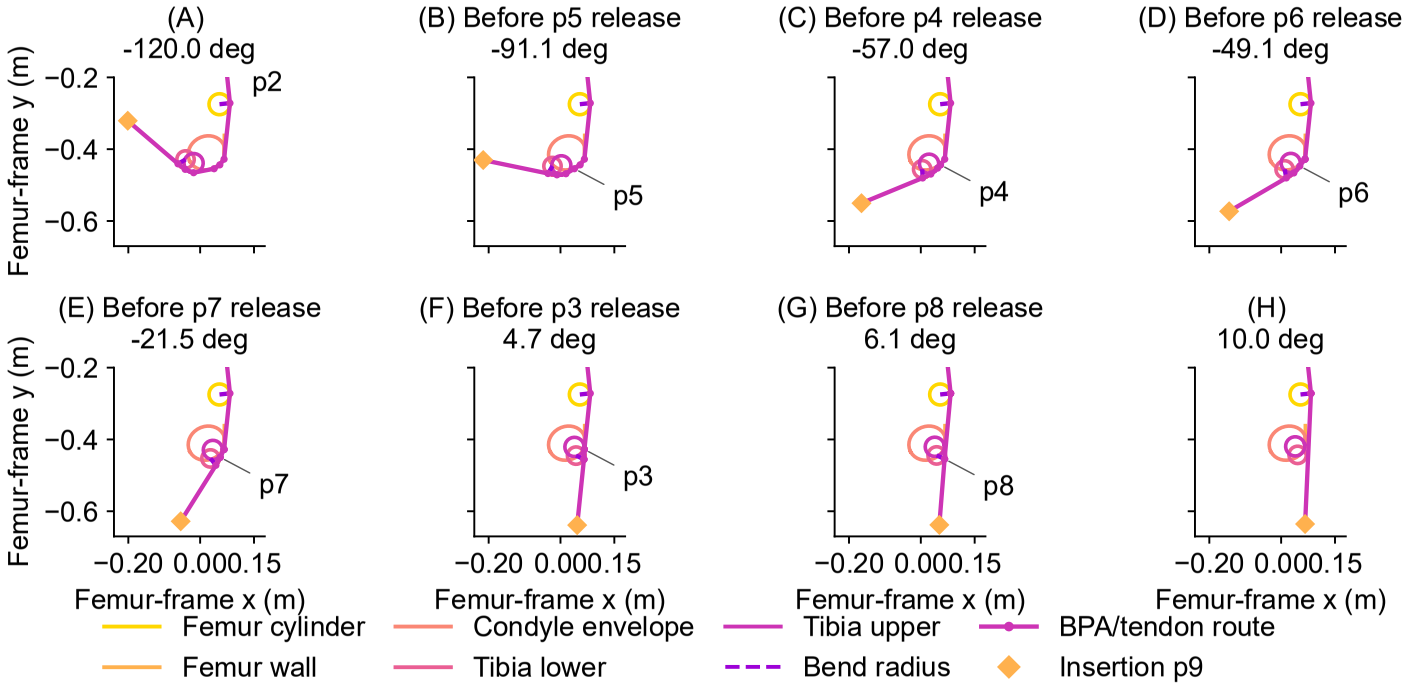


Figure C2: Saved extensor route at eight contact-release poses, enlarged in the femur frame. Titles identify the next released contact. The proximal origin lies outside this close-up; Figure C1 supplies anatomical context.

$\mathcal{A}(\theta_{\mathbf{k}}) \subseteq \{2, \ldots, 8\}$ denote the active interior rows at knee angle $\theta_{\mathbf{k}}$. The sweep over the design grid, here 100 poses from -120° to 10° at a spacing of 1.31°, proceeds from full flexion toward extension, each pose inherits the active set of the previous pose, and rows are only ever removed, so $\mathcal{A}$ shrinks monotonically as the knee extends.

At every pose, every active interior row is tested between its nearest active anchors,

$$i_j^- = \max\{i < j : i \in \mathcal{A} \cup \{1\}\}, \qquad i_j^+ = \min\{i > j : i \in \mathcal{A} \cup \{9\}\}, \tag{C6}$$

with the anchors expressed in the candidate's native frame through the per-angle joint transforms. Write A, B, and C for the planar, sagittal positions of $p_{i_j^-}$, $p_{i_j^+}$, and p_j, and let $s(x, y)$ denote the signed angle from vector x to vector y, positive counterclockwise, evaluated as a single atan2 of the planar cross product and the dot product so that no principal-value seam can corrupt the comparison. The elimination margin of row j is

$$\mu_j(\theta_{\mathbf{k}}) = \begin{cases} s(C - A,\ B - A)\,, & j \leqslant 5, \\ \\ s(A - B,\ C - B)\,, & j \geqslant 6, \end{cases} \tag{C7}$$

a signed angle between the through-contact leg and the bypass chord, measured in the native wrap sense of the candidate's side of the crossing segment; the two cases differ only in that sense of rotation, because the femur-side contacts turn the route one way around the distal femur and the tibia-side contacts turn it the opposite way around the proximal tibia. The rotation rule releases the contact once the margin turns positive, the orientation at which the chord, rather than the contact, provides the outer boundary of the route on

that side. The gate is opened slightly early,

$$\mu_j(\theta_{\mathbf{k}}) > -\Delta, \qquad \Delta = 1.5^\circ, \tag{C8}$$

as a release hysteresis: in the last poses before the contact becomes collinear with its chord, the computed moment arm of the wrapping segment collapses, and releasing the contact slightly before that orientation, with the collision gate below still controlling validity, removes the artifact without changing the route family.

Opening the rotation gate is necessary but not sufficient. A removal additionally requires the straight bypass chord $\overline{AB}$ to clear the exclusion envelopes of the candidate's side,

$$\operatorname{dist}\big(\overline{AB}_{\text{trim}},\, \mathcal{E}_j\big) \;\geqslant\; g, \qquad g = 0.5\,\text{mm}, \tag{C9}$$

where dist is the planar clearance between the trimmed chord and the envelope and tangency is allowed. The chord is trimmed by 2 mm at each end, capped at 45 % of its length, because a legitimate contact departs from the clearance boundary itself. On the femur side the envelopes $\mathcal{E}_j$ are the femoral-cylinder clearance circle, the condyle clearance polygon, and the femoral-wall span; on the tibia side they are the two tibial-cylinder clearance circles and the wall band between them. Every envelope is the normal offset of a bone profile by the fully inflated BPA radius R_{BPA},

$$\mathbf{q}(\vartheta) = \mathbf{e}(\vartheta) + R_{\text{BPA}}\,\hat{\mathbf{n}}(\vartheta), \tag{C10}$$

where $\mathbf{e}(\vartheta)$ traces the physical profile and $\hat{\mathbf{n}}(\vartheta)$ is its outward normal; the offset curve

shares the tangent of the physical profile, its radius of curvature is $\rho(\vartheta) = \rho_{\mathrm{phys}}(\vartheta) + R_{\mathrm{BPA}}$, and the candidate contact positions themselves are seeded at tangent points of this curve. Because the envelopes are drawn at the fully inflated radius while the routed BPA is contracted at the release angles, each envelope radius may be relaxed by $\delta_c = 3\,\mathrm{mm}$ inside the gate, floored at the bone profile plus $0.5\,\mathrm{mm}$; a rotation-gate opening that still fails the relaxed collision gate indicates that the seed placement, not the gate, must move.

At the first pose of the sweep, full flexion, the seed route is settled to a fixpoint: every contact whose two gates already open there is removed, and removals repeat until no further gate opens, so that the sweep starts from a valid topology; a colinearity guard may additionally remove a seed contact that starts the sweep already aligned with its neighbors, and its tolerance is zero in the design of record, so every seed row of Table C2 starts active. At every later pose at most one contact is released: if several rows open both gates in the same pose, the row with the largest elimination margin, the one whose rotation rule is most decisively open, is released first, and the remaining rows are retested at the next pose. The release angle of row j is recorded as the first grid pose at which the row is inactive. The activation angle is read from the same record with the sweep reversed, as the last pose at which the row is active, which is the pose at which the contact re-engages when the route is driven from extension toward flexion.

The same active-set record drives the geometric bend measure that scales the fictive-strain wrapping loss of Equations (4.25) and (4.27) inside the stiffness-aware force balance. Each wrap family, the femoral-cylinder contact, the femoral-condyle curve, the tibial contact curve, and the crossing-point curve at the ICR, contributes an arc length over its

effective radius, and the total

$$b(\theta_{\mathbf{k}}) = \sum_{f} R_f(\theta_{\mathbf{k}})\, \alpha_f(\theta_{\mathbf{k}}) \tag{C11}$$

is the bend signal handed to the force model. Each family's turn angle α_f ramps with knee angle between the two route-state events that bound it, holding zero before the first event and its plateau after the second, so the topology state and the bending signal are two readings of one record. The formulation is joint-agnostic: it requires only a fixed route representation with one crossing segment, per-angle transforms between the two body frames, and clearance envelopes on each side, so it carries unchanged to the other wrapped routes of this program.

C.8 Supporting Route Searches and Transmission Sweeps

The transmission studies extend the route search by allowing an artificial tendon to pass through a tackle before insertion. A redirect changes the force direction; a multi-part tackle also changes the relation between actuator travel and insertion force. The comparisons below re-solve actuator sizes for each transmission configuration and evaluate insertion-side torque against the same human reference. They remain model predictions; physical tests of the redesigned flexor and cam extensor are reported in Section 4.3.4.

For the knee-flexor route, the pulley variant converged to a collision-feasible layout with zero bound violations and a peak predicted flexion torque of 85.56 N m on the insertion-side measure, the torque applied at the insertion after the tackle's force loss. At unity transmission, the knee-flexor pulley route increased peak predicted knee-axis

torque from 81.23 N m to 85.74 N m, while limiting off-axis torque to 0.803 N m. This comparison measures the effect of the two optimized routing configurations; their different objective functions do not isolate the pulley alone. The pulley serves as a redirect in this configuration rather than a force multiplier.

The robot-body model includes an adductor magnus attachment update to $\mathbf{P}_2 = (-0.1357, -0.0929, 0.0591)$ m. At this location, the model delivers 34.28 N m of flexion torque, 1.12× the human target of 30.61 N m, and 27.62 N m of adduction torque, 0.82× the human target of 33.60 N m. The OpenSim model retains its eight original moving-path-point vias for rectus-femoris and vastii patella tracking; the candidate pulley routes were not added to those paths, and the model contains no pulley entities. The MuJoCo conversion used for the simulation work predates this attachment update and the pulley batch. Physical tests of the redesigned knee routes are reported in Section 4.3.4.

The attachment-route redesign was extended separately from the knee to all 27 muscle-actuators of the robot-body model and remains work in progress. The first pass reached 19 actuators. All but one selected route violated the feasibility constraints; the exception was the biceps femoris long-head pair, with a worst-angle margin of +0.03 %. The formulation was then restructured to enforce each route's joint-angle task envelope as a hard feasible set rather than a penalty.

The restructured pass encountered a route-builder concatenation error for twelve actuators, leaving fifteen finalized routes. Eight were constraint-feasible: semimembranosus, biceps femoris long head, tensor fasciae latae, soleus, lateral gastrocnemius, tibialis anterior, iliacus, and psoas. Six also met the torque requirement at every angle, with worst-angle margins from +0.25 % for biceps femoris long head to +250.7 % for tensor fasciae latae.

Semimembranosus and soleus satisfied the geometry but remained below their torque targets. Completing the extension requires the route-builder repair and sufficient actuator capacity or relaxed routing corridors for these two routes.

The unadjusted cam optimizer solution had $s_{\mathrm{req}} = +31.30\,\%$ and $m_h = +42.34\,\%$; the adjusted design in Chapter 4 has $+30.83\,\%$ and $+39.77\,\%$, respectively. Equation (4.38) distinguishes surplus normalized by peak human torque from the pointwise human-reference margin.

Figure C3 compares predicted insertion-torque magnitude with the maximum-isometric OpenSim reference. Colored curves show five constant pressures from the saved pressure grid for two BPAs through a two-part tackle (force gain 1, travel gain 2), with actuator sizes re-solved. The five muscles retain their original evaluation conditions: medial gastrocnemius has the ankle fixed at zero, and rectus femoris is evaluated at the hip.

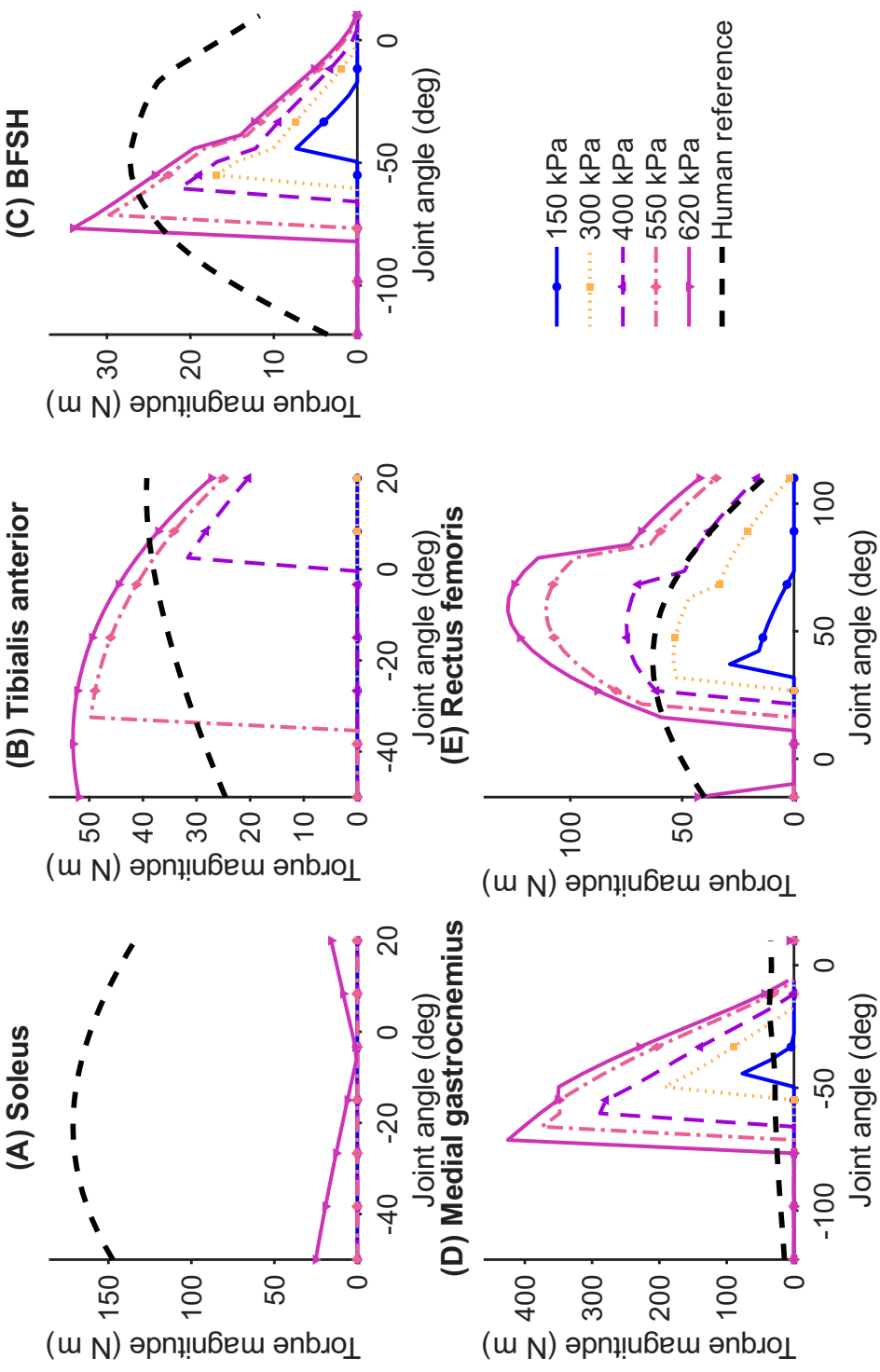


Figure C3: Predicted torque at five saved pressure levels for two BPAs through a two-part tackle, with sizes re-solved. Black dashed curves show the maximum-isometric human reference.

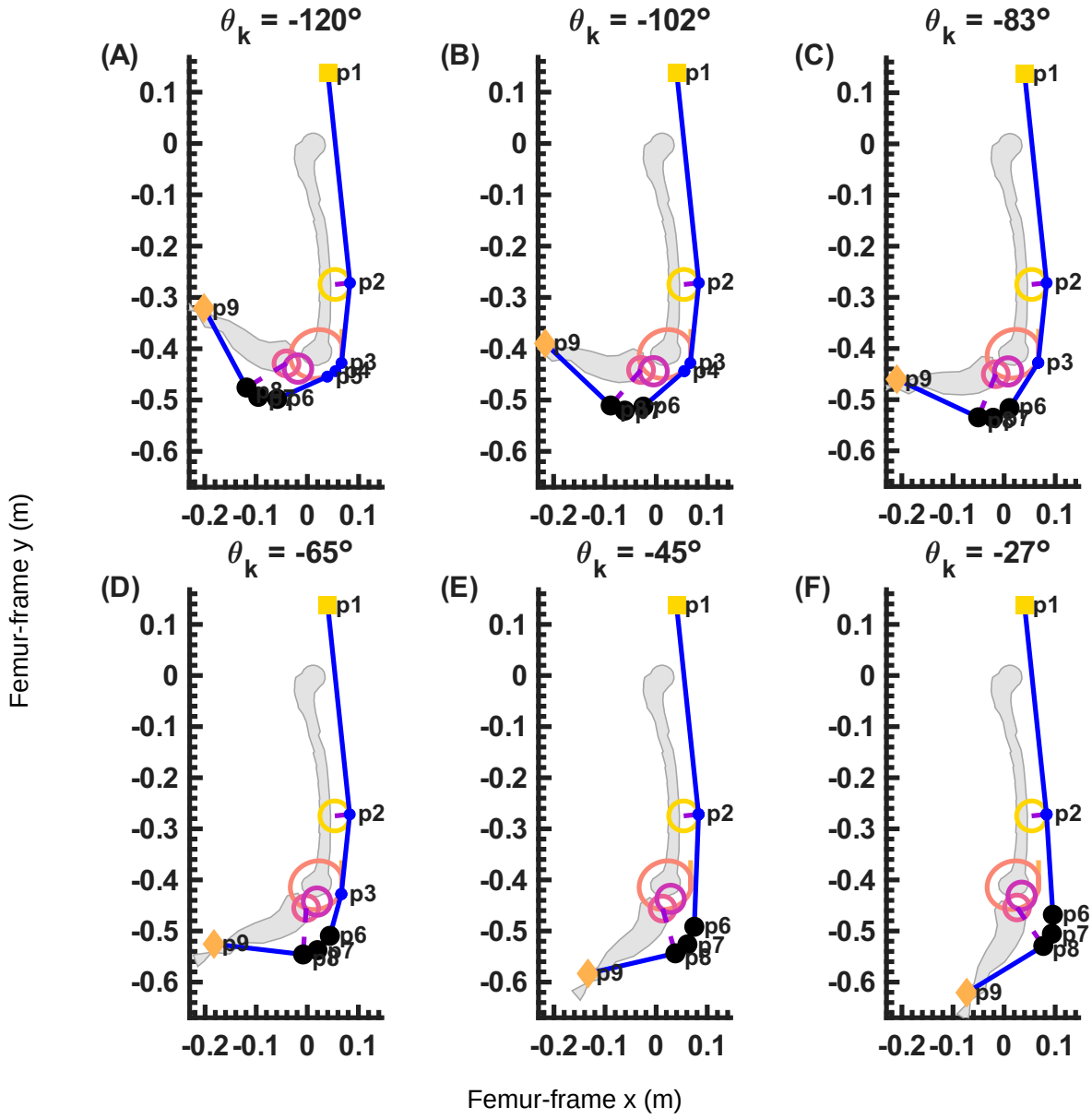
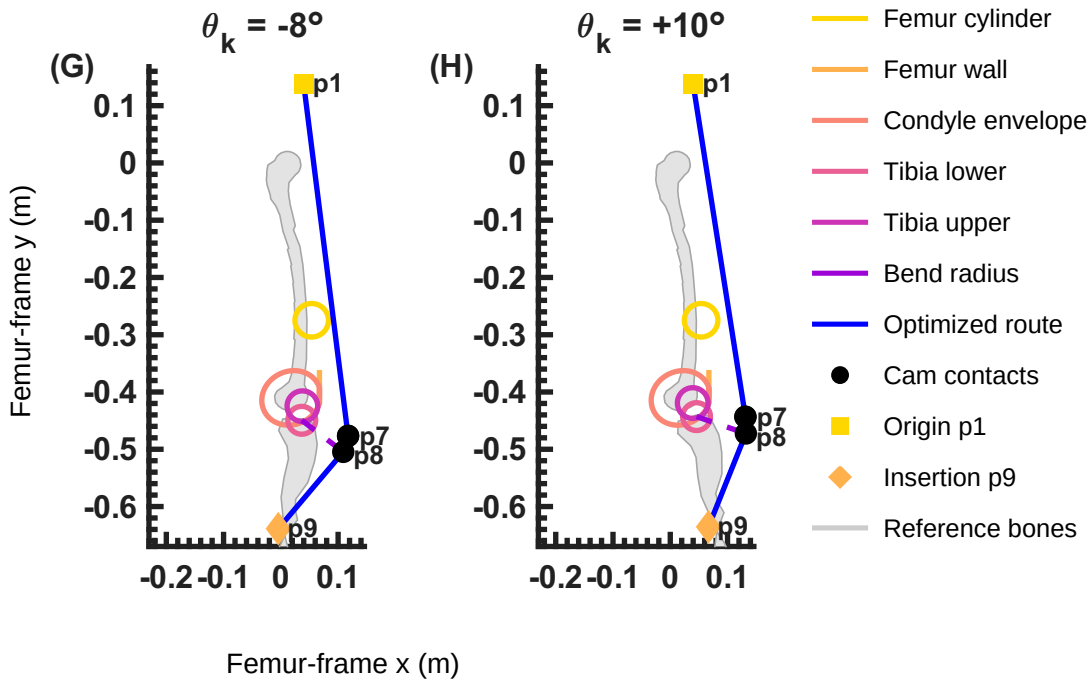


Figure C4: Cam-extensor poses A–F, spanning −120° to −27°. Blue lines show the route; colored outlines show the clearance envelopes. Gray silhouettes show the registered OpenSim anatomical reference; no foot is drawn. The remaining poses and legend are in Figure C5.



Cam-drum radius: 96.9 mm

Release angles: p5 -110.8 deg; p4 -89.8 deg; p3 -54.3 deg;
p2 -21.5 deg; p6 -18.9 deg. Gray bones are anatomical references.

Figure C5: Cam-extensor poses G and H at -8° and 10°, with the legend, 96.9 mm cam-drum radius, and contact-release schedule. Panel letters continue Figure C4.

C.9 Master's Robot Model: Joint Coordinates and Actuator Sizes

This comparison extends the 27 actuator sites of my master's robot-leg model [Bolen and Hunt, 2019]. Table C13 lists the two joint coordinates evaluated at each site and the nominal BPA sizes. The evaluated RoM is the joint-coordinate domain sampled on the 5×5 grid used to compare each route with its same-named human-muscle reference; a route may cross additional joints beyond the two evaluated coordinates. At matched pressure, braid geometry, and relative contraction, ideal BPA force scales with diameter squared [Chou and Hannaford, 1996], so four parallel 20 mm BPAs have the same theoretical maximum force as one 40 mm BPA. The baseline uses this substitution at the nominal 40 mm sites and one actuator of the listed diameter at the other sites.

C.9.1 Torque and Active-Force Coverage

Table C14 expands the means reported in Table 4.12. Torque coverage requires the reference torque direction and magnitude; active-force coverage requires positive insertion tension. Both exclude configurations outside the evaluated joint-coordinate domain, and slack or infeasible configurations do not count as active. For each site, domain coverage is estimated from the saved grid by

$$C = 100 \frac{\sum_{i,j} w_i w_j D_{ij} M_{ij}}{\sum_{i,j} w_i w_j D_{ij}}, \tag{C12}$$

where D_{ij} indicates inclusion in the evaluated domain and M_{ij} indicates that the human torque requirement is met. On each uniformly spaced axis, trapezoidal weights are one-

Table C13: Joint coordinates and nominal BPA sizes for the 27-site comparison of my master's robot-leg model. At nominal 40 mm sites, the baseline uses four parallel 20 mm BPAs; the updated configurations retain the inherited diameter.

Site	**Primary DoF**	**Secondary DoF**	**Dia. (**mm**)**	**Baseline (**mm**)**
ercspn_r	lumbar ext.	lumbar bend.	40	4×20
intobl_r	lumbar ext.	lumbar bend.	20	1×20
extobl_r	lumbar ext.	lumbar bend.	20	1×20
bifemlh_r	knee	hip flex.	20	1×20
semimem_r	knee	hip flex.	40	4×20
sar_r	knee	hip flex.	10	1×10
tfl_r	knee	hip flex.	10	1×10
rect_fem_r	knee	hip flex.	20	1×20
grac_r	knee	hip flex.	10	1×10
bifemsh_r	knee	hip flex.	20	1×20
vas_int_r	knee	hip flex.	40	4×20
soleus_r	ankle	knee	40	4×20
med_gas_r	ankle	knee	40	4×20
lat_gas_r	ankle	knee	20	1×20
tib_post_r	subtalar	ankle	40	4×20
tib_ant_r	ankle	subtalar	20	1×20
per_brev_r	subtalar	ankle	10	1×10
per_long_r	subtalar	ankle	20	1×20
per_tert_r	subtalar	ankle	10	1×10
flex_dig_r	MTP	ankle	10	1×10
flex_hal_r	MTP	ankle	10	1×10
ext_dig_r	MTP	ankle	10	1×10
ext_hal_r	MTP	ankle	10	1×10
glut_max_r	hip flex.	hip add.	40	4×20
add_mag_r	hip add.	hip flex.	40	4×20
iliacus_r	hip flex.	hip add.	20	1×20
psoas_r	hip flex.	hip add.	20	1×20

half at the outer endpoints and one at interior samples; the common angular increments cancel in the ratio. Active-force coverage uses the positive-tension indicator in place of M_{ij}. For single-DoF routes, this becomes interval coverage; the repeated fixed coordinate cancels. These are grid-based quadrature estimates, rather than exact continuous-domain boundaries. Each site's fraction is normalized before the equal-site mean is calculated.

Table C14: Torque and active-force coverage at each actuator site. Percentages are trapezoidal estimates of the included angular-domain fractions for that site.

	Baseline (%)		Selected (%)	
Site	**Torque met**	**Active force**	**Torque met**	**Active force**
ercspn_r	100.00	100.00	100.00	100.00
intobl_r	6.25	98.44	78.12	98.44
extobl_r	0.00	100.00	84.38	95.31
bifemlh_r	76.56	98.44	76.56	100.00
semimem_r	15.62	100.00	32.81	87.50
sar_r	25.00	100.00	50.00	100.00
tfl_r	18.75	65.62	37.50	100.00
rect_fem_r	7.81	68.75	20.31	100.00
grac_r	12.50	100.00	31.25	100.00
bifemsh_r	87.50	87.50	100.00	100.00
vas_int_r	12.50	100.00	12.50	100.00
soleus_r	37.50	100.00	100.00	100.00
med_gas_r	0.00	100.00	12.50	100.00
lat_gas_r	0.00	100.00	12.50	100.00
tib_post_r	98.44	100.00	98.44	100.00
tib_ant_r	59.38	100.00	100.00	100.00
per_brev_r	6.25	15.62	92.19	100.00
per_long_r	3.12	100.00	87.50	100.00
per_tert_r	87.50	98.44	98.44	100.00
flex_dig_r	62.50	82.81	98.44	100.00
flex_hal_r	46.88	100.00	62.50	100.00
ext_dig_r	4.69	95.31	7.81	100.00
ext_hal_r	87.50	100.00	92.19	100.00
glut_max_r	52.27	100.00	56.82	100.00
add_mag_r	31.82	100.00	65.91	100.00
iliacus_r	29.55	100.00	81.82	100.00
psoas_r	38.64	100.00	86.36	100.00
Mean	37.35	93.00	65.81	99.31

Appendix D: Supplementary Neuromechanical Verification

This appendix supplies supporting model records and evaluation sets for Chapter 6. It separates implementation checks, gait comparisons, pathway-removal tests, and spiking conversions because they test different claims. Neuron equations and calibration tables are provided in Appendix B.

D.1 Model Construction and Recording

The archived AnimatLab projects provide reference circuits for implementation comparisons. Figure D1 shows their neural organization. A later reorganized bilateral project added commissural and biarticular pathways; it completed simulation runs but did not sustain alternation under the tested drives. That assembly result is not evidence of stepping. The phase-1 recordings below concern an earlier instrumented configuration, not the reorganized network.

A 10 s instrumented run recorded joint, neural, sensory, and body-motion channels at a 0.2 ms interval. Contact-sensory recordings are membrane voltages, not contact forces. The initial external horizontal force and the limited window prevent the joint traces from establishing stable walking. The platform was present, but no timed vertical release was recorded; these traces therefore do not establish a drop or touchdown response.

Figure D2 contrasts the converted Gait2392 robot geometry with the reduced-order knee model used in the SNS-Simscape demonstration.

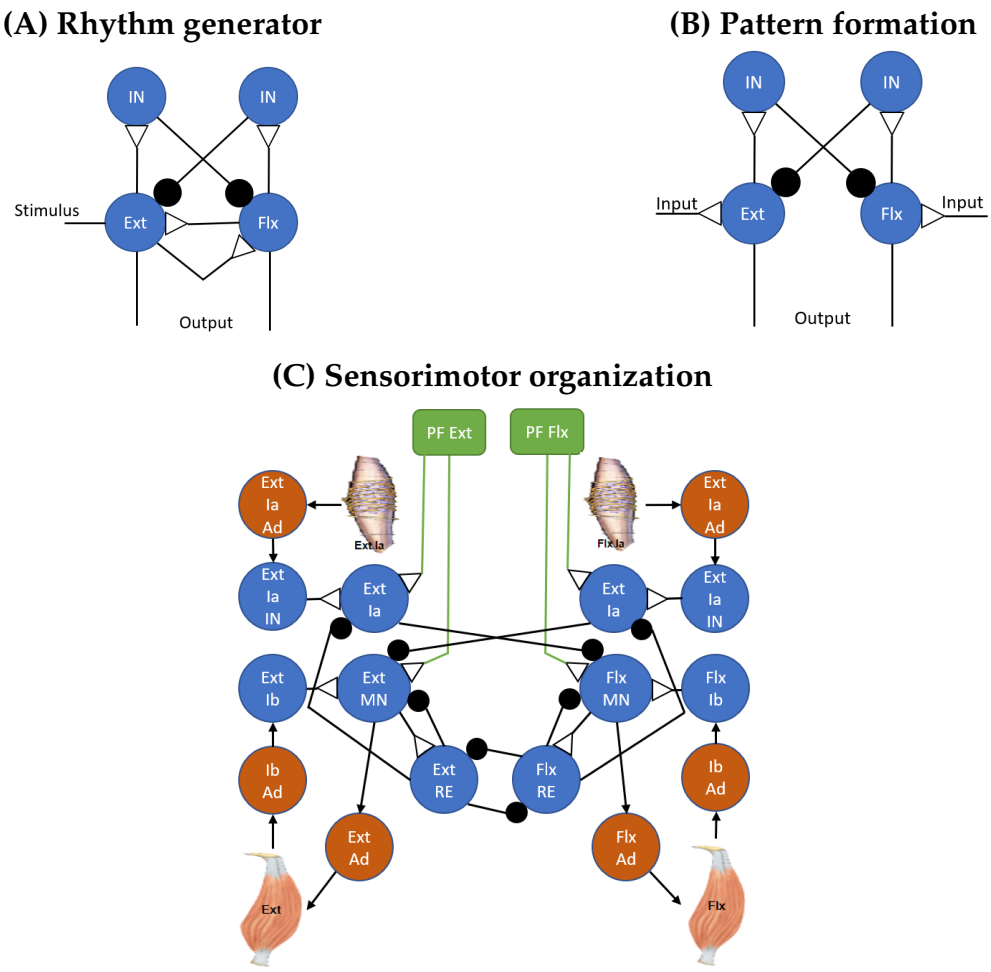
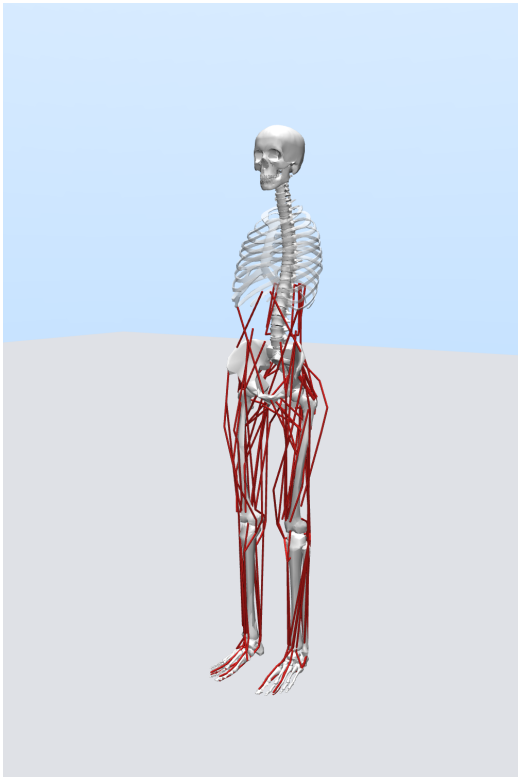


Figure D1: Archived AnimatLab neural subsystems: **(A)** rhythm generation, **(B)** pattern formation, and **(C)** sensorimotor organization. Ext and Flx denote antagonist pathways; IN denotes interneurons, MN motor neurons, and Ia and Ib muscle afferents. Triangles and circles distinguish excitatory and inhibitory connections.

(A) Converted robot



(B) SNS-Simscape knee model

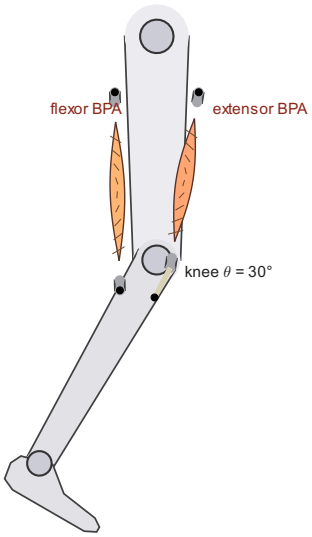


Figure D2: Converted Gait2392 robot (A) and the antagonist-BPA knee illustration used in the current SNS-Simscape demonstration (B). The two panels represent different models; scales differ.

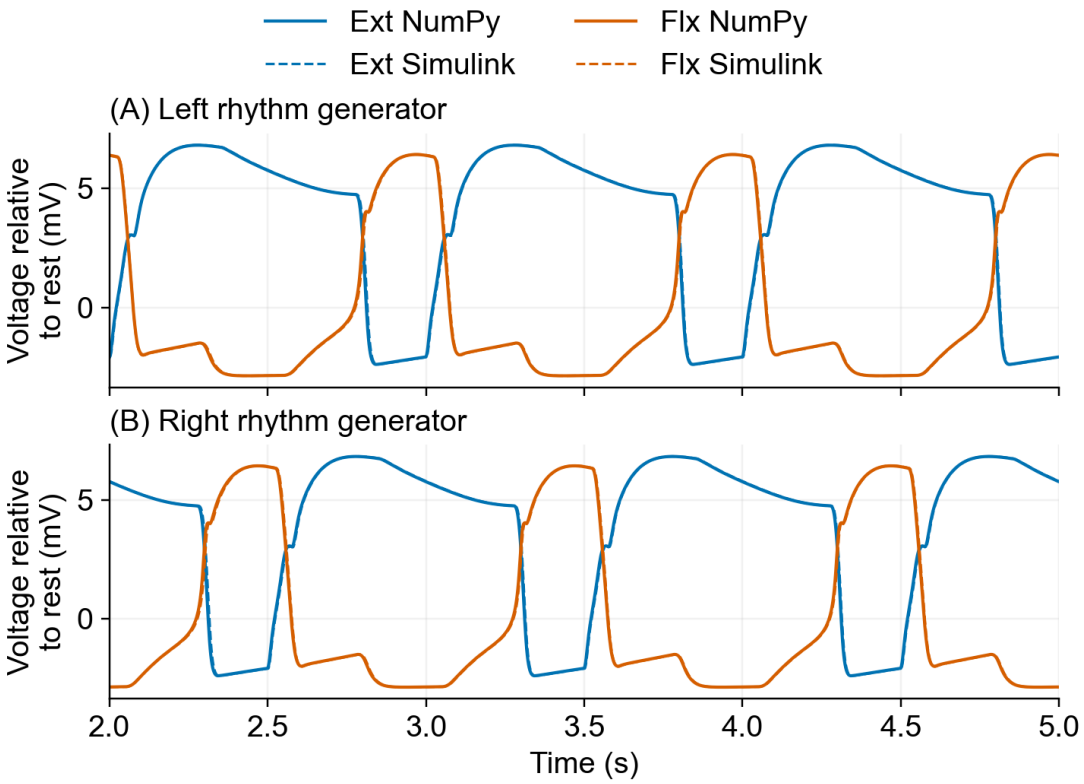


Figure D3: W2L rhythm-generator voltages under matched inputs in NumPy and Simulink. Burst periods and counts agree; full-trace correlation is at least 0.99996. Extensor and flexor traces are paired by leg over a common three-second interval; voltages are relative to the resting reference. The residual 2 ms lag is one discrete solver step. The comparison checks the driven rhythm core.

The suspended AnimatLab recording provides the neural and mechanical context for the motivating ground test in Section 6.1. Alternating activity propagates through RG, PF, and MN layers over the same short interval as the joint motion.

D.2 Controller Ports and Reference Checks

The matched W2L test compares implementations from a common circuit specification under identical drive and contact inputs, using burst period, bilateral phase, burst count, and voltage correlation. The Deng controller port was checked separately: a current-pulse test produced a 1.938 s period against a 1.94 s NumPy reference, with antagonist correlation $r = -0.86$. Agreement on these tests does not establish long-term phase agreement in every network.

The archived original AnimatLab model produced a period of approximately 0.221 s, compared with 1.938 s in the translated test. Both alternated antagonists, but the original bilateral rhythm was in phase. Different parameterizations prevent quantitative equivalence. In both implementations an RG perturbation disturbed activity without sustained suppression; comparing perturbed and baseline sequences, rather than counting peaks in a short window, establishes that qualitative response.

The full exported network agreed with its NumPy reference to 4.2×10^{-6} mV over the early transient across 410 neurons. Its subsequent divergence and the tonic-drive fixed-point discrepancy reported in Chapter 6 limit parameter transfer. The OpenSim-derived Simscape mechanism contains 1,446 blocks and 149 joints; mechanism compilation is distinct from completing its neural and muscle coupling.

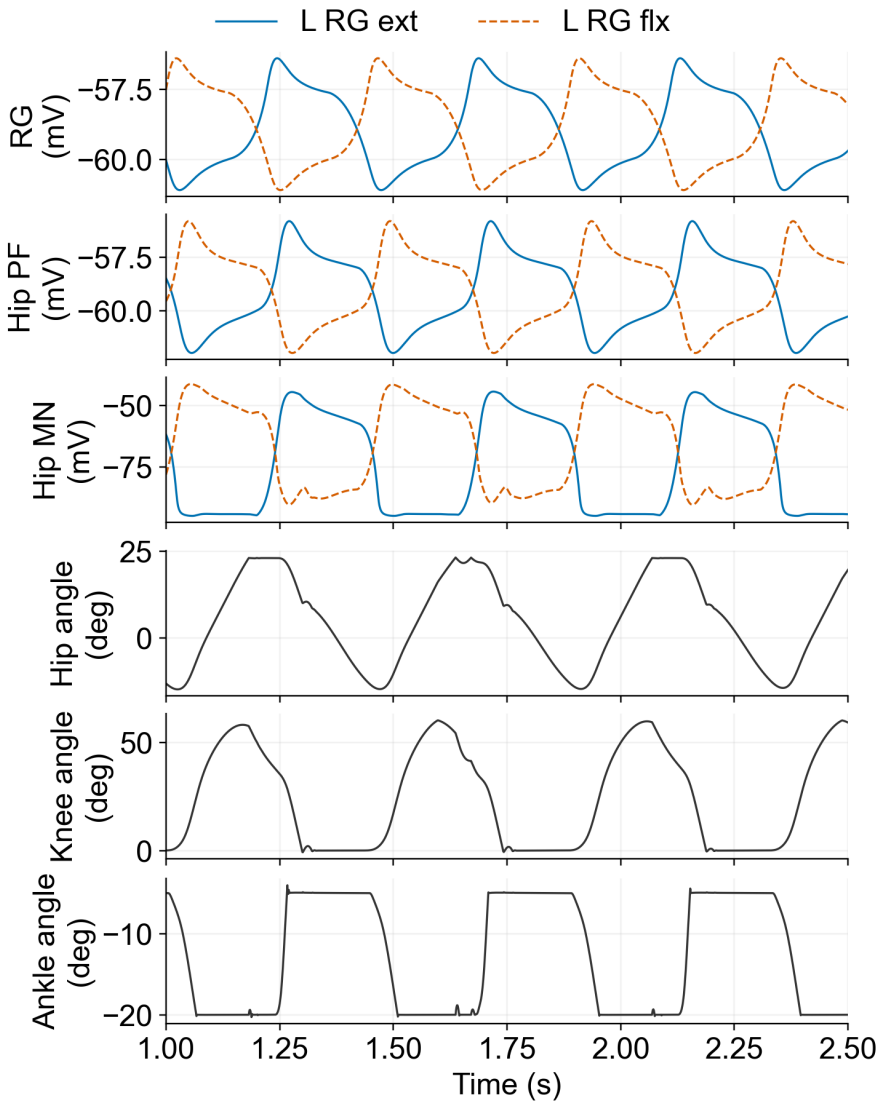


Figure D4: Recorded graded activity in the rhythm generator, hip pattern-formation subsystem, and hip motor neurons, followed by left hip, knee, and ankle angles from the same suspended AnimatLab trial. Extensor and flexor pathways are blue solid and orange dashed, respectively. The 1 s to 2.5 s interval shows approximately three cycles. Joint angles retain the original AnimatLab conventions, including positive knee flexion. The neural traces are graded membrane voltages.

D.2.1 Transient Knee Sensory-Current Response

The knee block model was tested from a settled operating point using a rectangular current pulse added after the spindle transducer and before the extensor Ia sensory-neuron membrane input. Stimulus and no-stimulus cases start from the same initial conditions and run continuously for 50 s; the 2 nA pulse acts from 30.000 s to 30.030 s. The original configuration did not settle: its knee angle spanned 0.712° during the 25 s to 30 s baseline window. Its former startup comparison is therefore not used to demonstrate transient reflex recovery.

The graded-response comparison changes exactly six parameters: the two MN output ramp widths from 1 to 10 mV, and the four Ia synaptic ramp widths from 0.5 to 10 mV. Equations, thresholds, conductances, wiring, mechanics, tonic drives, and activation dynamics are retained. This reduces loop gain as an explicit engineering comparison, rather than fitting physiological parameters. The simple actuator-force law is distinct from the characterized BPA model of Chapter 3. The flexor activation is zero at baseline, so the experiment does not establish suppression of a tonically active antagonist.

The modified configuration passes baseline settling tests for angle, velocity, all six membrane states, and both activations. Its paper-coordinate baseline is -42.6392°; positive excursions indicate extension. Extension peaks at 1.0908° after 111 ms. Recovery requires displacement within 0.01° of baseline and absolute angular velocity below $1 \times 10^{-4}\,\mathrm{rad\,s^{-1}}$; both remain within those bounds from 6.355 s after stimulus onset. The late residual is approximately 9.3°. A fresh-process replay reproduces the resampled pulse arrays exactly. Halving the ODE45 maximum step from 1 to 0.5 ms, at unchanged tolerances,

changes the angle trace by at most 5.21°. Comparisons use linear interpolation onto a 1 ms grid, without smoothing. These checks establish numerical recovery in the model, not physiological reflex fidelity. Figure D5 shows the stimulus, matched kinematics, neural signals, activations, and forces.

D.2.2 A Published Neural Mechanism as a Verification Target

The reduced Python implementation of the two-level circuit of Rybak et al. [2006] passed 13 checks covering drive-dependent activity, repeatability, and the distinction between resetting and non-resetting deletions. Stimulating the pattern-formation extensor population suppressed flexor motor output while the rhythm generator continued, so flexor activity resumed at its expected phase. Stimulating the rhythm-generator extensor population also suppressed flexor output but shifted the subsequent rhythm phase. This comparison tests the functional separation between rhythm generation and pattern formation: similar missing motor bursts can arise through different neural mechanisms. It is a neural-only verification of the translated circuit, independent of a mechanical model and separate from the Simulink demonstrations.

D.2.3 RG, PF, and MN Stimulus Tests

The fourteen-neuron Deng-derived implementation follows the circuit used in SNS-Toolbox demonstrations [Deng et al., 2019, Nourse et al., 2023]. A 0.1 ms Euler step reproduced the existing neural integrator's complete 24 s baseline voltage trace exactly in the tested process. The baseline period was 1.938 s. Neural output drove the existing

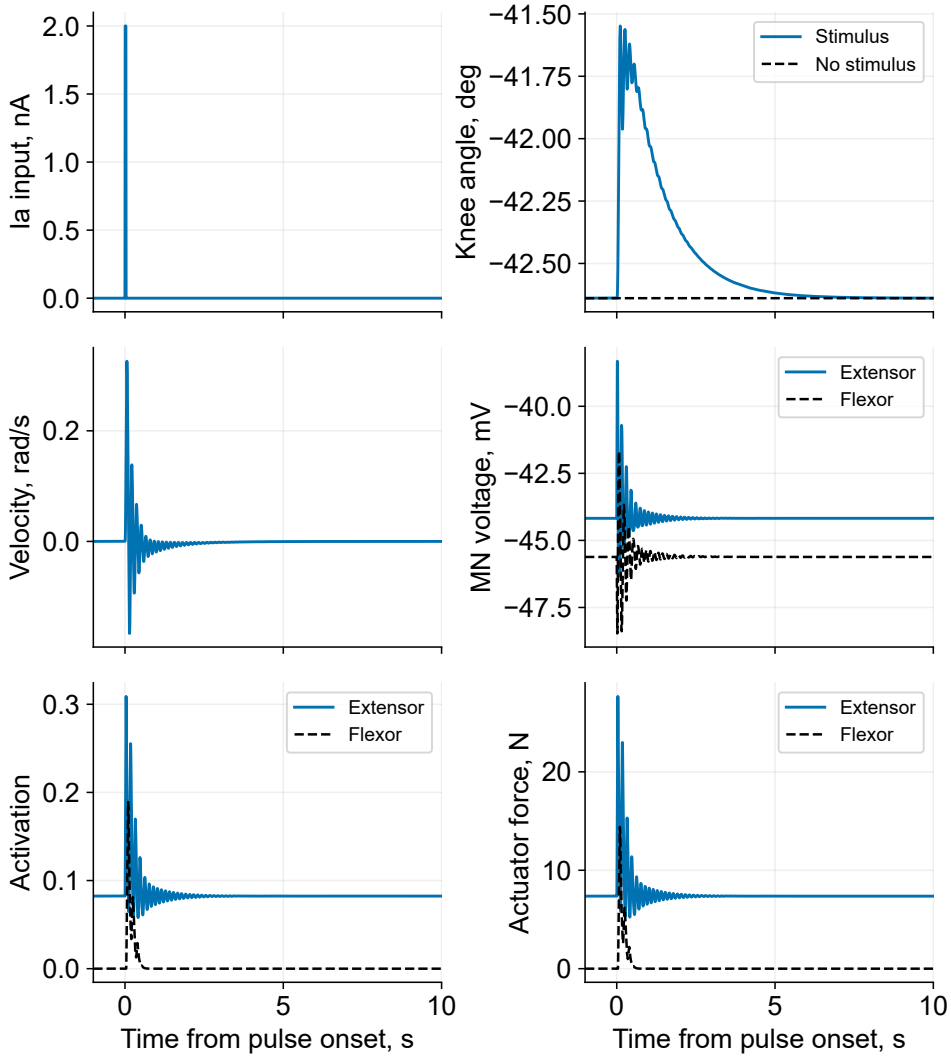


Figure D5: Stimulus, knee angle, angular velocity, MN voltages, activations, and actuator forces in the graded-response knee model. The pulse enters the extensor Ia sensory neuron after settling. Knee angle is compared with the no-stimulus control; extensor and flexor signals are distinguished in the other paired panels. The display applies only the sign conversion to flexion-negative coordinates.

suspended, unilateral W2L mechanical model without afferent return. Hip commands came from the MN outputs; knee and ankle commands used the existing PF-to-activation mapping. This configuration tests transmitted circuit effects, not the original rat mechanics or bilateral walking.

The initial sweep applied 2 nA and 10 nA to each layer for 0.1 s and 0.5 s, together with two direct MN inhibitory controls. None met the deletion criterion. Subsequent stronger-current control series, including stimuli lasting two baseline periods (3.876 s), retained the same circuit and mechanical parameters (Table D1). RG and PF stimuli began one-quarter cycle after a baseline RG onset; MN inhibition began 0.05 s into a baseline motor burst. Positive currents target the extensor RG or hip-PF population; negative currents target the hip-flexor MN. These currents are mechanistic controls, not experimentally calibrated physiological stimuli.

A deletion required at least 20 ms of baseline motor activity in the stimulus window and suppression of at least half of that activity, using an activation threshold of 0.5. Phase residuals compare the final four recovered onsets with the nearest baseline onset, normalized by the baseline period. Residuals at or below 0.05 cycles were classified as non-resetting, and those at or above 0.15 as shifted; intermediate values remained indeterminate. RG and motor residuals were assessed separately. The two-cycle stimuli suppressed both expected flexor burst onsets at all three layers. Baseline and stimulated neural and mechanical trajectories were identical before stimulus onset. The 1 s PF pulse shifted the motor pattern without resetting the RG clock, whereas the two-cycle PF pulse preserved both schedules. The duration dependence cautions against treating every motor phase shift as an RG reset. The coupled mechanical traces contained large joint-limit

violations, so these tests are interpreted through their neural responses in Table D1.

Table D1: Stronger-current controls in the reduced Deng-derived circuit. Phase residuals are fractions of the baseline cycle. "No deletion" means the specified suppression criterion was not met.

Target	nA	s	RG phase	MN phase	Outcome
RG	+50	0.5	0.052	0.052	Brief suppression; phase indeterminate
RG	+50	1.0	0.307	0.307	Suppression; RG and MN shifted
PF	+50	0.5	0.000	0.000	Brief suppression; schedule preserved
PF	+50	1.0	0.000	0.208	Motor pattern shifted; RG preserved
MN	−100	0.5	0.000	0.000	No deletion
MN	−500	0.5	0.000	0.000	Brief suppression; schedule preserved
RG	+50	3.876	0.211	0.211	Resetting deletion
PF	+50	3.876	0.000	0.000	Non-resetting deletion
MN	−500	3.876	0.000	0.000	Non-resetting deletion

Hip-PF inhibition altered the left motor pattern and joint motion while bilateral RG activity continued (Figure D6). This separate W2L model has 108 graded neurons, 216 chemical connections, and twelve Hill-type muscles, with its trunk fixed in air. A -10 nA pulse acted on the left hip-extensor PF neuron from 4.534 s to 4.634 s. Neural and angular recordings were identical to the unperturbed baseline before injection. The hip plateaus occur at the native joint stops near 20°; they represent limit engagement. This bilateral test is distinct from the fourteen-neuron deletion tests above. Figure D6**(A)** and Figure D6**(B)** show the left and right legs, respectively. Blue solid and orange dashed neural curves denote extensor and flexor cells; gray curves show the matched baseline, and dotted lines mark the hip limits.

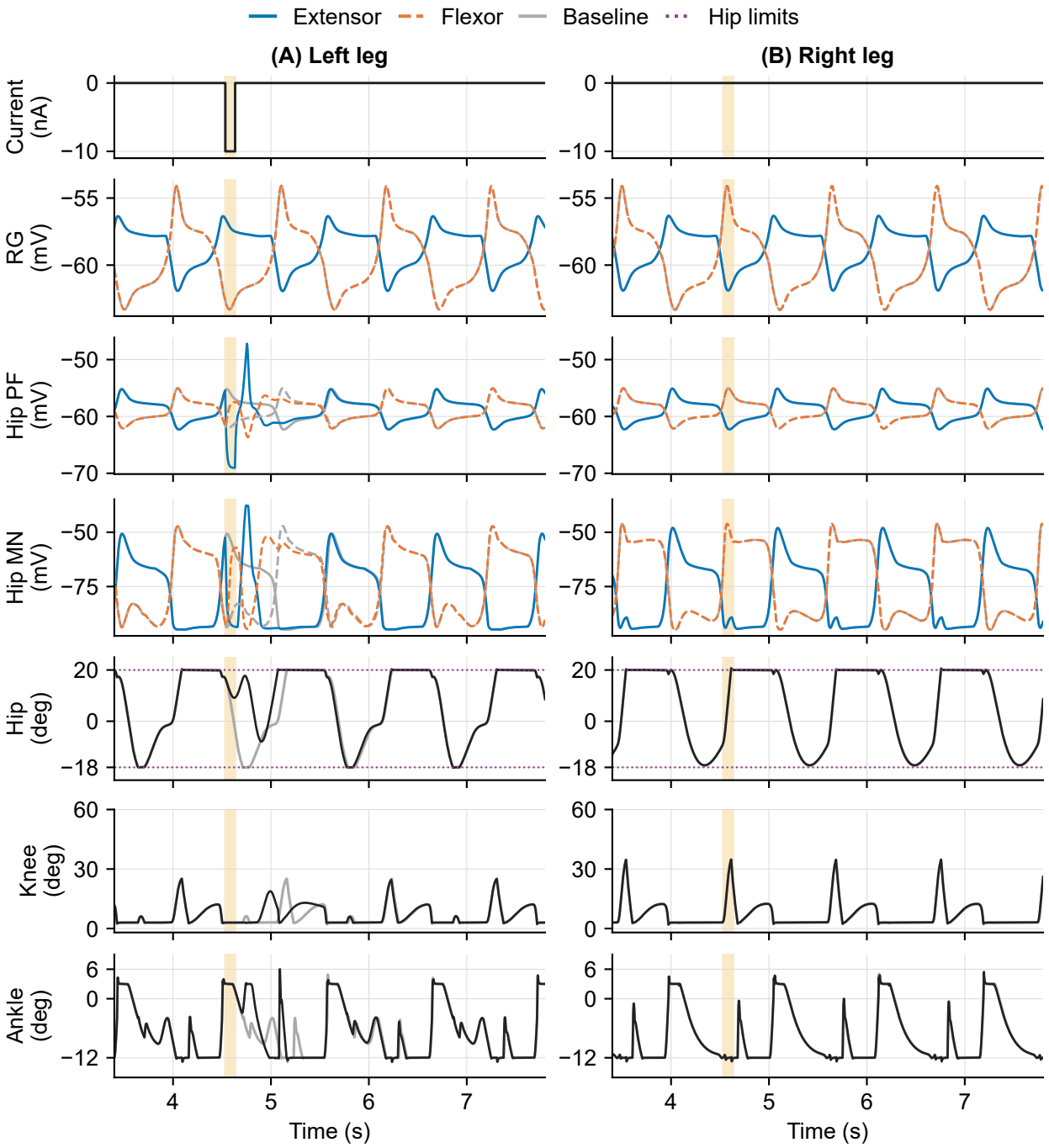


Figure D6: Response of the bilateral W2L model to left hip-PF inhibition.

D.2.4 Archived AnimatLab Models

Headless reruns tested whether archived models reproduced their reference behavior in the original simulator. Eleven targetable rhythm and gait metrics passed their acceptance bands. The bilateral contact-driven model's rhythm generators had periods of 0.450 s and 0.467 s; the modern W2L model showed joint oscillations at 2.25 Hz. The Li-model rerun reached 0.637 m s^{-1} with step periods near 1.3 s [Li et al., 2017]. These are reference-model checks, not results for the combined Deng–Li model or evidence of unsupported walking.

A separate MuJoCo translation of the Li controller and body did not reproduce that original-simulator walking result. In a 20 s ground test, mean forward speed was 0.040 m s^{-1}, with eleven detected right-foot stance episodes and none on the left. The right-side period of 1.294 s alone therefore did not establish bilateral stepping. This failure distinguishes reproduction of the source model from successful transfer to another simulator.

Additional execution checks. Repeated controller searches benefit from reducing neural and mechanical computation separately. I accelerated the neural equations with compiled kernels and checked their outputs against the reference implementations. In a 980-neuron timing test, the sparse kernel was 11.23 times faster than the preceding preallocated neural backend, with identical voltages. A complete 16 s moving-leg simulation decreased from 69.51 s to 15.86 s of wall time while preserving every saved trajectory array. A separate afferented simulation decreased from 113.74 s to 34.59 s under concurrent machine load. These measurements distinguish neural-kernel acceleration from the runtime of the

coupled model.

D.3 Tuning and Supported Evaluations

The Gait2392-based evaluations use the same reference-library and pathway-removal procedures to compare alternative neural organizations. The full graded, pruned, six-synergy, and W2L-layout configurations show different responses to feedback removal and different agreement with the tuning and library references. Removing Ia feedback eliminated detected cycles in the tested production configuration, whereas other removals and other network organizations had different effects. The library comparisons also exposed asymmetric activity and limited transfer beyond the tuning reference (Appendix D.3.1). These tests distinguish how each controller uses feedback and how its motion changes when the reference is changed.

A replay of the saved human-model controller produced ten detected cycles per leg. Figure D7 compares the joint trajectories directly with the Gait2392 inverse-kinematics reference and its associated ground-reaction forces. The model's mean stance fractions were 0.71 and 0.65 on the right and left, compared with 0.61 in the reference. Substantial hip and knee timing differences remain visible despite the detected cycles. The overlay therefore complements the scalar gait score by showing which parts of the movement disagree. Appendix D.3.2 reports the angle errors and alignment method.

Free network gains were searched with a tree-structured Parzen estimator, with each candidate evaluated through simulation. The staged procedure separates suspended rhythm generation, sensory feedback, standing, and ground evaluations (Table 6.3). Final-

ists were checked under 1 % multiplicative parameter perturbations and changes in drive. Such checks probe the neighborhood of a selected configuration; they do not establish robustness to untested loads or removal of the harness.

A run without detectable cycles returns -25 as a reserved failure value. Because this value is numerically greater than the genuine-gait scores below, failed runs must be identified before ranking scores. That value itself is not a valid measure of gait quality.

The curriculum is shown in Table 6.3 in Chapter 6.

D.3.1 Reference-Library Comparison

The 43-cycle library contains 32 running trials from eight subjects at four speeds, ten walking references, and a predicted-walking cycle. The predicted-walking reference was generated by optimizing muscle controls in a musculoskeletal simulation [Falisse et al., 2022]; it is a simulated comparison trajectory rather than a participant recording. The walking references span prescribed speeds of $0.5\,\mathrm{m\,s^{-1}}$ to $2.0\,\mathrm{m\,s^{-1}}$ and self-selected trials. The full graded controller, a pruned version with fewer connections, a six-synergy controller, and a controller using the W2L organization were compared on their tuning reference and the broader library. Library scores were substantially more negative (Table D2), and the evaluations showed unilateral or asymmetric gait. The joint trajectories and contact patterns identify which aspects of the movement transferred to each reference.

Table D2 reports the saved kinematic comparison scores. Less-negative values indicate closer agreement within a scoring configuration; these scores are not percentages or a stand-alone criterion for successful walking. The tuning-reference column uses each

controller's selected reference, while the library mean averages its scores over all 43 cycles. The direct angle overlays and errors in Section D.3.2 provide the more readily interpretable joint-level comparison.

Table D2: Four controller configurations evaluated on their tuning reference and the 43-cycle reference library. Larger recorded scores indicate closer agreement; absolute values depend on the evaluation conditions.

	Tuning-ref. score	**Library mean score (43 refs)**	**Most negative reference (score)**
Full graded	−186.4	−8776	Falisse et al. (2022), simulated walking (−8841)
Pruned graded	−176.1	−6924	Falisse et al. (2022), simulated walking (−6986)
Six-synergy	−200.4	−9961	Falisse et al. (2022), simulated walking (−10040)
W2L organization	−144.5	−5808	Subject 8 running trial (−5896)

A later matched human-model comparison also varied controller organization and feedback gains. The full graded baseline generated nine detected right-leg cycles and no left-leg cycles, while the W2L organization generated five and seven, respectively. Halving the Ia, II, and Ib gains into its central pathways reduced these to three and four. These are separate configurations from the reference-library sweep above; their value is the controlled comparison of controller dependence, not a replacement ranking of the earlier scores.

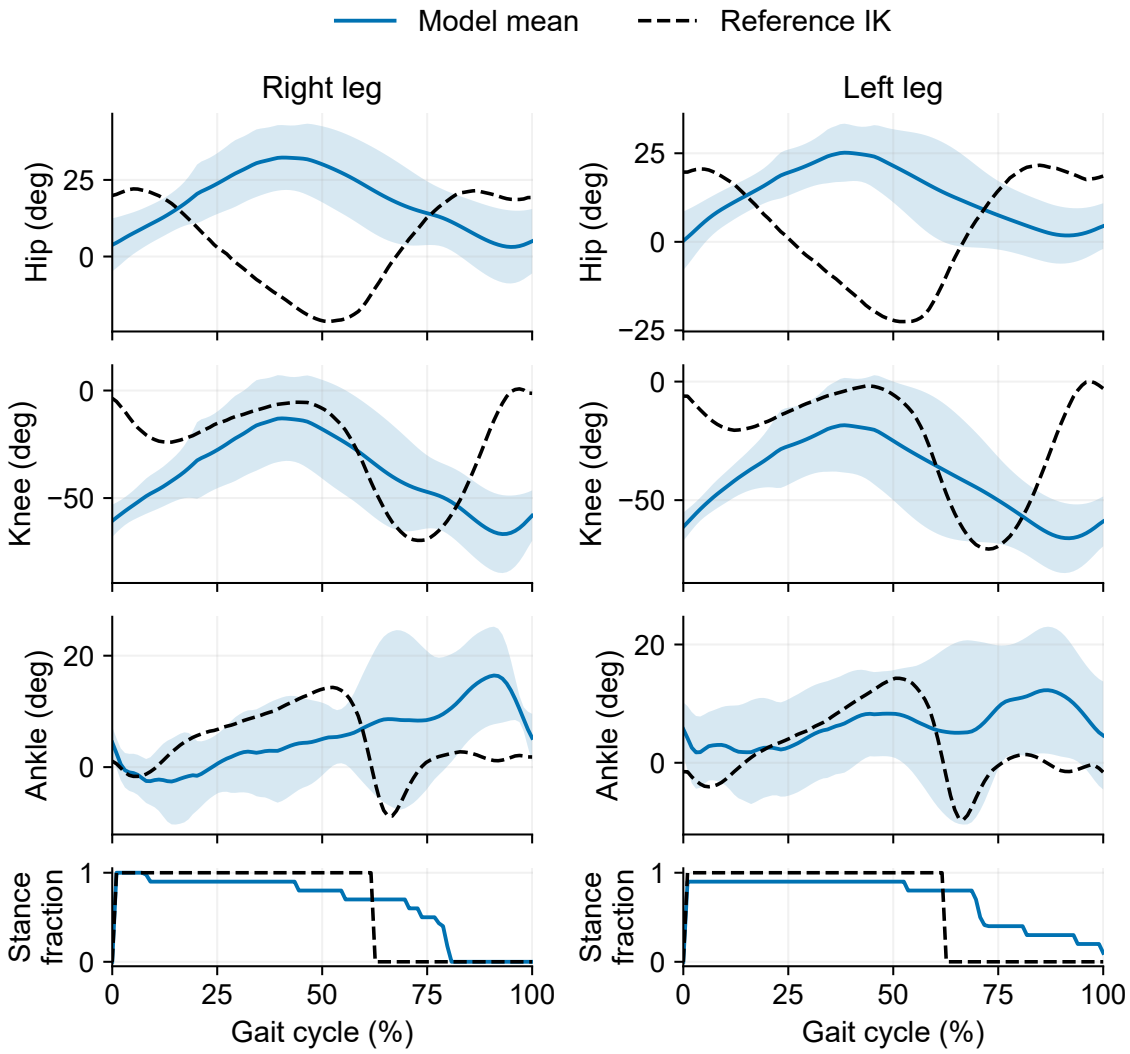


Figure D7: Supported human-model joint motion compared with the Gait2392 inverse-kinematics reference. Model curves show the mean and one standard deviation across ten cycles per leg. Each leg is aligned to its own foot-loading onset; no angle offset, amplitude adjustment, or fitted phase shift is applied. Bottom strips compare model contact probability across cycles with reference stance and swing from ground-reaction forces, using a 50 N threshold. Legs are aligned independently; bilateral phase is evaluated separately.

D.3.2 Direct Kinematics Overlay

The saved human-model controller reproduced its cycle-shape score of -145.07 in an independent process. The direct overlay in Figure D8 uses the Gait2392 walking inverse-kinematics trajectory and its paired ground-reaction-force file. Model cycles begin when vertical foot loading crosses 50 N; reference stance is defined using the same force threshold. Ten cycles per side, with durations between 0.35 s and 2.5 s, were retained using the existing cycle detector. Each leg was aligned to its own onset and resampled over the gait cycle, without fitting phase, subtracting the mean angle, or scaling amplitude.

Mean cycle periods were 1.285 s on the right and 1.254 s on the left, compared with 1.233 s and 1.213 s in the reference. Model stance fractions were 0.710 and 0.649, compared with 0.610 in the reference. Table D3 reports absolute-angle errors between mean cycles. These errors retain posture offsets and are distinct from the mean-removed cycle-shape errors used in tuning. The hip and knee show substantial differences in posture and timing, while the ankle trajectories agree more closely. Similar cycle periods therefore do not imply close agreement of the joint trajectories.

Table D3: Unshifted absolute-angle RMSE between mean model and reference cycles, in degrees.

Joint	Right	Left
Hip	28.03	24.65
Knee	27.44	28.45
Ankle	8.71	7.38

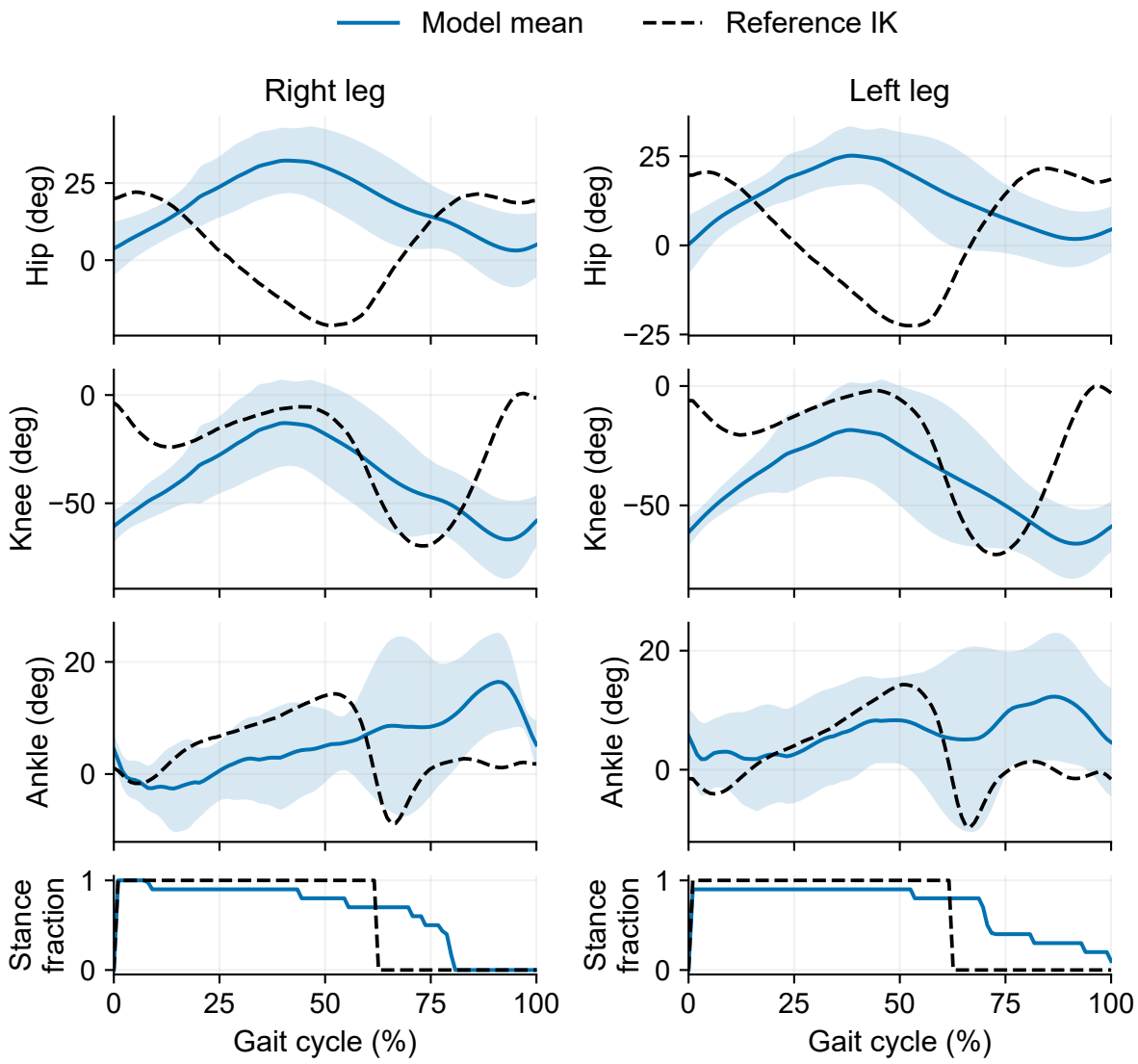


Figure D8: Supported Gait2392 human-model motion compared with the inverse-kinematics reference. Columns show the right and left legs. Blue curves and bands show the model mean and one standard deviation across ten cycles per leg; black dashed curves show the reference. Each leg is aligned to its own foot-loading onset, without an angle offset, amplitude adjustment, or fitted phase shift. The bottom row compares model stance fraction across cycles with reference stance and swing, using a 50 N loading threshold. Absolute-angle errors are reported in Table D3.

D.3.3 Pathway-Removal Tests

Individual pathway classes were removed from each selected configuration and the model re-evaluated on its tuning reference (Table D4). Removing Ia feedback from the production configuration eliminated detected cycles. Other removals changed the score without necessarily eliminating movement. The different response of the six-synergy variant shows that these dependencies belong to the tested network configurations, rather than constituting a general claim about biological necessity. The systematic pattern-formation and motoneuron stimulation series for these full Gait2392 networks was not completed. The reduced-circuit tests in Section D.2.3 address a separate, smaller model.

Table D4: Pathway-removal tests on the tuning reference. Full-model rows give unchanged baselines. A score of -25 is the fixed no-cycle fallback and denotes failure, despite exceeding the genuine-gait scores.

Configuration	Removed pathway	Score
`s3k` (full model)	none	-186.4
	Ia afferents	-25 (no cycles detected)
	all afferent classes	-267.8
	Renshaw cells	-267.3
	group-II afferents	-255.2
	Ib afferents	-180.3
	contact channels	-183.9
	interleg commissural pathways	-174.9
`syn6` (full model)	none	-200.4
	all afferent classes	-190.5
`w2lvar` (full model)	none	-144.5
	Ia afferents	-273.5
	weak rhythm-generator mutual excitation	-279.9
	Renshaw cells	-142.4

D.4 Spiking-Neuron Comparisons

The spiking variants replace afferent, rhythm-generator, and interneuron populations while retaining graded motor neurons. Synapses onto motor neurons are calibrated by mean conductance, preserving the continuous activation interface. This construction does not require the network dynamics or resulting movement to remain unchanged. Calibration and full conversion tables are in Appendix B.

Action potentials introduce voltage fluctuations absent from a graded signal, so raw membrane-voltage variability is not an equivalent measure of controller output. Figure D9 compares filtered neural readouts, muscle activation, and joint motion under the same drive, support, and selected parameters. For a hardware comparison using pulse-width modulation (PWM), both signals would need to pass through the same command encoder and valve limits.

D.4.1 MuJoCo Network and Ground Evaluation

Under constant drive, the non-spiking and spiking rhythm periods were 0.399 s and 0.621 s, respectively. The spiking period is 55.6 % longer. Timing agreement therefore requires more than copying the selected parameters. The untuned spiking variant produced bilateral cycling in a ground evaluation where the non-spiking winner cycled predominantly on one side (Table D5; Figure D9). This individual comparison motivates further controlled testing; it does not establish superiority of spiking control.

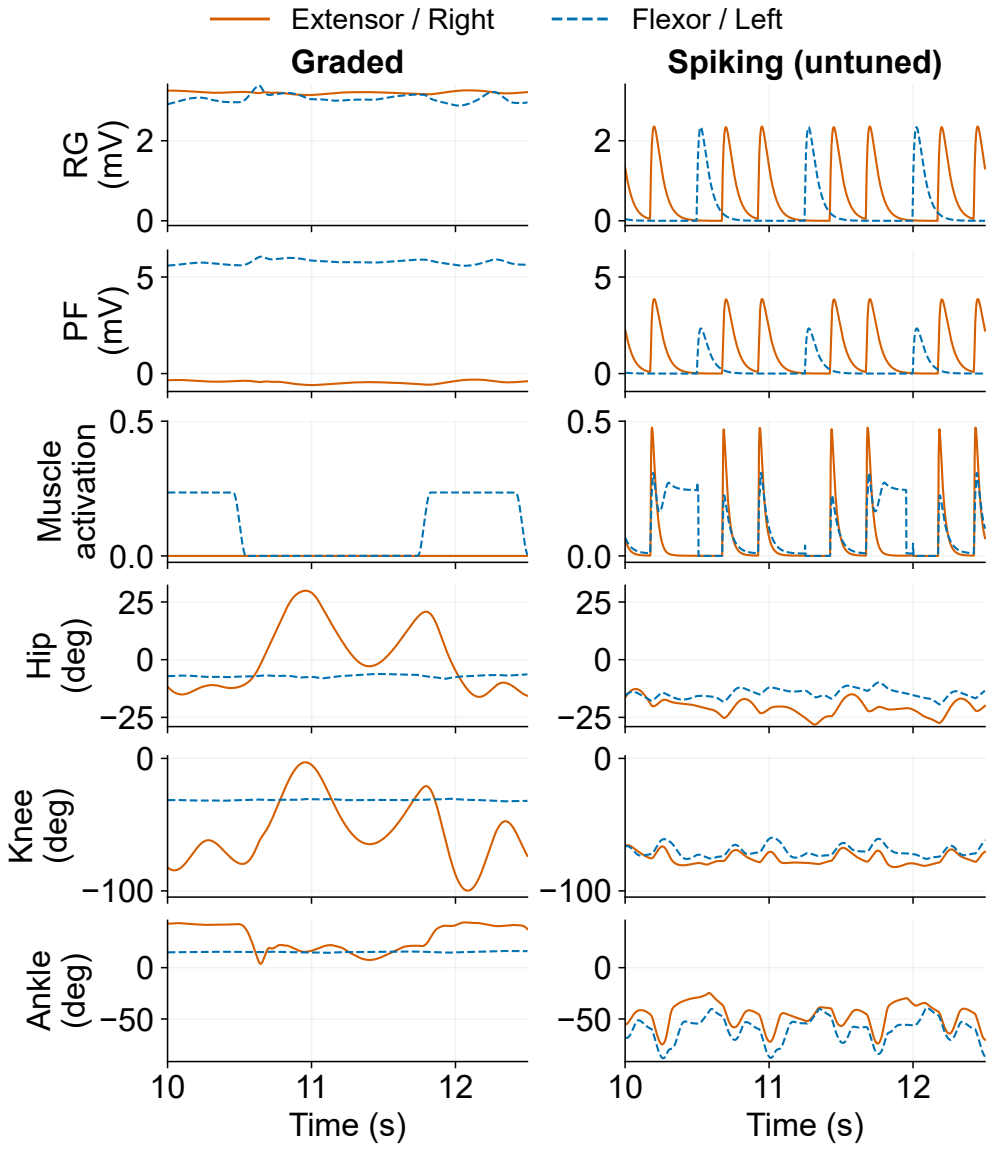


Figure D9: Matched 10 s to 12.5 s ground-trial window: graded controller (left) and untuned spiking controller (right), at the same selected parameters and support settings. Rows show right-side RG and PF readouts, vastus-lateralis and semimembranosus activation, and bilateral hip, knee, and ankle angles. Neural rows pair extensor (orange solid) and flexor (blue dashed) outputs; joint rows pair right (orange solid) and left (blue dashed) legs. Spiking signals are filtered readouts, and muscle activation is the actuator state.

Table D5: Supported ground evaluation at the same selected parameters (16 s), with no spiking re-tuning. The knee minimum is the minimum of the right leg's mean detected cycle. Contact fraction is the fraction of samples with foot normal force above 20 N during the post-startup evaluation window, whether or not cycles were detected. Thus, the non-spiking left leg can remain in contact for a fraction of 0.985 while completing no detected cycles. The recorded kinematic score summarizes the cycle comparison; less-negative scores indicate closer agreement within this matched test.

Metric	**Non-spiking winner**	**Spiking mirror (untuned)**
Kinematic score	−161.57	−236.35
Gait cycles, right / left	9 / 0 (left frozen)	8 / 9 (bilateral)
Cycle knee minimum	−64.3°	−79.8°
Contact fraction, right / left	0.637 / 0.985	0.799 / 0.856
Maximum pelvis tilt	22.1°	19.2°
Non-finite states	no	no
Fell	no	no

D.4.2 Simulink Demonstrations

The earlier 5 s knee startup comparison used a graded baseline and a spiking twin. Its final-window statistics describe startup activity; the settled, matched-control test in Section D.2.1 instead measures recovery from a brief sensory-current pulse. The earlier startup figure and its numerical table are retained with the supporting source records rather than repeated here.

In the beverage example, the spiking implementation retained approximately the same sag, 2.02° versus 2.00° in the graded implementation, compared with 7.85° without the reflex. This comparison demonstrates transfer of the tested response after calibration.

D.4.3 AnimatLab Conversions

The tested spiking conversions did not preserve the graded models' rhythm (Table D6). Converting thresholds while retaining graded synapses preserved near-baseline oscillations, whereas converting synaptic amplitudes and decay changed or eliminated alternation. These results identify the synapse conversion as a limitation of this mapping. They do not demonstrate that a suitably calibrated spiking implementation cannot reproduce the reference behavior.

Table D6: AnimatLab spiking conversions versus their graded baselines (5.1 s simulations; joint ranges in degrees where the rhythm survives). The thresholds-only ablation converts the neurons but keeps the graded synapses.

Variant	**Baseline (graded)**	**Spiking copy**
Two-layer walker	0.443 s rhythm; hip -15 to $+23$, knee -1 to $+60$	no sustained rhythm (2 spikes; hip pinned near $+6$)
Bilateral variant	0.462 s; antiphase -0.42 cycle	sustained unilateral firing; one side at 442 Hz to 450 Hz
Reorganized project (standalone export)	no alternation under tonic drives	synchronous 11.8 Hz co-bursting; same-leg envelopes $r = 1.000$ at lag 0
Thresholds only (graded synapses)	hip -15 to $+23$, knee -1 to $+60$	oscillates: hip -15.1 to $+23.2$, knee -0.8 to $+47.9$

D.5 Supported BPA-Robot Evaluations

An early supported software-interface experiment recorded neural activity, actuator commands, and multi-joint motion. Its nominal vertical-support setting was 70 % of the 411 N body weight. The recorded period was 1.542 s, knee RoM was 63.5°, and peak foot-contact

forces were 880 N and 1372 N, at a peak commanded pressure of 372 kPa.

Component tests now check tensile-force application against native MuJoCo actuators. Together with the characterized force law, this interface supports simulated BPA control and provides a basis for future work on dynamic actuator models, sensory compensation, and physical BPA control.

With split left/right drives and retuned feedback, the W2L robot progressed 1.72 m over 30 s under the support protocol. It retained progression at 60 % drive and passed 12 of 12 perturbed reruns, but did not satisfy the strict contact-duty walking criterion; below that drive the gait ceased. The mechanical support was not weaned.

In the separate human-model drive-weaning study, the selected score improved from -161.57 to -145.07, with ten contact cycles per leg and contact duties of 0.71 and 0.65. It passed eight of eight perturbed reruns and retained bilateral stepping down to zero external neural drive. The motion remained in place under support. These two studies distinguish reduced neural assistance from reduced mechanical assistance.

D.5.1 Independent Replays and Remaining Mechanical Assistance

A separate W2L configuration was evaluated after correcting the delivery of saved per-target gains to the evaluator. Four independent 30 s replays produced identical trajectories. Over the 4.5 s to 30 s analysis window, the run had 28 left and 21 right contact events, a median bilateral phase of 0.496 cycles, forward displacement of 2.246 m, and fitted speed of 0.0918 m s^{-1}. All twelve independent reruns with 1 % parameter perturbations passed the supported-stepping criterion. These are results for this configuration, distinct from the

drive-weaning study above.

Removing nominal constant weight support did not remove the position-control harness: its mean vertical force still supplied approximately 16.76 % of body weight. Contact duty remained below the strict walking criterion. These trials demonstrate repeatable stepping with that remaining assistance. An accelerated neural backend preserved the complete reference trajectories and reduced the isolated neural-step cost by a factor of 1.425; this does not establish the same speed improvement for the complete neuromechanical simulation.

D.5.2 Pathway Deletions and Stimulus Tests

The recovered W2L configuration was evaluated with nominal weight support set to 0.25 and the remaining controller settings fixed. Mild and strong drive reductions, forward and backward 40 N pushes, and separate Ia, Ib, Renshaw, crossed-heel, and heel-cascade deletions retained the supported-stepping criterion. Their effects differed: Renshaw deletion reduced forward displacement to 0.081 m, while commissural deletion failed bilateral stepping. Disabling all sensory feedback retained joint cycling and 1.996 m progression but shifted median contact phase to 0.223 cycles, outside the accepted alternation range. The intact comparison produced 2.239 m and a phase of 0.498 cycles. These experiments distinguish rhythm generation, progression, and bilateral contact coordination, rather than treating them as interchangeable outcomes. Figure D10 compares the drive, push, and pathway interventions.

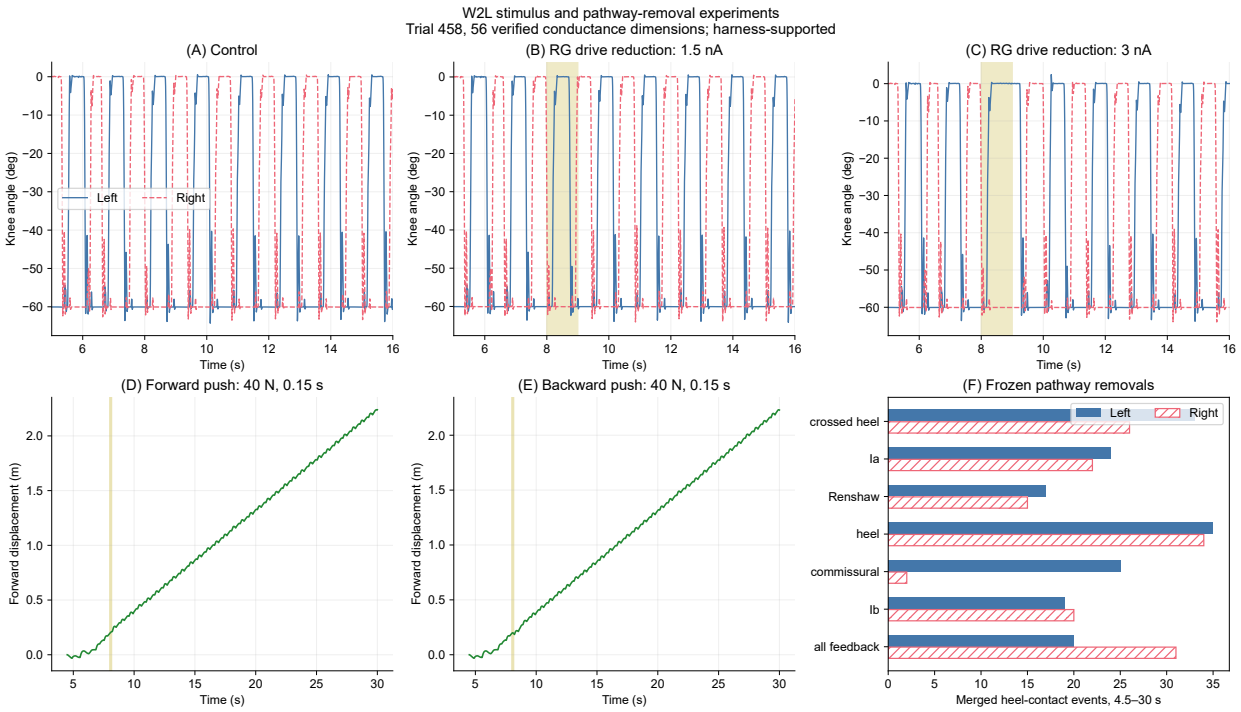


Figure D10: Recovered W2L controller responses to drive changes, pushes, and pathway deletions. The same effective controller gains are used across conditions, with the specified intervention applied in each panel.

D.5.3 Di Russo Controller Replication Attempt

The original Di Russo controller uses a reduced planar Delp/Thelen model with nine active coordinates and 18 Thelen muscles, rather than the full Gait2392 model. The source mechanical model loaded and its OpenSim system initialized successfully; this checks model construction, rather than gait. Its optimized parameter file contains 256 parameters, including individual reflex connections, muscle delays, and balance terms. Recovering that source clarified differences from the exploratory adaptation used here. Corrections to the adaptation addressed phase units, synchronous updates, output saturation, sensor ordering, contact-force direction, and initialization. Its compiled neural equations agreed with the corrected Python equations within 3.34×10^{-16} over 2000 steps, establishing agreement between those two implementations.

A 960-trial exploratory duration-and-speed search nevertheless selected a tumbling trajectory, with pelvis rotation reaching approximately 223°. That candidate was rejected as walking, and further optimization of the incomplete adaptation was stopped. The original controller's custom runtime was unavailable for a native baseline replay. The useful outcome is therefore the recovered source and the identified implementation requirements; replication of the published walking behavior remains incomplete. Source inventories, corrected-equation checks, and the rejected trajectory are retained in `Code/MuJoCo_SNS/spinal/reports_20261027_codex/`.

D.5.4 CAD-Derived Robot Integration

A separate CAD-derived robot model retained the four-bar knees and spherical hips, with 27 selected actuator routes per side. Its anatomical frame is x anterior, y superior, and z right; explicit frame transformations distinguish it from older converted models. The controller drives simulated reference tensions rather than calibrated BPA forces, so this model does not yet validate the complete actuator-to-controller integration proposed in Chapter 7.

The model completed a 20 s supported crouched-stance test. During the final 5 s, knee excursions were 1.35° right and 0.95° left, while the vertical rig carried 94.68 % of body weight and angular stabilization remained active. Disabling muscle forces under the same support settings caused lumbar failure after 0.202 s. This comparison establishes a contribution of the muscle controller under strong assistance, not independent balance. A separate run with rhythmic drive completed its horizon, but alternating gait and progression after settling have not been verified.

A subsequent matched 40 s comparison retained the same geometry, initial pose, reference-tension capacities, and external assistance in both conditions. During 15 s to 40 s, vertical assistance supplied 80% of body weight. The additional control combines muscle-mediated separation feedback (160 $\mathrm{N\,m^{-1}}$, capped at 20 N) with a 6% activation floor on currently abducting routes, compared with the 5% global floor (Table D7). Full recorded trajectories reproduce bit for bit between fresh reference and accelerated runs. This is numerical replay within the specified model, rather than a robustness test.

The 262-neuron controller receives 230 inputs and drives 54 reference-tension actuators.

Table D7: Matched CAD-model comparison over 15 s to 40 s. Vertical assistance remains 80% of body weight in both conditions.

Measure	Control off	Control on
Reversed ankle ordering (%)	67	0
Minimum ankle separation (mm)	−38.3	142.4
Median ankle separation (mm)	−6.2	159.0
Mean right TFL activation (%)	22.3	81.8
Mean left TFL activation (%)	99.6	99.8
Maximum sampled knee-loop error (mm)	4.485	0.387
Maximum sampled interleg penetration (mm)	2.700	0

Its MN target-current compensation cancels incoming PF and reflex synaptic currents to track the force allocator's targets. Consequently, the effective comparison concerns RG timing, engineered force allocation, and mechanical response, rather than normal PF/reflex authority over activation. Toe-out and small post-ramp knee excursions remain, and the stepping acceptance tests fail.

The foot attachment is an unverified assembly datum. A separate hypothesis translates each whole foot 60 mm toward the proximal shank, retaining tibial geometry and mass. With otherwise unchanged hybrid control, median ankle separation changes from 159.0 to 57.3 mm, and sampled interleg penetration reaches 1.316 mm. A separate W2L-organization timing probe has 544 neurons and 230 inputs; the knee and ankle pattern-formation outputs share four neurons, which are counted once in this total. Its shifted-foot condition passes bilateral unloading and lifting counts but fails repeated alternating foot apices and forward progression. The inherited MN compensation remains active, so this is not a full PF-driven controller comparison.

Route sites can lie outside CAD surfaces and transmit force through their named parent

bodies, equivalent to ideal rigid offset brackets or guides. Bracket mass, compliance, packaging, and collision-aware wrapping remain unmodeled. These attachment assumptions are separate from verification of the source moving-point functions and complete path derivatives. The older 20 s stance test and newer 80%-assistance comparison are different configurations, not a matched support-reduction experiment.

The W2L replay metrics and support accounting are archived in `Code/MuJoCo_SNS/spinal/reports_20261027_codex/verification.json` and the accompanying support-ladder records. CAD-model stance metrics are in `Code/MuJoCo_SNS/robot_torso/controller_assessment.json`. These records retain the configurations and evaluation conditions needed to distinguish the two models.

D.6 OpenSim-to-MuJoCo Conversion Repairs

The conversion work in Section 6.3 involved two distinct problems: importing the robot's artificial-muscle routing with human muscle parameters, and repairing the converted human reference used for controller development. The first combination failed muscle-equilibrium checks because the routed lengths were incompatible with the retained human parameters. The subsequent human-model repairs described below belong to that reference-model loading path; they are not evidence that the BPA-routed model reproduces every human muscle force.

Joint directions and coordinate conventions. Single-actuator tests exposed reversed hip-flexion and hip-adduction actions. The corresponding hinge-axis directions were negated so that the muscle actions agreed with the adopted OpenSim conventions. The knee

required a different treatment: its converted flexion-negative convention was retained. Rotating the knee without updating its coupled translational coordinates gives a misleading geometry comparison, so those coordinates must follow the prescribed coupling during a pose or actuator test. These are specific joint and path checks, rather than a change from a left-handed to a right-handed model.

Small muscles and force scales. The quadratus femoris, gemelli, and piriformis actuators were disabled on both sides. Their six entries remain in the 92-actuator model, with no force contribution. A separate check found eight muscle-actuator entries with an active maximum-force scale of only 1 N. Their active-force scales were restored from the source-model values. Table D8 lists the corrected values; each applies separately to the left and right actuator.

Table D8: Active maximum-force scales restored in the converted human model.

Muscle	Converted scale (N)	Restored scale (N)
Erector spinae	1	2500
Internal oblique	1	900
External oblique	1	900
Extensor hallucis longus	1	162

These are source-model maximum-force parameters, not measured BPA capacities or the realized force at every pose. The passive-force scales already contained the source values. The converted length ranges for these entries remain a limitation, so restoring the active scale alone does not establish full force–length agreement.

Patella-related paths and coupled joints. The converted moving muscle-path points use joints coupled to knee angle. Their sampled limits conflicted with the coupling constraints away from the sampled pose, producing large constraint forces and restricted motion. The coupled path-point joints were therefore left without those conflicting limits while retaining their prescribed trajectories. The anatomical driver joints were treated separately: limits were re-enabled on 14 hinges using the converted ranges. The rectus femoris route also lacked the patella tracking needed for knee-extension action; its intermediate path point was replaced by the vastus medialis patella-tracking point. The force comparison below quantifies the remaining differences at matched poses and activations.

Contact and virtual geometry. The converted model contained 19 explicit ground–body contact pairs, independent of geometry contact flags. Suspended tests remove these pairs; ground-contact tests retain only the six foot–ground pairs. Path-marker geometry is noncolliding. These settings isolate intended foot loading from unintended contact with other model components.

These checks verify specific joint actions, coupled paths, and intended contacts. They retain the rigid-tendon approximation and do not establish full OpenSim equivalence or walking; the matched-pose force comparison below quantifies the remaining discrepancies.

D.6.1 Cross-Platform Force Comparison

The comparison script reads OpenSim static-optimization forces and activations and assigns the corresponding poses and activations to the converted MuJoCo model. It calls the forward calculation at each pose; it does not assign the trajectory velocities or integrate

muscle activation and fiber dynamics through time. OpenSim static optimization likewise solves frame-by-frame force allocation rather than a compliant-tendon forward simulation [OpenSim Project, n.d.]. The comparison therefore tests force predictions along sampled poses at matched activation, not dynamic equivalence or tendon-energy storage.

Figure D11 shows representative force sequences normalized to each curve's peak. Table D9 retains the saved per-muscle statistics: $R^2 = 1 - \mathrm{SSE}/\mathrm{SST}$, squared Pearson correlation, and the best-aligning lag in analysis frames. The calculation and saved values are in `Code/MuJoCo_SNS/spinal/_muscle_force_compare.py` and `muscle_force_compare.csv`.

Table D9: MuJoCo vs. OpenSim muscle force along the subject01 walk at matched activations. Lag in analysis frames (≈ 17 ms each).

Muscle	R^2	shape corr^2	lag [frames]
soleus	0.742	0.800	1
med. gastrocnemius	0.888	0.894	2
lat. gastrocnemius	0.884	0.887	2
tibialis anterior	0.945	0.945	0
vastus lateralis	0.941	0.945	1
rectus femoris	−1.15	0.384	0
semimembranosus	0.565	0.621	0
biceps femoris short	−0.26	0.868	0
psoas	0.927	0.934	1
gluteus maximus (2)	0.926	0.943	1
tensor fasciae latae	0.933	0.935	0
peroneus brevis	0.977	0.989	0

Eight of the twelve representative muscles have R^2 between 0.87 and 0.98 and mean-force ratios within 10 %. Soleus and semimembranosus have lower agreement, and the negative R^2 values for rectus femoris and biceps femoris short head indicate errors larger

than predicting the corresponding mean reference force. Shape correlation alone does not remove those errors. The rectus femoris path was altered during conversion, while the biceps femoris short-head result has a substantial force-scale difference. These observations identify muscles requiring closer inspection; they do not isolate routing, force-law, and velocity effects from one another.

The frame-lag calculation describes alignment of these saved sequences. It cannot establish a physical delay caused by tendon compliance because the comparison does not simulate that mechanism. The results support selected pose-wise force checks, with the reported exceptions, rather than a general claim of equivalence for neural-control studies. A dynamic comparison would require matched velocities, activation histories, and explicitly specified tendon dynamics.

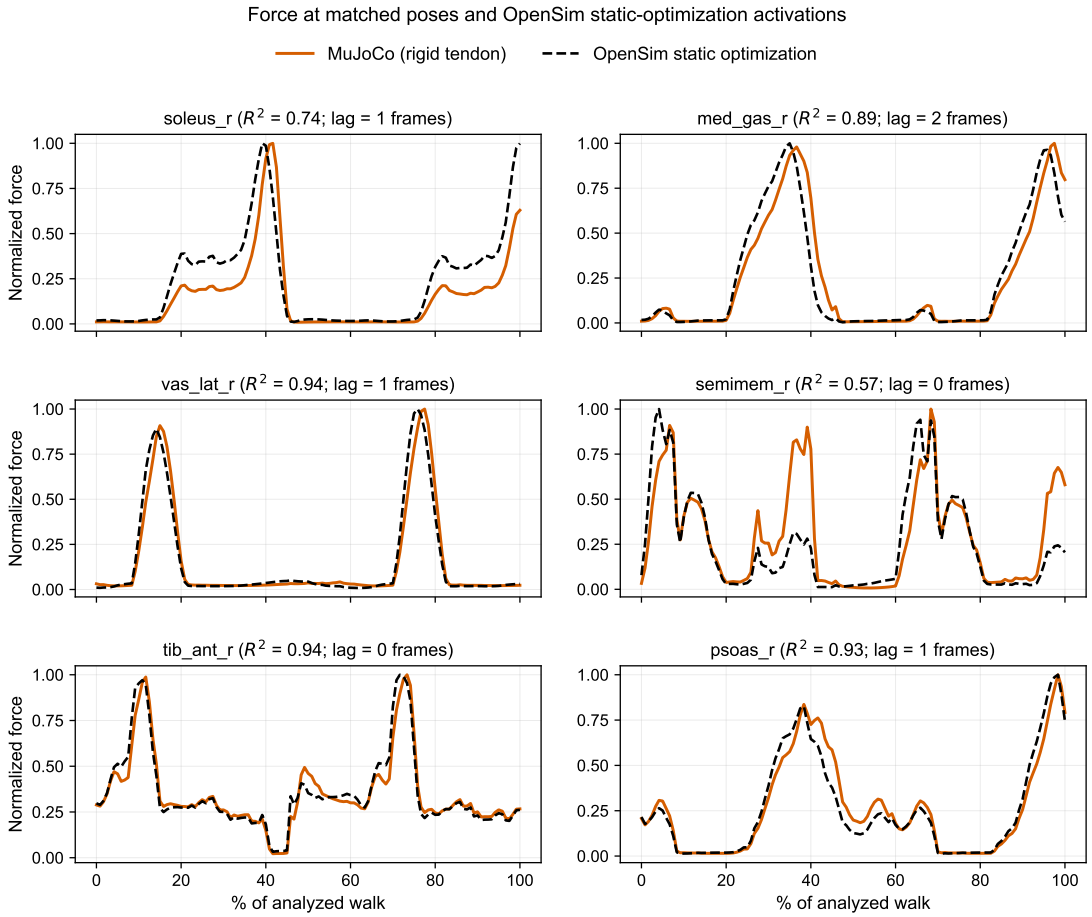


Figure D11: Force sequences at matched poses and activations, normalized to each curve's peak. Absolute-force agreement is reported in Table D9.

Appendix E: Supporting SADb Records

This appendix records the curation snapshot and pathway examples supporting Chapter 5. Counts refer to the stated audited or retained set; they are not estimates of the size of the neuroscience literature.

E.1 Record Sets and Coverage

The application contains 1,393 records from the five-table base: 943 Papers, 247 Models, 157 Review Papers, 42 Feedback pathways, and 4 Recorders. Of these, 436 are in the app but not in Airtable due to their 1000 entry limitation. The retained working base in Airtable contains 957 records: 561 Papers, 244 Models, 109 Review Papers, 39 Feedback pathways, and 4 Recorders. The application table includes all record types, while its pivot and citation views use the paper set.

Coverage of the application's Papers category is reported in Chapter 5. The following counts describe the retained maintenance subset and use a different denominator.

Among the 561 retained papers, 523 carry curation notes, 426 preparation tags, 240 notes assessing animal-study potential, 118 links to named feedback pathways, 86 an afferent classification, and 258 robot/simulation translation notes. There are 550 DOIs. These fields overlap and should not be summed as disjoint categories.

Chapter 5 reports the full citation-graph counts and shows the citation view (Figure 5.1); Figure 5.3 shows a filtered record and its details. Citation proximity is a navigation aid,

not mechanistic evidence.

E.2 Afferent Classes and Linked Pathways

Table E1 summarizes the afferent classes, receptors, and signals available as candidate model inputs. Table E2 connects representative feedback pathways to their reported actions in the step cycle and the linked studies. Their record names are database labels; pathway action depends on the experimental preparation and locomotor state described in the cited sources. Translating these records into a controller requires an explicit encoding and verification step.

Table E1: Afferent classes, encoded signals, and pathway roles represented in the Sensory Afferent Database. References link to corpus records; Ia, II, and Ib signal summaries follow Prochazka and Gorassini [1998].

Class	**Receptor**	**Signal carried**	**Recorded pathway roles (anchors)**
Ia	muscle spindle primary	muscle length and velocity	homonymous stretch excitation [Eccles and Lundberg, 1958]; reciprocal inhibition of the antagonist pool [Jankowska et al., 1967]
II	muscle spindle secondary	static muscle length	midlumbar disynaptic relay onto motor nuclei [Edgley and Jankowska, 1987]; stance-phase support of extensor activity [Mazzaro et al., 2006]
Ib	Golgi tendon organ	whole-muscle force	autogenic inhibition at rest, stance-phase reversal to excitation, and the load gate on swing onset [Dietz and Duysens, 2000, Domínguez-Rodríguez et al., 2020]
III/IV	free nerve endings	metabolic and nociceptive signals, converging with cutaneous and group II–III afferents as flexor-reflex afferents	phase-dependent resetting of the locomotor rhythm [Schomburg et al., 1998]
Cutaneous	skin and foot-pad receptors	contact and perturbation events	flexor excitation and phase-modulated stance correction [Sherrington, 1910, Forssberg, 1979]

Table E2: Principal feedback pathways, named by afferent, action, and phase, with linked demonstrating papers. Counts are identical in the audited 943-paper corpus and retained 561-paper corpus.

Pathway record	Class	Recorded action in the step cycle	Linked papers	Demonstrating references
Ia monosynaptic excitation	Ia	stretch excitation of the homonymous motor pool, with the pathway gain modulated over the step cycle	7	[Eccles and Lundberg, 1958, Capaday and Stein, 1986]
Ia reciprocal inhibition	Ia	inhibition of the antagonist motor pool through the Ia inhibitory interneurons, rhythmically active in fictive locomotion	6	[Jankowska et al., 1967, Pratt and Jordan, 1987]
Ib disynaptic inhibition	Ib	autogenic non-reciprocal group I inhibition of the loaded muscle's own pool, reduced during stance	5	[Jankowska et al., 1981, Stephens and Yang, 1996]
Ib disynaptic excitation	Ib	the locomotor reversal: group I input excites synergist extensor pools during the extensor phase, replacing the resting inhibition	6	[McCrea et al., 1995, Pearson et al., 1998]
Ib stance to swing	Ib	the load gate: Ib input from ankle extensors prolongs stance, so swing is not initiated while the leg is loaded	17	[Dietz and Duysens, 2000, Domínguez-Rodríguez et al., 2020]
Ia or II stance to swing	Ia, II	flexor-length gate on stance termination: lengthening of flexor muscles in late stance ends extension and resets the cycle	10	[Hiebert et al., 1996, Akay et al., 2014]
type II excitatory	II	stance-phase support of ankle-extensor activity through the midlumbar group II relay	5	[Edgley and Jankowska, 1987, Mazzaro et al., 2006]
Cutaneous flexor excitation	Cutaneous	flexion withdrawal and the stumbling corrective reaction, phase-modulated over the cycle	16	[Sherrington, 1910, Forssberg, 1979]